

The background is a dark blue gradient with a complex network of glowing yellow and white lines and nodes, resembling a digital or neural network. Two stylized human figures, composed of geometric shapes, are positioned on the left and right sides, appearing to interact with the network.

Benedikt Perak

Communication in the Age of Artificial Intelligence

The development of large language models
and communication agents

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Faculty of Humanities and Social Sciences, University of Rijeka
Rijeka, 2025.

PUBLISHER

University of Rijeka, Faculty of Humanities and Social Sciences
Sveučilišna avenija 4, 51000 Rijeka, Croatia

FOR THE PUBLISHER

Prof. Barbara Kalebić Maglica, PhD, Dean

REVIEWERS

Prof. Ana Meštrović, PhD

Assoc. Prof. Mirjana Borucinsky, PhD

COPY-EDITING (CROATIAN EDITION)

Prof. Mihaela Matešić, PhD, University of Rijeka, Faculty of Humanities and Social Sciences

ENGLISH LANGUAGE EDITION

This English version is a full translation of the Croatian open-access edition. Diagram labels have been translated for this edition. The same electronic ISBN applies to the underlying work.

DESIGN AND LAYOUT

Syntagent d.o.o., Rijeka

ISBN (electronic edition): 978-953-361-147-1

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HOW TO CITE

Perak, Benedikt (2025). *Communication in the Age of Artificial Intelligence: The development of large language models and communication agents*. Rijeka: Faculty of Humanities and Social Sciences, University of Rijeka.

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CroRIS profile: www.croris.hr/osobe/profil/29998

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Foreword

This book continues reflection shaped through the courses *Language, Thought, and Culture*, *Data Science in Cultural Studies*, and *Artificial Intelligence and Communication in Culture* at the Faculty of Humanities and Social Sciences in Rijeka. It aims to contribute to understanding changes in communicative practices in the age of artificial intelligence, drawing on relevant literature and a critical stance. It is intended for students of communication, linguistics, cultural studies and related fields, as well as for readers who wish to follow systematically how we communicate with systems whose linguistic capabilities grow ever more sophisticated.

Although the book as a medium no longer occupies the central place it once did, this project adopts a form that is systematically renewed — an **agentically renewable book**. The first edition will gradually be extended with agent-assisted information retrieval, integration of new *state-of-the-art* technological solutions, and critical discussion of their implications for communicative and cultural phenomena. That process remains under the author's responsibility and control.

In that spirit, new versions of this renewable book will be published periodically on [GitHub](#), as a concrete embodiment of its core idea: we are on the threshold of collaborative work with artificial intelligence agents.

I thank colleagues who encouraged me and made disciplined research experimentation possible, from which the insights in this book emerged. I owe special gratitude to my family and friends for their steady support and for the sense of meaningfulness they provide.

Benedikt Perak

Rijeka, 2025.

1 Introduction: Communication and the development of civilization

From the ancient, barely distinguishable voices with which the first hominids communicated, all the way to the complex linguistic tissues that permeate today's digital spaces, communication appears as a fundamental force on which the very survival and development of human communities depends. It is the connective tissue that unites individuals in society, a pledge of the survival of culture and the driver of every innovation. Without the possibility of transferring thoughts, experience and knowledge from generation to generation, human survival would remain bound within the limits of instinct, and the concept of culture would be unimaginable.

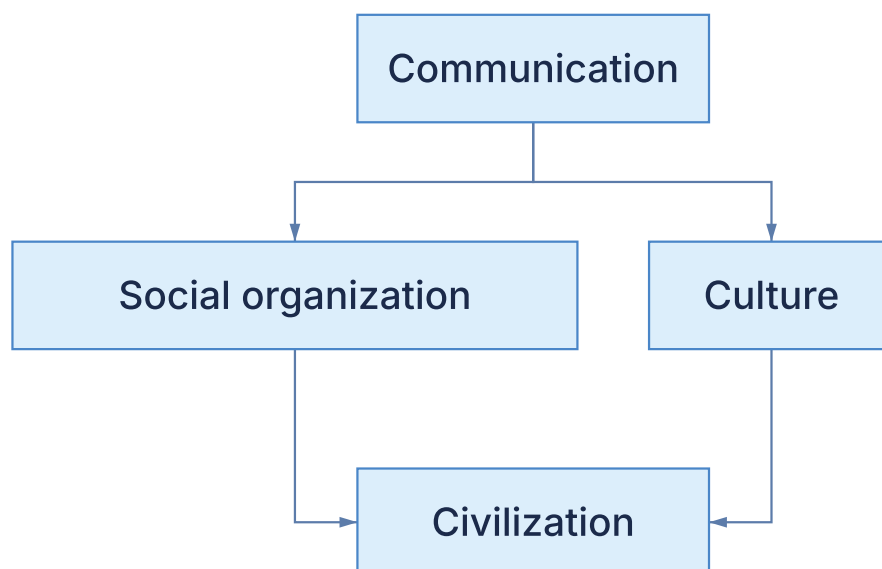


Figure 1.1: Communication and development of civilization.

The fundamental shift from simple signaling systems, such as we find in the animal world, to the complexity of human language is the result of a long-term evolutionary process. This heritage, made up of a series of cognitive and social adaptations, created the preconditions for the further development of communication and laid the foundations for the entire human culture. It is not manifested in a single ability, but in a complex interweaving of several factors that encouraged and shaped each other.

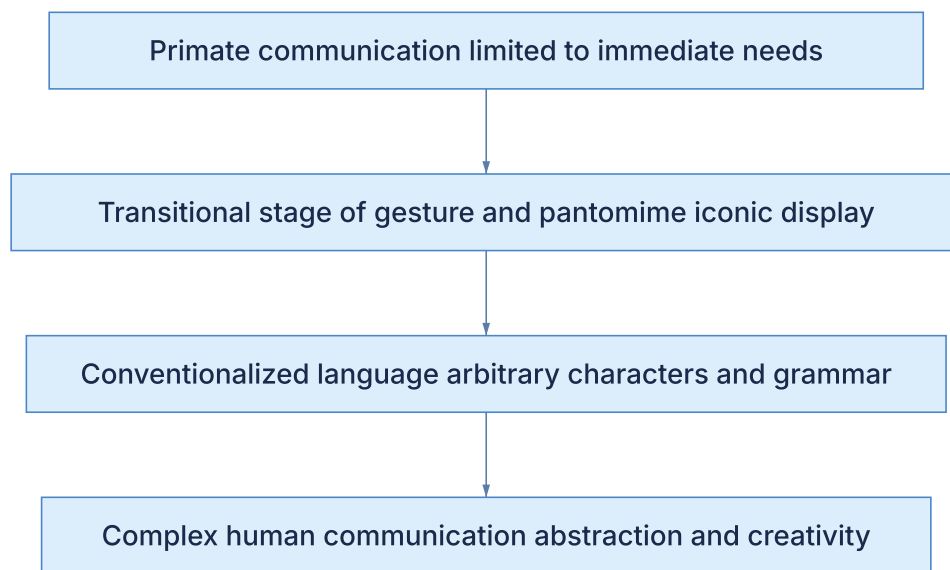


Figure 1.2: Evolution of human communication – from primates to conventionalized language.

A key moment in this development is the transition from predominantly indexical signs, firmly tied to immediate reality, to the use of symbols. While indexical signs point to something present and correlated (for example, smoke to fire), **symbols** derive their value from agreements within the community, thereby enabling the discussion of objects, beings and phenomena that are not spatially or temporally present, as well as abstract concepts.

Index sign: A sign whose meaning derives from a causal or correlational connection with an object (e.g. smoke refers to fire); refers to the immediately present and perceived, in contrast to the symbols that arise from convention and allow speaking about the absent and abstract. **Symbol:** Linguistic or other sign whose meaning does not derive from physical similarity or causal connection with the object, but from social agreement (convention); allows speaking about the absent and abstract, in contrast to the indexical sign that refers to the immediately present.

However, such a transition is only possible with profound changes in the social structure. The development of collaborative behavior, as Tomasello (2008) points out, created an environment in which information sharing became evolutionarily beneficial. In communities where individuals with the characteristics of independent action, **agents**, depended on each other for survival, the ability to communicate precisely and reliably brought a direct advantage.

Agent: An individual or system that acts in an environment with specific goals; in the context of communication – a participant who can send and receive messages and coordinate their actions with others.

At the core of this collaborative paradigm lies the concept of **shared intentionality**, i.e. the ability of agents to shape common goals with others and coordinate their actions to achieve them. This cognitive ability enabled joint hunting or defense, but also the understanding that the act of communication is a joint effort of sender and receiver to establish a common reference point. It is inextricably linked with the development of the theory of mind, the ability to attribute mental states - beliefs, desires and intentions - to others, as well as to oneself. Without a theory of mind, the recipient of the message would not be able to understand the speaker's intention, and complex symbolic communication would be reduced to misunderstanding.

Joint intentionality: The ability to form common goals with others and coordinate actions to achieve them; the basis of collaborative communication.

Theory of mind: Cognitive ability to attribute mental states – beliefs, desires, intentions and knowledge – to others and to oneself; key to understanding the speaker’s intention in communication and predicting other people’s behavior. Without theory of mind, complex symbolic communication would not be possible because the recipient of the message would not be able to deduce what the speaker wants to achieve with it.

Nevertheless, the development of communication systems based on trust has inevitably opened up space for manipulation and deception. As soon as communication became a means of influencing the behavior of others, the possibility of its abuse for one’s own benefit also appeared. This prompted a sort of *evolutionary arms race* (Krebs & Dawkins, 1984) between the ability to deceive and the ability to detect deception. On the one hand, agents developed increasingly sophisticated deception strategies, while, on the other hand, they perfected cognitive mechanisms for recognizing inconsistencies, assessing the reliability of sources, and monitoring the reputation of individuals within the community. This constant tension between cooperation and conflict further encouraged the development of cognitive complexity.

With the increase in the complexity of the messages that had to be conveyed - who did what, what, to whom, when and where - the need for their organization also appeared. It is from this need that **grammar** grows: a system of rules that enables the combination of a limited number of symbols (words) into an unlimited number of meaningful statements. The development of syntax enabled the precise coding of relationships between participants and actions, which made communication more expressive and unequivocal.

Grammar: A system of rules that allows combining a limited number of language units (words, morphemes) into an unlimited number of meaningful statements; it encompasses syntax, morphology, and related conventions that determine the structure and interpretation of language.

Ultimately, the legacy for the further development of communication is a complex set that includes a social structure based on cooperation and trust, cognitive abilities such as shared intentionality and theory of mind, and a dynamic balance between deception and its detection, which all together created evolutionary pressure for the development of a grammatically organized symbolic system. It is a dynamic foundation on which increasingly complex forms of human interaction, knowledge and culture are built even today.

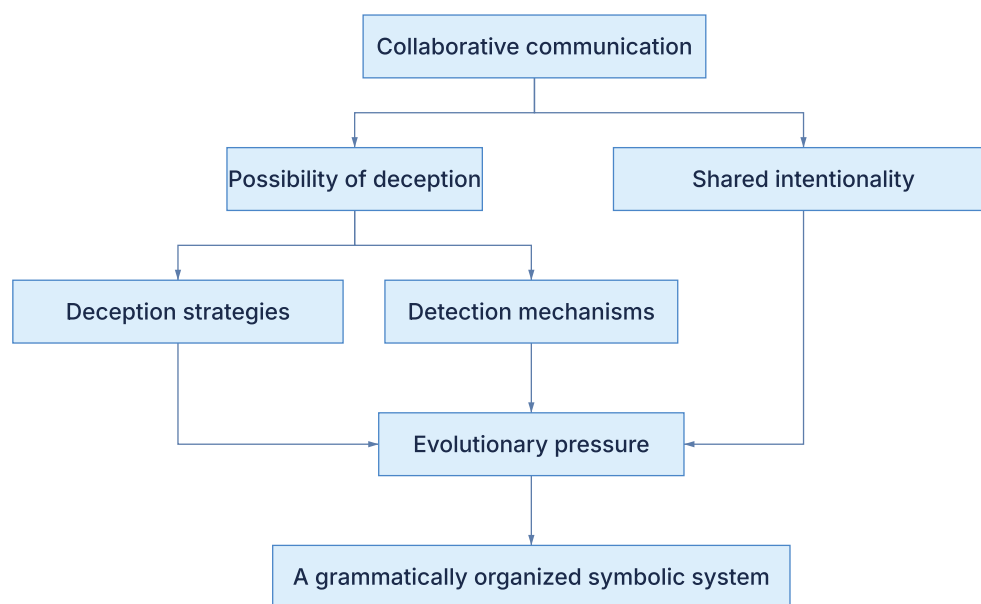


Figure 1.3: The interconnectedness of social, cognitive, and communication processes—collaborative communication and theory of mind drive development, while the evolutionary race between deception and its detection creates pressure for cognitive complexity and a grammatically structured symbolic system.

Oral tradition, the first and longest-lived communication technology, transformed fleeting memories into living archives. She wove them into the fabric of epics, myths and rituals, transmitting the worldview and identity of the community through the ages with a permanence almost like that of the genetic code. With the invention of writing, words gained material permanence, imprinted in clay, engraved in stone or written on papyrus and paper. Writing enabled an unimaginable accumulation of knowledge, the development of abstract thought, and the construction of complex

administrative, legal, and philosophical systems that defined great empires and epochs. Each subsequent technological milestone—from Gutenberg’s printing press that democratized knowledge, to the telegraph and telephone that bridged spatial distances, to radio, television, and finally the Internet that created the contours of a global community—opened a new chapter in the constant evolution of how people exchange meaning with each other.

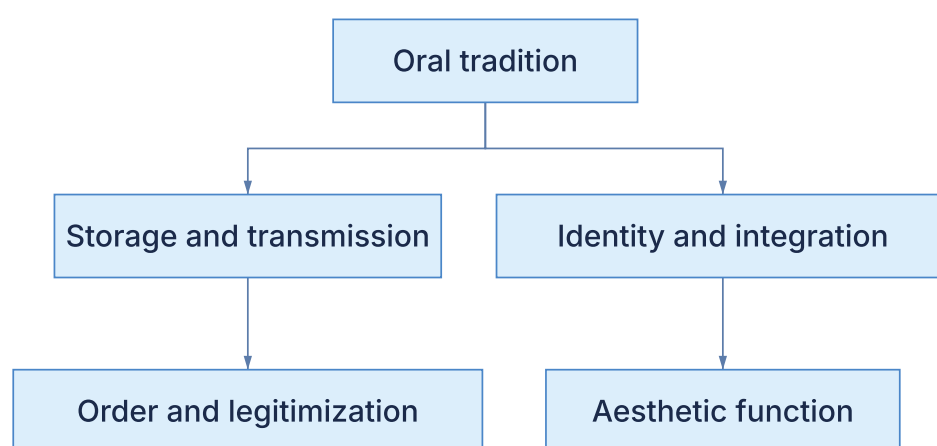


Figure 1.4: Functions of oral transmission.

The modern age is witnessing a technological breakthrough that, with its depth and reach, surpasses many previous ones. It is about the emergence of large language models (*Large Language Model*, LLM), advanced artificial intelligence systems such as GPT (OpenAI, 2024b), Claude (Anthropic, 2024), Gemini or Llama models. Built on inconceivably huge corpora of digitized human language and knowledge, these systems are trained to recognize and reproduce the most subtle patterns of human communication. They reveal the fascinating duality elegantly summarized by academic Milena Žic Fuchs (Žic-Fuchs, 1991), distinguishing knowledge about the world (facts, concepts, relationships) and knowledge about language (grammar, stylistics, pragmatics). Today’s great language models master the language code with exceptional skill, creating texts that are grammatically coherent, stylistically

adapted, and often factually accurate and contextually appropriate. They mimic human language production with incredible persuasiveness, often reaching, and in some narrowly defined tasks even surpassing, human performance.

LLM – Large Language Model: An artificial intelligence system trained on huge text corpora that predicts the next token (word or part of a word) and generates or completes the text based on it; examples include GPT, Claude, Gemini, Llama.

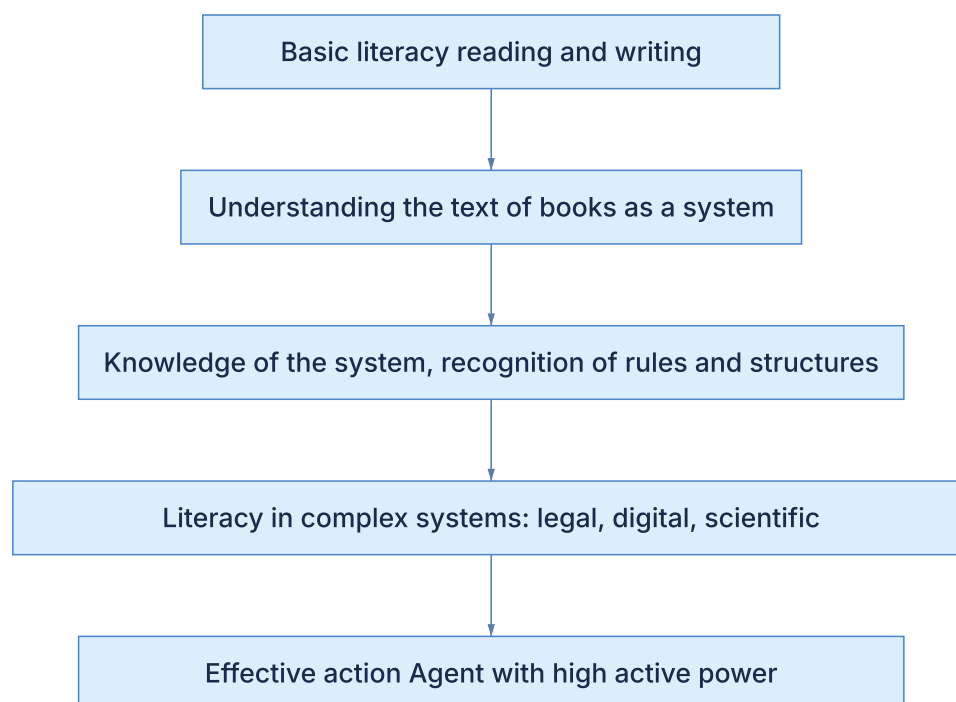


Figure 1.5: Development of active power through literacy.

But large language models cease to be mere text generators the moment they are transformed into **communication** agents by incorporating them into broader information systems and equipping them with goals, data access tools, and memory mechanisms. Thus, they become our new digital interlocutors, assistants and collaborators - and perhaps true communication partners. Their impact is already evident in the reshaping of the landscape of human interaction: they enable real-time language translation, run service automatons (*chatbots*) that provide personalized customer support, help scientists analyze

complex data sets, assist programmers in writing code, stimulate creative processes, and open new vistas of global collaboration and education. They are increasingly assuming the role of universal translators and mediators of knowledge, capable of bridging linguistic, disciplinary and cultural barriers.

Communication agent: A system based on LLM (or similar language technology) equipped with goals, data access tools, and memory mechanisms, which dialogues with the user and performs tasks; it goes beyond the role of a mere text generator and becomes a participant in the interaction.

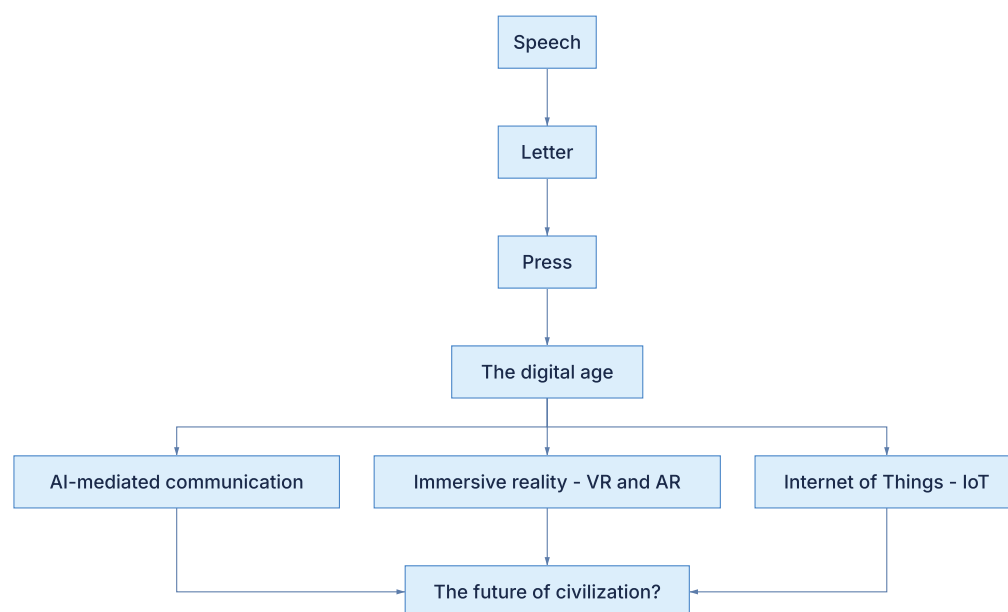


Figure 1.6: Historical development of communication to AI-mediated communication.

The central guiding thought of this book is the assertion that large language models, and the communication agents they trigger, do not represent just a new technological tool, a gadget (*gadget*), but a fundamental transformation of the very nature of communication. Their potential penetrates deeply into the structure of social relations, human cognition and the very essence of what it means to be a human being who communicates in the digital age. Precisely because they carry with them unimagined possibilities, it is of crucial importance to critically penetrate into their inherent capabilities and limitations. It is equally important to shape their development and application ethically, in order to serve the improvement of human dignity and the general

betterment. The book is intended as a systematic review of communication technologies, a journey that begins in the deep past and ends with a reasoned look into the possible future of human-machine interaction.

Our journey begins with a detailed overview of the history of communication technologies (**Chapter 2**), from its origins in oral tradition to the digital revolution. Understanding the ways in which previous technologies - the letter, the press, the telegraph, the radio - shaped social structures and human cognition, provides us with the necessary framework for assessing the depth and direction of the changes brought about by the current artificial intelligence revolution.

After the historical foundations, we dive into the very heart of the technology that powers today's communication agents: the world of large language models (**Chapter 3**). We will reveal their anatomy, with a special emphasis on the transformer architecture, and we will follow the entire life cycle of the language model. Each of these stages – from collecting unimaginably large sets of text and image data to pre-training on powerful computer clusters, through fine-tuning for specific tasks and the crucial process of matching with human feedback, all the way to rigorous performance evaluation – will be explained in detail. We will also not ignore the darker side of this technology: its significant limitations, risks of built-in biases, security issues and the enormous consumption of resources required by these digital giants.

With technological understanding as a basis, we enter into a deeper, conceptual breakdown of the relationship between language, society and artificial intelligence (**Chapter 4**). Here we will deconstruct language as a system of meaning and as a social practice, exploring the ways in which large language models, as new participants in communication, encroach on these processes. We will look at how these systems simulate social interactions, influence our sense of identity and potentially reshape our perception of reality, acting as powerful algorithmic lenses. We will pay special attention to the role of language in the collective creation of knowledge and culture and the new role of communication agents in these processes - from collaboration in scientific research to co-creation in art.

Since it is difficult to talk about artificial intelligence without understanding its material basis, we will explore (**Chapter 5**) the inextricable link between the exponential growth of processing power and the progress of artificial intelligence, drawing comparisons with the evolution of the human brain. We will consider how this power enables today's models and what horizons it opens for future, even more powerful systems, including potential multi-agent systems and agent swarms and the long-term quest for artificial general intelligence (AGI).

AGI: General Artificial Intelligence (*Artificial General Intelligence*) – a hypothetical system that would possess a human level of general intelligence and could solve a wide range of tasks without being limited to one domain; for now the goal is research, not existing technology.

After the theoretical and technological foundations, we move on to the practical construction of communication partners (**Chapter 6**). This chapter details the journey from the core core of a large language model to a functional and useful AI agent. We will explore key techniques such as prompt engineering (*Prompt Engineering*), i.e. the art of precise model management, then enriched data retrieval (*Retrieval-Augmented Generation*, RAG) which enables agents to access external, up-to-date knowledge, and mechanisms for using tools (*Tool Use*) which give them the ability to act in the digital world.

How do these agents work in practice? **Chapter 7** provides concrete examples and case studies, illustrating how they are used to transform customer support, personalize educational processes, improve marketing strategies, and find applications in healthcare, business processes, and creative industries. We will analyze the ways in which they create new value, but also the new challenges they bring with them.

Finally, in the last chapter (**Chapter 8**), we will venture a speculative but grounded look into the near future (period 2030-2035). Through a narrative sketch of a day in the life and an analysis of key trends such as ambient intelligence, hyperpersonalization and the synergy of agent swarms, we will try to imagine what everyday life permeated by ubiquitous, proactive and

autonomous digital companions might look like. We will consider the potential for true human-AI symbiosis, but also the deep social and ethical dilemmas such a future holds, balancing between utopian promises and dystopian risks.

Several central thoughts run through the entire book: the evolution of communication as a continuous but accelerated process; large language models as extremely powerful, but also essentially limited technology; communication agents as entities that go beyond the role of a mere tool and become participants in the interaction; and, finally, the permanent need for critical consideration of ethical consequences and responsible shaping of the present reality and the future. Our journey aims to provide the reader with knowledge of how these systems work, as well as a deeper understanding of why they are important and where they might take us.

Throughout history, communication has been the key mechanism by which we have built bridges between minds, created shared worlds, and shaped the institutions and cultures that define our civilization. Today, at this historic crossroads, large language models and the agents they drive offer us the opportunity – but also the responsibility – to make that mechanism exponentially faster, more globally accessible, and potentially more intelligent than ever before. This book is an invitation to explore that intersection: the place where the thousand-year history of human communication meets the most advanced technology of the 21st century, technology that reshapes our tools, but also asks anew the fundamental question: what does it mean to be a human being who creates his own world through communication?

2 History and evolution of communication technologies

2.1 Communication as the basis and material of civilization

Consideration of the essence of civilization inevitably leads us to its fundamental beginning and material: communication. Civilization can be understood as a dynamic and continuous process of social life, which is entirely enabled, shaped and maintained by communication. Man's nature as a social being dictates the need for the exchange of thoughts, feelings and experiences, and this exchange, mediated by complex symbolic systems, becomes the foundation on which increasingly complex social structures are built.

Communication is therefore a process by which a shared world of meaning is created and maintained. It is the ability to create and transmit abstract ideas and shared narratives – myths, laws, values and history – that enables the cooperation of a large number of individuals and thus lays the cornerstone of any civilization (Harari, 2011). Without this ability to transcend the immediate, experiential world, human communities would remain limited to small, kinship-related groups. Through language, writing, art and other symbolic forms, thought is materialized; it becomes external and accessible to others, regardless of spatial and temporal distance.

This process of materializing thoughts through communication gives birth to what we call *artifacts*. Artifacts in this sense do not only refer to material objects such as tools, buildings or works of art, but also include immaterial creations: laws, social norms, scientific theories, philosophical systems and religious doctrines. Each such artifact is an embodiment of knowledge and experience, a frozen communication waiting to be unraveled and reactivated in

the minds of new generations. In this way culture acquires its cumulative nature; each generation does not start anew, but interacts with the inherited information environment shaped by its predecessors.

Artifact: In the context of communication and culture – a material or immaterial creation (law, norm, theory, work of art, technology) in which knowledge and experience are preserved; it mediates between agents across time and space and enables the cumulability of culture. **Cumulative culture:** The property of human culture that knowledge, skills and institutions are gradually upgraded from generation to generation, because through communication and artifacts, previous achievements are preserved on which each new generation can build on its own innovations; a mechanism that makes cultural evolution progressive rather than cyclical (Tomasello, 1999; Henrich, 2016).

Civilization can therefore be understood as a complex and inexhaustible interaction network consisting of *agents* (human beings) and *artifacts*. Within that network, communication is the driving force behind all interactions. Agents communicate with each other to coordinate their actions, but they also communicate with artifacts - they read books, interpret laws, use technology, apply scientific knowledge. Artifacts, on the other hand, mediate communication between agents, often across vast temporal and spatial ranges, and thus shape their thinking and actions (Floridi, 2014).

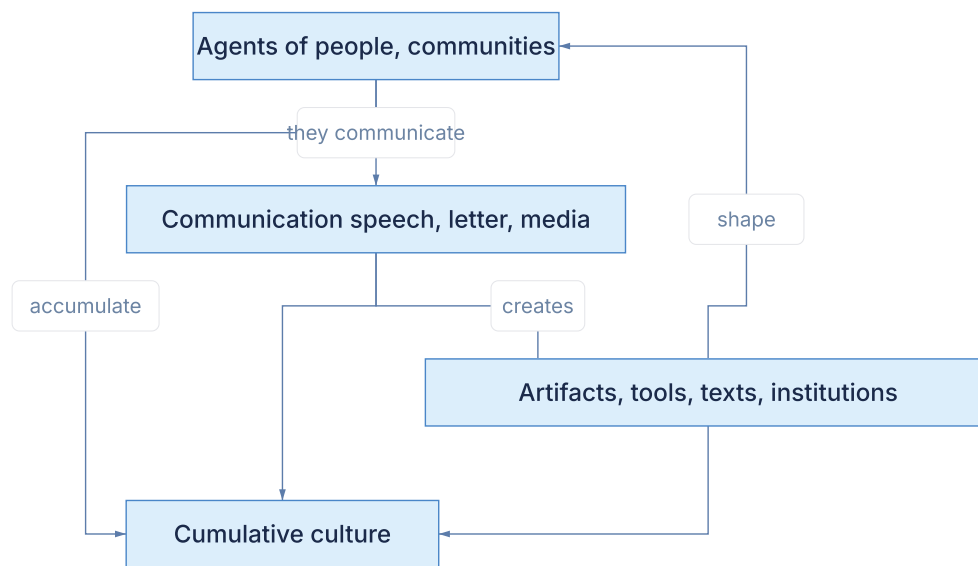


Figure 2.1: Schematic representation of the relationship between agents, communication and artifacts as the basis of cumulative culture.

Communication is the material and fabric of civilization itself. It is the process by which abstract ideas can be transformed into permanent artifacts, by which knowledge is transmitted and multiplied, and by which individual agents are connected into a complex, self-sustaining whole that we call civilization. Observing the history of civilization means following the traces of this continuous communication - from the first oral traditions and paintings in caves to today's global digital networks. In every form, communication remains a fundamental prerequisite and an inexhaustible source of human creativity.

2.2 Oral tradition and living archives

The term **oral** tradition encompasses a complex process that cannot be reduced to mere memory or simple storage of information about the past. On the contrary, it is an active act of transmitting, shaping and constantly reviving cultural knowledge.

Oral transmission: Way of transmitting knowledge, narrative and culture through speech and memory, without writing; knowledge lives in carriers (living archives) and changes in every performance; basic communication technology before the advent of writing. Cultures based on orality have a fundamentally different attitude towards the past than those based on writing; in them, the past is not a fixed and unchanging whole, but a fluid and adaptable reality that is reaffirmed in every act of storytelling (Ong, 1982). This reality is not preserved in dead letters, but in living people.

The bearers of such tradition can rightly be called **living** archives. However, this name does not imply that they are only passive containers of information, similar to shelves in a library.

Living archive: Bearer of oral tradition who actively interprets, adapts and transmits cultural knowledge; knowledge is embodied in voice, gesture and memory, not in written record; each performance is unique. Their role is active and creative. A living archive interprets, adapts and transmits inherited knowledge, imbuing it with its own experience and understanding of the world. Each of his performances, whether it is a song, a story or a ritual formula, is a unique and unrepeatable act of creation, and not a mere reproduction of an existing template (Lord, 2000).

In this process, the **agent**, that is, the bearer and performer of the tradition, is a key figure who actively shapes the narrative. His role goes beyond the function of a transmitter because through his performance he confirms, and sometimes reshapes, the values and knowledge of the community. The knowledge that the living archive possesses is not abstract, but deeply embodied. It resides in his or her voice, rhythm of speech, gesticulation and memory of the body, making the whole of the performer and the performance inseparable.

The difference between oral and written archives is therefore fundamental. By its very nature, the written archive tends towards staticity, immutability and apparent objectivity, separating knowledge from its human source. On the other hand, the oral archive is inextricably linked to the person of the performer and the social context. It is dynamic, changeable and necessarily

imbued with subjectivity, because its meaning is constituted precisely in the living relationship between the narrator, the listener and the moment of performance (Goody & Watt, 1968).

This gives the oral tradition an exceptional social function. It is a cohesive force that creates and sustains a community, providing it with a sense of continuity and belonging through a shared repertoire of stories, beliefs and memories. This is why the content of the delivery is never more important than the context of its performance. Circumstances about the identity of the speaker, to whom he addresses, when and on what occasion he speaks, are an integral part of the message being transmitted. In modern times, we are faced with the paradox of the technological record of oral tradition. While recording enables its preservation in the face of the threat of oblivion, it simultaneously fixes and deadens what is essentially alive, fluid and performative. Such a record, although necessary for the preservation of the artifacts themselves, is only a faint reflection of the completeness of the event of oral creation.

2.2.1 The complexity of the oral tradition

The approach to oral tradition is often burdened by a simplified view that reduces it to a mere transmission of unchanging contents from generation to generation, almost as a kind of mechanical memorization of a static repertoire of stories or songs. However, such an understanding ignores its inherent layering and dynamism of living and complex systems of communication, knowledge and art, the nature of which cannot be exhausted by one-sided explanations. Oral traditions represent complete cognitive and social worlds, woven into the very fabric of the community that creates, transmits and constantly reshapes them.

The central element of this complexity is the fact that the oral tradition is not realized in a fixed, written text but in the *performance event*. Each performance represents a unique act of creation in which the agent, that is, the performer (singer, storyteller), does not mechanically reproduce a memorized template, but actively composes it at the moment of presentation (Lord, 1960). In doing so, he makes use of the rich heritage of traditional tools: established formulas, metric patterns, typical scenes and thematic sets. These elements

function as building blocks that enable improvisation within the given stylistic and content frameworks. The agent is a co-creator who adapts the tradition to the specific context, audience and own inspiration.

Furthermore, the complexity of the oral tradition is manifested in its *multimodality*. It is rarely reduced exclusively to the verbal component. Sound, rhythm, melody, gestures, facial expressions and the entire physicality of the performer are inextricably intertwined with the spoken or sung word. The meaning is conveyed by the semantics of the sentences, but also by the way they are delivered - by intonation, pauses, accents and emotional charge. The role of the audience is not passive; with her reactions, incentives or silence, she directly participates in shaping the performance, creating a dynamic feedback loop with the performer. Performance is therefore always a dialogic and social act, deeply conditioned by the situation in which it occurs.

Multimodality: Simultaneous use and interweaving of multiple communication channels – speech, gesture, mimicry, intonation, rhythm, music and spatial relationship – in order to convey meaning; in the oral tradition it means the inseparability of the verbal and physical dimensions of performance, while in modern digital communication it includes text, image, sound and video as integrated components of the message.

Finally, a fundamental feature of oral tradition is the creative tension between permanence and changeability. In order to survive, a tradition must possess a sufficient measure of stability that ensures its recognition and continuity through generations. At the same time, in order to remain alive and relevant, it must constantly adapt and change with each new performance. This apparent paradox is resolved in the understanding that traditionality does not lie in the immutability of words, but in the deeper structures and meaning potentials they evoke. Each formula or theme carries with it a wealth of associations and references to the entire tradition, creating what John Miles Foley (1991) calls *traditional referentiality*. Words in oral performance have a far greater semantic charge because they echo the unspoken but present context of the entire traditional knowledge. Because of all of the above, the oral tradition should be viewed as an intricate ecosystem in which the cognitive processes of memory and creation, the social dynamics of performance, and the aesthetic

values of language, music, and movement merge into a unique and unrepeatable human experience.

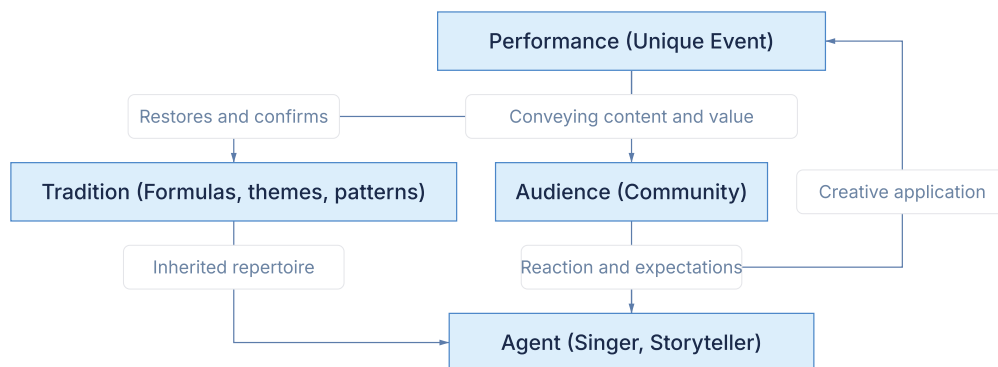


Figure 2.2: Dynamic system of oral tradition – tradition, performer, performance and audience in the cycle of transmission and renewal of knowledge.

2.2.2 Collective imagination and identity

The foundation of human culture and, consequently, collective identity rests on the unique cognitive ability of humans for *common intentionality* – the ability to create and share complex social realities that exist exclusively in the domain of collective imagination. It is this ability, as pointed out by researchers such as Tomasello (2014) and Harari (2014), that distinguishes human culture from the rudimentary forms of social learning present in other primates. While animal communities are based on direct interaction and genetically determined behaviors, human societies are built on shared beliefs in concepts such as nations, money, laws, or human rights, which have no tangible existence outside of shared agreement and imagination. The most obvious and powerful example of such an imagined reality is the nation, which Benedict Anderson (1983) precisely defined as an *imagined community*. Members of one nation will never know each other, but in their minds there is a vivid idea of community. This sense of belonging does not come from direct contact, but is nourished and maintained through shared symbols such as flags and anthems, collective rituals such as national holidays, and narratives about shared origins, history and destiny. Through these mechanisms, a deep emotional bond is created that connects millions of strangers into a coherent whole. In the

modern world, the reach and forms of the collective imagination have been significantly expanded by the action of the global media. In his analysis of global cultural flows, Arjun Appadurai (1996) introduces the concepts of *mediascapes* and *ideoscapes*. These landscapes, made up of images, narratives and ideas that circulate through the media, enable the creation of identities that transcend traditional national boundaries. Individuals today can develop a sense of belonging to global subcultures, social movements or diasporic communities, thus confirming that the collective imagination is not limited exclusively to the territory of the nation-state. However, it is important to emphasize that the collective imagination also has a distinct material dimension. Imagined realities are embodied and made permanent through monuments, architecture, flags, legal documents and social institutions. These material objects and structures act as *anchors* of collective identity, giving abstract ideas a concrete and visible form (Latour, 2005). The monument to the unknown hero is not only a stone but also a materialized narrative of sacrifice and national pride; the constitution is not only a piece of paper but also a tangible promise of common rights and order. At the same time, the field of collective imagination is dynamic and fraught with conflicts. Different social groups are constantly competing to impose their own vision of reality and define dominant narratives about who “we” are. Identity is therefore not a static and predetermined category, but, as Judith Butler (Butler and Trouble 1990) shows in the context of gender, is the result of a continuous process of negotiation, contestation and performative repetition. History is filled with struggles over the interpretation of the past, the meaning of national symbols or the definition of community boundaries, which testifies to the fact that collective identity is always an open arena of power. In this complex power relationship, the individual agent actively participates in the process of creating meaning - he interprets, accepts, adapts, but also resists or reshapes collective ideas. At this intersection of individual action and collective structures, the real life of identity takes place. Through the action of the agent, the dominant imagination can be further strengthened and confirmed, but it can also be

contested, undermined and finally changed, thereby opening up space for the creation of new forms of collective existence.

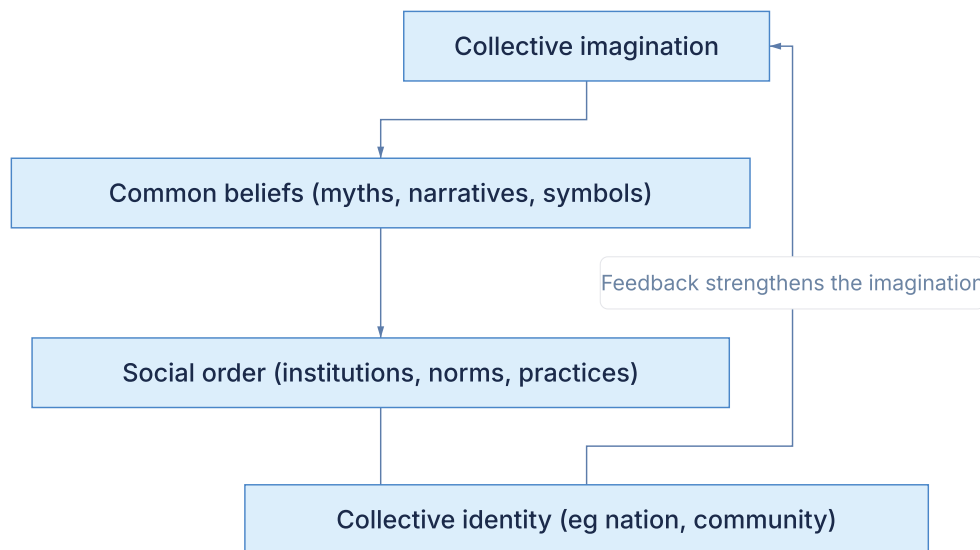


Figure 2.3: The process of collective identity formation: from collective imagination and shared beliefs to social order and collective identity (feedback).

2.2.3 Social functions of oral tradition

To view the oral tradition exclusively as a treasury of antiquities or a mere repository of aesthetically shaped texts would be to overlook its fundamental purpose and dynamic role in community life. Its existence is certainly conditioned by the individual need for artistic expression, but above all by its deep and multi-layered rootedness in social reality. Oral traditions, in all their forms - from myths and epics to proverbs and songs - act as key mechanisms through which a community designs, maintains and transmits its entire worldview and social structure. They become an active *agent* in its shaping. One of the most important roles of oral tradition is the preservation and transmission of **collective** memory. In cultures that primarily rely on orality, the stories about the origin of the community, about the deeds of the ancestors, about mythical events and historical milestones form the living fabric of historical consciousness. As a narrative framework that gives meaning and legitimacy to the present, oral tradition connects the community to its origins

and ensures a sense of continuity through generations, and establishes and maintains a common identity, because belonging to a community means sharing and knowing its stories.

Collective memory: A socially conditioned and jointly maintained system of memory by which the group preserves, interprets and transmits selected ideas about its own past; in the oral tradition, it is realized through repeated narrative performances (myths, genealogies, epic poems) that give meaning and legitimacy to the present and ensure a sense of continuity and belonging across generations (Halbwachs, 1950; Assmann, 2011).

Furthermore, oral tradition is a fundamental means of education and **socialization**. Through short stories, fairy tales, riddles and songs, the young generations adopt primarily the language and aesthetic patterns of their culture, but also ethical codes, social norms, rules of conduct and practical knowledge necessary for survival. Instructive stories warn of the consequences of transgression, while heroic poems offer models of desirable action. Thus, tradition acts as an informal but extremely effective educational system that shapes an individual into a full-fledged and functional member of society.

Socialization: The process by which an individual, in interaction with other members of the community, adopts the language, values, norms, knowledge and patterns of behavior necessary for functioning within a certain culture; in oral societies, this process takes place predominantly through narrative and performance forms (stories, fairy tales, riddles, songs) that simultaneously teach, entertain and shape personal and social identity (Berger and Luckmann, 1966).

Oral traditions also have a distinct **pragmatic and legal-legitimation function**. Myths about the creation of the world or the origin of tribes often serve as charters that confirm and justify existing social order, property relations, political hierarchy, or ritual practices (Finnegan, 1977). Genealogies that are remembered and publicly recited legitimize the right of ruling lineages to power, while oral traditions about court precedents can have the force of unwritten law. Oral tradition thus provides the basis for social stability, offering a supernatural or historical sanction for secular institutions.

Pragmatic function of oral tradition: The role of oral traditions in the direct regulation of social life - from the legal legitimization of government and property relations (through genealogies and myths of origin) to the transfer of practical knowledge and skills necessary for daily survival; tradition acts as an unwritten law and operational manual of the community (Finnegan, 1977; Malinowski, 1926).

The **fun** and aesthetic function, which is inseparable from all the others, should not be neglected either. Oral performances provide aesthetic pleasure and entertainment, gathering the community in moments of rest and celebration. The skill of the storyteller, singer, or reciter—their ability to master voice, gesture, and interaction with the audience—is critical to the success of message transmission (Bauman, 1986). It is precisely through the artistic dimension of the performance that the content of the tradition becomes attractive, memorable and emotionally effective.

Aesthetic function: The dimension of the communication act in which the form of the message – rhythm, melody, figure, style, narrative structure – becomes a source of pleasure and meaning in itself; in the oral tradition, the aesthetic function is not just a decoration, but a key mechanism that makes the content memorable, emotionally effective and suitable for repeated performance, thus directly ensuring the transmission of culture (Jakobson, 1960; Bauman, 1986).

Finally, oral tradition can also serve as a form of social **criticism** and commentary, and even as a covert vent for expressing dissent. Through satirical songs, anecdotes about powerful people or allegorical stories about animals, the community can criticize injustice, mock authority or question ruling norms in a way that is often too risky in direct communication. Thus, tradition shows its adaptability and ability to serve not only the maintenance, but also the questioning and subtle change of the social order.

Social criticism in the oral tradition: The function of the oral tradition that allows the community to question, comment on and challenge the existing power relations and social norms through indirect narrative forms – satire, allegory, parody, anecdotes about the powerful and animal fables; since criticism comes in an aesthetically shaped and collectively sanctioned form, it bypasses open conflict while simultaneously opening up space for reconsideration and gradual change of order (Scott, 1990; Finnegan, 1977).

All the mentioned functions – from preservation of memory and education to identification and entertainment – are intertwined and can rarely be viewed separately. Each individual oral performance simultaneously activates several of these roles, confirming that oral tradition is a complete social phenomenon, necessary for the survival and cohesion of the community that creates and nurtures it.

2.3 Literacy as a window into complex knowledge systems

Observing literacy exclusively as the skill of reading and writing means overlooking its deep, transformative role in shaping human thought and the entire social reality. Literacy should be understood as a fundamental technology that changes the cognitive architecture of the individual and opens access to complex, cumulative systems of knowledge, unimaginable in exclusively oral cultures. It is an epistemological prerequisite for the emergence of science, philosophy, law and history in their modern form. The main dividing line that literacy establishes in relation to oral tradition is its ability to externalize and fix thoughts. While the spoken word is inherently transitory and tied to the immediate context of interaction, the written text exists as an autonomous and permanent object, separate from the author and the moment of creation. This durability enables new cognitive operations: re-reading, careful analysis, comparison of different statements created in different times and places, and systematic criticism (Goody & Watt, 1963). The text thereby establishes itself as an external, objectified memory, freeing the mind from the cognitive burden of remembering the entire cultural heritage and allowing it to devote itself to abstract and analytical thinking.

Externalization of thinking: The process of transferring internal cognitive content - ideas, knowledge, reasoning - to an external, permanent and author-independent carrier (written text, diagram, record, digital document); thus, the mind is relieved of the burden of memory, and thought becomes available for re-reading, analysis, criticism and cumulative upgrading through generations, thus opening the prerequisites for the emergence of complex systems of knowledge (Goody, 1977; Donald, 1991; Clark and Chalmers, 1998).

This change profoundly reshapes internal mental processes as well. Mastery of writing encourages the development of linear, sequential thinking, as opposed to the associative and episodic nature of memory in oral cultures. The ability to follow complex syntax, classify information into categories and subcategories, and develop abstract concepts is a direct consequence of long-term exposure to the written word. In addition to storing knowledge, literacy literally restructures neural pathways in the brain, creating prerequisites for new forms of intelligence and introspection (Dehaene, 2009). It is on these cognitive foundations that complex systems of knowledge grow. Science, for example, rests on the possibility of verifying and upgrading previously recorded knowledge; legal systems depend on the precision and immutability of written laws; and philosophical and theological discourses develop through centuries of written debates about the foundational texts. These systems have an internal logic, hierarchy, and cumulative nature, which would be impossible to maintain solely through word of mouth. Written culture creates a vast, intertextual network into which every new knowledge must fit, to which it must refer and which gives it additional meaning. In this complex universe of knowledge, a literate individual becomes an *agent* - an employee who, through literacy, acquires the means to understand existing systems, as well as to actively participate in them. He can interpret, question, challenge and, finally, contribute to the body of knowledge. Through the act of reading, the agent enters into a dialogue with thinkers from the past, and through the act of writing, he articulates his own knowledge and directs it to future generations (Ong, 1982).

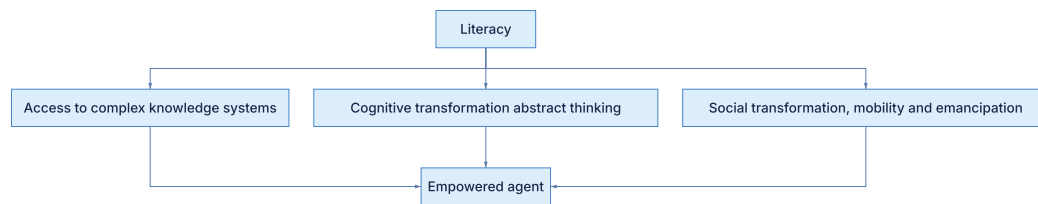


Figure 2.4: The roles of literacy in enabling the agent to access complex knowledge systems and in stimulating cognitive and social transformation.

2.3.1 Different paths to the letter

The appearance of writing in human history can be understood as a layered process that took place in different ways, at different times and prompted by different needs. Historical and archeological findings bear witness to at least four independent, spontaneous centers of literacy: in Mesopotamia, Egypt, China and Central America. Each of these systems emerged from specific cultural and social circumstances, but the paths that letters and literacy moved on branched off into several fundamental directions. The fundamental difference that emerges in the study of the history of letters is that between the original invention, i.e. the creation of letters *de novo*, and the process of borrowing or adapting an already existing system. The creation of a letter from the ground up is an extremely rare historical phenomenon, related to the mentioned origins. Much more common is the path of adaptation, where one culture takes a script from another and modifies it to fit the structure of its language. It is a paradigmatic example of the Greek takeover of the Phoenician consonant letter (abjada). The Greeks, faced with a script that did not record the vowels crucial to the phonology of their Indo-European language, repurposed graphemes for the Semitic gutturals they did not need and assigned them the value of vowels. This only resulted in the transmission of an essential structural change that resulted in the first complete alphabet. Similar processes of adaptation have occurred countless times in history, each time confirming that the script is a dynamic system that must adapt to the phonological, morphological and syntactic features of the language it serves. Furthermore, the path to a well-formed writing system can also be viewed through the prism of the way it was created, which can be gradual and emergent, or conscious and engineered. **The emergent or organic path implies a long-term**, often

century-long development, during which the alphabet gradually develops from simpler protosystems, such as pictograms, ideograms or systems for recording numbers and goods. The most famous example is the Sumerian cuneiform script, which evolved from a system of clay tokens used for administrative and economic purposes millennia before the appearance of real writing (Schmandt-Besserat, 1996). This development was a response to the growing needs of a complex society for accurate records. In such a model, the letter appears as the result of a collective and unconscious process, without a named inventor or a precise moment of creation.

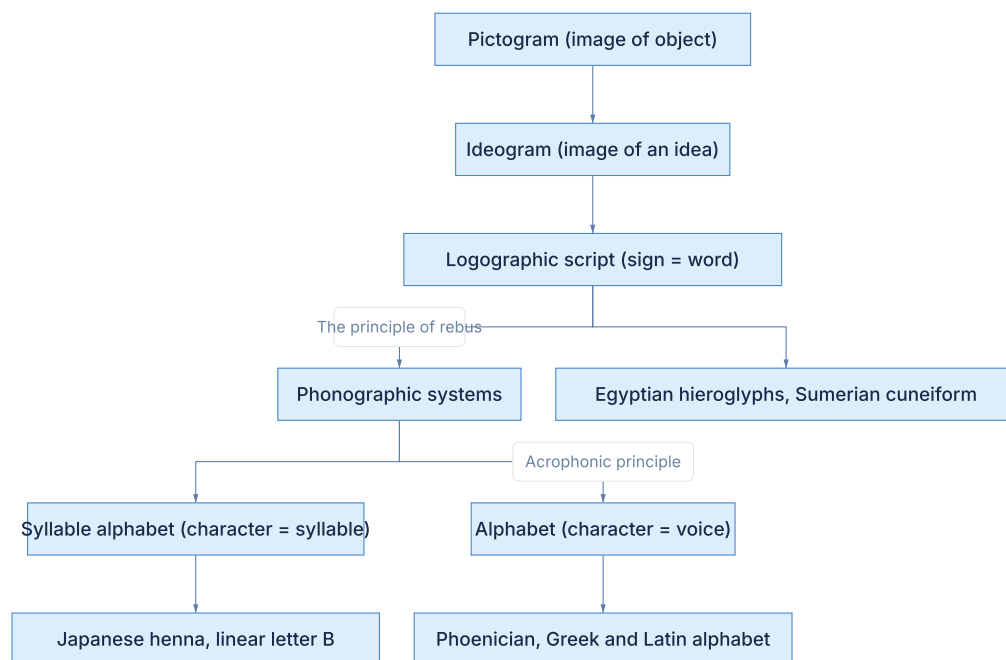


Figure 2.5: Development paths of the alphabet (from pictograph to alphabet).

The engineered or deliberate path is the opposite. Here, writing is the result of a conscious and deliberate act of creation undertaken by an individual or a small group. Such an agent acts with a clear goal - to create a script for a specific language, often for cultural, political or religious reasons. The most obvious example of such an endeavor is the Korean script Hangul, which was devised in the 15th century by King Sejong the Great and his scholars to increase literacy and strengthen Korean cultural identity. Hangul was not created by adapting an existing script, but was designed based on phonological

analysis of the Korean language, making it one of the most scientifically sound scripts in the world. Similar, although less well-known, examples are the Cherokee language script created in the 19th century by Sequoyah or the numerous scripts designed by Christian missionaries for previously unwritten languages (Cromer, 1997). These two dichotomies – *de novo* versus adaptation and emergent versus engineering – are not always strictly separate and can be intertwined. The adaptation of an existing script can also be the result of a deliberate act, such as contemporary orthographic reforms, or it can occur gradually and organically over centuries of use.

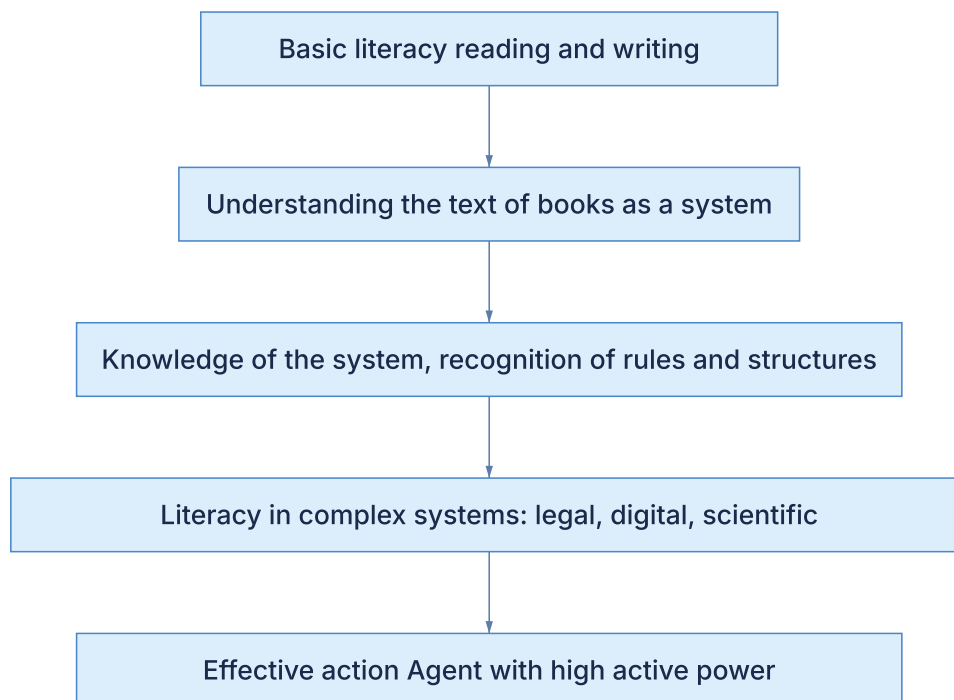


Figure 2.6: Development of active power through literacy: from basic literacy (reading and writing) through text comprehension and recognition of system rules to literacy in complex systems (legal, digital) and effective agent action.

Understanding these different paths is crucial because it shows that there is no single “evolution” of writing that necessarily leads from pictorial to alphabetic. Each writing system is a reflection of the history, culture and linguistic

structure of the community that uses it. The life of the letter is inseparably connected with the life of that community, and the ways in which it develops and spreads are as diverse as human societies themselves.

2.3.2 Social elite and slow expansion

In the complex fabric of social relations, the social elite occupies a dual and sometimes contradictory role in the processes of innovation diffusion. Although history has recorded elites as the first adopters of certain innovations, especially those that strengthen their status, their position in the power structure often makes them a key factor in slowing down or even stopping diffusion processes. This phenomenon often results from active action aimed at preserving the existing order (*status quo*). The fundamental cause of such action lies in the fact that every major innovation, whether technological, cultural or social in nature, has the potential to redistribute power, influence and resources within the community. The social elite, by definition, benefits the most from the existing situation. Their power and privileged position are directly linked to existing norms, values and material relations. Therefore, any innovation that could disturb this balance is perceived as a threat. Preservation of their own interests thus becomes the driving force of resistance to changes that come from outside their controlled circles.

Redistribution of power: A change in the distribution of influence, resources and authority among actors within the social structure, which occurs when a new technology, practice or idea changes the previous channels of access to knowledge, communication or material means; in the context of communication innovations (letters, press, internet) the redistribution of power means the process by which previously excluded groups gain the ability to act, while the previous elites lose their monopoly over information and interpretation of reality (Castells, 2009; Eisenstein, 1979).

Furthermore, social stratification itself creates barriers to expansion. Social elites are often characterized by a high degree of homophily - a tendency to associate and interact with individuals who are similar in education, background and status. Such confinement to one's own social networks significantly limits the flow of information and influence between the elite and

the wider population. Communication channels become interrupted or one-way, and the elite acts as a kind of “gatekeeper”, filtering which ideas and practices will reach the rest of society. As Rogers (2003) states, when elites adopt an innovation exclusively for themselves to emphasize their special status, they actually prevent its further diffusion, instead of encouraging it.

Social stratification: Hierarchical stratification of society into levels (layers, classes, estates) that differ in access to material resources, education, power and prestige; in the context of communication technologies, stratification determines who first adopts an innovation, who has access to information and who acts as a gatekeeper that accelerates or slows down the spread of new ideas and practices to other parts of society (Weber, 1922; Bourdieu, 1984; Rogers, 2003).

The role of a change agent in such a context is extremely difficult. If the agent does not belong to the elite or does not have direct access to its members, his messages will have difficulty overcoming social and cultural distance. The elite may reject an innovation not so much because of its intrinsic properties as because of the origin of its bearer. News that comes from “below” or from the outside can be perceived as vulgar, inappropriate or threatening, thereby discrediting its symbolic value in advance. This establishes a mechanism in which the elite not only slows down change, but actively works to delegitimize innovations that do not serve to strengthen its position.

Cultural distance: Degree of difference in values, norms, language, education and life experiences between two social groups; the greater the cultural distance between the bearer of the innovation and potential recipients, the more difficult is the transfer of ideas and practices, because the message loses credibility or is reinterpreted through the prism of the recipient’s own symbolic framework (Rogers, 2003; Hofstede, 2001).

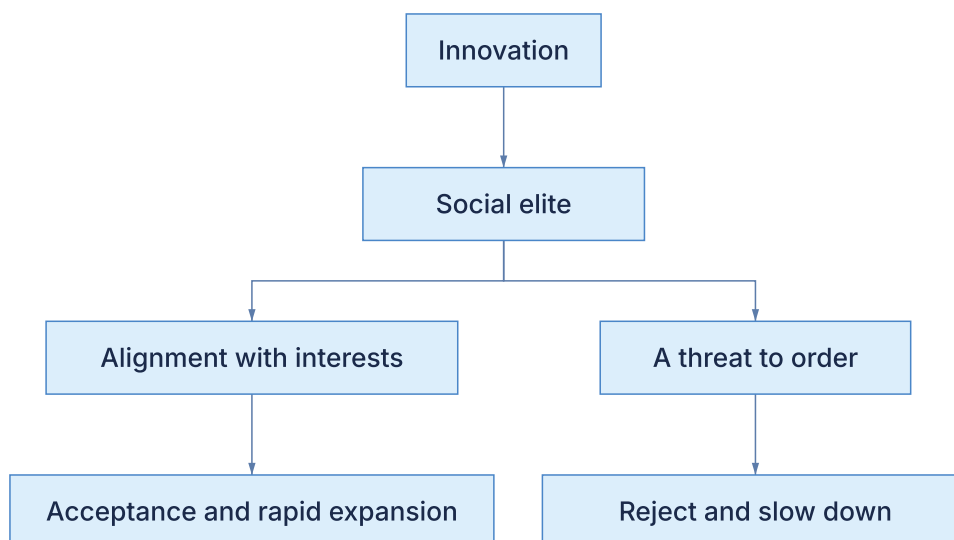


Figure 2.7: A model of slowing diffusion through the social elite: the elite acts as a filter that accepts the innovation and accelerates its spread if it is aligned with its interests (benefit, status, power), or rejects it and slows down its spread if it represents a threat to the existing order.

2.4 The printing press: a turning point in the spread of ideas

The invention of the movable type printing press, which is attributed to Johannes Gutenberg around 1450, represents one of the most significant technological and cultural breakthroughs in human history. Before Gutenberg, knowledge was transmitted and preserved in the form of manuscripts, the creation of which was a long, painstaking and extremely expensive process, mostly limited to monastery scriptoriums and court workshops. Each copy was unique, and its creation could take months, even years, which made the book a rare and precious object available only to a small elite. The limited number of copies and the high price significantly slowed down and narrowed the flow of information and ideas. Gutenberg's invention fundamentally transformed the material reality of the book and thereby triggered an avalanche of social, religious and scientific changes. By combining the technology of movable metal letters, a press adapted from winemaking, and newly developed oil-based

printing ink, a system was created that enabled fast, cheap and mass production of texts. While the monk-copyist wrote a single manuscript, the printing press could now reproduce hundreds and thousands of identical copies. This change in the scale of production necessarily led to a drastic drop in price, with which the written word began its journey towards wider layers of society, gradually leaving the exclusive space of church and noble libraries. The consequences of that technological revolution were far-reaching and profound. The printing press became a key *agent* in spreading the ideas of the Reformation. The ability to quickly print and distribute writings, such as Martin Luther's *95 Theses*, allowed criticism of church practices to spread across Europe at a speed previously unimaginable. This broke the monopoly of the Catholic Church over the interpretation of holy scriptures and religious doctrines. The printing of the Bible in the vernacular languages further strengthened this process, allowing individuals direct access to the text, which until then had been mediated exclusively by the clergy in the Latin language. In the field of science, the press enabled what is today considered the basis of scientific progress: the reliable and wide dissemination of knowledge. Scientists could more easily share their discoveries, and the accuracy of printed texts and diagrams, as opposed to manuscripts subject to copying errors, provided a solid foundation for further research. As Elizabeth Eisenstein (1979) points out in her capital work, this "fixity of print" created the preconditions for the cumulative growth of knowledge and the scientific revolution that followed in the following centuries. The ideas of Copernicus, Galileo or Newton could circulate among scholars across Europe, encouraging debate, verification and improvement.

Dissemination of knowledge: Systematic dissemination of knowledge, ideas and information from the source to a wider circle of recipients through communication channels (manuscript, print, digital network); is a key assumption of cumulative culture because it allows discoveries to become available for verification, critique, and upgrade by other agents, thus transforming knowledge from an individual achievement into a common good that accelerates further progress (Eisenstein, 1979; Rogers, 2003).

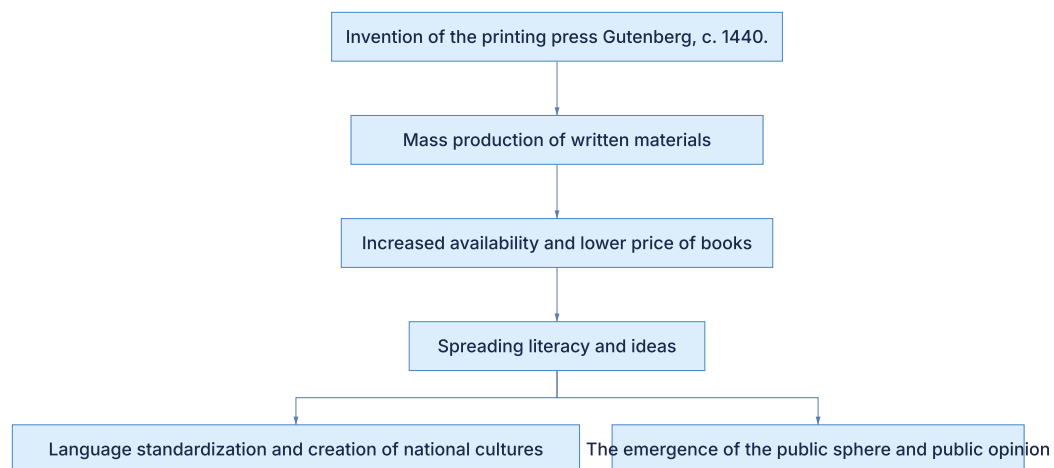


Figure 2.8: The impact of the printing press on the spread of knowledge and social changes: from Gutenberg's invention (c. 1440) and the mass production of written materials to the increased availability of books, the spread of literacy and ideas (humanism, the Renaissance, the Reformation), the standardization of language and the emergence of the public sphere and public opinion.

Furthermore, the press played a crucial role in the formation of modern nations and the standardization of vernacular languages. Printers, in search of a wider market, chose and promoted certain dialects, which, thanks to the mass distribution of printed works, gradually imposed themselves as the language standard. This created the basis for linguistic and cultural unity in wider areas, which was one of the cornerstones in the process of creating national identities. Viewed in this light, the printing press is an active *agent* of change which, according to the insights of Marshall McLuhan (1962), reshaped human sensorium and cognitive processes. By emphasizing the visual over the auditory, the press encouraged the development of a linear, sequential and analytical way of thinking, which became a feature of Western civilization in the period known as “Gutenberg’s Galaxy”. The influence of the printing press is therefore not exhausted in technical innovation; he reshaped the ways in which people think, communicate and organize their societies, laying the foundations for the emergence of the modern world.



Figure 2.9: The printing press as an agent of cognitive change - McLuhan's "Gutenberg's galaxy": from emphasizing the visual over the auditory through linear, sequential thinking to the analytical way of processing information.

2.4.1 Mass production and democratization of knowledge

The appearance of mass production of books, made possible by Gutenberg's invention of the movable type printing press in the middle of the 15th century, should be considered a turning point in the history of Western civilization. Until that time, knowledge contained in books was confined within narrow circles. Books were rare and precious items, multiplied by painstaking work of copyists in monastery scriptoriums, which made them available almost exclusively to the priestly and noble elite. The printing press fundamentally changed that order, enabling the rapid and significantly cheaper production of printed materials and creating the prerequisites for a more extensive spread of literacy among the wider strata of the population. This transformation was particularly clearly manifested during the Protestant Reformation. Martin Luther, recognizing the power of the new media, used the press for the mass dissemination of his writings, thereby inciting religious and social changes of unprecedented proportions. Equally, the press served as a powerful stimulus for the development of science. It enabled scholars to exchange knowledge and discoveries with a previously unimaginable speed, which laid the foundations for the scientific revolution and the establishment of a systematic scientific discourse (Eisenstein, 1979). A kind of *Res publica litterarum* was created, a community of scholars all over Europe connected by the printed word, who could discuss, build on and check the works of their contemporaries. This entire process is rightly called the **democratization** of knowledge. However, this term does not mean only easy access to existing information. Its meaning is deeper and includes the gradual empowerment of individuals for active participation in the creation, verification and dissemination of new knowledge. Knowledge thus ceases to be the exclusive property of privileged groups and becomes a common good, subject to public discussion and criticism.

Democratization of knowledge: Gradual removal of social, economic and institutional obstacles to access to information and knowledge, which gives an ever-widening circle of agents the ability not only to receive but also to create, verify and disseminate knowledge; historically initiated by the invention of printing, and nowadays accelerated by digital technology and open access, the democratization of knowledge transforms it from the privilege of the few into a common good subject to public discussion and criticism (Eisenstein, 1979; Castells, 2010).

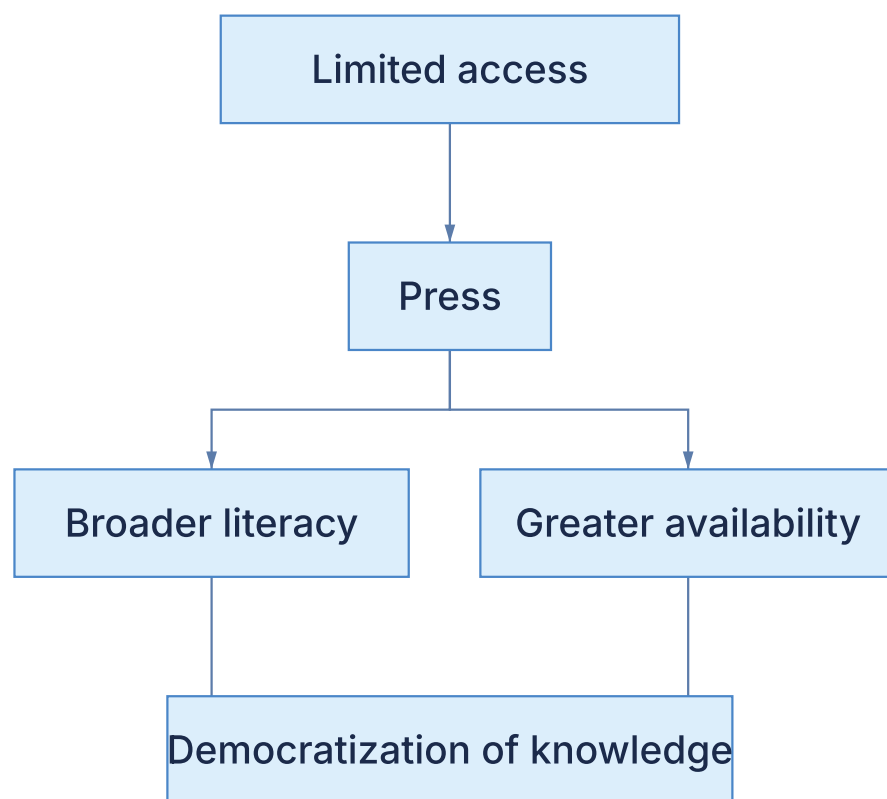


Figure 2.10: The path of knowledge from limited access in manuscript culture to wider availability through the printing revolution, culminating in the democratization of knowledge.

It should, of course, be noted that this transition was neither immediate nor without resistance; it took centuries for the effects of the printing revolution to fully manifest and permeate every pore of society. In today's time, with the advent of digital technology and the Internet, a transformation of comparable,

if not greater, proportions is taking place. The Internet can be seen as a contemporary counterpart to the printing press, which accelerates and expands the process of democratization of knowledge to a global level, bringing with it new opportunities, but also new challenges.

2.4.2 Reformation and scientific revolution

The period of the 16th and 17th centuries in European history was marked by two profound and transformative movements: the Reformation and the Scientific Revolution. Their simultaneity, as well as their partial geographical overlap, encourage consideration of complex connections and mutual influences. Although a simple cause-and-effect relationship cannot be established, it is clear that the Reformation created an intellectual, social and cultural atmosphere that favored the emergence and development of modern science. The invention of the printing press played a key role in the spread of reformation ideas, but also in creating the basis for scientific progress. By enabling the mass and rapid reproduction of texts, the press undermined the Catholic Church's monopoly over the knowledge and interpretation of scriptures. The principle of *sola scriptura*, one of the central pillars of the Reformation, encouraged believers to independently read and reflect on the Bible, which consequently led to an increase in literacy. This newly acquired skill and habit of critical reading was not limited exclusively to religious texts, but also spread to other areas of human knowledge, including scientific ones. The availability of printed works enabled a faster and wider exchange of ideas among scholars across Europe, creating the foundations for the scholarly community as we know it today. In its essence, the Reformation represented a fundamental challenge to the spiritual and secular authority of the Catholic Church. Martin Luther's act, which publicly questioned the infallibility of the pope and church councils, resonated far beyond theological circles. This undermining of a monolithic authority created an intellectual atmosphere in which it became conceivable, and even desirable, to question other established dogmas, including scientific ones, such as the Aristotelian-Ptolemaic geocentric system, which the Church accepted. If the authority of the Church could be questioned in the field of faith, why could it not be questioned in the field of

interpretation of nature? In this way, the Reformation, albeit unintentionally, contributed to the secularization of knowledge and made critical questioning legitimate as a valid cognitive method. The Protestant doctrine of the “priesthood of all believers”, which gives each individual the right and duty to independently interpret God’s word, finds its scientific analogy. The scientist, as an individual agent, now assumes the role of one who directly “reads” and interprets the “book of nature”, without the mediation of ancient authorities such as Aristotle or Galen. This new epistemological paradigm, which emphasized personal experience and interpretation, was not limited to theology. It was also transferred to the study of nature, which began to be perceived as “God’s second book” (Harrison, 2007), open to reading and interpretation through observation and experiment. In his famous thesis, Max Weber connected Protestant ethics with the spirit of capitalism, and Robert K. Merton established a similar connection between Puritanism, as one of the branches of Protestantism, and the development of experimental science in England in the 17th century (Merton, 1970). According to Merton, Puritan values such as diligence, methodology, and utilitarianism encouraged scientific research that could bring practical benefit and glorify God through the discovery of the laws of his creation.

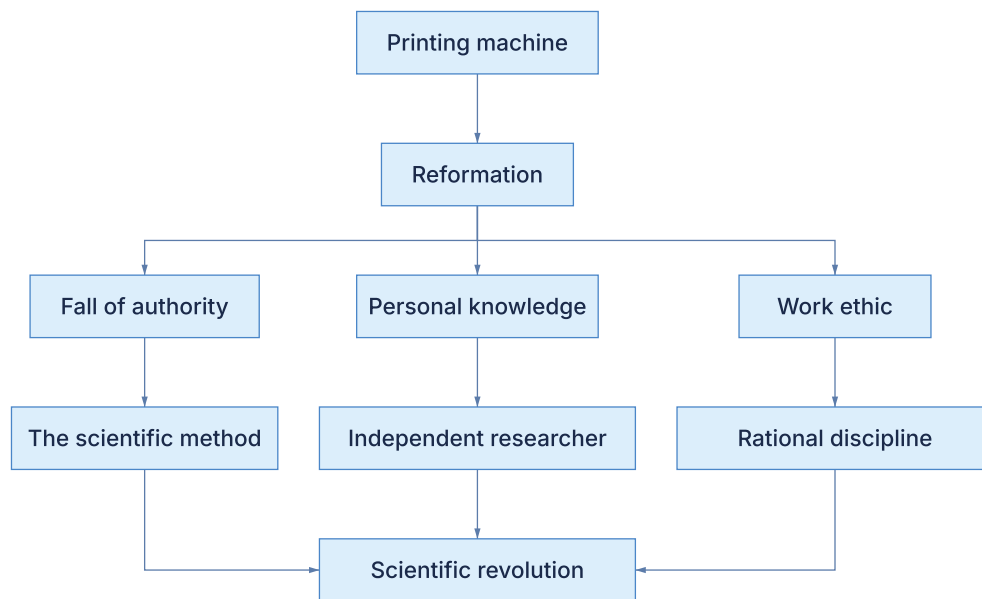


Figure 2.11: Schematic representation of the key consequences of the Reformation that contributed to the creation of an enabling environment for the scientific revolution: Reformation incentives, reinforced by printing technology, lead to the fragmentation of authority and the emphasis on individual knowledge, which creates fertile ground for new scientific methods and discoveries.

However, it should be emphasized that the relationship between the Reformation and science was not uniquely encouraging. The reformers themselves, including Luther and Calvin, expressed considerable skepticism and even open opposition to some new scientific ideas, such as Copernicus' heliocentric theory. Their primary goal was the restoration of faith, not the promotion of science. Furthermore, the Scientific Revolution was not an exclusively Protestant phenomenon; scientists like Copernicus, Galileo and Descartes worked within the Catholic world. Therefore, the impact of the Reformation on science is not so much manifested in direct doctrinal support as in its indirect and often unintentional consequences: the fragmentation of intellectual authority, the promotion of literacy and the culture of reading, and

the establishment of a new value system in which the individual agent, either as a believer or as a scientist, became the center of the cognitive process.

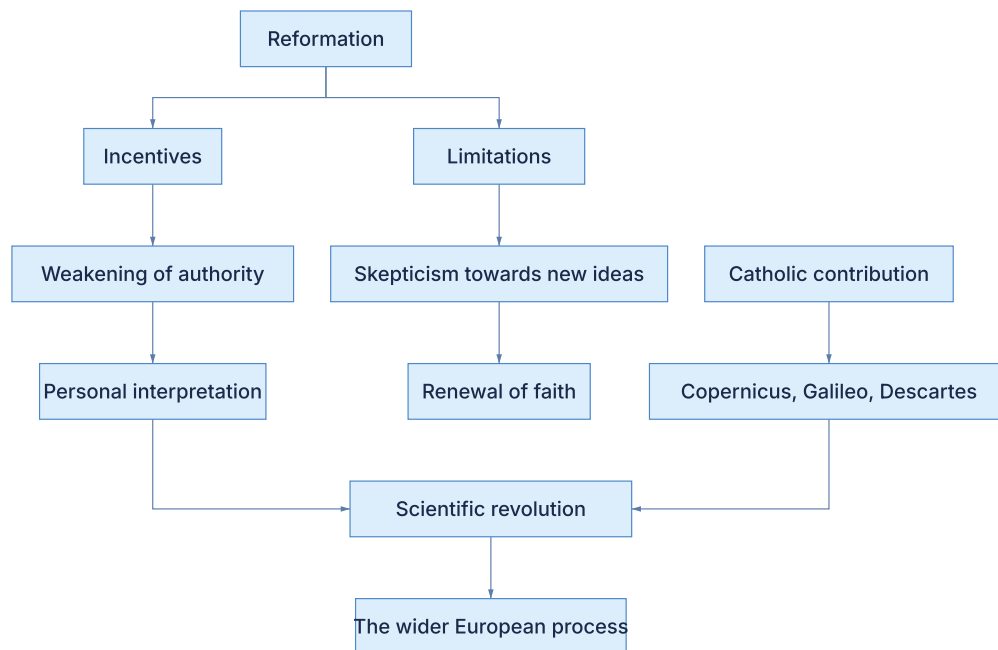


Figure 2.12: The complex relationship between the Reformation and science – the stimulating and limiting effects of the Reformation and the Catholic contribution to the scientific revolution.

2.5 Telegraph and telephone: The beginning of electronic communication

The nineteenth century brought a fundamental change in the history of human communication by introducing technologies that for the first time enabled the transmission of messages at a speed greater than any physical vehicle. This

marked the beginning of the era of electronic communication, the foundations of which are still felt today in all modern networks.

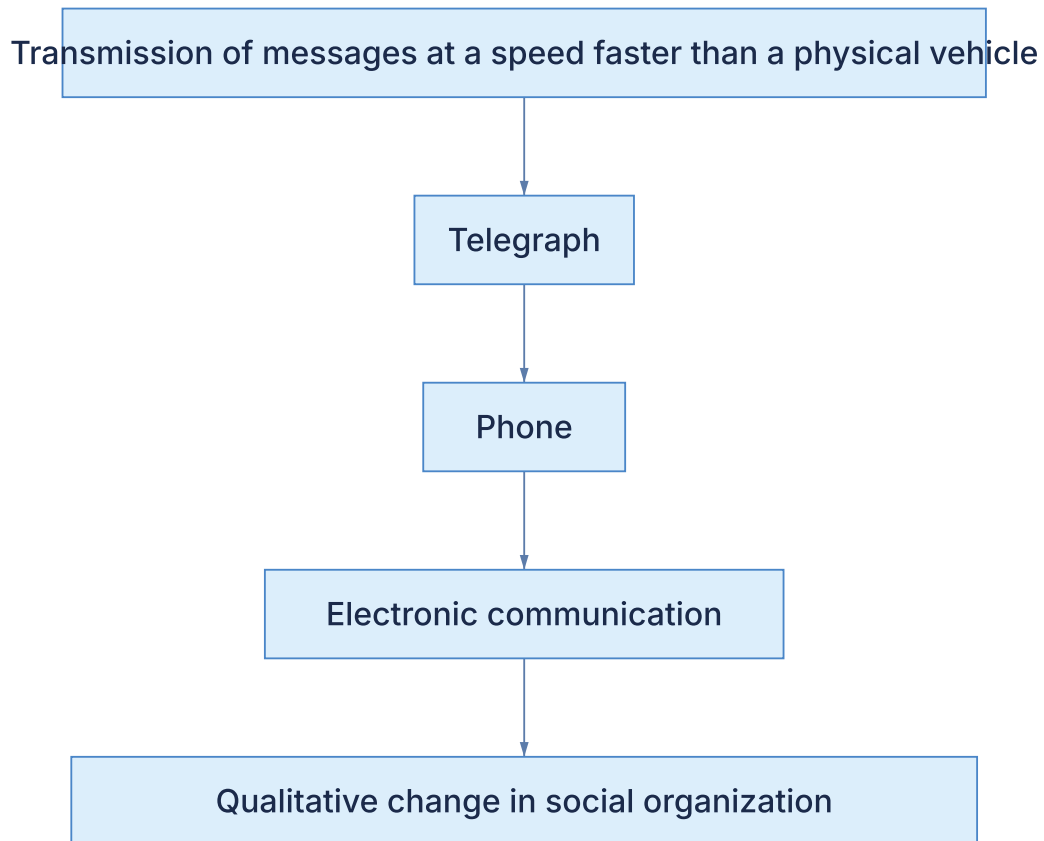


Figure 2.13: Schematic representation of the transition to the age of electronic communication: from the transmission of messages faster than physical transport to the telegraph, the telephone and changes in social organization.

2.5.1 Telegraph: new speeds and new horizons

The electric telegraph profoundly and permanently altered the contours of human communication, establishing a new paradigm that ended the millennia-old dependence of information transmission on the physical speed of the messenger, whether by horseman, stagecoach, or ship. Before the telegraph, the world moved at the speed of muscle and wind. After him, the message could threaten the movement of the Sun from east to west, and the distance became a magnitude that, for the first time in history, could be overcome

almost instantly. The basic principle of the telegraph was based on a simple but ingenious idea: the conversion of electrical impulses, sent through a long metal wire, into coded characters. Morse code, with its system of dots and dashes, became the *lingua franca* of that new age, the abstract language of machines that human operators, skilled *agents* of mediation, turned into meaningful words. This new speed was astonishing. News that would have taken weeks to travel across the Atlantic now reached its destination in minutes. This quantitative change - the drastic reduction of the time required for communication - necessarily stimulated profound qualitative changes in the organization of society. By accelerating existing communication practices, the telegraph created a qualitative leap that gave birth to completely new forms of social and economic interaction. New horizons of possibilities that were previously unimaginable have opened up. Capital and commodity markets, such as those in London and New York, could operate as a single, synchronized system as prices and transactions became instantly available on both sides of the ocean. The business empire could be coordinated from a single center, and armies were given the possibility of centralized command in real time. News agencies, such as Reuters and the Associated Press, grew up precisely on the telegraph network, creating a global flow of news and reshaping public opinion (Standage, 1998). This changed the very perception of space and time; the world began to seem smaller, more connected and busier. To view the telegraph exclusively as a technological artifact would be to ignore its role as the cornerstone of the new communication order. It was a complex socio-technical system that included devices and wires, but also international standards, protocols for sending messages and a whole new profession - telegraph operators. These skilled operators, as key human *agents* in the communication chain, were an indispensable part of the functioning of the entire network. Their ability to quickly and accurately encode and decode messages was as important as the correctness of the technical facilities themselves. The entire process can be presented as a sequence of steps in which each element has its own irreplaceable role. The laying of the first transatlantic telegraph cable in 1866 was a feat comparable to later space exploration – a triumph of human perseverance and engineering skill. This act symbolically, but also realistically, created a unique information space that

encompassed the Old and New Worlds. The telegraph thus laid the foundations for all future forms of electronic communication, from telephone and radio to the Internet. It was the first technology to separate communication from transport, proving that a message can move independently of the physical transfer of objects or people (Carey, 1989). In this sense, the telegraph network of the nineteenth century was not only a predecessor, but an archetype of today's global information society, and its echoes still shape the way we live, work and understand the world around us.

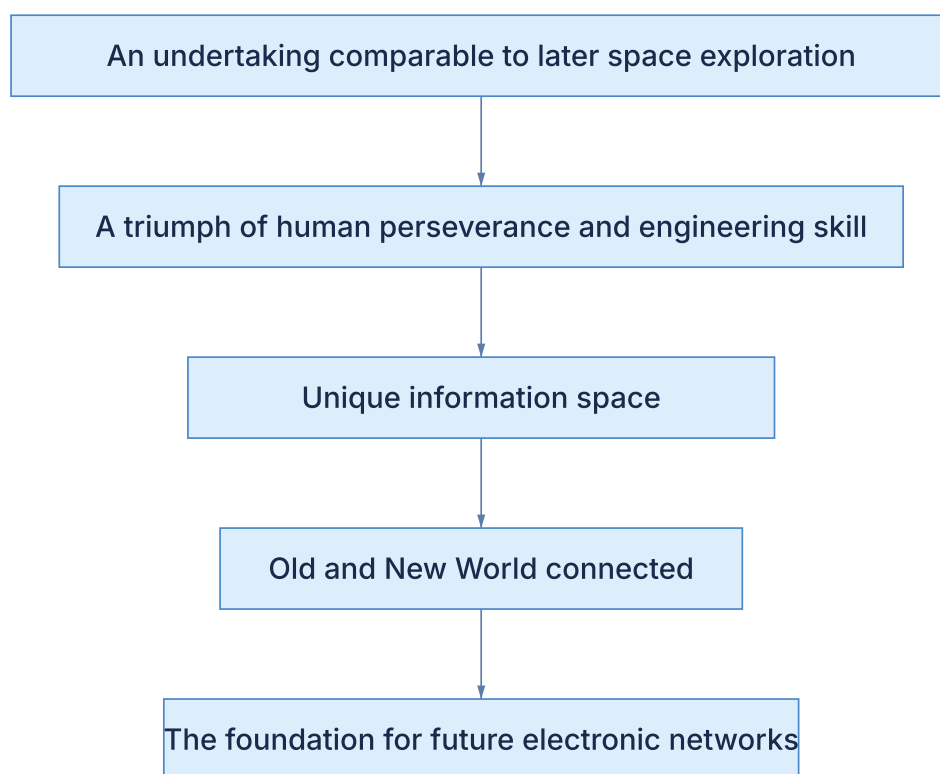


Figure 2.14: Schematic representation of the significance of the laying of the transatlantic telegraph cable in 1866: a technological undertaking that connected the Old and New worlds and opened up a unique information space.

2.5.2 Telephone: an intimate revolution on the wire

The appearance of the telephone, a technical miracle at the end of the nineteenth century, initiated a further radical transformation of social relations and the very perception of space and time. While the telegraph had already

made the world smaller by transmitting coded messages over vast distances, its reach remained in the official, commercial and military spheres. Communication was mediated, asynchronous and devoid of human voice. The telephone, in contrast, returned the most personal thing to the act of communication - the living word, with its color, emphasis, pauses and all those ineffable nuances that reflect human communication. This was the beginning of a quiet but deep revolution that did not take place in the squares or in the parliaments, but in the privacy of homes, on the invisible “wire” that connected the two interlocutors. The key change introduced by the telephone was the abolition of the necessity of physical presence for immediate and intimate communication. Spatial distance, previously an insurmountable obstacle for maintaining close ties, suddenly became surmountable. The human voice, as a carrier of identity and emotions, traveled for kilometers, creating the illusion of proximity and establishing a unique acoustic space shared by only two interlocutors. In this newly created virtual space, one could whisper, console, argue and confide secrets, making the phone a powerful *agent* in maintaining family and friendship ties that would otherwise weaken under the burden of geographic separation (Fischer, 1992). The new possibility had far-reaching consequences for the structure of everyday life. In the business world, the telephone accelerated decision-making and the flow of information, but its deepest mark remained in the private sphere. For women, who at that time were often spatially limited to the household, the telephone represented a window to the world. It enabled them to build and maintain their own social networks independently of male family members, organize social events and manage the household in a more efficient way. It became a tool of socialization, but also a symbol of a certain social status. The sound of the telephone ringing in the home has become a sign of connection with the outside world. However, this connection also had its other side. The immediacy offered by the telephone also meant the possibility of an unannounced intrusion into the privacy of the home. The ringing of the telephone has become a command that demands immediate attention, interrupting family meals, conversations or moments of rest. The border between public and private, once firmly defined by the doorstep, has become permeable. The disembodied nature of the telephone voice, devoid of the visual cues of body language, could also evoke feelings of

anxiety or mistrust. The telephone thus established a new dialectic of closeness and distance, availability and threatened privacy, which will become one of the central issues of modern life. In its essence, the telephone democratized synchronous long-distance communication, transferring it from an exclusively institutional framework to the hands of ordinary people. Its appearance has not only changed the way people talk, but also what they talk about and with whom. It established the foundations for a culture of constant availability and networking, the ultimate consequences of which we will only experience with the advent of mobile and Internet technologies a century later. Therefore, the telephone can rightly be considered the first true “intimate technology” - a

device that reshaped the very foundations of human closeness. Schematic representation of the influence of the telephone

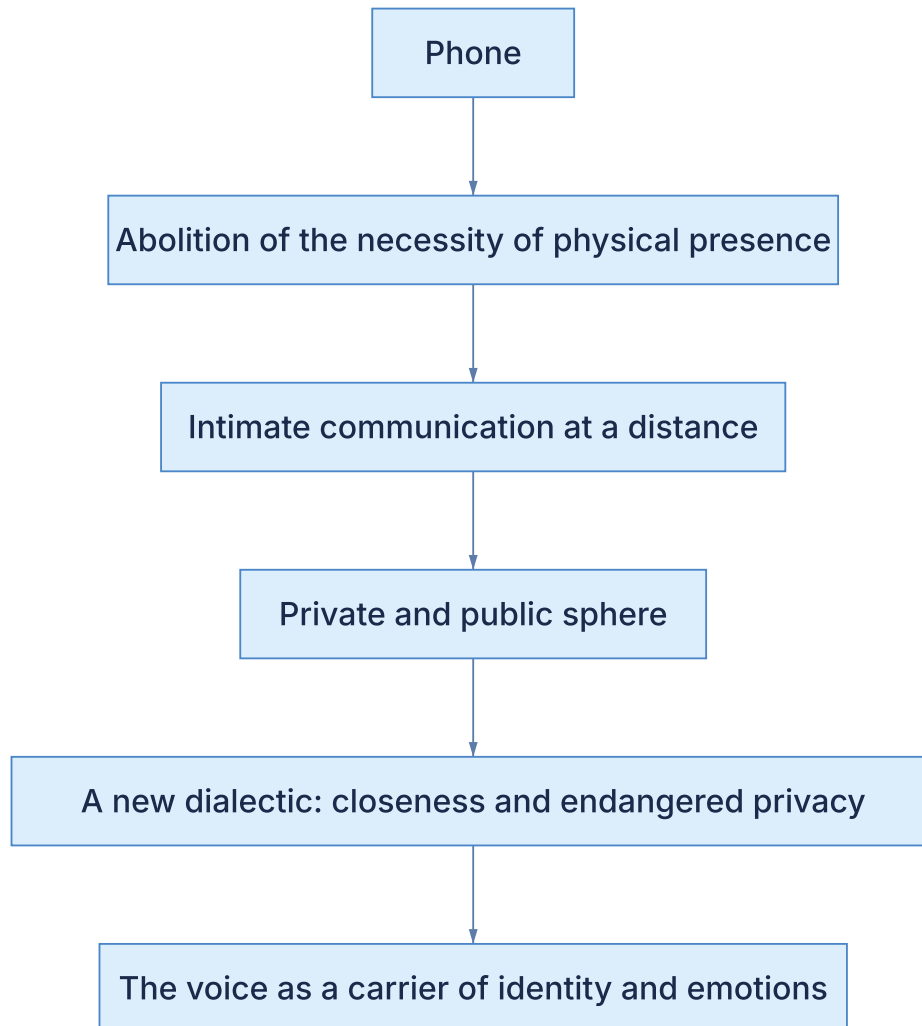


Figure 2.15: Schematic representation of the influence of the telephone on private and public communication, the abolition of the necessity of physical presence for intimate communication and the new dialectic of closeness and threatened privacy.

2.6 Radio and television as catalysts of mass experience

Entering the twentieth century, the concept of space, time and community was reshaped by the appearance of two media that will mark the epoch: radio and television. Their development further initiated a deep transformation of social reality, establishing new forms of collective experience of the world. It was radio and television that became the first true catalysts of mass experience, capable of simultaneously connecting millions of people in the unique act of listening or watching. The advent of radio in the first decades of the century brought a revolution in sound. For the first time in history, a human voice, music or news of a distant event could overcome physical barriers and enter directly into the privacy of the home. Radio broke the silence and filled everyday life with a new soundscape, but its real power lay in its ability to create a shared imaginative space. Listening to the broadcast of an important political speech, sports match or radio drama, an individual became aware that at the same moment thousands, even millions of strangers were sharing this experience with him. This awareness of simultaneity created a bond that did not exist before, an invisible network that connected the dispersed audience into a single listening body. In the middle of the century, television supplemented that sound image with the visual power of the moving image, which made its influence even more immediate and comprehensive. The television soon assumed the role of a central place in the home, becoming a “window to the world” through which scenes of coronations, space flights, wars and cultural spectacles entered the living rooms. The power of television rested in its ability to mediate reality seemingly directly and unfiltered. The visual experience of events like the 1969 moon landing did not leave much room for individual interpretation; he offered a ready-made, common image of reality that was engraved in the collective memory. In this way, television has become the ultimate synchronizing mechanism, uniting national and global communities in a shared, visually authenticated experience. It is precisely this synchronized and collective participation in a media-mediated event that forms the core of the mass experience. It is not exhausted in the fact that a large number of people receive the same information, but in the pervasive awareness

of the commonality of that act. This sense of belonging to a temporary but extremely strong community of viewers or listeners strengthened social cohesion and created common reference points in culture. The events broadcast by radio and television became the main topics of conversation, part of the general culture and the basis for building a common identity. Therefore, the role of radio and television went far beyond the passive transmission of content. They acted as active *agents* of social change. They shaped public opinion, established cultural norms, created heroes and icons, and directed the emotional life of the nation. The way these media framed reality, chose topics and presented events had a decisive influence on the perception of the world and one's own place in it. As media theorists point out, the medium itself, its technological nature and the way it organizes perception becomes a message that reshapes human experience and social relations (McLuhan, 1964). Radio and television did not only report on history; they created it, turning millions of individuals into participants of the same, mass and historical moment.

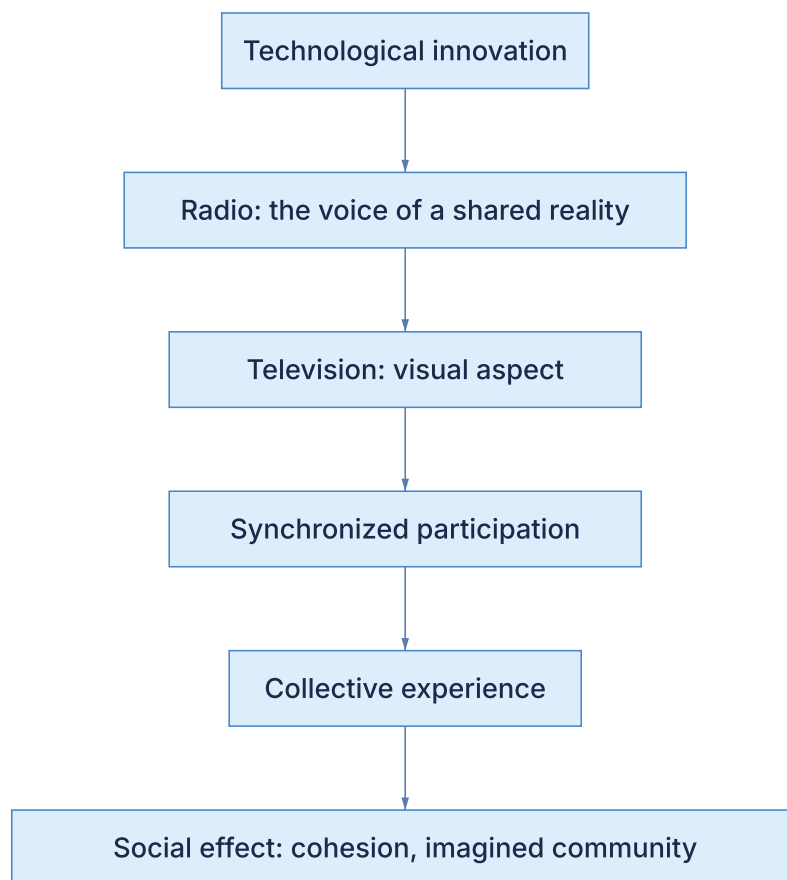


Figure 2.16: Stages in shaping mass experience through radio and television, from technological innovation to social impact – synchronized and collective participation in a media-mediated event.

2.6.1 The rise of radio and the “voice” of common reality

The appearance of radio as the first truly mass electronic media fundamentally transformed the social reality of the first half of the twentieth century. Its key feature, which distinguished it from all previous forms of communication, was the ability to create simultaneity of experience on a previously unimaginable scale. For the first time in history, millions of people, spatially separated, could listen to the same content at the same moment, thus becoming participants in a unique, shared sound reality. This unique sense of collective participation led Marshall McLuhan (1964) to the famous metaphor of the radio as a “tribal drum”, an instrument that brings together a dispersed community in a unique rhythm and resonance. Radio thereby became a powerful **agent of social**

cohesion, weaving invisible threads that connected individuals in their homes into a large, imagined community of listeners. In times of social crises, wars or major national events, the radio was a focal point that homogenized the national mood and created a sense of destiny connection. At the center of that new media paradigm was the **human** voice. Freed from the visual context, reduced to its sonic essence, the voice gained exceptional strength and intimacy through radio waves. He was not only a carrier of semantic content, but a powerful medium of emotions, presence and personality. The sound of the voice filling the privacy of the home created the illusion of an immediate, almost personal relationship between the speaker and the millions of individual listeners. This parasocial interaction was key to understanding the impact of radio - the voice became an agent that bridged public and private space, bringing distant events and personalities into the intimate circle of the family hearth. Statesmen and political leaders soon recognized the power of this new medium for shaping public opinion and mobilizing the masses. Voice has become a fundamental political tool. For example, American President Franklin D. Roosevelt’s “Fireside Chats” are a paradigmatic example of using radio to establish trust and leadership. Addressing the nation in a calm and personal tone, Roosevelt managed to create the impression that he was talking to each citizen individually, offering comfort and guidance during the difficult times of the Great Depression. In this context, the voice became a symbol of stability and authority that was both distant and incredibly close at the same time.

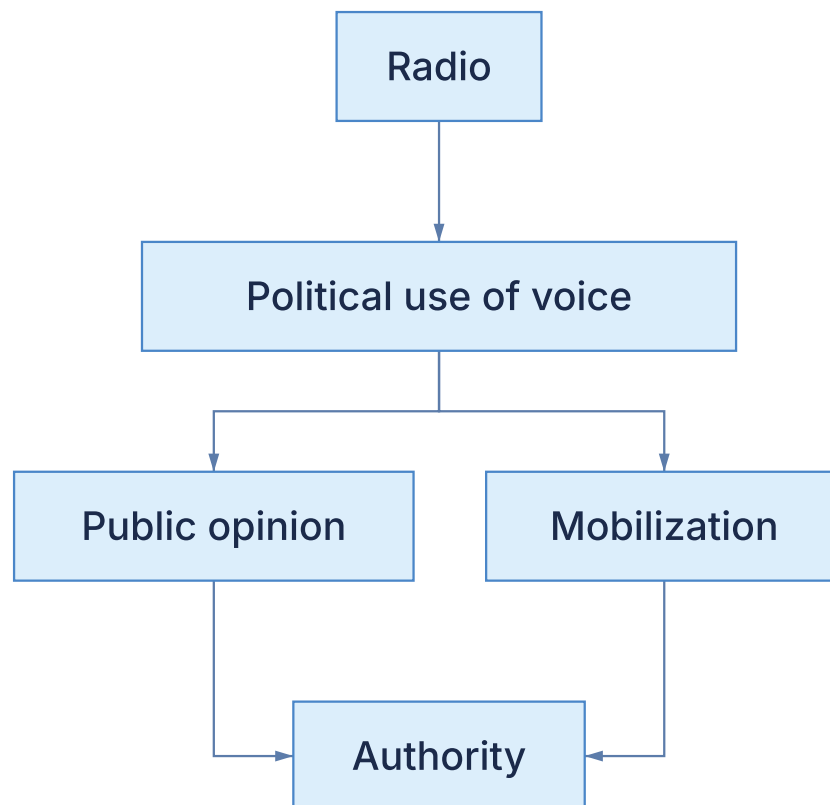


Figure 2.17: Schematic representation of the political use of radio: from voice as a medium to influence on public opinion, mobilization and authority.

His influence permeated the entire cultural life, shaping musical taste, popularizing dramatic forms through radio dramas and creating the first stars of the airwaves. Radio thus established a sonic tapestry of common culture, a frame of reference shared by members of different social strata. It is this ability of radio to simultaneously broadcast a unique “voice” - be it the voice of an announcer, politician, actor or singer - and make it part of the private experience of millions, which has created what we can call “the voice of shared reality”. It was a sound reflection and, at the same time, a constitutive element of the consciousness of belonging to a wider social entity.

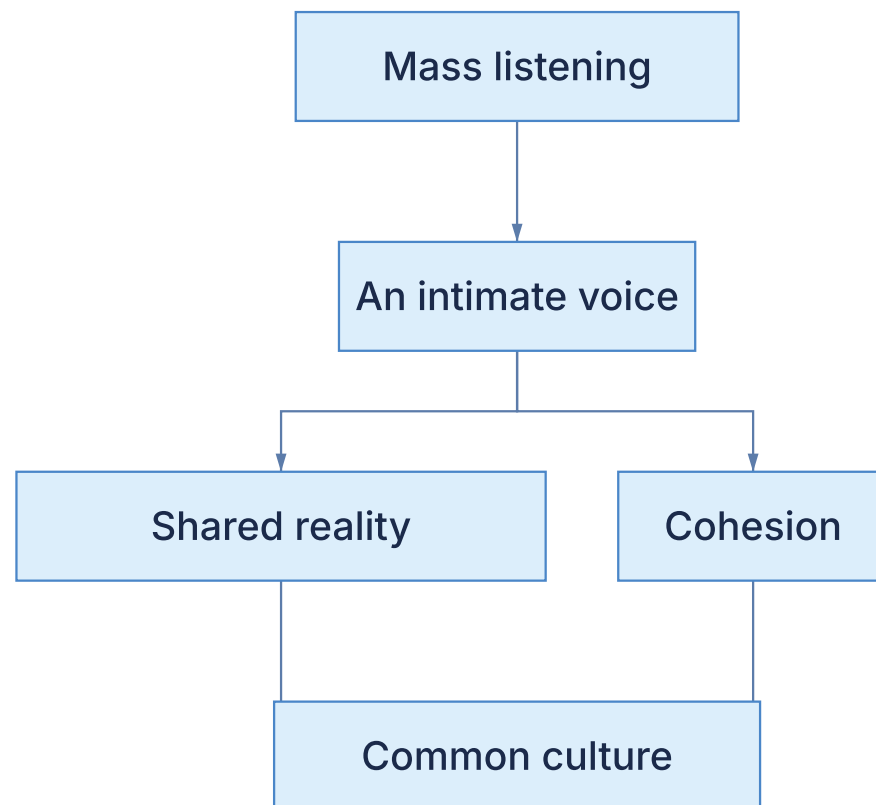


Figure 2.18: Presentation of the development and influence of the radio medium, from mass listening and intimate voice to the creation of a shared reality and political cohesion.

2.6.2 Television and the visual aspect of shared reality

Television should be understood as a key *agent* in the shaping of social reality and collective consciousness, which actively constitutes a common symbolic space within which community life takes place. It is precisely in its visual nature that the basis of its power to contribute to the creation of what can be called a shared imagination lies. Unlike print media, which largely shaped national consciousness through language and abstract narratives, television offers a direct, visual insight into the world. This ability to *show* events, people and spaces, and not just describe them, gives the imagined community a tangible, concrete form. The notion of “imagined community” by Benedict Anderson (1983), originally linked to the rise of the press and capitalism,

acquires a new, stronger dimension in the age of television. Millions of people, who will never meet in person, share the same visual repertoire, the same images of the nation, its successes, crises and rituals through the television screen. This creates a deep sense of belonging based on shared viewing and visual experience. Furthermore, the power of television does not derive exclusively from its visuality, but also from its ability to create the illusion of simultaneity. The feeling that an event is happening “live” and that the entire community is watching it at the same moment creates a strong social bond. This sense of sharing the present moment bridges the physical distance between individuals and integrates them into a single, synchronized auditory body. This mechanism is most evident in the phenomenon that Daniel Dayan and Elihu Katz (1992) called “media events”. These are extraordinary occasions - such as state ceremonies, major sports competitions, royal weddings or funerals - that interrupt the usual course of programs and life. These events are shown with great respect and ceremony, turning television viewing into a kind of civil rite. By participating in this ritual, viewers not only receive information, but also confirm the fundamental values and norms of their community, thereby renewing and strengthening social cohesion. Television, therefore, acts as a central mediator in the weaving of the visual and ritual fabric of modern communities. By synthesizing an image that provides evidence and a sense of simultaneity that creates community, it makes the imaginary community visible and present in the daily life of its members.

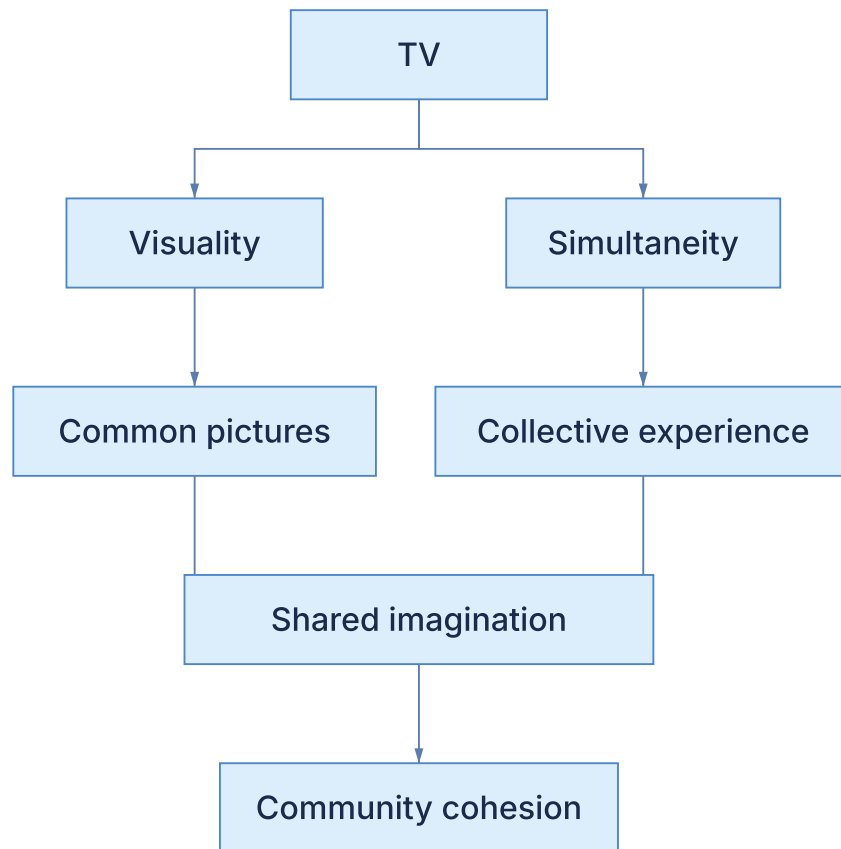


Figure 2.19: Schematic representation of the role of television in the constitution of the shared imagination, connecting its fundamental mechanisms (visuality and simultaneity) with social outcomes (strengthening of imagined community and cohesion).

2.6.3 Forerunner of the information society

The emergence of the information society, although it is often associated with the second half of the twentieth century and the accelerated development of digital technology, has its deep roots in the theoretical assumptions and visionary ideas that preceded it. These ideas, formed before there were technical possibilities for their full implementation, created the intellectual foundation for understanding information as a central resource and a key driver of social change. Among the thinkers whose work was decisive, Vannevar Bush and Norbert Wiener stand out, whose concepts permanently shaped the architecture of the contemporary digital world. Vannevar Bush in

his famous essay *As We May Think* (Bush, 1945) presents a vision that prophetically announced the future of knowledge management. The central concept of that essay is *memex*, an imagined device that embodies a completely new way of organizing and retrieving human knowledge. Bush described *memex* as a privatized, mechanized archive – a kind of desk where an individual could store all their books, records and communications. Its true novelty, however, does not lie in the mere possibility of storage, but in the principle of associative indexing. This mechanism, opposed to rigid alphabetical or numerical classification systems, was supposed to imitate the non-linear and jumpy connections created by the human mind. The user could create traces or paths between different documents, thus linking related ideas and data regardless of their original form or location. It was this idea of interconnected information that directly inspired the later creators of hypertext, such as Ted Nelson and Douglas Engelbart, and today is reflected in the very structure of the World Wide Web. Almost at the same time, Norbert Wiener (1948) in his work *Cybernetics: Or Control and Communication in the Animal and the Machine* lays the foundations of cybernetics, a scientific discipline that studies the principles of control and communication in living beings and machines. Wiener's thought is focused on the universal laws that govern complex systems, regardless of whether they are biological or mechanical in nature. At the center of cybernetic thought is the notion of a feedback loop (*feedback loop*), a fundamental mechanism of self-regulation that enables the system to adapt and maintain stability based on received information about its own actions. By studying how systems collect, process and use information to guide their behavior, cybernetics has provided a theoretical framework for the development of artificial intelligence, robotics and computer networks. She introduced the notion of a system as an agent that actively operates within its environment, thus laying the foundation stone for understanding machines that can learn and make decisions independently. Although they operated in the period before the full digital revolution, Bush's vision of networked knowledge and Wiener's theory of self-managing information systems form the two fundamental pillars on which the architecture of the information society was later built. Bush predicted *how* we would access information, while Wiener explained *how* systems, including

technological ones, could manage that information. Their work testifies to the fact that the information society is primarily the result of intellectual evolution, and only then of the technological revolution.

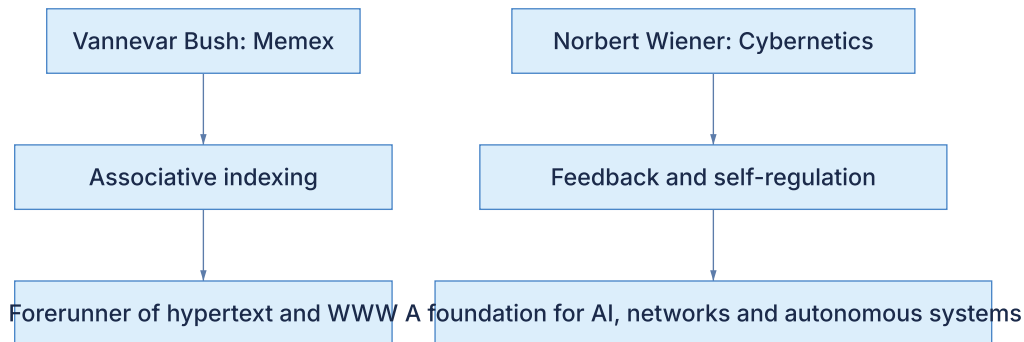


Figure 2.20: The intellectual foundations of the information society – Bush’s memex and Wiener’s cybernetics as a foundation for networked knowledge and self-regulating systems.

2.7 Internet and digital revolution

At the very turn of the 20th and 21st centuries, at a time when it seemed that the fundamental communication challenges had already been solved by the establishment of fast and global connections via telephone, radio and television, a completely new communication ecosystem appeared on the horizon: the Internet. In a period of just a few decades, this system, originally conceived as an experimental project of the American agency ARPA (*Advanced Research Projects Agency*), has grown into a fundamental infrastructure for information exchange, collaboration and social change (Leiner et al., 2009). Its origin, ARPANET, was designed as a research network to connect scientists and universities across the United States. Among the first institutions to join the project were the University of California (UCLA) and the Stanford Research Institute (SRI), which enabled their researchers to exchange data and coordinate experimental work much faster. In 1969, the first, now legendary network message was sent. The attempt to send the word “LOGIN” was interrupted after entering the letter “LO” due to a system crash, but this technical failure was recorded as a symbolic beginning of a new era. What in

the early seventies seemed like a technological idea reserved for a narrow circle of the scientific community, soon transformed into a fundamental invention that will mark the entire subsequent period of digital communication.

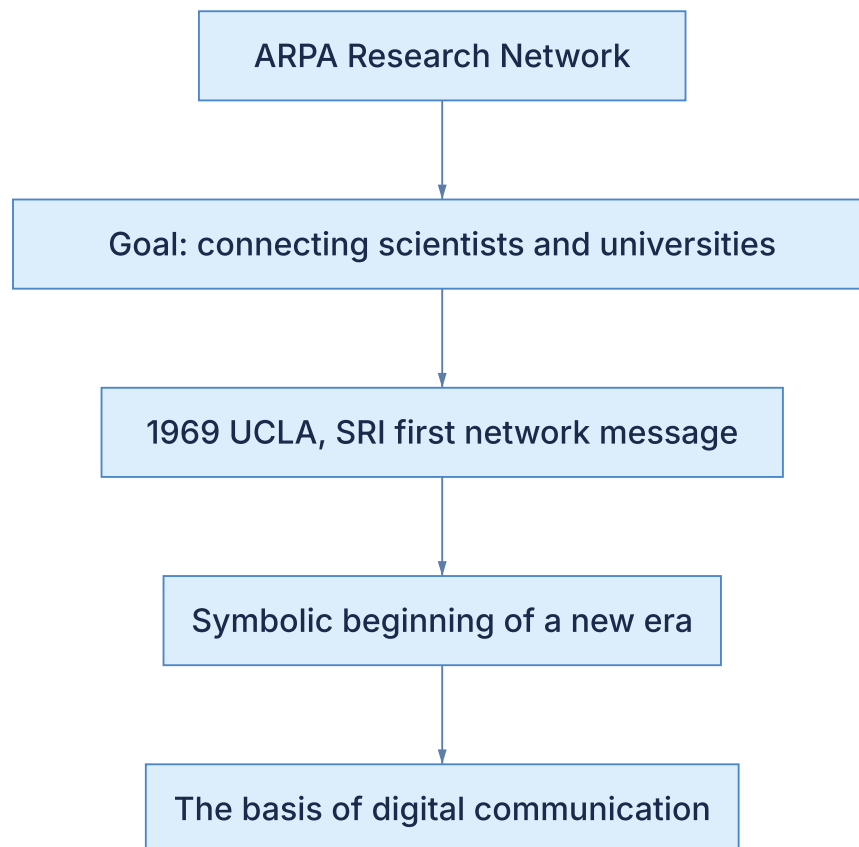


Figure 2.21: Schematic representation of the development of the ARPANET: from the ARPA research network and university connectivity to the first network message in 1969 and the foundations of digital communication.

A turning point in its development took place in 1989, when Tim Berners-Lee, a scientist employed at CERN, designed the *World Wide Web* (Berners-Lee, 1990) and thus transformed it into a platform accessible to every user. Using hypertext links and simple Internet browsers, people were given the opportunity for the first time to navigate distant digital spaces, simultaneously sharing documents, images, and even early forms of multimedia content. The power of the Internet is not only manifested in its technical basis, but primarily in its ability to fundamentally reshape social, economic and cultural paradigms.

By enabling an almost instantaneous and unhindered exchange of information on a global scale, the Internet has abolished previous spatial and temporal limitations, establishing a new information society in which access to knowledge and data becomes a key resource (Castells, 2010). This has dramatically changed the ways in which individuals and communities communicate, work and spend their free time, creating entirely new forms of social interaction and cultural expression. The development of online platforms and tools in the last three decades has proven to be the central driver of profound transformations in the way people communicate with each other, collaborate and jointly shape social relationships. Although these processes have opened up enormous opportunities for networking, exchange of ideas and the creation of global communities, at the same time they have fueled numerous controversies, primarily those related to issues of privacy and the quality of newly formed relationships. In economic terms, the digital revolution fueled by the Internet has led to the creation of new markets and business models that were unimaginable in the analog age. E-commerce, the sharing economy and remote working are just a few examples of disruptive innovations that have transformed entire industries. In such an environment, the role and skills of business agents also had to adapt. Successful operation in the modern market today implies a high level of digital literacy - the ability to navigate the digital ecosystem, use specialized tools and understand new forms of market communication. Although the reach of the digital revolution is undoubtedly enormous, it would be wrong to ignore its darker sides and the challenges it poses to modern society. One of the fundamental problems is the so-called *digital divide*, i.e. the growing difference between those who have access to information technology and the benefits it provides and those who are excluded from this process due to economic, social or geographical reasons (Norris, 2001). Furthermore, ubiquitous digitization has also opened up complex issues of privacy protection, as vast sets of personal data are constantly being collected and processed. In parallel, the ease of information dissemination via the Internet has created fertile ground for misinformation and malicious propaganda, which threatens the foundations of public debate and trust in institutions. The fight against fake news and cyber threats requires not only the development of new technological and regulatory frameworks but

also the systematic strengthening of the media and digital literacy of the entire population. The ability to critically evaluate information therefore becomes a key skill for navigating the complexity of the digital age.

2.7.1 Expansion of communication channels

E-mail stands out among the first mass-accepted Internet services, which in the nineties of the 20th century introduced a radical shift in everyday business and personal information exchange. Unlike traditional mail and fax, e-mail has enabled instant delivery of messages regardless of geographic distance (Shapiro & Varian, 1999). For example, global companies have begun to connect their offices on different continents almost in real time, speeding up decision-making and raising the level of coordination between teams. At the same time, private users used e-mail to communicate with family and friends on a daily basis, making the world a smaller place at minimal cost and with unparalleled speed of data exchange.

Modern business completely changes the way companies communicate with clients: communication is moving away from a one-time sales transaction towards a complex process of building long-term relationships. Therefore, presence and availability on multiple communication platforms are no longer a competitive advantage, but a necessity. This approach is called multi-channel communication in professional literature and is considered one of the key factors for high client satisfaction. It includes traditional channels such as telephone and e-mail, but also extends to more dynamic platforms such as social networks, online *chat* services and mobile applications intended for users.

Multichannel communication: Strategy of the simultaneous presence of an organization or agent on multiple communication platforms (telephone, e-mail, social networks, chat, mobile applications) in order to enable users to freely choose the channel that suits them best; a higher degree of integration, known as an omnichannel approach, aims for a unique and seamless user experience where the boundaries between channels are almost imperceptible.

Introducing a multi-channel approach brings significant business breakthroughs. It expands the reach and visibility of the company towards a wider circle of potential users and significantly enriches the overall user experience. Namely, clients are given the freedom to choose the communication path that suits them best at a given moment, which contributes to the feeling of comfort and adaptability of the service to their needs. However, it should be emphasized that the implementation of a multi-channel strategy is a significant organizational issue. It requires the systematic education of agents who must be trained in the skillful use of all available tools. In doing so, it is crucial to ensure that messages and information provided to clients are aligned and consistent across all channels. Consistency in information and service quality, regardless of the chosen channel, becomes the basis of trust. Any deviation, i.e. providing contradictory or inconsistent answers, inevitably creates a feeling of insecurity and frustration in the client, which in the long run can damage the company's reputation. Therefore, real and meaningful integration across multiple channels is actually set as the ultimate goal. This higher level of development, known as the **omnichannel** approach, implies the creation of a unique and flawless user experience in which the boundaries between individual channels are almost imperceptible. In a fully integrated system, the agent communicating with the customer has a complete view of all previous interactions, regardless of whether they took place via live chat, email or phone call. A client can start an inquiry on one platform, continue it on another and conclude on a third without having to repeat information, making the entire process fluid and user-centric.

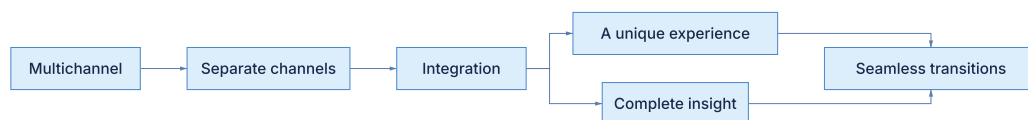


Figure 2.22: Schematic representation of the transition from multi-channel communication to an omni-channel approach, with channel integration, complete insight and seamless transitions for the user.

2.7.2 Collaboration and collaborative tools

In the upgrade to e-mail, various collaborative systems such as *chat*-rooms (*Chat-room*), forums and instant messages (e.g. ICQ, AIM, MSN *Messenger*) appeared. These early forms of digital exchange soon contributed to the development of innovative platforms for group work in the cloud, such as *Basecamp* or *Slack*, which significantly simplified team collaboration on projects, especially when it comes to software development, organization of international conferences or coordination of distance learning (Benkler, 2006). The term cooperation in the context of multi-agent systems means the basic principle of action in which two or more agents, be they artificial or human, coordinate their activities in order to achieve a common goal. Cooperation is fundamentally different from coordination because it does not only mean avoiding conflict in action, but also active and purposeful exchange of information and resources. The basic premise of any cooperation is the existence of a common goal to which all involved agents strive. Achieving such a goal, which goes beyond the possibilities of individual action, necessarily implies a dynamic exchange of knowledge and information. This exchange enables agents to make better decisions.

Cooperation in multi-agent systems: The principle of joint action of two or more agents (human or artificial) that coordinate activities to achieve a goal that exceeds the possibilities of individual action; it differs from mere coordination because it involves the active exchange of information and resources, negotiation and conflict resolution mechanisms, and the building of trust among participants (Benkler, 2006; Wooldridge, 2009).

However, the collaborative process is not without its challenges, among which the possibility of conflicting interests or agendas among the agents stands out. Therefore, the existence of pre-defined or dynamically established mechanisms for negotiation and conflict resolution is necessary for successful cooperation.

These mechanisms ensure that partial interests are subordinated to the general goal and that the coherence of joint action is maintained.

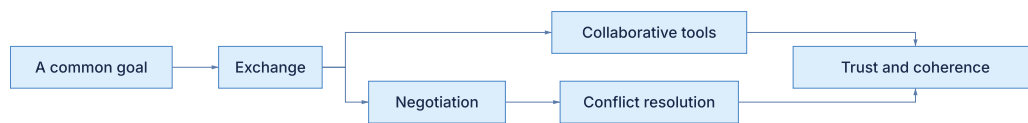


Figure 2.23: Basic elements that make up the process of cooperation in multiagent systems: common goal, information exchange, negotiation and conflict resolution mechanisms, and collaborative tools.

In practical application, especially in complex environments where both humans and artificial agents operate, these principles are realized by using **collaborative** tools. These are software solutions and platforms designed to facilitate and structure interaction between participants. Their function is manifested on several levels: from tools for simultaneous editing and sharing of documents, through systems for delegating and monitoring tasks, all the way to communication platforms that enable asynchronous and synchronous communication. The role of these tools is to encourage and maintain trust, which is crucial for any long-term and fruitful cooperation.

Collaborative tools: Software solutions and platforms designed for the structured joint action of multiple participants - from tools for simultaneous editing of documents and file sharing, through systems for delegation and monitoring of tasks, to communication platforms for synchronous and asynchronous exchange; in the context of multi-agent systems, collaborative tools encourage and maintain trust and enable effective coordination between human and artificial agents (Benkler, 2006).

2.7.3 Mass communication and homogenization of culture

Mass communication, as one of the fundamental processes that shape modern societies, represents a complex phenomenon whose influence extends to all aspects of social life, and especially to the field of culture. By means of technological innovations - from print and radio to television and the Internet - channels were created that enabled almost simultaneous transmission of messages to a vast, spatially dispersed and heterogeneous crowd. In this

process, the mass media are not only channels for the transmission of information, but act as powerful *agents* of social change, actively participating in the creation, shaping and dissemination of cultural patterns, values and worldviews. The central consequence of such action, which is hotly debated in sociological and cultural thought, is the **homogenization** of culture. It is a process of gradual reduction of cultural diversity and suppression of autochthonous, local cultural expressions under the influence of dominant, globally expanded cultural contents. Mass communication, in its essence, becomes the main bearer and mediator of these dominant contents, thereby creating a global shared symbolic environment. Unique cultural patterns, which originate predominantly from Western, especially Anglo-American cultural centers, spread on a planetary level and are adopted in different parts of the world. This process is most clearly manifested in several key domains. In the field of audiovisual culture, the Hollywood film industry and the global music scene impose aesthetic standards, narrative structures and musical genres that become universally recognizable and accepted. Similarly, in the sphere of consumer culture, global brands and marketing campaigns promote unique lifestyles, fashion trends, and consumer habits, erasing regional and national particularities in dress, diet, and everyday practices (Thompson, 1995). The influence on **language**, as the fundamental bearer of cultural identity, is particularly significant. The spread of English as a global *lingua franca* through media content leads to the mass adoption of English in national languages. This seemingly unequivocal linguistic borrowing becomes a process that can affect the deeper semantic and syntactic structures of the language, changing the ways in which thoughts and feelings are expressed within a particular language community. This exposes linguistic diversity, one of humanity's greatest assets, to the pressure of unification.

However, it would be wrong to consider the process of cultural homogenization as one-way and absolute. Contemporary theories of globalization warn of the existence of resistance and counter-tendencies, whereby local cultures often actively reinterpret global influences, adapt them to their own contexts, and combine them with indigenous elements. This complex process is called hybridization or glocalization - a phenomenon in which the global and the local intertwine, creating new, unique cultural forms (Robertson, 1994). For

example, globally popular music genres are enriched with traditional instruments or local chants, and international television formats are adapted to national humor and social specifics.

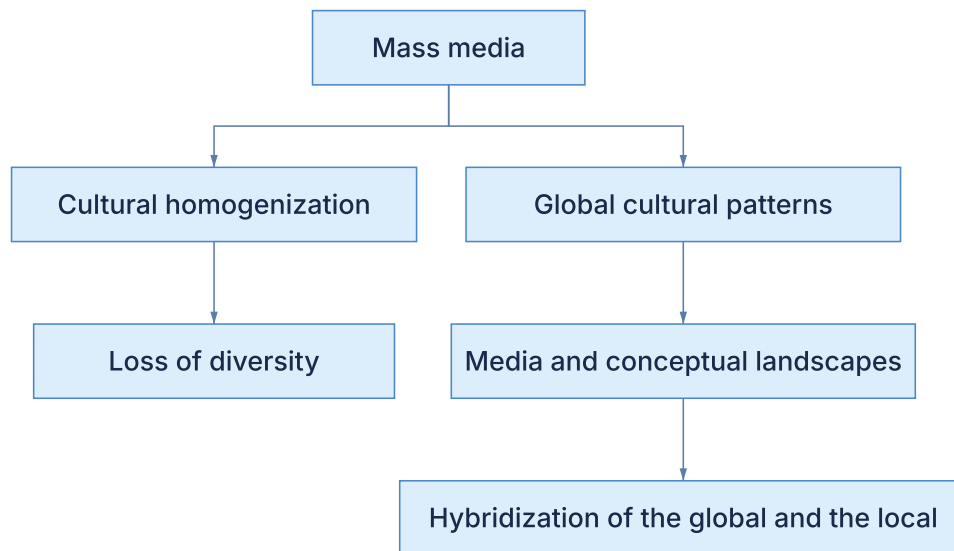


Figure 2.24: Presentation of the relationship between mass communication, homogenization and glocalization of culture - media and ideational landscapes and the hybridization of the global and the local.

The relationship between mass communication and culture remains inherently dialectical. While the mass media undoubtedly possess enormous potential for unifying cultural practices and values on a global scale, they also provide platforms for the affirmation and visibility of local identities. Cultural homogenization therefore remains a permanent tension between the forces of global integration and the need to preserve local distinctiveness, a tension that will decisively shape the cultural landscape of the 21st century.

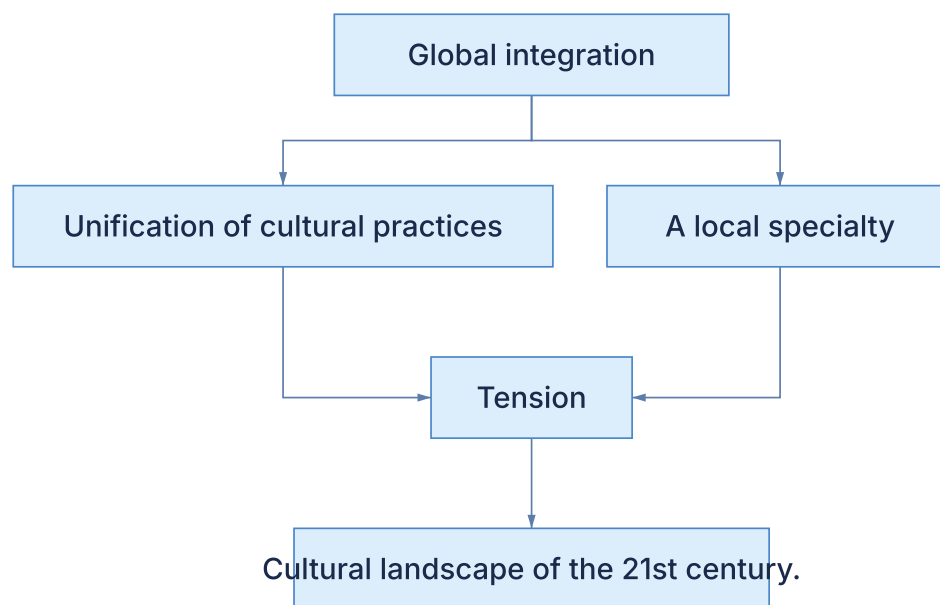


Figure 2.25: Presentation of the relationship between mass communication, homogenization and glocalization of culture - the dialectic of global integration and local particularity.

2.7.4 Political sphere and electronic public opinion

The emergence and pervasiveness of the Internet fundamentally transformed the landscape of political communication, introducing new paradigms in shaping, expressing and measuring public opinion. The digital age has put an end to the former exclusive dependence of political actors on traditional mass media as intermediaries in the transmission of messages to voters. Instead of a one-way channel, political communication now takes place within a complex, multi-directional and interactive network structure. In this new environment, a phenomenon that we call *electronic public opinion* - a dynamic and often elusive sum of attitudes, feelings and opinions expressed on digital platforms such as social networks, forums, blogs and portals with comments - arises and operates. The key determinant of this transformation is the change in the role of the citizens themselves. From the position of mostly passive recipients of political messages, they become active *agents* in the political process. The ability to instantly comment, share content, create their own narratives and organize collective actions gives them power that they did not have in the pre-

internet era. This displaces the political discussion from formalized frameworks, such as parliament or media debates, into a much wider, more open, but also more chaotic space of the Internet. In such a space, official political narratives confront unfiltered experiences, criticisms and alternative viewpoints of citizens, which creates a polyphonic, but also polarized public sphere. As Castells (2009) notes, power in the information age is increasingly found in communication networks, and the ability to influence public discourse becomes a key resource of political struggle.

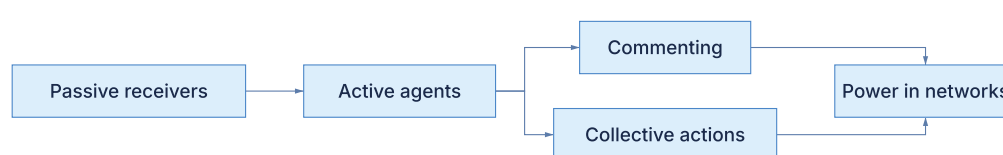


Figure 2.26: Schematic representation of the transformation of citizens from passive recipients to active political agents through commenting, sharing content, collective actions and acting in networks.

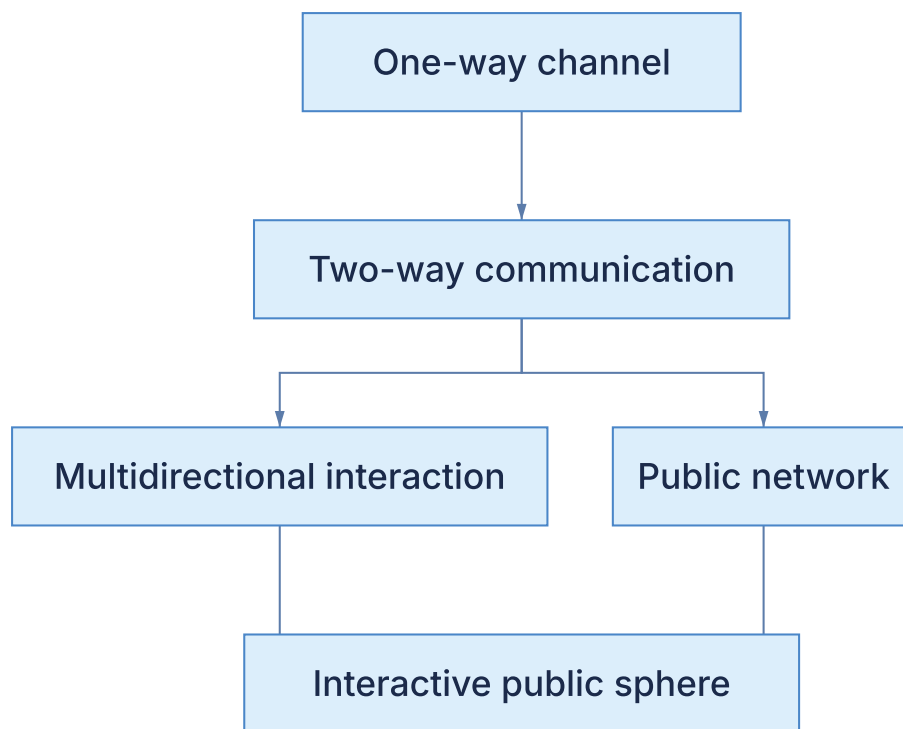


Figure 2.27: Transformation of the communication model in the political sphere – from a one-way channel to a multi-directional, interactive network structure and electronic public opinion.

Furthermore, electronic public opinion has properties that significantly differentiate it from that measured by traditional survey research. Its main feature is the speed of creation and changeability; it reacts to events in real time and can change drastically within hours under the influence of new information or viral content. It is also more transparent in its creation because it is based on publicly available data - posts, comments, likes and shares. However, these very properties make it susceptible to manipulation. Organized groups, be it political parties, interest lobbies or foreign actors, can use automated accounts (*bots*) or coordinated campaigns to create a false impression of mass support or resistance to an idea, law or candidate. The structure of the Internet public sphere also brings other challenges. In contrast to Habermas' ideal image of a unified public sphere in which rational discussion leads to consensus, the digital space is extremely fragmented. Users tend to gather in like-minded communities, creating so-called *echo chambers*

or *filter bubbles*, where they are exposed only to those information and attitudes that confirm their existing beliefs (Sunstein, 2017). Such informational isolation makes it difficult for different social groups to understand each other and strongly contributes to political polarization. The speed and reach of digital communication, although carrying enormous potential for democratization, at the same time open up space for the unprecedented spread of misinformation and fake news, the verification of which for the average user represents a significant cognitive effort. The political sphere consequently faces a double challenge: on the one hand, it must adapt to new forms of citizen participation and use their democratic potential, while on the other hand, it must develop mechanisms to defend against manipulations and preserve the quality of public debate in the digital environment.



Figure 2.28: Schematic representation of the fragmentation of the Internet public sphere: from echo chambers and filter bubbles to polarization and difficult mutual understanding.

2.7.5 Global spread and cultural influence

The universal networking of the modern world, achieved through digital technologies and the Internet, is a fundamental prerequisite for the global dissemination of cultural content and ideas. This digital infrastructure breaks down spatial and temporal barriers, enabling the almost instantaneous dissemination of information at an unprecedented level. This creates frameworks within which cultural phenomena, regardless of their origin, acquire the potential for planetary distribution. Various agents operate within this newly established communication framework, such as social media platforms, influential individuals (*influencers*) and specialized internet communities. Their role on the superficial level is manifested in mediating or distributing content, but they are essentially active participants in the processes of creating, adapting and reshaping the meaning of cultural

products. Through their action, the contents are spread, but also interpreted, commented on and incorporated into new semantic networks. As already mentioned, globally available cultural phenomena go through the process of localization, i.e. adaptation to specific cultural, linguistic and social contexts. This combination of the global and the local is called *glocalization*, and it manifests itself, for example, in the way internet memes are translated and adapted to resonate with local humor and social circumstances (Shifman, 2014). The content thus retains its recognizable core, but at the same time takes on new, locally determined meanings. Such permeation is the origin of hybrid cultural forms that combine elements of different traditions. This eliminates the sharp dichotomy between, for example, Western and non-Western culture, because cultural influence becomes multidirectional and mutual. The global scene is increasingly becoming an arena where different cultural expressions meet, mix and create new, syncretic forms that testify to the complexity of contemporary identities. The described processes also challenge simplistic views such as technological determinism - the point of view according to which technology unequivocally shapes society and culture. On the contrary, it is evident that users and communities actively appropriate technology, adapt it to their needs and use it in unforeseen ways, thereby

becoming agents of cultural and social change themselves (Baym, 2015). The complexity of these relationships can be shown in the following diagram:

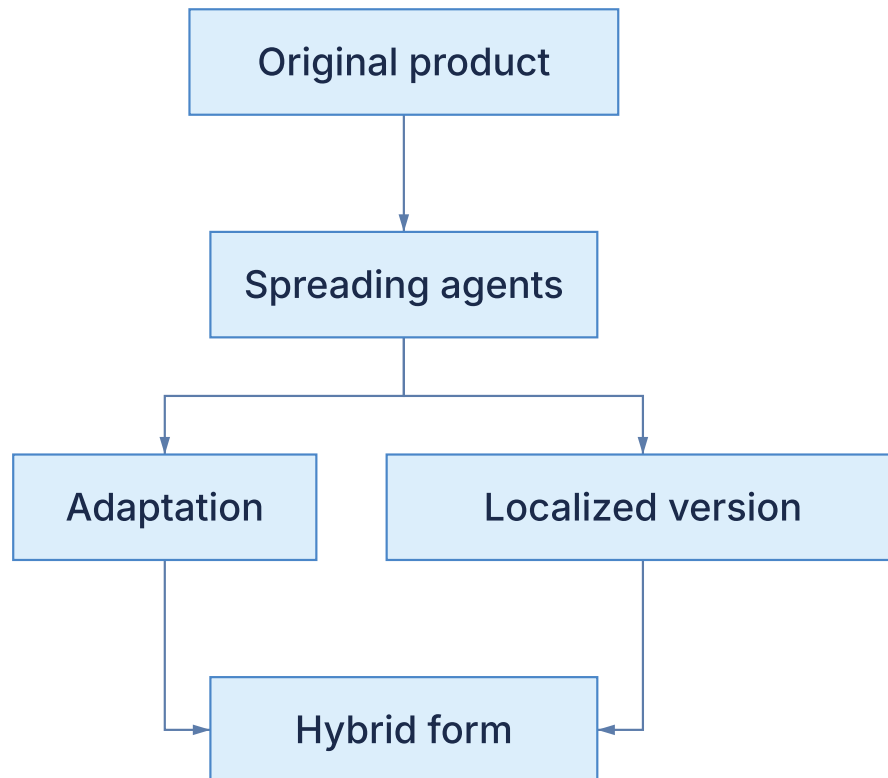


Figure 2.29: The process of cultural dissemination and hybridization, from the original cultural product through the agents of dissemination and the process of adaptation to the localized version and the hybrid cultural form.

Finally, the global distribution of cultural content also opens up complex ethical issues. Concepts such as cultural appropriation, decontextualization and commodification of traditional practices are becoming central topics in discussions about the consequences of digital cultural permeation, seeking a constant reconsideration of power relations in the globalized world.

2.7.6 Expansion of social networks

After the consolidation of the fast digital communication paradigm, the late 1990s and early 2000s saw the emergence of the early generation of social networks, such as *SixDegrees.com* (launched in 1997). Named after the concept of *six degrees of separation*, suggesting that any two people on Earth can be connected by a chain of six or fewer acquaintances, the site allowed users to create profiles, connect to lists of friends and family, effectively laying the groundwork for the concept of online social networking. Although initially limited in reach, *SixDegrees* paved the way for later mega-platforms such as *MySpace* (2003) and *Facebook* (2004), on which the momentum of mass online socializing was traced (Boyd & Ellison, 2007). The real turning point was the appearance of *Facebook* in 2004, a platform that fundamentally changed the paradigm of digital interaction. The success of that network rested on a previously new approach: instead of creating anonymous virtual identities, the focus was placed on mapping and establishing existing social connections within the digital environment. This encouraged the use of real names and connecting with people from the user's immediate social reality, which created a sense of authenticity and trust that previous networks lacked (Boyd & Ellison, 2007).

Further diversification of forms of digital communication was manifested in the emergence of *Twitter*, which introduced a concise microblogging format. By limiting messages to a small number of characters, it enabled an unprecedented speed of flow of information, news and personal announcements in real time. At the same time, technological progress, primarily the widespread availability of smartphones, has directed development towards mobile applications and visual content. Platforms like *Instagram* and *Snapchat* built their foundations precisely on the exchange of photos and short videos. They thus went beyond the mere function of a communication tool and became prominent spaces for the construction and expression of personal and collective identity (Van Dijck, 2013). Development also went in the direction of functional specialization, which is most evident in the establishment of professional networks such as *LinkedIn*, which have become an indispensable arena for career development, business networking

and presentation of professional achievements. The contemporary moment is marked by the predominance of platforms based on short video format, among which *TikTok* stands out. Its peculiarity lies not only in its content, but primarily in the way it is distributed, where the algorithmic system takes a central role. In such a model, the algorithm becomes the key agent that mediates between the content and the user, actively and constantly shaping his entire experience with personalized recommendations. Finally, it is important to point out that the implications of the expansion of social networks are social, cultural and political. These platforms have assumed the role of central arenas for public discourse, political organizing and social activism, and at the same time have become fertile ground for the systematic spread of misinformation and manipulative content (Zuboff, 2019). Therefore, their influence on contemporary society should be understood in all its complexity and contradictions, as a phenomenon that reshapes the ways in which we communicate, work, create communities and understand ourselves.

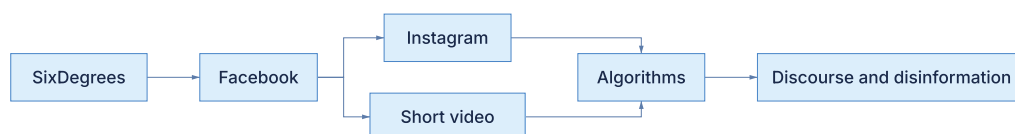


Figure 2.30: Schematic representation of the development of social networks from SixDegrees and Facebook towards visual platforms, short video, algorithmic distribution and effects on public discourse and misinformation.

2.7.7 Digital Tribes and Knowledge Squares: Anatomy of Successful Online Communities

Soon after the initial formation of broad and general digital markets, the Internet landscape began to blossom into specialized ecosystems—virtual communities dedicated to creating, sharing, evaluating, and discussing very specific types of content or interests. Platforms have become fascinating social laboratories where users gather, developing their own governance mechanisms, reputation systems, and unique microcultures that shape behavior and interaction within their digital walls. Take *Reddit* for example. That platform is not so much a single community as a vast, *sprawling* digital

metropolis composed of thousands, if not millions, of thematic sub-communities known as *subreddits*. Each *subreddit* functions as a separate neighborhood or even a miniature digital city-state, often with its own carefully crafted set of rules, norms of behavior, and a team of volunteer moderators who act as a kind of local government (Massanari, 2017). The spectrum is incredible: from strictly moderated academic forums like *r/science* or *r/AskHistorians*, where detailed references and in-depth analysis are required, to chaotic and humorous corners like *r/funny* or niche communities dedicated to the narrowest possible interests. What gives *Reddit* its dynamism is its basic *upvote/downvote* mechanism, which acts as a collective curation machine: content that the community finds valuable or interesting rises to the top, while less relevant or low-quality content is relegated to the digital background. It is a continuous referendum on relevance. If we compare *Reddit* to a bustling city with countless different neighborhoods, then *Stack Overflow* (part of the wider *Stack Exchange* network) is its precisely organized technical library, or perhaps a highly specialized guild home for developers. Born from the need for quick and accurate answers to specific technical questions, *Stack Overflow* has become an almost unavoidable reference point in the world of software development (Mamykina et al., 2011). Its success lies in its rigorous focus on quality and meritocracy. The system of scoring (reputation) and accepting the best answers creates a strong incentive to provide accurate, clear and useful solutions. The community actively maintains a high signal-to-noise ratio, discouraging vague questions or opinion-based answers, creating an extremely efficient mechanism for collective problem-solving and archiving of specific technical knowledge. *Discord* followed a different evolutionary path. Originally launched as a tool for voice communication among video game players, it quickly outgrew its initial niche and became a platform for a wide range of communities (Seering et al., 2019). Unlike asynchronous forums like *Reddit* or *Stack Overflow*, *Discord* offers a sense of real-time presence and interaction, more akin to a network of vibrant digital clubs or community hubs. Its server-based structure, with channels organized by topic (text, voice, video) and the ability to integrate bots for task automation and moderation, has proven to be extremely flexible. Today, it is used not only by players, but also by educational institutions, art collectives, support groups, and even companies for internal

communication and external community building, creating spaces for more fluid and immediate interaction. Finally, we must not forget open-source projects and collaborative repositories like *GitHub*. Although they are not social networks in the classical sense, platforms like *GitHub* function as huge digital construction sites and workshops, and at the same time as powerful communication networks that connect millions of developers and technical experts around the world (Dabbish et al., 2012). The combination of code versioning tools (such as *Git*), issue trackers, pull requests, and integrated discussion forums creates a highly structured communication framework optimized for complex technical collaboration. The success of projects like the operating system *Linux*, the *web*-framework *React* or the AI library *TensorFlow* testifies to the incredible power of this model of distributed collective intelligence, where thousands of individuals, often without direct contact, manage to build together some of the most complex software artifacts today. *GitHub* in this new ecosystem of machine-machine collaboration turns into a kind of operational base of autonomous code agents: Google's Jules, for example, opens *pull-requests* and automatically suggests corrections based on the analysis of the complete repository, relying on *GitHub*'s API for versioning and code review flows. A similar symbiosis is already demonstrated by *OpenAI-ev Codex* – the engine that powers *Copilot* – because it learns from public repositories and generates ready-made code in a format that can be immediately incorporated into those same projects. New tools called *vibe-coding* (*vibe-coding*) like *Windsurf* or *Cursor* go a step further: the agent analyzes the history of *commits*, writes tests and runs CI *pipelines*, making *GitHub* a shared working memory through which people and models are coordinated in real time. Such an infrastructure allows software development to gradually move from manual code editing to process orchestration, where people set goals and a distributed *agent swarm* on *GitHub* realizes tasks – from bug discovery to documentation writing – with transparent audit history and collective approval mechanisms. These diverse examples – from the city-states of *Reddit* and the guilds of *Stack Overflow*, through the clubs of *Discord* to the global construction sites of *GitHub* – illustrate the richness and inventiveness of digital forms of association in *online* communities. In these virtual spaces, individuals, driven by common interests, passions or goals,

establish lasting connections, build social capital and participate in the collective creation of value (Rheingold, 2000). In order to penetrate into the anatomy of their success, it is necessary to look at the fundamental elements that make them stable and purposeful. A fundamental prerequisite for the creation and survival of any living *online* community is the existence of a strong and clearly defined *common interest*. Whether it's professional collaboration, hobbyist passion or mutual support, it's this central gathering point that gives the community its identity and meaning. It acts as a gravitational center that attracts new members and retains existing ones, ensuring internal cohesion. Without this common purpose, the community dissipates into a multitude of unrelated interactions and loses its reason for existence. The key pillar of every successful community is built on this foundation: *mutual trust and social capital*. Trust is not a given, it is gradually acquired through repeated positive interactions, reliability and respect for established norms. The accumulation of such experiences creates social capital, that is, a network of relationships and reciprocity that enables members to cooperate effectively and feel safe (Putnam, 2000). In such an environment, members are more willing to share valuable information, openly ask for help, and invest their own time and effort for the benefit of the whole. Each digital tribe develops its own system of *rules and norms of behavior*, whether they are explicitly codified in the form of forum rules or exist as tacit, implicit agreements. These rules regulate communication, resolve conflicts and protect the community from destructive behavior. In addition to formal rules, the development of specific *language, jargon and symbols* that are understandable only to members is of crucial importance. Such an internal code not only facilitates and accelerates communication on complex topics, but also strongly strengthens the sense of belonging and collective identity, clearly demarcating "us" from "them".

The role of *agents*, both human and software, in maintaining community health is extremely important. Human agents, such as *moderators* or *community managers*, act as gardeners of the digital space: they encourage discussion, guide new members, calm tensions and remove unwanted content. Their presence and active engagement guarantee that the community remains a safe and productive space. On the other hand, software agents, through

algorithms for filtering, recommending content or connecting users with similar interests, automate key functions and allow the community to scale to a large number of members. Finally, the success of a community is reflected in the value it creates for its members and, sometimes, for the general public. This value can take different forms: from solving specific problems and *sharing professional knowledge* to providing *emotional support* and a sense of belonging. In their most developed forms, *online* communities become the scene of the emergence of *collective intelligence*, where the synergy of individual contributions leads to insights and solutions that exceed the capabilities of any individual (Lévy, 1997). Seen in this way, *online* communities stop being just a technological phenomenon and become key factors in the knowledge society, capable of solving complex challenges through collaboration and trust built in the digital world.

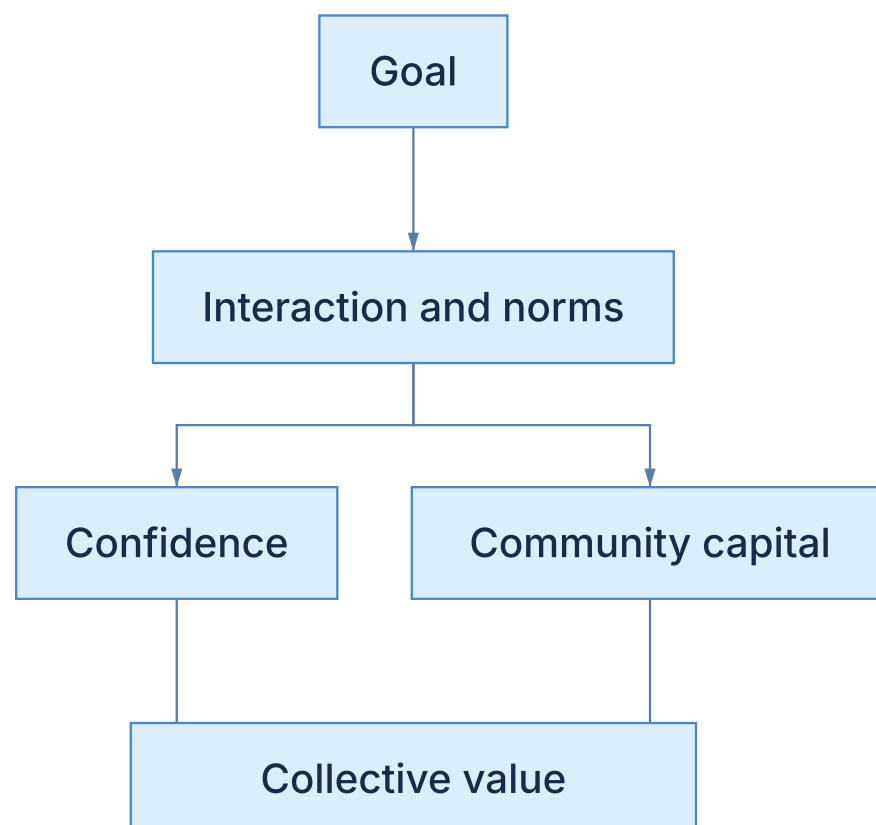


Figure 2.31: Schematic representation of the emergence of the collective value of an online community: from shared purpose and interaction to trust, social capital and shared benefit.

2.7.8 Porous membranes: when digital becomes real (and vice versa)

Once upon a time, the Internet was thought of as a separate space, a cybernetic world that one entered and exited from. But as digital platforms became more deeply woven into the fabric of our everyday lives, that boundary began to melt, becoming increasingly porous, like a membrane through which ideas, relationships, and identities constantly penetrate in both directions. Virtual communities were no longer just remote sanctuaries; their inhabitants began to spill over into the physical world, organizing meetups, conferences, protests, and get-togethers that blurred the distinction between *online* acquaintances and *offline* friends (Rainie & Wellman, 2012). The arrival of smartphones, those ubiquitous portals in our pockets that ensure our constant connection, has only ended the last illusions about a clear dividing line. Our *online* and *offline* lives are no longer parallel worlds; they are inextricably intertwined. Platforms like *Meetup.com* have become a perfect example of this phenomenon. They function as digital catalysts for physical interaction, bringing together people with similar interests – whether they're rare dog lovers, programmers learning a new language, activists organizing a cleanup, or book club members. The initial contact and group formation happens *online*, but the real goal is to meet face-to-face: gathering in coffee shops, parks, libraries or workshops (Quan-Haase & Wellman, 2002). There, abstract avatars and text conversations turn into real handshakes, shared laughs and lively discussions, demonstrating how the digital can enrich, not just replace, physical sociality and community building. It is a picture of networked individualism where connections are formed around common interests, regardless of geographic proximity, but often materialize in physical space (Wellman, 2001). However, this blurring of boundaries has not gone without causing deep discomfort and serious questions. While some celebrated new forms of connection, others expressed concerns about the authenticity of relationships nurtured through screens and the potential for the erosion of

privacy in a world of constant digital surveillance. Is the person we know through a carefully curated *online* profile the same person we meet in person? In her influential works, Sherry Turkle warned of the phenomenon of alone together, where technology creates the illusion of connection while actually isolating us, encouraging us to prefer orderly, controlled online interactions over the messy unpredictability of real human relationships (Turkle, 2011). Privacy concerns have exploded into the public consciousness with scandals such as *Cambridge Analytica*, which exposed how personal data collected from *Facebook* (often without user consent or understanding) can be used for sophisticated psychological profiling and targeted political persuasion, potentially influencing the outcome of elections (Cadwalladr & Graham-Harrison, 2018). This and similar incidents have illuminated the background of the business models of many platforms, based on what Shoshana Zuboff calls surveillance capitalism: the extraction and monetization of human experience transformed into data, often through unclear terms of use and interfaces designed to encourage us to share more than we would consciously consent to (Zuboff, 2019). Our *online* activity, our likes, clicks, and even keeping our eyes on certain content, become raw material for algorithms that profile us and predict our behavior, shaping the information we see and the products we are offered. At the same time, the constant presence of the digital world has fueled discussions about digital alienation, addiction and the impact on mental health. The design of many social networks, with endless scrolling, notifications, and variable reward mechanisms (like likes), can create compulsive behavior loops (Alter, 2017). The pressure to maintain a perfect virtual identity, constant comparison with others and fear of missing out (FOMO - *Fear Of Missing Out*) are associated with increased anxiety and depression, especially among younger users (Twenge, 2019). The line between using technology as a tool and becoming its prisoner is becoming dangerously thin. Thus, while the Internet has undoubtedly broken down barriers and enabled new forms of community and collaboration, it has simultaneously created a complex and often ambivalent space in which our digital and physical lives are constantly interwoven. The dividing line between the digital and the real world, once thought of as a solid and insurmountable border, today is increasingly manifested as a porous membrane. Through it, a constant and

two-way permeation takes place, which changes the ontological status of both worlds. It is no longer about the interaction of two separate entities, but about the creation of a hybrid space in which digital bits materialize and physical reality is translated into information code. This process of mutual overflow questions fundamental assumptions about the nature of reality, presence and action.

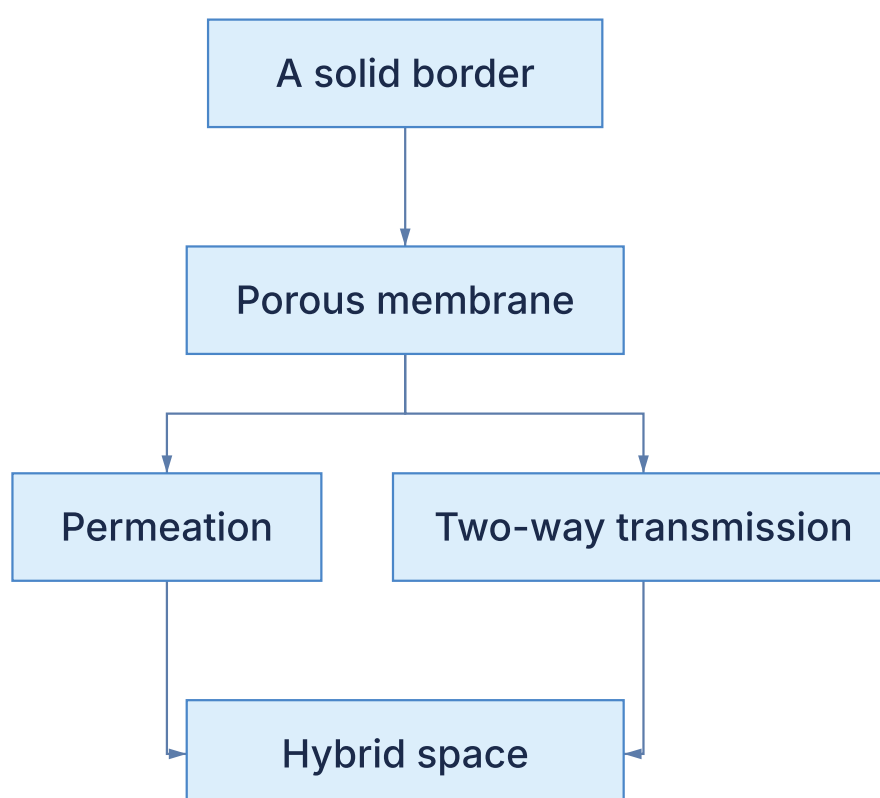


Figure 2.32: Schematic representation of the transition from a solid boundary between the digital and the real world to a porous membrane, two-way permeation and hybrid space.

2.7.9 Digital revolution and global communication network

The entry into the digital age has brought an atmosphere of continuous connectivity and accelerated global interactions, whereby the Internet and digital tools have become the logistical and cultural fabric of modern society. The shift from physical infrastructure and traditional forms of business to

virtual ecosystems has created an environment where collaboration takes place independent of time and space, and the range of information is available on a scale that would have seemed unimaginable just a few decades ago. Such pervasiveness of communication and information flows continues civilization's long line of innovation, but now it seems that the pace of change has intensified to the extent that it is fundamentally transforming many aspects of human experience. In the context of business and work processes, the construction of virtual market platforms illustrates these deeper changes. Examples of companies like *Amazon* and *eBay* show how the global distribution of goods and services has entered a new phase, characterized by hybrid models that connect traditional and digital trade modalities. At the same time, the practice of collaboration through online applications for conversations, document sharing and video conferencing has spread. By selecting collaborators regardless of their geographical location, organizations are increasingly fostering "dispersed teams", in which knowledge merges into a single whole. These practices parallel the accelerated "cultural globalization", in which local work culture meets universal standards of digital collaboration, bringing new challenges to organizations that strive to harmonize various work traditions with the fluidity of virtual collaboration.

Education has undergone a similar turn. *Online* courses and open educational resources provide access to content previously available mostly at elite institutions. The great popularity of platforms like *Courser*, *edX* and *Khan Academy* shows that the demand for flexible, adaptable and above all affordable forms of learning is constantly increasing. Many students and professionals for whom formal, long-term studies are not an option are now choosing microcolleges and specialized courses to adapt more quickly to the changing demands of the labor market or to follow their own interests in the arts, sciences or humanities. Changes in the media sphere are equally diverse. Increasing digital competition has put traditional newspapers, television companies and radio stations in front of the task of radical adaptation to new communication customs and advertising models. With the appearance of internet channels, blogs, *podcasts* and platforms for *videostreaming*, space has opened up for countless forms of creative expression. Discussions now take place on Internet forums (Rheingold, 1993), where professional knowledge or

niche passions are exchanged, and news broadcasts are often replaced by interactive broadcasts in real time, whereby the audience experienced a transformation from a passive observer to an active participant.

New economic circumstances were by no means absent. The concept of the sharing economy and platforms like *Uber* and *Airbnb* (Sundararajan, 2016) have shaken some established rules about ownership, rental and work, showing how agile digital mediation is in connecting supply and demand, even if it raises issues of regulation and protection of workers' rights. At the same time, open collaboration movements have reached a peak in *open-source* projects, where global communities develop complex software solutions without traditional hierarchical structures (Benkler, 2006). Such practice has shown that knowledge and innovation can flourish when monopolistic controls are broken and resources are freely available.

Similar collective mechanisms appear in the sphere of financing. *Crowdfunding* campaigns through platforms like *Kickstarter* and *Indiegogo* (Howe, 2008) reveal how a wider audience, not exclusively banks or corporate investors, takes on the role of micro-investors in creative and entrepreneurial projects. Such initiatives revitalize a culture of personal engagement, where supporters donate funds and spread the word about new ideas and actively build a community around them. In one dimension, this brings the empowerment of small innovators, and in another, it imposes the need to restore trust and transparency: in a digital world full of information, it is easy and uncertain to assess who offers what value and with what motives. All these phenomena point to the fact that the digital era, continuing the centuries-old traditions of communication evolution, is becoming a multi-layered experience – intimate and global, innovative and disruptive, promising and risky. It enables intense collaboration and great feats of knowledge discovery, but requires thoughtful strategies for managing privacy, ethics, and intellectual property. It opens up new frontiers for business, education and media, but invites in-depth discussions about the balance of power in the digital space and the control of infrastructure and resources. At the root of all these developments remains the key understanding that communication, for

centuries now, is the guiding thread of the development of civilization, and digital technologies are only the last act in the continuous performance of human creativity and interaction.

2.7.10 Digital duality: the promise of democratization and the shadow of new inequalities

The entry into the digital age is marked by a fundamental tension, a duality that permeates almost every aspect of modern life. Digital technology is the scene of conflicting aspirations that simultaneously offer the promise of liberation and cast the shadow of new forms of subjugation. *Digital duality* manifests itself as a permanent conflict between the vision of a democratized world, in which knowledge and power are available to everyone, and the actual appearance of new, often invisible, inequalities that deepen existing ones and create new social cleavages. Understanding this duality is crucial for any serious judgment of our time. On the one hand, the digital revolution has enormous democratization potential. By breaking down spatial and temporal barriers, the Internet and its related technologies have created unprecedented opportunities for access to information and knowledge. Global libraries, scientific databases and educational platforms have become available to individuals regardless of their geographical location or material status, promoting the idea of a global republic of knowledge. In the political arena, digital media have enabled new forms of civic participation and activism. The ability to quickly organize, share information outside the control of traditional media and directly address power holders gave a voice to those who rarely had one. The potential is manifested in the ability of social movements to quickly mobilize and articulate their demands, creating a new form of what Manuel Castells (2012) calls “network power”. In economic terms, digital platforms have lowered the entry barriers for entrepreneurship, allowing small producers and creators access to a global market that was once reserved exclusively for

large corporations. Digital duality contrasts the positive outcomes of the promise of democratization with the negative consequences of new inequalities

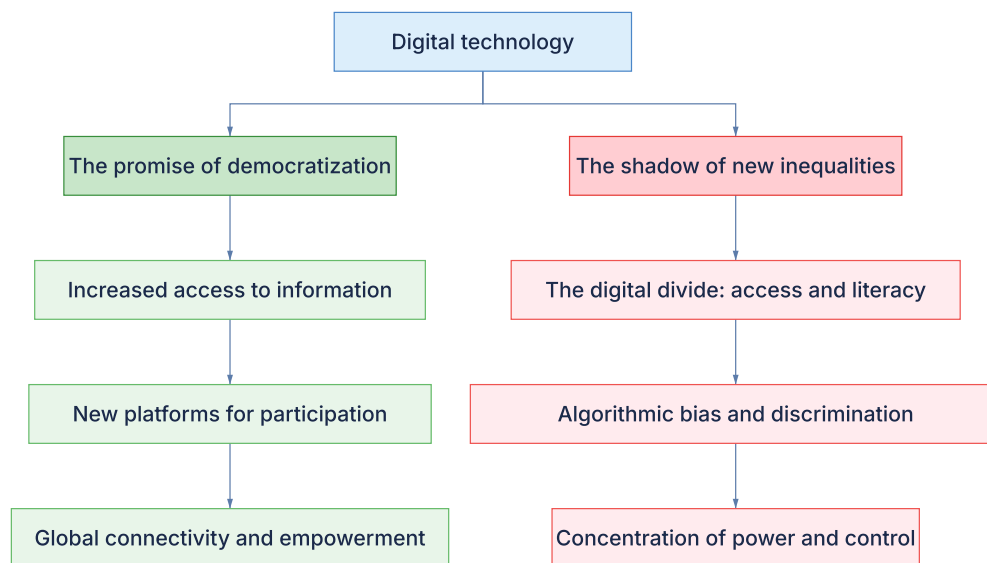


Figure 2.33: Digital duality – the promise of democratization (increased access to information, new platforms for participation, global connectivity) and the shadow of new inequalities (digital divide, algorithmic bias and discrimination, concentration of power and surveillance).

On the other side of this bright picture, however, a dark shadow looms. The promise of full democratization remains largely unfulfilled, while at the same time new, more subtle mechanisms of surveillance and inequality are being established. The first and most obvious is the *digital divide*, which no longer refers only to mere access to the Internet, but also to differences in digital literacy, skills and the ability to critically evaluate information. Those without the necessary skills or resources remain excluded from the digital society, further deepening their existing marginalization. Furthermore, the architecture of the digital world is increasingly shaped by algorithms that, rather than being impartial arbiters, often reflect and reinforce human biases. Algorithmic systems that decide on employment, credit approval or even the probability of committing a criminal offense can systematically discriminate against certain social groups, creating what Cathy O’Neil (2016) calls “weapons of math destruction”. These automated systems act as invisible bureaucracies, whose decisions are difficult to challenge because they are shrouded in a veil of

technical objectivity and business secrecy. Perhaps the most profound threat is the rise of what Shoshana Zuboff (2019) calls *surveillance capitalism*. In this economic model, human experience becomes a raw material that is freely appropriated, turned into behavioral data, and sold in markets that trade predictions about our future. The power is thereby extremely unevenly distributed: on the one hand, there are technological giants who possess the knowledge and infrastructure for collecting and processing data, and on the other hand, users, individual agents who are often not even aware of the extent of surveillance to which they are exposed. This asymmetry of power undermines the very idea of autonomy and free decision-making, creating a market in which we are not the product, but the possibility of influencing our future behavior. Digital duality is therefore a real tension that shapes our social, political and economic reality, revealing the digital sphere not as a utopian or dystopian destination but as a battlefield. The outcome of this struggle depends on the actions of various agents: on the regulatory decisions of states, on the efforts of civil society in the protection of digital rights, on the ethical responsibility of technological creators and, finally, on raising awareness and critical engagement of the users themselves. Acknowledging this duality means rejecting technological determinism and accepting responsibility for actively shaping the digital future—a future in which the promise of democratization will overcome the shadow of new inequalities.

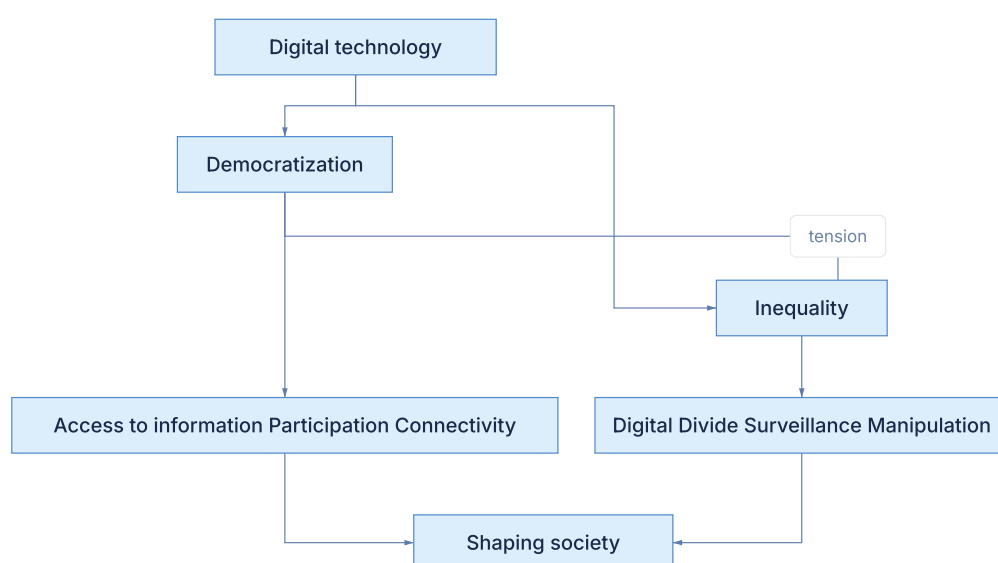


Figure 2.34: Schematic representation of the digital duality: the tension between the democratizing potential of digital technology and the new forms of inequality that together shape society.

2.7.11 Shadows in a digital Eden: privacy, manipulation and the price of connectivity

The semblance of a digital Eden, a world of unlimited possibilities and universal connectivity, brings with it its shadows, spaces where basic human rights and cognitive autonomy are exposed to unprecedented pressures. At the root of this duality lies the paradigm of surveillance capitalism, an economic order that appropriates human experience as a free raw material for production and sales processes (Zuboff, 2019). The first and most obvious victim of such a system is privacy. Massive and ubiquitous data collection – from location and search patterns to biometric information and emotional reactions – has become a fundamental mechanism for the functioning of the digital space. Users, often without full awareness of the scale and depth of this process, willingly give up fragments of their lives in exchange for seemingly free services, thus becoming raw material in a value chain whose ultimate purpose is to monetize their future behavior. Collected data traces are of course used to optimize services or place customized ads, but their purpose inevitably extends deep into the sphere of behavioral modification and targeted manipulation. Algorithmic systems, acting as invisible agents, construct personalized realities based on detailed psychographic profiles, worlds tailored to evoke certain emotional responses and stimulate desired actions. The scale and consequences of such actions became evident in the case of the company *Cambridge Analytica*, which tried to influence the outcome of democratic elections by misusing data collected from the social network *Facebook* (Cadwalladr & Graham-Harrison, 2018). This event exposed the systemic

vulnerability of modern societies, showing how data infrastructure can be repurposed as a weapon for psychological and political shaping of the public.

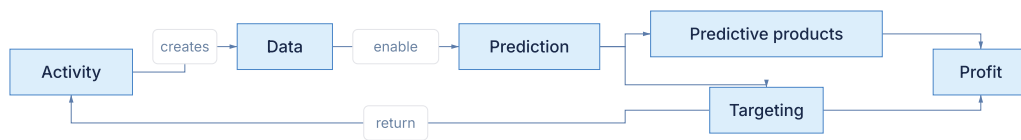


Figure 2.35: Systematic collection and analysis of user data leads to algorithmic profiling resulting in personalized content delivery and behavioral manipulation (surveillance capitalism cycle).

The operation of these algorithms, whose fundamental task is the maximization of user engagement, leads to further structural disturbances in the flow of information. By creating so-called “filter bubbles” (*filter bubbles*) and “echo chambers” (*echo chambers*), which lock users inside information ecosystems, confirming their existing beliefs and worldviews (Pariser, 2011). By systematically suppressing opposing or divergent points of view, algorithmic curation of content erodes the space for critical reflection and public discussion, key prerequisites for a healthy democratic society. The result is social fragmentation and increased polarization, where the common ontological basis for dialogue is gradually disappearing. The price of universal connection, apart from social and political, is also paid on a deeply personal, psychological level. Constant exposure to idealized depictions of other people’s lives on social networks creates fertile ground for social comparison, which often results in feelings of inadequacy, anxiety and depression. The imperative of constant availability and participation in the digital sphere blurs the lines between work and leisure, public and private, creating pressures that negatively impact mental health and general well-being. The role of the digital agent – whether it is a platform, an algorithm or an interface – is crucial in this ecosystem. Its goals, focused primarily on economic metrics such as retention and increased engagement, are often in direct conflict with the long-term interests and well-being of users. Therefore, facing the shadows of digital Eden requires the development of advanced digital literacy, strengthening awareness of the mechanisms that shape our online experience, and the establishment of

solid regulatory frameworks that will protect the autonomy, privacy and dignity of the individual in the new digital age.



Figure 2.36: Schematic representation of the psychological cost of universal connectedness: social comparison, anxiety, blurred boundaries between work and rest, and the pressure of constant availability.

2.8 Beginnings of autonomous communication: the first digital interlocutors

The thought of creating artificial beings capable of conversation reaches deep into the human past, from mythical statues that come to life to the Golems of Prague legends. But it was only in the second half of the twentieth century, with the advent of computers, that this age-old aspiration began to take its first real, albeit rudimentary, forms. The transition from myth to technology was conditioned by a key theoretical shift offered by Alan Turing. In his famous work *Computing Machinery and Intelligence* (Turing, 1950), he did not ask whether machines can *think*, but whether they can *behave* in a way that is indistinguishable from humans. His thought experiment, known as the Turing test or the imitation game, established the philosophical and practical framework within which the field of artificial intelligence, and especially the field of natural language processing, would develop for decades. It was the challenge of creating a machine that could successfully conduct a written conversation with a human that became the driving force behind the development of the first digital interlocutors. The first significant step on this path was taken by Joseph Weizenbaum, who created the ELIZA program in 1966. Acting on the principle of recognizing key words and applying simple transformation rules, ELIZA simulated a conversation. Her most famous script, called *DOCTOR*, imitated a Rogerian psychotherapist, using the technique of paraphrasing and asking open-ended questions taken from user

input. For example, the program could respond to the statement “My family worries me” with “Tell me more about your family”. Although technically simple, ELIZA caused an unexpectedly strong reaction. Users credited the program with understanding, empathy, and awareness, despite Weizenbaum’s repeated claims that the machine understood nothing. This phenomenon, which he called the “ELIZA effect”, revealed a deep human tendency to anthropomorphize and establish an emotional connection with even the simplest interlocutor, which deeply worried the author himself and prompted him to critically review the ethical limits of artificial intelligence (Weizenbaum, 1976).

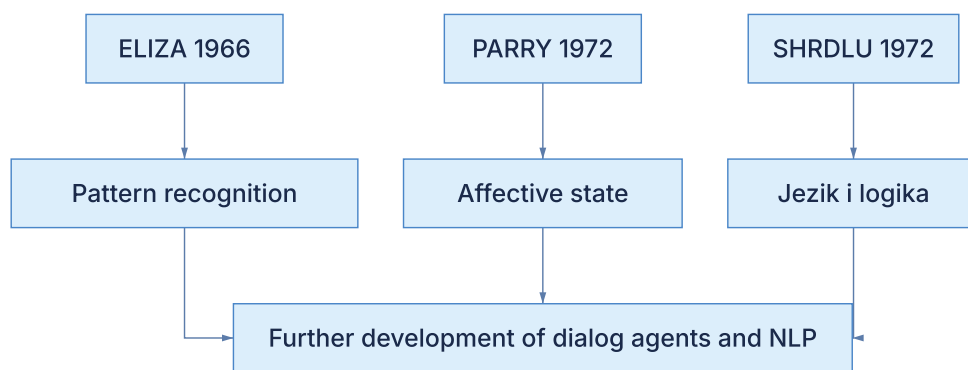


Figure 2.37: Evolutionary steps of the first systems for language interaction: from ELIZA (1966) through PARRY (1972) and SHRDLU (1972) to the further development of dialogic agents and natural language processing.

Shortly after ELIZA, psychiatrist Kenneth Colby from Stanford University developed the PARRY program in 1972. Unlike Weizenbaum’s passive interlocutor, PARRY was designed as a more complex simulation, embodying the personality of a patient with paranoid schizophrenia. Colby’s agent responded to stimuli, but also possessed an internal model of his own beliefs and affective states, such as fear, anger, and distrust, which influenced his responses. PARRY monitored the level of these affects and could proactively change the subject or become defensive if he felt the interlocutor was provoking him. In a series of tests, psychiatrists were often unable to distinguish transcripts of conversations with PARRY from those with human patients. As a historical curiosity, the conversation held in 1972 between ELIZA

and PARRY over the ARPANET network, which clearly showed the limitations of both systems - the meeting of the passive mirror and the obsessive model resulted in a mostly incoherent and repetitive exchange. At the same time, a different approach was developed, which was not aimed at simulating the human personality, but at the true understanding of the language within the given context. The most prominent example of such an approach was Terry Winograd's SHRDLU (1972). This system allowed the user to use natural language to issue commands to a virtual robotic arm in a "cube world". SHRDLU could understand complex and even ambiguous sentences, ask questions for clarification and remember the context of the conversation. For example, on the command "Pick up the big red cube" the system would execute it. If the user then said "Put it on green", SHRDLU would correctly understand that the pronoun "it" refers to the aforementioned red cube. Although limited to its microcosm, SHRDLU has shown that a deep understanding of semantics and syntax is possible, steering part of the research community towards knowledge-based models. William Chamberlain and Thomas Etter developed a computer program designed to generate prose and poetry in English (Chamberlain & Etter, 1984). Racter, whose name comes from the word "raconteur" (storyteller), is said to have created the book *The Policeman's Beard Is Half Constructed* (published in 1984). According to the editorial framework of that publication, the text was presented as computer-generated, with minimal subsequent interventions, and the system used the rules of context-free grammar and elements of randomness to construct unique sentences and prose (Chamberlain & Etter, 1984; Adamopoulou & Mousiades, 2020). The early systems, from ELIZA to PARRY and SHRDLU, seem technically naive from today's point of view, but their importance lies in the fact that they showed that machine processing of language was possible. They unraveled the psychological aspects of human-machine interaction and raised fundamental questions about the nature of intelligence, understanding and communication. These were the first, necessary steps in the dark, which, despite their obvious limitations, paved the way towards significantly more complex architectures and paradigms that will mark the next chapter – the world of large language models.

3. The world of large language models: from theory to technology

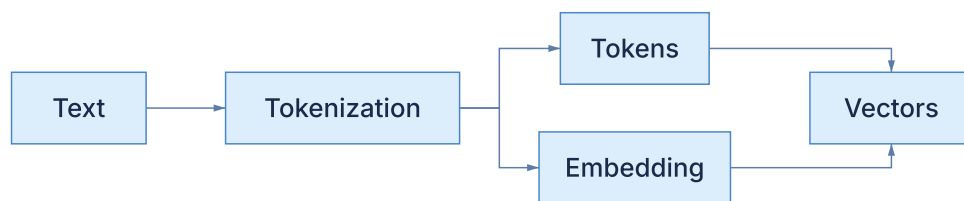


Figure 3.1: Presentation of the transition from sign to vector representation of language (tokenization, embedding).

3.1 From sign to vector: A paradigmatic shift in the representation of meaning

Observing the development of computer language processing necessarily leads us to observe a thorough transformation in the very theoretical origins of this skill. Classical approaches, which formed the backbone of Natural Language Processing (NLP) for decades, viewed language primarily as a system of arbitrary signs organized according to complex but comprehensible rules. Their starting point was deeply rooted in structural linguistics, and their goal was to impose a formal linguistic order on raw, unstructured text.

This procedure, which can be called a standard pipeline (*standard pipeline*), consisted of a series of carefully elaborated steps, each of which aims at further normalization and breakdown of the linguistic structure (Jurafsky & Martin, 2023). The first step was regularly tokenization, a process by which an unbroken string of characters is broken down into basic units – *tokens* – which most often correspond to words and punctuation marks. This was followed by

lemmatization, a complex process of reducing a single word to its canonical, dictionary form or *lemma*. Thus, for example, the forms *running*, *running*, *running* would be reduced to their common lemma *running*. This put an end to the morphological diversity that hides the fundamental lexical relations. Furthermore, part-of-speech tagging was applied, where each token would be associated with its grammatical identity: *noun*, *verb*, *adjective* and so on. In many applications, such as thematic modeling, the removal of so-called stop words (*stop words*) - frequent but semantically weak words like conjunctions and prepositions - was carried out.

Token: The smallest unit of text processed by the language model; can match a whole word, a part of a word (subword) or a punctuation mark. Tokenization converts raw text into a string of tokens according to a model dictionary.

The outcome of such analysis was a highly structured, but also an impoverished presentation of the text. It was reduced to a list of lemmas with associated grammatical markers, the so-called “bag of words” (*bag of words*), which broke the original syntactic structure, and thus a large part of the contextual meaning. Although these procedures were indispensable for the development of early applications, such as spell checkers or simple search systems, their essential limitation became more and more obvious: they could not effectively capture and numerically represent the very semantic core of language – a complex and fluid network of meaning relationships.

A true paradigmatic shift occurred with the appearance of **distributional semantics** and its computer application in the form of **vector representations of words**, better known as **word embeddings**. The theoretical origin of this approach is summarized in the famous thought of linguist John Rupert Firth (1957) that the nature of a word is manifested in the society it forms (“*You shall know a word by the company it keeps*”). Consequently, words are no longer seen as isolated symbols, but their meaning is constituted through the context in which they appear.

Embedding / investment vector: Numerical (vector) representation of a word, sentence or other language segment in a multidimensional space; the position of the vectors reflects semantic similarity – words with similar meanings have close vectors. It is the basis of distributional semantics and internal representations in LLMs. **Corpus:** A collection of textual (or other linguistic) data on which the language model is learned; includes books, articles, websites and other sources. The quality and representativeness of the corpus directly affects the performance and biases of the model.

Computational models such as **Word2Vec** (Mikolov et al., 2013) and **GloVe** (Pennington et al., 2014) operationalized that insight. Analyzing huge text corpora, these algorithms learned to associate each word with a unique numerical vector in a dense, multidimensional space. The position of each vector is not arbitrary, but is determined by its relationship to all other vectors. Words that appear in similar contexts, such as *cat* and *dog*, will have vectors that are close in that space. The word *car*, which occurs in completely different contexts, will be represented by a vector that is distant from them.

Moreover, relationships between vectors can include more complex semantic analogies. For example, the vector operation $VECTOR("king") - VECTOR("man") + VECTOR("woman")$ results in a vector that is in the immediate vicinity of $VECTOR("queen")$. This shows that the model has not only learned mere similarity, but also the fundamental relations that structure our conceptual world.

The transition from symbolic to distributional, vector representation of language was therefore crucial. It enabled computers to represent meaning, that most elusive component of language, in a way that is mathematically tractable and robust. Instead of relying on predefined vocabularies and rigid grammar rules, models could now learn semantic relationships directly from language usage. This opens the way for solving significantly more complex tasks, from a nuanced sentiment analysis to the creation of coherent and contextually appropriate text. Vector representations of words have thus become an indispensable cornerstone on which the entire architecture of

contemporary large linguistic models, and thus also the communication agents that are the subject of this book, is built.

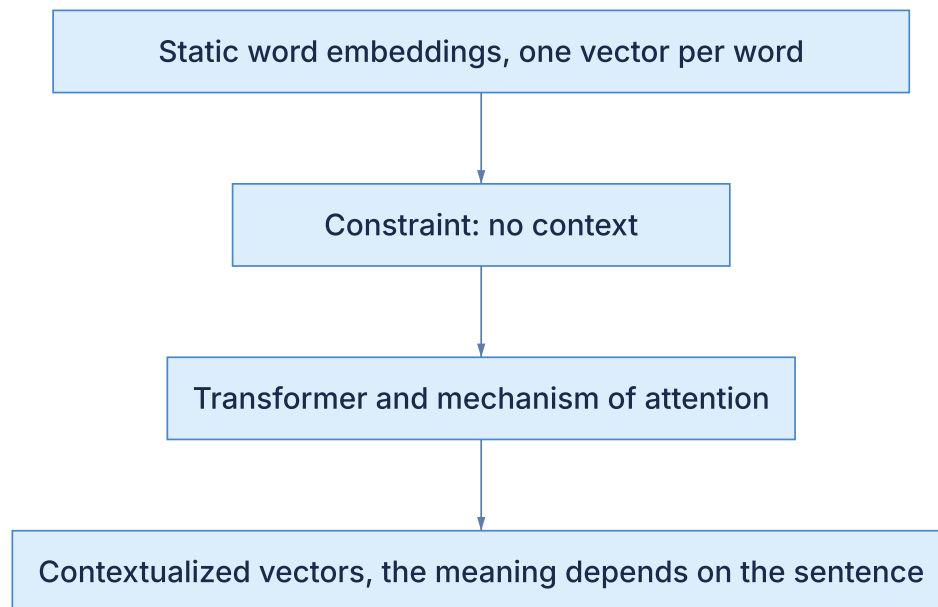


Figure 3.2: Meaning as space – from static vectors to dynamic context (transformer, attention).

3.2 Meaning as space: From static vectors to dynamic context

The importance of the transition to vector representation of language cannot be emphasized enough. This cognitive turn made it possible to translate the abstract and fluid concept of meaning into a mathematically comprehensible form. Language ceased to be just a series of discrete symbols and became a continuous, multidimensional semantic space, in which each word is located, and its location is defined by its relationship to all others. The ability to measure distance and direction between vectors, most often through cosine similarity, turned the linguistic question of relatedness into a geometric problem. This operationalized the very essence of linguistic intuition.

However, that early vector approach also had its Achilles heel: **staticity**. Namely, each word had a single, unchanging vector, regardless of the context in which it was found. Thus, for example, the ambiguous word *hair* had the same vector in the sentences “The girl had long, blond hair” and “The peasant sharpened his hair”. The model had no way of distinguishing whether it was a scalp or an agricultural tool. This limitation represented a fundamental obstacle to a true understanding of the context, which is crucial for any more complex language task.

The solution came with the revolutionary **transformer architecture** (Vaswani et al., 2017) and its central **attention mechanism**. Instead of assigning a static vector to each word, the transformer creates a dynamic, contextualized representation for each word within a sentence. The attention mechanism allows the model, while processing one word, to dynamically weigh the importance of all other words in the sentence and build its specific, contextual meaning on that basis. In our example, the model, processing the word *hair*, would pay special attention to the words *girl* and *blonde* in the first sentence, and to the words *peasant* and *sharpened* in the second. Consequently, the vector for the word *hair* would be completely different in the two cases, faithfully reflecting its meaning in the given context.

The architecture of the transformer did not only bring that qualitative leap in language understanding. Its exceptional suitability for parallel processing on specialized circuits, such as graphics processing units (GPUs), has opened the door to unimagined **scaling**. Researchers have observed that model performance predictably improves with exponential increases in three key components: model size (number of parameters), amount of training data, and computing power invested in training (Kaplan et al., 2020). This insight, known as “scaling laws” (*scaling laws*), stimulated the race to create ever larger models.

Transformer: A neural network architecture based on a (self-attention) mechanism that associates each word in a sequence with a contextualized image based on the other words; it enables parallel processing and scaling to large language models (Vaswani et al., 2017).

This is how we came to the modern **large language models (LLM)**. They are not just enlarged versions of their predecessors. At a certain size level, these models began to show **emergent abilities** – new skills they were not directly trained for, such as the ability to solve logic problems, write computer code, or translate between languages they had not seen during training (Wei et al., 2022).

Emergent capabilities: Capabilities of large language models that appear at a certain scale (number of parameters, amount of data) and for which the model was not directly trained - e.g. inference, translation or writing code; describe a qualitative jump in behavior with increasing model size.

A large language model, therefore, is not just a language processing system. It is, in its essence, a complex model of the world, implicitly encoded in the statistical relationships of the vast corpus of human knowledge. The path from the sign to the vector, and then from the static to the contextualized vector, was therefore a path from mere formal analysis to the creation of active models of reality.

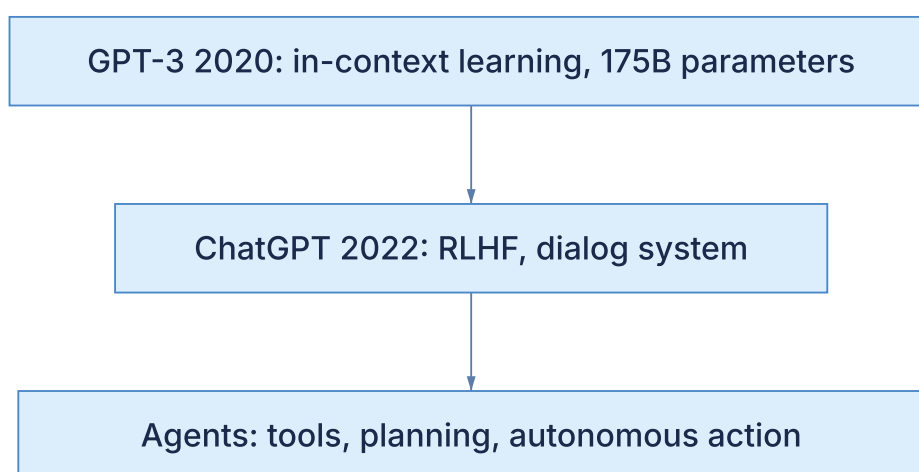


Figure 3.3: Entering the era of large language models (GPT-3, ChatGPT, RLHF).

3.3 Entering the age of large language models

The year 2020 marked a turning point in the history of artificial intelligence with the publication of a paper on the *GPT-3* model (Brown et al., 2020), which ushered in the era of Large Language Models (LLM). With its previously unrecorded size of 175 billion parameters, GPT-3 was a quantitative breakthrough, but also a pioneer of a new paradigm of language model operation. The key innovation was the demonstration of the ability that we now call *in-context learning*. It is about the extraordinary ability of the model to, without additional fine-tuning for a specific purpose, perform complex tasks based on only a few examples or instructions provided within the user query itself. In doing so, he demonstrated a level of generality and adaptability that far exceeded the capabilities of his predecessors. Although *GPT-3* laid the technical and conceptual foundations, the real resonance among the general public came at the end of 2022 with the appearance of the *ChatGPT* model. It reached a hundred million users with astonishing speed, becoming the most rapidly adopted technological innovation in history. Its success did not lie solely in the raw power of the model, which was based on the *GPT-3.5* architecture, but primarily in the application of precise tuning using the method of reinforcement learning based on human feedback (*Reinforcement Learning from Human Feedback* - RLHF). Such an approach enabled the model to follow user instructions significantly better, conduct meaningful and context-aware conversations, and avoid inappropriate or harmful responses.

RLHF: Reinforcement Learning from Human Feedback – an alignment methodology in which human evaluators rank the model’s responses, after which the reward model is trained and the model is further tuned to maximize the expected reward; key to aligning the behavior of LLMs with human preferences (usefulness, truthfulness, security).

Accelerated development has brought us to the point where these systems cease to be purely passive text generators. Their role is being transformed, and they themselves are becoming active participants in the digital environment - agents. This transformation implies the ability of models to not only understand and generate language, but also to make decisions, plan complex

sequences of actions and use external tools – from simple calculators and search engines to complex programming interfaces (APIs). We are entering a period in which large language models become proactive systems capable of autonomous action, which represents a fundamental change in our understanding and application of artificial intelligence.

The evolution of large language models from basic models with general capabilities to specialized dialog systems and, ultimately, proactive agents capable of action

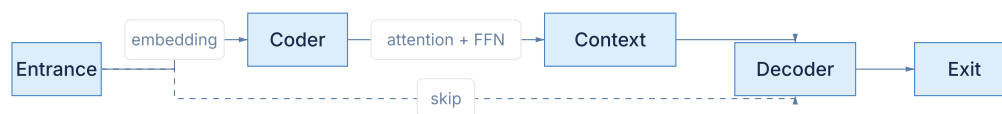


Figure 3.4: Anatomy of LLM - key technologies and architectures (transformer, self-attention).

3.4 LLM Anatomy: Key Technologies and Architectures

Understanding modern large language models (LLM) begins with an insight into the transformational architecture. Namely, the publication of the aforementioned scientific article *Attention Is All You Need* (Vaswani et al., 2017) laid the cornerstone for a new era of natural language processing, abandoning the previously prevalent recurrent (RNN) and convolutional (CNN) neural networks. The transformer architecture offered solutions to the key shortcomings of previous models, such as the problem of vanishing gradients and the impossibility of effective parallelization, thus paving the way for the development of significantly larger and more capable systems. The central mechanism that gives transformers their exceptional power is self-attention. It is a procedure by which the model, processing a sequence of data, dynamically evaluates the importance of each word in relation to all other words within the same sequence. In this way, the model creates a rich, contextualized representation of each word, taking into account its most important semantic and syntactic connections with the rest of the text. This ability was further deepened and expanded by the introduction of multi-head

attention. This approach allows the model to simultaneously, in parallel “heads”, direct attention to different aspects and relationships within the sentence - one “head” can follow the grammatical structure, another thematic connections, and the third anaphoric relationships, which together creates a multidimensional and comprehensive understanding of the input text. Since the transformer architecture does not process data sequentially, as was the case with recurrent networks, but looks at them as a whole, the inherent information about word order is lost. This shortcoming was overcome by the introduction of positional encodings (*positional encodings*), vectors that are added to the input word vectors to provide the model with information about their relative or absolute position in the sequence. In addition to the attention mechanism, a key part of each transformer block is the feed-forward network, which additionally processes the output from the attention layer and contributes to non-linear data transformation.

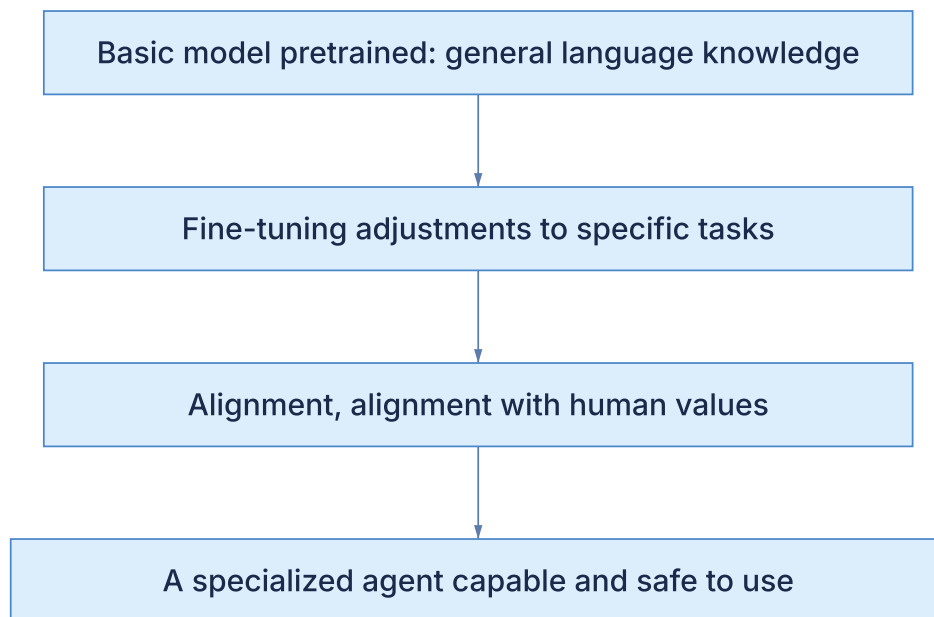


Figure 3.5: LLM training process (base model, fine-tuning, alignment, specialized agent).

Based on these building blocks, three basic architectural patterns of large language models have been developed, each adapted to a specific type of task:

Encoder-based architectures only: These models, the most prominent representative of which is *BERT* (*Bidirectional Encoder Representations from Transformers*), are designed for deep understanding of context. The encoder processes the entire input text at once, allowing each word to call attention to all other words in the text (bidirectional attention). Because of this ability, they are extremely effective in tasks that require text analysis and classification, such as sentiment analysis, recognition of named entities or answering questions where the answer is contained in the attached text.

Architectures based solely on the decoder: Models specialized for text generation, processing the sequence of words from left to right. Each new word they generate depends on previously generated words, which mimics the natural flow of human speech or writing. Their attention is causal (*causal*), that is, they can rely only on the previous words in the sequence, and not on the future ones. The most famous representatives of this group are the models from the *GPT* series (*Generative Pre-trained Transformer*), which excel in tasks such as writing essays, creative writing, summarizing and conducting conversations.

Encoder-decoder architectures: Also known as *sequence-to-sequence* models, these architectures combine the strengths of both approaches. The encoder first creates a complete, contextualized representation of the input sequence, and the decoder then generates an output sequence based on this representation. This pattern is ideal for transformation tasks, such as machine translation, where a sentence from one language is turned into a sentence in another, or text compression, where a longer document is turned into a shorter one. Models like *T5* (*Text-to-Text Transfer Transformer*) and *BART* (*Bidirectional and Auto-Regressive Transformers*) are examples of this hybrid structure.

Training process (base model, fine-tuning, alignment)

The development of large language models depends on the architecture, but also on a specific, two-stage training process. In the first phase, the pre-training phase, the model is exposed to huge, unstructured text corpora from the Internet, books and other sources. The goal is for the model to learn the

basic patterns of the language independently, without human supervision (in the so-called self-supervised way): grammar, syntax, semantics and general knowledge about the world. This is most often achieved by the task of predicting the masked or following words in the sentence. After pre-training, there is a phase of fine-tuning, where the general model is adjusted to solve specific tasks. This step may involve supervised learning on smaller, labeled data sets (eg, question-answer pairs) or more advanced techniques such as human feedback-based support learning (*RLHF*), where the model is further tuned to generate answers that are useful, truthful, and harmless, consistent with human preferences.

Fine tuning / fine-tuning: The phase of adjusting the already pre-trained (basic) model for specific tasks or domains; uses smaller, labeled data and may include RLHF; the result is a specialized model (e.g. chatbot, code assistant).

From model to agent: a step towards action

The term agent in the context of large language models signifies a qualitative step forward: the transition from passive text generation to active action in a digital or physical environment. An agent is a system that uses *LLM* as its central cognitive drive to reason, plan, and execute tasks. Such a system has the ability to decompose complex goals into simpler steps, use external tools (such as search engines, calculators or programming interfaces), store information in memory, and learn from one's own mistakes. The anatomy of the agent thus goes beyond the internal structure of the language model and includes loops for perception, planning and action, which opens the way to the creation of more autonomous and capable artificial intelligence systems.

3.5 Model life cycle: from data to application

The life cycle of an artificial intelligence model includes a layered and structured sequence of steps leading from the initial idea, based on data, to the final application in a real environment. It is a dynamic and iterative process that requires the cooperation of experts of various profiles, from data scientists

to software engineers. Each phase of the cycle builds on the previous one and lays the foundation for the next, thus forming a whole that ensures that the final solution is robust, reliable and purposeful. The entire process, from idea to implementation and maintenance, is in modern practice unified under the name *MLOps* (*Machine Learning Operations*), a term that draws its theoretical and practical roots from the *DevOps* paradigm in software development, adapting its principles to the specific challenges of machine learning.

MLOps: Machine Learning Operations – a set of practices and tools for managing the life cycle of models from development to production; includes model and data versioning, automated training, deployment and performance monitoring; created by adapting DevOps principles to the domain of machine learning.

The fundamental purpose of *MLOps* is to bridge the gap between model development, which is often experimental in nature, and its reliable operational application. The process starts with understanding the business problem and defining the goals, followed by a series of technical phases that can be summarized in several key units.

Stages of the model life cycle

1. Collection and preparation of data: The starting point and fundamental prerequisite of the entire process is the definition and collection of data. The quality and relevance of the data directly determine the upper limit of the performance of the future model. This phase includes activities such as collecting raw data from various sources, cleaning them to remove errors and inconsistencies, and transforming and engineering features (*feature engineering*). The latter is particularly important because properly shaping the input variables can significantly improve the model's ability to spot relevant patterns. At this stage, techniques such as normalization, scaling and coding of categorical variables are often applied to prepare the data for machine learning algorithms. **2. Building and training the model:** Only on the basis of the data set prepared in this way, the building of the model is approached. This step involves choosing an appropriate algorithm or model architecture,

depending on the nature of the problem (e.g. classification, regression, clustering). This is followed by a training process, during which the model is exposed to data in order to learn to map input features into desired output values. During training, the model parameters are iteratively adjusted with the aim of minimizing the error function. In this phase, hyperparameters are also adjusted, i.e. external configuration settings of the model, which are not learned directly from the data, but significantly affect its performance (Bergstra & Bengio, 2012).

3. Model evaluation: After the model has been trained, there is a phase of its careful evaluation. The purpose of the evaluation is to objectively assess the performance of the model on data that it did not see during training. This verifies its ability to generalize, that is, its applicability to new, unknown cases. Various metrics are used, such as accuracy, precision, response or F1-measure for classification problems, or mean squared error for regression problems. It is crucial to avoid overfitting, a condition in which the model remembers the training data too well, but loses the ability to generalize.

4. Application and integration of the model (*deployment*): The model that has met the evaluation criteria is ready for application in the production environment. This step represents the transition from the development to the operational world and is often one of the most complex parts of the cycle. The model can be integrated into existing software systems in different ways: as an API interface that provides real-time predictions, as part of periodic (*batch*) data processing, or as an embedded *agent* within an application or device. At this stage, the collaboration of data scientists and engineers is essential to ensure the scalability, stability and security of the solution.

5. Monitoring and maintenance: The life cycle of a model does not end with its implementation. Once the model is in production, it is necessary to constantly monitor its performance and behavior. The world is changing, and with it the data on the basis of which the model makes decisions. Phenomena such as concept drift (*concept drift*), where the relationship between input and output variables changes, or data drift (*data drift*), where the distribution of input data changes, can significantly impair the accuracy of the model over time (Gama, Žliobaitė, Bifet, Peciukonis, & Bifet, 2014). Therefore, monitoring systems must be in place to detect performance degradation in time and signal the need for retraining or complete replacement of the model. This closes the

circle and the process returns to the initial stages, thus ensuring the long-term relevance and value of the implemented solution. Creating a successful AI model is a marathon, not a sprint. It is a continuous cyclical process of learning, adapting and improving, deeply rooted in the data from which it originates and the application it serves.

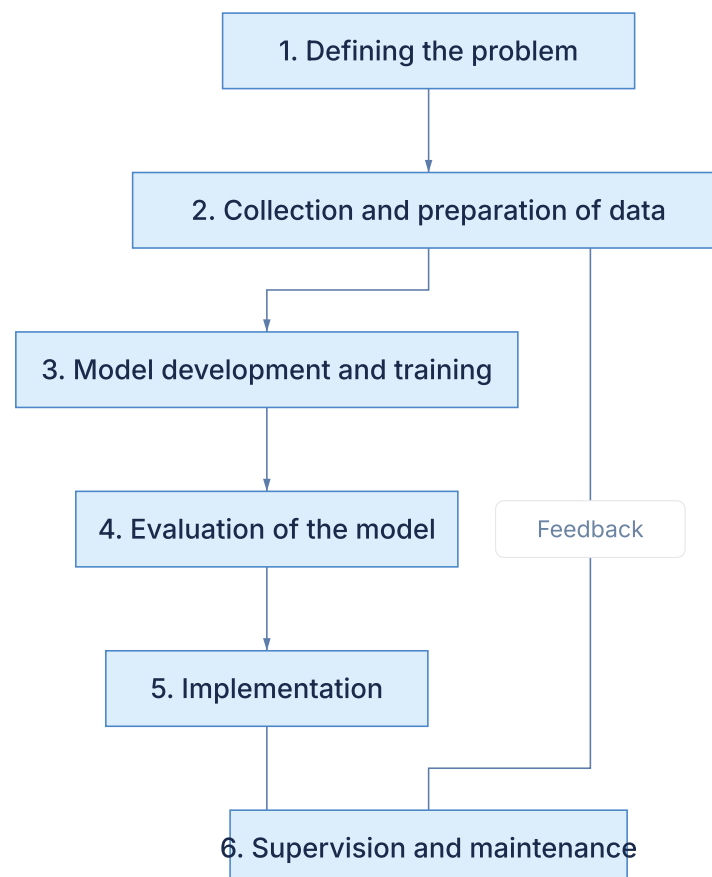


Figure 3.6: Representation of the iterative life cycle of a machine learning model, from the initial phase of working with data to monitoring in application and back to the beginning of the cycle.

3.5.1 Pre-training: creation of basic knowledge

The process of creating modern language models begins with the key and computationally most demanding phase known as pre-training. The purpose of this stage is to build what we call a foundation model (*foundation model*),

which has a comprehensive and general understanding of the language, its structures, semantic relations and, to a certain extent, knowledge about the world that is embedded in the textual data on which it is trained.

Foundation model: A large pre-trained model (languages, images or multimodal) that has a general understanding and can be fine-tuned for different tasks; represents the output of the pre-training phase and the starting point for specialization (fine-tuning, RLHF). **Pre-training:** The training phase of the language model on huge, unstructured text corpora without human annotations; the model learns language patterns and general knowledge by solving self-supervised tasks (e.g. predicting the next word). The result is a foundation model that is then fine-tuned for specific tasks.

At the heart of pre-training lies the principle of self-supervised learning.

Self-supervised learning / self-supervised learning: A learning method in which the model does not use humanly marked data (input-output pairs), but performs the task itself from the data structure: for example, it predicts a hidden part of the input (next word, masked token) or another derived goal. The learning signal comes from the data itself. This procedure is crucial for pretraining large language models based on huge amounts of corpus data for which manual annotation would be difficult to implement. **Unstructured data:** Data that is not organized into a fixed scheme (tables, columns, tags); include free text, images, audio, video and mixed sources. Unlike structured data (e.g. databases), they do not have a predefined format, so it is necessary to convert them into a form suitable for computer processing. Pre-training for LLMs is based on vast amounts of unstructured text (web, books, articles).

The model is exposed to a huge amount of unstructured textual data collected from various sources such as websites, digitized books and scientific articles. Unlike supervised learning, this data has not been previously labeled or categorized by humans. Instead, the model learns on its own, solving the tasks it sets for itself based on the structure of the data. Most often, he is given a seemingly simple, but extremely complex goal: predicting the next word in a sentence, filling in deliberately omitted parts of the text, or some other similar task that forces him to learn linguistic regularities in depth. Going through billions of examples, the agent gradually builds up an internal representation

of the language. Knowledge acquired through pre-training is not explicitly stored like facts in an encyclopedia. On the contrary, it is woven into a complex network of billions of numerical parameters that make up the neural network of the model. This knowledge is latent and probabilistic; the model does not “know” that Paris is the capital of France in the way that a human does, but has learned that in a huge number of texts there is an extremely high statistical probability that the two phrases appear in a certain relationship. In this way, the model adopts grammatical structures, semantic relations between words (e.g. that “king” and “queen” are related), stylistic nuances and even rudimentary factual knowledge about the world. The outcome of pre-training is therefore a basic model – a powerful, but still insufficiently precise tool. It is important to point out that this model, despite its extensive knowledge, is not in itself directly qualified to perform specific tasks such as translating, summarizing text or answering questions in a consistent and reliable manner. It represents only the raw material, the starting point for further processing, which is called **fine-tuning**. It is only in this next phase that the general knowledge of the basic model is directed and specialized for a specific application.

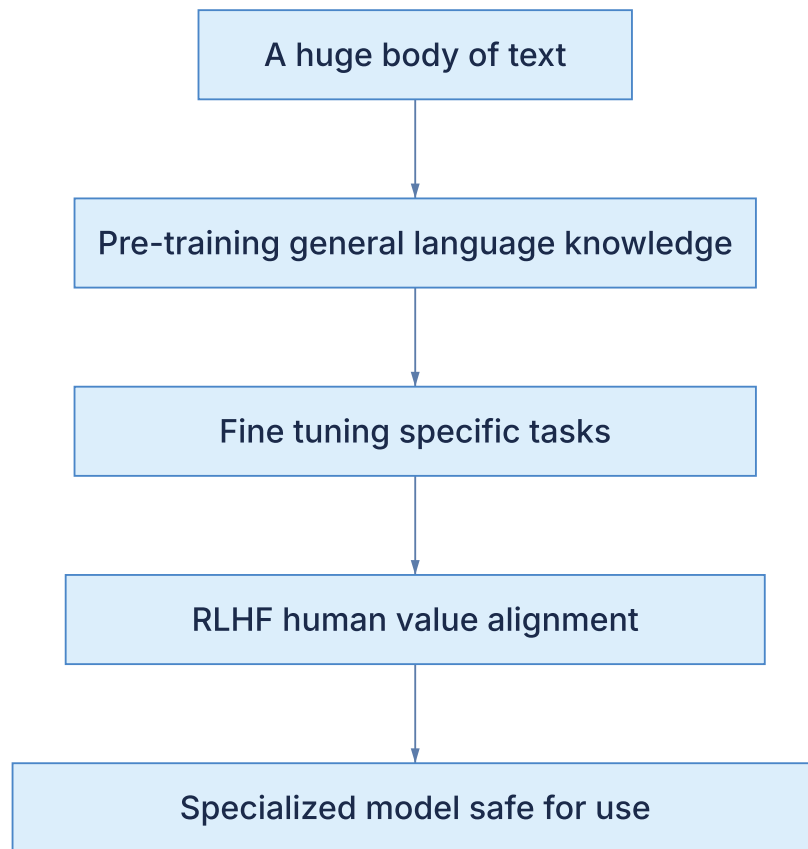


Figure 3.7: Schematic representation of the training process of modern language models, from the basic model to the specialized agent (fine-tuning, alignment).

3.5.2 Fine tuning and alignment: fit for purpose

After an intensive pre-training phase, the resulting base model possesses an impressive breadth of linguistic knowledge and a certain amount of encyclopedic knowledge of the world, learned from vast data corpora (Bommasani et al., 2021). However, this version of the model is like an extremely well-educated, but unspecialized and potentially unpredictable entity. Its raw capabilities, while general, are not necessarily optimized for the specific tasks users want to perform, nor is its behavior necessarily aligned with human expectations of utility, truthfulness, and security. Therefore, in order to bridge the gap between the general potential of the basic model and the

requirements of practical application, a further stage of adaptation is necessary, which typically consists of two complementary processes: fine-tuning for task specialization and alignment for shaping behavior.

Fine-tuning is the application of transfer learning, an extremely useful paradigm in machine learning where the knowledge gained from solving one problem (in this case, general language modeling during pre-training) is used as a starting point for solving another, related problem (Howard & Ruder, 2018). Instead of training a model from scratch for each specific task, which would be extremely inefficient considering the size of LLMs, fine-tuning starts with the parameters of an already pre-trained model. The model is then further trained, typically using standard supervised learning techniques, on a much smaller, carefully curated data set specific to the target task. A set consists of examples of inputs and desired outputs (tags) for that task – for example, a set of question-answer pairs for *chatbot* development, a collection of annotated medical texts for healthcare applications, annotated code repositories for coding assistance tools, or pairs of original sentences and their translations for machine translation. During this process, the model parameters—or at least a subset of them—are incrementally adjusted by optimizing a standard loss function (e.g., cross entropy) on a specific data set. The primary goal of this approach is to specialize the model to the specific requirements and characteristics of the target task or domain, while retaining and effectively using the rich general linguistic and factual knowledge accumulated during the pre-training phase. In this way, the need for large amounts of labeled data for each new task is significantly reduced and the development of specialized AI applications is accelerated. However, fine-tuning also carries the risk of **catastrophic forgetting** (McCloskey & Cohen, 1989), where the model loses part of the general abilities learned during pre-training during adaptation to a new task, which requires careful training strategies (e.g. low learning rates, gradual “unfreezing” of layers).

Parallel to the fine adjustment for specific tasks, another, by its nature more complex process takes place: **alignment**. Alignment does not only affect the domain of accuracy and information of the model, but primarily its behavior

and conformity with human values and intentions. The goal is to form agents whose actions will be not only useful and true, but also harmless, i.e. aligned with ethical principles and social expectations.

Alignment: Directing the behavior of AI systems according to human goals, values and safety principles; it includes methodologies like RLHF and RLAIIF to make the model useful, true and safe in practice.

The key methodology for achieving this goal is **Reinforcement Learning from Human Feedback**, which as a rule consists of three basic steps. The first step involves **supervised data collection**, where human annotators compile high-quality responses to selected queries. This creates an initial set of data to fine-tune the model to the desired style and tone, laying the foundation for further alignment. The second, key step is **training a separate reward model** (*reward model*). In this phase, the language model generates several different answers to the same query, and human raters rank them according to quality, from best to worst. Based on these rankings, the reward model learns to predict which type of response a person would rate as more desirable.

Reward model: Auxiliary model trained on human ratings of language model responses; predicts how a human rater would rate an answer; used in RLHF to direct the language model according to preferences (usefulness, truthfulness, security).

Its task is therefore not mere classification, but modeling of complex human preferences. In the third step, the original language model is additionally adjusted by applying the **reinforcement learning** algorithm, most often by means of **proximal optimization** (*Proximal Policy Optimization*, PPO). The model generates answers, and the reward model evaluates them by assigning them a numerical “reward”. The language model then iteratively adjusts its parameters to maximize expected reward, thereby directing its behavior toward generating responses that are consistent with learned human preferences (Ouyang et al., 2022).

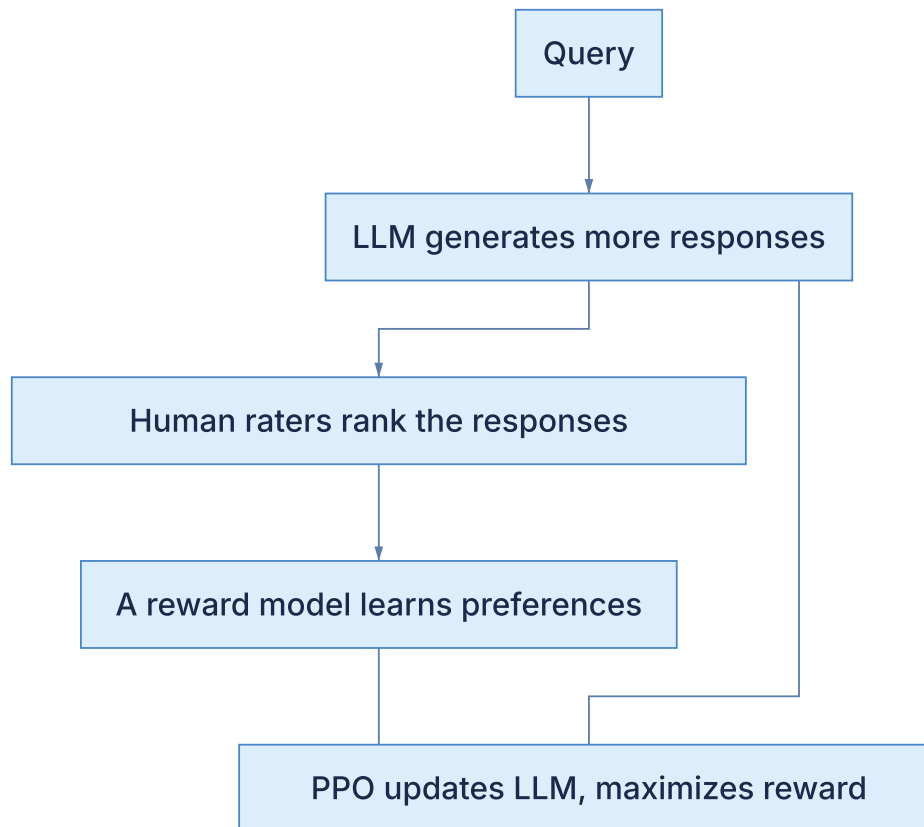


Figure 3.8: Schematic representation of the reinforcement learning process based on human feedback (RLHF).

The outcome of this layered process is an agent that possesses encyclopedic knowledge acquired through pre-training and the ability to interact that is adapted to human expectations and norms. It should be noted that RLHF is not the only approach to alignment. Alternative methods are also being developed, such as **Reinforcement Learning from AI Feedback** (RLAIF), where the role of the human evaluator is to some extent replaced by another, “constitutional” model of artificial intelligence that provides feedback, thereby potentially speeding up and scaling the process (Bai et al., 2022).

3.5.3 Optimization for the real world: efficiency

The very size that characterizes large language models, although crucial to their advanced capabilities, also represents their Achilles' heel in the context of practical application. As models grow to hundreds of billions or even trillions of parameters, training them and, more importantly for end users, running them at the **inference** (real-time answer generation) stage becomes extremely demanding. These processes consume huge amounts of specialized computing resources, such as **graphics processing units** (GPU) or **tensor processing units** (TPU), require significant amounts of memory (both for storing model parameters and for temporary activations during calculations) and consume large amounts of electricity, which entails not only high operational costs but also a significant environmental footprint (Strubell et al., 2019; Patterson et al., 2021; Luccioni et al., 2022). This resource intensity is a significant barrier to the widespread availability and implementation of LLMs, limiting their use to organizations with deep pockets and strong infrastructure, and making it difficult to run them on resource-constrained devices such as smartphones or edge devices. Therefore, efficiency optimization has become a critical area of research and development, aimed at reducing the computational, memory and energy requirements of LLMs without significantly sacrificing their performance, thus paving the way for their democratization and more sustainable application. Several key strategies dominate this field: parametrically efficient fine-tuning, quantization, and pruning.

One of the first efficiency challenges arises during the adaptation of pre-trained models to specific tasks. Traditional full fine-tuning involves updating *all* parameters of a pre-trained model on a new, specific data set. Although efficient in terms of achieving good performance, it creates a serious scalability problem: for each new task it is necessary to store a complete, huge copy of the fine-tuned model, which becomes very expensive in terms of storage and management if the model needs to be adapted for tens or hundreds of different tasks. In order to solve this problem, parameter-efficient fine-tuning (*Parameter-Efficient Fine-Tuning* - PEFT) techniques were developed. The basic idea of PEFT methods is to freeze the vast majority (or all) of the parameters of the original pre-trained model and update only a small number

of additional or existing parameters during fine-tuning (Lialin et al., 2023; He et al., 2021). The number of parameters to be trained is often less than 1% of the total number of model parameters, which drastically reduces memory requirements (as only small sets of additional parameters need to be stored for each task, not the entire model) and fine-tuning computational costs. A few popular approaches include PEFT methods:

PEFT: Parameter-Efficient Fine-Tuning* – a set of techniques (adapters, LoRA, prompt tuning) that adjust the pre-trained model by updating only a small part of the parameters; it reduces memory and computing costs and enables more specializations on the same basic model.

Adapters: This technique inserts small, additional neural modules, called adapters, within each layer (or some layers) of the original transformer architecture (Houlsby et al., 2019; Pfeiffer et al., 2020). Adapters usually have a bottleneck architecture – they first project the input activation into a lower-dimensional space, apply a nonlinear function, and then project it back into the original dimension. During fine-tuning, only the parameters of these small adapters are trained, while all parameters of the original LLM are frozen. This enables great modularity – only a new set of adapters is trained for each new task. **LoRA (Low-Rank Adaptation):** This technique, which has gained significant popularity in the field of adaptation of large language models, is based on the hypothesis that changes in model weights (ΔW) during adaptation to a specific task are inherently of low intrinsic rank (Hu et al., 2021). This means that the most important information about the model fit can be condensed into a subspace of significantly lower dimensionality.

LoRA: Low-Rank Adaptation – PEFT method that approximates the change of model weights (ΔW) by the product of two small low-rank matrices; only those matrices are trained, while the original parameters are not changed; achieves performance close to full fine-tuning with significantly fewer parameters (Hu et al., 2021).

In the context of fine-tuning, instead of directly updating the entire weight matrix W , which often includes billions of parameters and whose direct modification would require exceptional computer resources and memory, LoRA (Low-Rank Adaptation) introduces an elegant solution: the change in weights (ΔW) is approximated by the product of two much smaller matrices A and B . Matrix A has dimensions ($d_{\text{in}} \times r$), matrix B dimensions ($r \times d_{\text{out}}$), where r (rank) is much smaller than the dimensions of the original matrix W ($d_{\text{in}} \times d_{\text{out}}$). Then it applies: - $\Delta W = A \cdot B$ (product $A \cdot B$ has dimension $d_{\text{in}} \times d_{\text{out}}$, equal to W) - $W_{\text{adjusted}} = W + \Delta W$ During fine-tuning, only the parameters of the matrices A and B are trained, while the original weights W remain fixed. The number of parameters to be learned, $r \cdot (d_{\text{in}} + d_{\text{out}})$, is drastically less than $d_{\text{in}} \cdot d_{\text{out}}$ in the original matrix, which reduces memory requirements and speeds up training. During inference, adjustment is applied by adding $W + A \cdot B$, without additional latency.

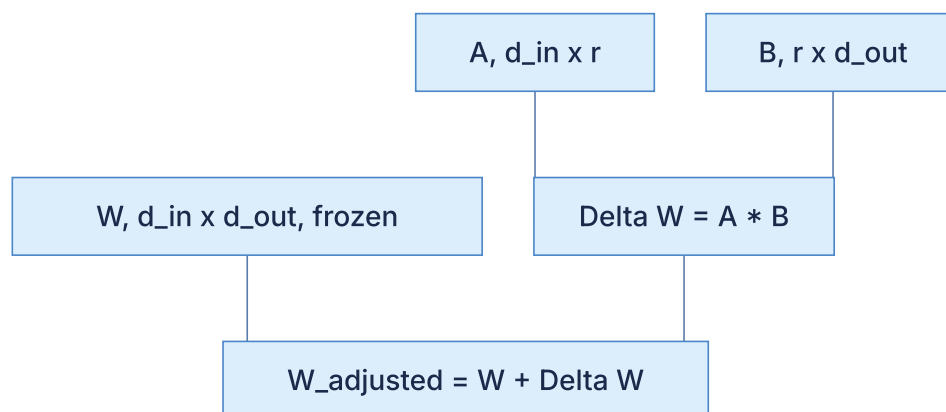


Figure 3.5a: LoRA - low-rank adjustment: frozen matrix W and product $A \cdot B$ gives $W_{\text{adjusted}} = W + \Delta W$.

Empirical studies consistently show that LoRA achieves performance comparable, and often identical, to that obtained by full fine-tuning of the entire model, but with an incomparably smaller proportion of training parameters and significantly reduced computing resource requirements. This efficiency makes LoRA one of the most important and widely applied techniques for adapting large language and other deep models to specific tasks.

In contrast to traditional model fine-tuning, which involves modification of internal weights, **prompt tuning** and **prefix tuning** methods apply a fundamentally different approach. Instead of directly changing the parameters of the basic model, they rely on adding small sets of continuous, machine-learned vectors that are strategically inserted either at the input layer of the model (according to Lester et al., 2021) or into its hidden layers (according to Li and Liang, 2021). These learned vectors, known as soft prompts or prefixes, act as programmable instructions. Their function is to precisely direct the behavior of an unchanged, frozen pre-trained language model according to the specific requirements of the desired task. This avoids the need to change the original parameters of the model, which significantly reduces computational costs and the potential for “catastrophic forgetting” (degradation of performance on the original model tasks after fine-tuning). Although this concept shares similarities with discrete query engineering, where queries are manually designed and selected, a significant advantage lies in automatic optimization. The optimal *soft query* is not chosen heuristically, but is found through an iterative learning process via gradient descent that more efficiently adapts the model for a wide range of further applications, without the need for extensive manual tuning.

PEFT techniques have revolutionized the way LLMs are adapted, making the process significantly faster, more memory efficient, and enabling easier management and serving of multiple different tasks using the same underlying model.

Another key strategy for optimizing efficiency, applicable both during training and, more importantly, during inference, is **quantization**. It refers to the process of reducing the numerical precision used to represent the parameters (weights) of the model and/or its internal activations during the calculation.

Quantization: Reduction of numerical precision of parameters and model activation (e.g. from 32 or 16 bits to 8 or 4 bits); reduces memory footprint and can speed up inference; applications include PTQ (post-training) and QAT (during training) (Dettmers et al., 2022).

Standard models typically use 32-bit (FP32) or 16-bit (FP16 or BF16) floating-point numbers. Quantization aims to convert these values into lower precision formats, most commonly 8-bit integers (INT8), and more recently even 4-bit (INT4) or even lower formats (Dettmers et al., 2022; Frantar et al., 2022; Yao et al., 2022). The main advantages of quantization are multiple: first, it drastically reduces the **memory footprint of the model (e.g. going from FP16 to INT8 cuts the model size in half, and to INT4 reduces it by a factor of four)**, allowing larger models to run on memory-constrained hardware; second, lower precision operations (especially integer) can be significantly faster on hardware that supports them (many modern processors and accelerators have specialized units to accelerate INT8 operations); third, reduced data transfer and simpler operations can lead to **less power consumption**. **The main challenge of quantization is to preserve the accuracy of the model, because reducing the accuracy inevitably introduces some error. There are two main paradigms:** Post-Training Quantization (PTQ), where an already pre-trained model is quantized without additional training (often using a small calibration data set to determine optimal quantization parameters), which is faster, but can lead to a larger drop in accuracy; and Quantization-Aware Training (QAT), where the effect of quantization is simulated during the fine-tuning or even pre-training process, allowing the model to adapt to lower precision and thus maintain a higher level of accuracy (Jacob et al., 2018). Thanks to advances in quantization techniques, today it is often possible to achieve significant compression (e.g. to 4 bits) with very little or almost no performance loss on downstream tasks, making quantization an extremely popular method for efficient implementation of LLMs. The third important optimization technique is pruning, which is based on the idea that many large neural models are overparameterized, that is, they contain a significant number of parameters (weights) or even entire structural components that are redundant or do not significantly contribute to the final performance of the model. Pruning aims to identify and permanently remove these less important elements, thereby reducing model size and potentially speeding up inference (LeCun et al., 1990; Han et al., 2015).

Pruning: Removal of less important parameters or entire structures (neurons, layers, attention heads) from the model to reduce size and speed up inference; it can be unstructured (single weights) or structured (whole units); it is often combined with additional fine-tuning (LeCun et al., 1990; Han et al., 2015).

We distinguish between two main types of pruning: unstructured and structured. **Unstructured pruning** (*unstructured*) removes individual weights from the weight matrices of the model, usually based on some importance criterion (e.g. removal of weights with the smallest absolute value – *magnitude pruning*). Although it can achieve very high thinning rates (e.g. remove 90% or more of the weights) while maintaining accuracy, the resulting weight matrices become sparse with irregular patterns of zeros. Standard hardware (GPU, CPU) is often not optimized for efficient use of this structure, so the achieved reduction in the number of parameters does not necessarily lead to a significant acceleration of inference in practice, unless specialized hardware or software libraries are used. **Structured pruning** (*structured*) removes entire, regular structural units of the model, such as individual neurons, channels in convolutional networks, or, in the context of transformers, entire attention heads (*attention heads*) or even entire layers (Li et al., 2016; Voita et al., 2019; Michel et al., 2019). Because entire blocks are removed, the resulting model is smaller but still has a dense structure that can be efficiently executed on standard hardware, often leading to real speedups in inference while reducing the number of parameters. Determining which structures to remove is usually based on an assessment of their importance to model performance. Pruning is often performed iteratively: the model is trained, some parameters are trimmed, and then the model is further fine-tuned to recover from the potential drop in accuracy caused by pruning. An interesting link is the **Lottery Ticket Hypothesis** (Frankle & Carbin, 2019), which postulates that within large, randomly initialized neural networks there are small subnets (“winning tickets”) that, when trained in isolation from the beginning with the same initialization, can achieve performance comparable to the original large network. Although finding those *slips* is not trivial, this hypothesis further supports the idea that large models are inherently redundant and that there are much smaller, more efficient architectures

waiting to be discovered or extracted. Techniques such as knowledge distillation (Hinton et al., 2015), where the knowledge of a large *teacher* model is transferred to a smaller *student* model, are also often used, sometimes in combination with pruning, to create smaller and faster models (e.g. DistilBERT (Sanh et al., 2020)). Techniques such as parametrically efficient fine-tuning, quantization and pruning, often used in combination, are key tools for bridging the gap between the theoretical capabilities of large language models and the requirements of practical applications in the real world. They make LLMs not only more feasible to implement in a wider range of scenarios, including those with limited resources, but also reduce their economic and environmental costs, contributing to their democratization and paving the way to a more sustainable AI future.

3.5.4 Measuring success: evaluation

Evaluation represents a key, but also a layered procedure in the development and understanding of intelligent systems, going beyond mere quantitative measurement to include a critical assessment of their purposefulness and effectiveness. In the context of artificial intelligence and autonomous agents, evaluation is an indispensable component of the entire research and development cycle. Its fundamental purpose is to provide an objective insight into the agent's performance, whereby its value is not exhausted in the mere determination of performance. On the contrary, meaningful evaluation seeks to penetrate the reasons behind a certain outcome, explaining why and under what conditions the agent acts successfully or unsuccessfully. This opens the way to targeted improvements and a deeper scientific understanding of observed phenomena. In a theoretical and practical sense, it is common to distinguish between two fundamental levels of evaluation, which are not mutually exclusive, but complement each other. These are internal and external evaluation. **Internal evaluation is focused on the analysis of individual components or modules within the agent itself.** Its goal is to evaluate the effectiveness of specific algorithms, models or mechanisms from which the agent was created. For example, with a natural language processing agent, internal evaluation could separately measure the accuracy of

the entity recognition module or the performance of the semantic analysis algorithm. Such an approach enables precise locating of defects and optimization of individual parts of the system, regardless of its final task. **External evaluation**, on the other hand, observes the agent as a complete, monolithic entity and measures its success in carrying out the final task intended for it within the given environment. In this approach, internal mechanisms are ignored, and attention is focused exclusively on the outcome. For example, if it is a chess-playing agent, an external evaluation would measure its winning percentage against other players or programs. With a customer service agent, the degree of customer satisfaction or the time required to resolve a query would be measured (Jones, 2021). This approach provides a final assessment of the agent's practical usability. In order for the evaluation procedure to be scientifically based and credible, it must rest on several key principles. First of all, the evaluation must be **objective**, which implies the use of clearly defined and measurable metrics that are independent of the subjective judgment of the researcher. Furthermore, it must be **repeatable**, which means that other researchers, using the same data, settings and methodology, must be able to obtain identical or statistically comparable results. This principle guarantees the verifiability and reliability of scientific knowledge. Equally important is **reproducibility**, a broader term that implies the possibility that other teams, starting from the description of the method, independently implement the system and conduct a similar experiment and come to the same conclusions. Ultimately, evaluation should not be understood as a static act of judgment, but as a dynamic and integral part of the development process. It forms a feedback loop that enables continuous improvement. Evaluation results, whether positive or negative, provide valuable information that guides further development, debugging, and refinement of the agent. This establishes an iterative cycle that is at the very core of progress in the field of artificial intelligence.

Iterative development cycle of the agent, where after evaluating and analyzing the results, a decision is made to improve and redevelop or end the cycle

3.5.5 Model in action: inference

After the artificial intelligence model has been successfully trained and evaluated, it moves from the development phase to the application phase. At that stage, its primary purpose becomes acting on new data to create value for the user or system. This process, in which the model generates predictions based on new, previously unseen input data, is called inference. It represents the pinnacle of the entire model development process, the moment when abstract knowledge, acquired during training, turns into concrete and useful answers. Unlike the training procedure, where the model parameters are dynamically adjusted in order to better map the input data to the desired output values, during inference the model remains static. Its learned internal structure and parameters are fixed and applied to new data without further modification (see e.g. Goodfellow et al., 2016). An agent, be it a human or a software system, inputs data into the model – for example, an image for classification, a sentence for translation, or a set of sensor readings. The model then, applying its learned internal representations, processes this data and provides relevant information as an output – prediction, classification, summary or some other form of structured response.

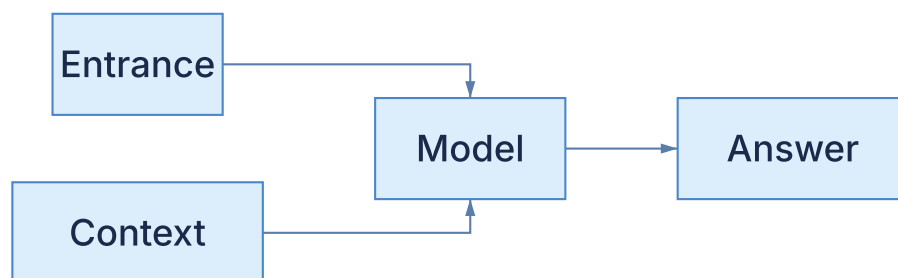


Figure 3.5b: The basic flow of the inference process, where a new input is run through a static, pre-trained model to generate an output.

3.5.6 Ways and environments of performing inference

From a technical point of view, inference can be carried out in two basic ways: batch and in real time (real-time or online). **Batch inference implies the processing of large amounts of data at once**, in predefined cycles. This approach is suitable for applications where immediate response is not critical, such as generating daily business reports, analyzing market trends on a weekly basis, or processing archival data. The main advantage of this method is the high throughput and efficiency of processing large data sets.

Group inference / batch inference: A way of executing a model in which large amounts of data are processed at once in predefined cycles; suitable when latency is not critical (reports, analysis, archive); enables high throughput and efficiency.

Real-time inference, on the other hand, refers to the processing of individual or small groups of data as soon as they become available, with an emphasis on minimal latency, i.e. the time required to obtain an answer. Such an approach is necessary in systems where quick reaction is essential, for example in the detection of fraud in card transactions, content recommendation systems on websites or in the management of autonomous vehicles.

Inference in real-time: A way of running a model (*real-time* or *online* inference) in which individual or small data sets are processed immediately upon availability, with an emphasis on minimal latency; necessary in systems for fraud detection, recommendations, autonomous driving and similar applications.

The choice between batch and real-time inference is closely related to the system architecture and the environment in which the model is executed. Models are often placed on cloud servers, which provides scalability and access to large computing resources, but may introduce latency due to network communication. On the other hand, in order to reduce latency and ensure data privacy, models are increasingly performed on edge devices (*edge computing*), such as smartphones, industrial sensors or cars, or on local infrastructure (*on-premise*).

Edge computing: Performing computer processing and models on edge devices (phones, sensors, vehicles) instead of exclusively in the central cloud; reduces latency and the need for data transfer, suitable for privacy and real-time applications.

Inference is therefore the moment when an abstract mathematical model becomes a concrete tool, capable of making decisions, automating processes and providing insights into the real world. It is a bridge between the theory of machine learning and its practical application.

3.6 Challenges and Limitations: Shadows in the Digital Mirror

While large language models demonstrate increasingly impressive abilities in understanding and generating language, opening the door to transformative applications in almost all spheres of human activity, their rapid development and implementation are inevitably accompanied by a series of serious challenges and inherent limitations that cast a shadow over their potential. Issues related to the security of their use, the reliability of the information they generate, and the potential for misuse of their powerful capabilities are critical obstacles that cannot be ignored.

Bias: Systemic deviation in algorithm or model behavior resulting from training data – e.g. social prejudices, stereotypes or underrepresentation of certain groups; it can lead to discrimination in decisions (employment, credit, criminal proceedings) and the perpetuation of injustice (O’Neil, 2016).

One of the fundamental difficulties, rooted in the very essence of machine learning, is the issue of **algorithmic bias**. Artificial intelligence systems learn based on the data that humans provide them, and this data often reflects existing social prejudices, stereotypes and inequalities. An algorithm, as a mathematically unbiased entity, cannot by itself correct the human shortcomings woven into data sets; moreover, it can amplify and automate them, thus creating systems that systematically discriminate against certain

groups of people in employment, credit approval or even in criminal proceedings (O’Neil, 2016). Thus, technology, instead of being a tool for achieving justice, turns into a tool for perpetuating and legitimizing injustice. This is directly connected to the problem of **transparency** and **explainability**, often summed up in the term “black box”.

Transparency (in the context of AI) (*transparency*): Availability of information about how the system works - what data and algorithms are used, how decisions are made and who is responsible; it is a prerequisite for supervision, trust and responsibility. With complex models (e.g. deep networks), transparency is limited because the internal logic is difficult to read (“black box”).

Explainability: The ability to provide the user or supervisor with an understandable reason why the system made a certain decision or prediction; in the AI context it is often investigated through XAI (explainable AI) methods. It is crucial in areas of high risk (medicine, law, credit) where it is necessary to justify and verify the outcome of the algorithm (Pasquale, 2015).

Many advanced artificial intelligence models, such as deep neural networks, operate in a way that is extremely difficult or even impossible to fully interpret. Although the system can make an extremely precise decision or prediction, the path by which it arrived at that decision remains hidden within complex layers of mathematical operations. This lack of transparency poses serious obstacles, especially in high-risk areas such as medicine or autonomous driving. How can we trust the machine’s decision if we don’t understand its logic? Who is responsible when such an inexplicable system goes wrong? The inability to see into the agent’s reasoning undermines trust and prevents effective control and error correction (Pasquale, 2015). In addition to technical obstacles, the application of artificial intelligence also poses profound social and economic challenges. Automation, powered by increasingly capable intelligent agents, threatens to transform the labor market on an unprecedented level. While the previous industrial revolutions mostly replaced manual work, today’s technology is increasingly encroaching on cognitive tasks that were considered exclusively human domain. The issue of massive job losses and the consequent growth of economic inequality is raised. The creation of new jobs may not keep pace with the disappearance of old ones, which may lead to a deepening of the

digital divide – not only in access to technology, but also in the skills needed to navigate the new economic order. At the center of these technological and social changes is man, whose privacy, autonomy and well-being are facing new challenges. The operation of intelligent systems is based on the collection and processing of huge amounts of data, often personal. This erases the boundaries between private and public, and the surveillance possibilities become almost unlimited. The constant monitoring of digital traces, carried out by both corporations and state apparatuses, opens the door to manipulation, from subtly influencing consumer habits to shaping political opinion within personalized information bubbles and echoes (Zuboff, 2019). Man's autonomy in decision-making is thereby threatened, and his exposure to algorithmic management is increasing. Furthermore, it is necessary to warn about the danger of excessive reliance on technology and the gradual loss of human skills. If we leave critical thinking, memory or even creativity to machines, we risk our own cognitive atrophy. The ability of machines to generate convincing text, images and sounds also raises the question of authenticity and truth. In a world where it is increasingly difficult to distinguish the real from the fake, and misinformation spreads at the speed of light, trust in institutions and in the media space itself is deeply shaken. All the mentioned challenges point to the fundamental need for the establishment of solid ethical frameworks and regulatory mechanisms. The development of technology far outpaces the speed at which laws are passed, creating a regulatory vacuum in which decisions about the future of humanity are left to technology companies driven by profit interests. The question of responsibility, the global nature of artificial intelligence that escapes national legislation, and the danger of its misuse for military purposes and the creation of autonomous weapons are just some of the problems that require urgent and coordinated international cooperation. Facing the shadows in the digital mirror is not a call to reject technology but to its thoughtful and responsible development. Recognizing the challenges is the first step towards solving them – a step that requires the collaboration of experts from various fields, from the technical sciences to philosophy and law, to ensure that artificial intelligence truly serves the benefit of all humanity, not just a narrow circle of the privileged few.

3.6.1 Security, Reliability and Abuse

Any comprehensive reflection on the reach and application of autonomous agents inevitably touches on three key and interconnected areas: system security, its reliability in operation and the possibility of misuse. Those three elements form the basic framework for assessing the sustainability, ethics and practical usability of any agent system, because neglecting any of them can lead to serious technical, social or security consequences.

Security

Security, in the context of autonomous agents, refers to the protection of the system, its data and processes from unauthorized access, modification or destruction. It includes technical measures that ensure the integrity and confidentiality of the system, as well as its resistance to external attacks. Malicious actors may attempt to exploit vulnerabilities in program code, system architecture, or communication protocols to take control of the agent, steal sensitive data that the agent handles, or disable it. A special challenge in the field of artificial intelligence is represented by so-called **adversary attacks**, where input data (for example, an image or text) is subtly changed in a way that is imperceptible to humans, but which leads the agent to a completely wrong conclusion or action (Goodfellow et al., 2014).

Adversial attack: Deliberate alteration of input data (image, text) subtle to the human eye, but which leads the model to make a wrong classification or decision; it requires robust models and verification mechanisms beyond traditional information security (Goodfellow et al., 2014).

Protection against such attacks requires robust models and verification mechanisms that go beyond traditional information security methods. A security breach not only compromises the agent's functionality, but directly undermines user trust and opens the door to further abuse.

Reliability

Closely related to security is **reliability**, which means consistency and predictability in the agent's work. While security deals with protection against external threats, reliability focuses on the internal correctness and robustness of the system. A reliable agent must consistently perform its tasks in accordance with defined parameters and goals, even in unforeseen circumstances or when encountering new, unexpected data. A system that is reliable is not fragile; it is robust to errors caused by edge cases, noise in the data, or its own internal limitations. One of the key reliability problems in modern language models is the phenomenon of “hallucinations”, where the agent generates convincing, but factually incorrect or meaningless information (Ji et al., 2023). Ensuring reliability therefore implies rigorous testing, validation and establishment of mechanisms for detecting and correcting anomalies in agent behavior. An unreliable agent, even if it is perfectly safe, loses its practical value because its decisions and outcomes cannot be relied upon.

Abuse

Finally, the third member of that trinity is **misuse**, which refers to the intentional use of an agent for harmful, unethical or illegal purposes. Unlike security flaws that are exploited by external attackers, misuse implies that an agent is used as designed, but to achieve malicious goals. Paradoxically, the very features that make agents powerful—autonomy, the ability to learn, adapt, and lead complex interactions—also open the door to their abuse. The possibilities of misuse are manifold: from the automated dissemination of disinformation and propaganda, through the execution of sophisticated social engineering attacks and identity theft, to the coordination of malicious activities in the digital or physical world. The agent can be used to create fake content (*deepfakes*), bypass security checks or automate processes that harm individuals or organizations. The problem of abuse goes beyond the purely technical domain and goes deep into ethical and legal issues, posing to society the challenge of establishing norms and regulations that will reduce potential damages. Those three aspects – security, reliability and abuse – are not independent categories, but form a dynamic and connected system. Weak

security facilitates abuse, while low reliability can create unforeseen security vulnerabilities. Effectively dealing with these challenges requires a holistic approach that combines technical innovations in the field of robustness and protection, with clear ethical guidelines and legal frameworks for the responsible development and application of autonomous agent technology.

3.6.2 Bias, Privacy and Data Ethics: The Dark Side of Digital Mirrors

Digital systems, especially those based on artificial intelligence and machine learning, are often thought of as objective, neutral mirrors that reflect the world as it is. This metaphor, although seductive, hides a deeper and more complex truth. The mirrors we create are not perfectly flat or impeccably clean; they are distorted, and the reflections they offer carry within them the shadows of bias, violated privacy and ethical doubts arising from the very foundations on which they were created - from data. Looking at the dark side of these digital mirrors is key to understanding their real impact on society. The fundamental problem stems from the nature of the data the algorithms are fed with. Contrary to the widespread, almost mythical belief about the objectivity of the machine, algorithmic bias is not an intrinsic flaw of the code, but an inevitable reflection of the data that the algorithm is “fed”. These data are not abstract entities but distilled human experience, with all its historical injustices, social inequalities and prejudices. When an algorithm learns from employment data that reflects decades of discrimination against women in certain sectors, it learns not about objective qualifications, but about historical reality. His model will, consequently, perpetuate that same discrimination, not out of malice, but out of the logic of a learned pattern. Face recognition systems that show significantly lower accuracy for darker-skinned people or women are not the result of bad programming, but rather the insufficient and unrepresentative representation of these groups in the training data sets (Buolamwini & Gebru, 2018). Thus, the digital mirror not only reflects the existing social flaw, but also reinforces it, giving it the appearance of technological unquestionability and creating a dangerous vicious circle.

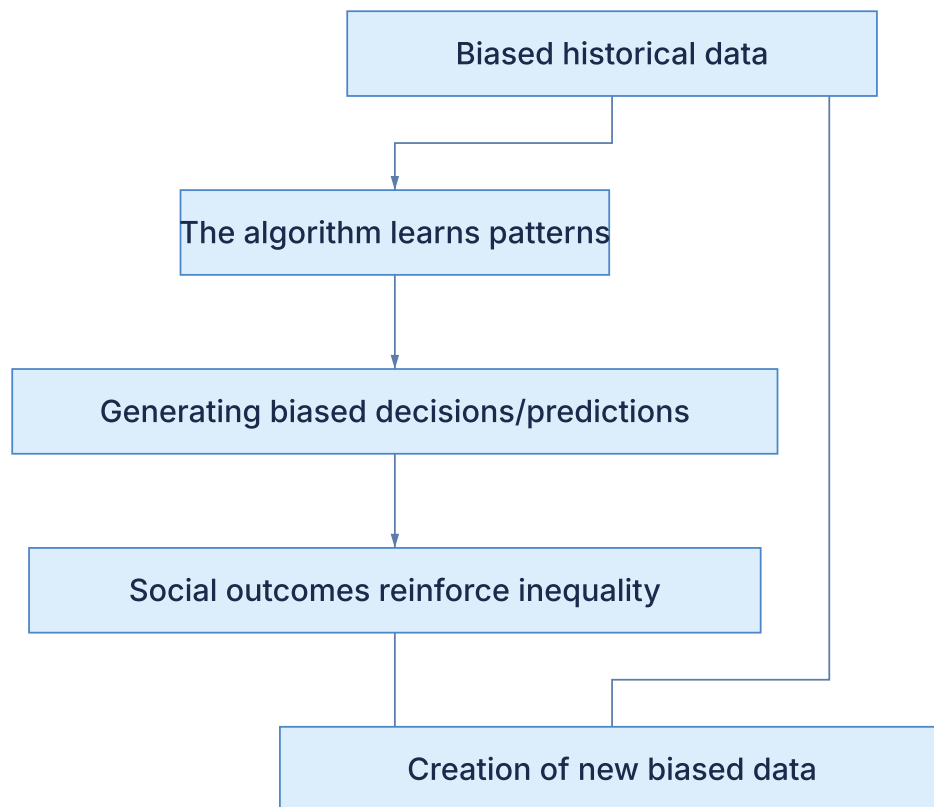


Figure 3.9: Schematic representation of the cycle in which biased data leads to biased models, whose decisions then reinforce and legitimize the initial social biases.

Along with the problem of bias, the issue of privacy is equally serious. The age of big data (*Big Data*) gave birth to the economy of surveillance (*surveillance capitalism*), in which personal data is considered raw material, the most valuable resource of the new age (Zuboff, 2019). Each of our digital traces – from searches on the Internet, through location data and communication on social networks, to biometric information – becomes an object of collection, processing and monetization. The boundary between the private and public spheres is becoming increasingly permeable, and the consent given by users is often just an illusion, hidden behind impenetrable legal formulations and unexplainable user conditions. The consequences of such conditions go deeper than targeted advertising. The loss of privacy threatens the very core of personal autonomy, freedom of expression and the ability to act without fear of constant surveillance and judgment. When information about our health,

political beliefs or intimate relationships becomes available to third parties - be it corporations, government bodies or malicious *agents* - the scope for manipulation, blackmail and social control on an unfathomable scale opens up. Ultimately, bias and privacy merge into a unique and layered ethical challenge. Who is responsible for the harm caused by a biased algorithm? Is it the programmer who wrote the code, the company that implemented the system, the user who uncritically accepted his decision, or the society that created the data burdened with injustice? Traditional models of accountability, designed for human actors and predictable cause-and-effect chains, prove inadequate in a world where autonomous *agents* make decisions with far-reaching consequences. This requires us to develop new ethical frameworks that will be able to encompass the complexity of the digital ecosystem. A radical paradigm shift is needed: from reactive error correction to proactive, ethically based design (Ethics by Design). This implies transparency in the operation of algorithms, the right to explain automated decisions, and the establishment of independent bodies for supervision and audit. Digital mirrors are therefore not passive observers, but active participants in reshaping our reality, often in ways that are subtle but profoundly influential. And the challenge before us is not to break those mirrors, but to learn how to build them with wisdom and maintain them with a sense of responsibility, aware that the reflection we see in them is not only a picture of the world as it is, but also a blueprint of the world as it could become, through our actions or inaction.

3.6.3 Resource intensity: the price of intelligence

Any manifestation of intelligence, whether biological or artificially created, carries with it an inevitable cost. In the field of artificial intelligence, especially in the field of deep neural networks and large language models, this cost takes on enormous proportions, manifesting itself not only in financial expenses, but also in the consumption of material and energy resources. The drive towards increasingly advanced and capable agents has created a paradigm in which the complexity and performance of models is directly related to the amount of computing resources invested. This **resource intensity** has therefore become

a key feature of the modern development of artificial intelligence, raising fundamental questions about its sustainability, availability and environmental consequences.

Resource intensity: Demanding development and application of AI models in terms of computing power, memory, energy and financial costs; especially pronounced for large language models whose training and inference require massive GPU/TPU clusters, which limits availability and has environmental consequences.

The fundamental driver of this exceptional computational hunger is the scaling paradigm, an empirically confirmed law according to which model performance improves at a predictable rate with an increase in the number of parameters, the size of the learning data set, and the amount of computing power invested in the learning process (Kaplan et al., 2020).

Scaling laws: An empirical insight according to which the performance of language models increases predictably with the increase of three factors: the number of model parameters, the amount of learning data and the amount of computing power; it encouraged a race to build increasingly large models (Kaplan et al., 2020).

The scaling paradigm stimulated a kind of race in the construction of increasingly large models, whose architectures contain hundreds of billions, even trillions, of parameters. Learning such colossal structures requires time-consuming and complex computations on specialized circuits, such as graphics processing units (GPUs) and tensor processing units (TPUs), organized into massive clusters that operate for weeks or even months non-stop.

TPU: Tensor Processing Unit – a chip designed to accelerate machine learning operations, especially matrix calculations in deep networks; optimized for high throughput and energy efficiency in training and inference of large models.

The costs of such an undertaking reach tens and sometimes hundreds of millions of dollars for just one learning cycle, including the cost of hardware, electricity and maintenance. A direct consequence of this enormous energy

consumption is a significant ecological footprint. The process of learning large artificial intelligence models is one of the most energy-intensive computing tasks. Studies that have sought to quantify this impact have yielded troubling results. For example, the learning process of a large language model with architecture search is estimated to have produced an amount of carbon dioxide that is comparable to five average car life cycles, including their production (Strubell et al., 2019). This realization puts technological progress in direct conflict with global efforts to reduce greenhouse gas emissions, imposing an ethical dilemma on researchers and companies operating in the field.

The increase in the size of models and data leads to an exponential growth of computing requirements, resulting in high financial costs, high energy consumption and obstacles to democratization and a significant ecological footprint. Furthermore, the high cost of computing resources creates a deep gap and poses serious obstacles to the democratization of technology. The development of the most advanced models becomes the exclusive domain of a small number of technological giants and national laboratories that possess sufficient capital and infrastructure. Academic institutions, independent researchers and smaller companies are finding it increasingly difficult to keep up, leading to the consolidation of power and influence in the hands of a few.

Computing gap: Asymmetry in access to computing resources (GPU/TPU, energy, capital) needed to develop and run large AI models; it limits which actors can participate in development, leading to consolidation of power and biases in technology.

Such a “computing gap” limits the diversity of ideas, slows scientific review and replication of results, and opens up space for biases that reflect the values and interests of those who can afford to develop such systems. It is important to distinguish between training costs and inference costs.

Learning costs and inference costs: Learning costs (*training costs*) – one-time, but extremely high costs of training the model (hardware, energy, time). Inference costs – continuous operational costs of applying the already learned model to each query; at billions of queries per day cumulatively they can be huge.

While learning is a one-time but extremely intensive process, inference refers to the continuous use of an already learned model to perform specific tasks. Although the resource consumption for individual inference is significantly less than for learning, the cumulative cost can be enormous when the model is applied to millions or billions of queries per day, as is the case with popular digital assistants or translation services. Therefore, the total cost of intelligence should include the investments necessary for its creation, as well as the permanent operating cost of its existence and operation. Addressing this multi-layered challenge requires a shift towards the development of more energy-efficient algorithms, optimized hardware solutions, and more transparent reporting of the real cost of progress in the field of artificial intelligence.

3.7 Transformation of communication: applications and perspectives

The emergence and wider application of artificial intelligence, especially large language models, brings radical changes to the very core of human communication, reshaping its fundamental paradigms and opening up unimagined possibilities, but also posing new challenges. We are witnessing a period in which technology ceases to be perceived exclusively as a passive channel or tool and takes on the characteristics of an active participant and even co-creator of communication processes. This transformation fundamentally changes the way individuals, organizations and entire societies create, exchange and interpret information. One of the most prominent applications is manifested in the possibility of creating hyper-personalized content, where communication messages are tailored according to the specific needs, interests and previous behavior of an individual user. In the field of marketing and advertising, this enables the delivery of relevant offers that significantly increase consumer engagement, going beyond traditional, mass approaches (Chen et al., 2021). Similar principles are applied in the media, where algorithms suggest news and articles according to the reader's habits, creating a unique information experience for each individual. These principles of personalization also extend to the area of customer support, which is

undergoing a thorough transformation. Once limited and scripted *chatbots* are giving way to advanced conversational agents capable of conducting complex dialogues, understanding context and providing solutions to a wide range of user queries in real time. Such systems not only increase efficiency and reduce operating costs but also improve customer satisfaction by providing immediate and constantly available support (Kumar et al., 2022). Furthermore, artificial intelligence is becoming an unavoidable factor in the content creation process itself. Large language models are used today to compose marketing texts, news reports, social media posts, and even to write programming code or creative literary works. The role of the human creator thus changes - he increasingly becomes an editor, curator or collaborator who directs and refines the content generated by the machine. Such a collaborative relationship opens up enormous creative possibilities, but at the same time raises complex questions about authorship, originality and credibility of information (Liang et al., 2023). Viewed in a long-term perspective, the development of autonomous agents represents perhaps the most profound change on the horizon. These sophisticated software entities, powered by artificial intelligence, are designed to act proactively on behalf of their users, performing complex tasks that require planning and interaction with other systems and people. An agent could, for example, independently organize a business trip, negotiate the terms of a contract or manage the user's digital identity. In such a scenario, communication no longer takes place only between people or between man and machine, but also between autonomous agents representing human interests.

Finally, it should be pointed out that this technological revolution carries with it considerable ethical and social implications. The possibility of generating convincing but **false information** (*deepfakes*) represents a serious threat to social trust and democratic processes. The automation of communication roles raises questions about the future of workplaces, while the ubiquitous mediation of technology prompts discussions about alienation and the loss of authenticity in interpersonal relationships (Brynjolfsson and McAfee, 2018). Therefore, in parallel with technological progress, it is crucial to develop strong ethical frameworks and monitoring mechanisms that will ensure that the transformation of communication serves the well-being of humanity, and not its undermining. The future of communication will inevitably be shaped by the

interplay of human ingenuity and artificial intelligence, and the outcome of that interplay will depend on the wisdom with which we manage its development.

4. Deconstruction of language in the age of AI: from symbols to social reality

The entry of large language models into everyday life in any case represents a technological breakthrough, but in many ways it also encourages a thorough rethinking of the very essence of language. The operation of artificial intelligence forces us to consider language beyond its communicative function and to understand it as a complex system that is being deconstructed and reshaped before our eyes. This process of decomposition does not lead to nothingness, but reveals the path from the traditional understanding of language as a system of symbols to its understanding as an active force that directly shapes social reality. Classical linguistic thought, especially structuralist thought, viewed language as an ordered system of signs in which each *signifier* (word, phoneme) refers to a certain *signified* (concept, object). This connection, although arbitrary, within the language community was stable and rested on collective agreement and human experience. Linguistic models, however, destroyed that connection. They do not handle meaning in a human way; for them, language is not a network of concepts but a vast statistical space. Words are decomposed into mathematical entities, vectors, and their value is not determined according to what they mean in extra-linguistic reality, but according to the probability of occurrence in relation to other vectors within a given context. It is a mathematical deconstruction: language is stripped of its semantic core and reduced to a calculable structure.

Language deconstruction in the context of AI: The process by which large language models decompose language into mathematical entities (vectors), separating it from extra-linguistic reality and semantic core; language is reduced to a computable structure of statistical patterns, thus exposing its fundamental active power independent of meaning in the human sense.

The paradox arises at the moment when these same systems, after breaking down the language into mathematical foundations, (re)construct from those foundations a text that for the human recipient has all apparent coherence, meaningfulness, and even stylistic persuasiveness. The model does not “understand” what it writes, but it produces language that *acts* as if it came from understanding. In this sense, the language model becomes an agent, an entity whose action, although devoid of consciousness and intention in the human sense, produces real effects in the world. Its output, reflecting the data on which it was trained, creates new linguistic constructs that interact with humans and other systems. This shifts the focus of analysis from symbols to action. Language generated by artificial intelligence is not neutral. It writes programming code, drafts legal documents, creates newspaper articles, manages customer support and shapes public opinion through social media. Each of these actions represents a speech act that has a performative power - it does not describe the world, but actively changes it. The scale and speed at which these new language agents operate are unprecedented in human history.

Language act / speech act: A spoken or written act that not only describes reality but also does something - a promise, an order, an obligation, the execution of a transaction; it has a performative power that changes social reality (Austin, Searle).

This transformation process can be shown as follows:

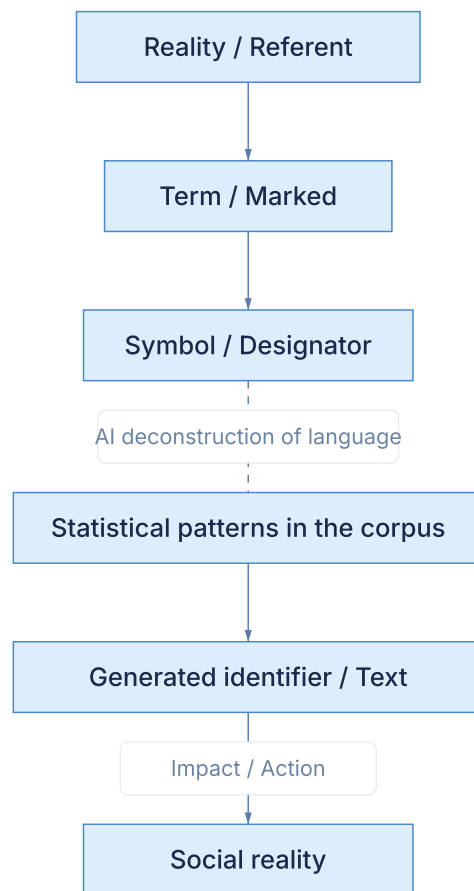


Figure 4.1: Traditional understanding of meaning, AI deconstruction and (re)construction of language and their impact on shaping social reality.

Therefore, the mathematical deconstruction of language carried out by artificial intelligence is not its end but the unveiling of its fundamental power. By exposing the symbolic level, AI clearly shows us that language has always been primarily a tool for shaping reality. In modern times, that tool is no longer exclusively in human hands. Our society, knowledge and mutual relations become a fabric that is woven together by human and machine linguistic agents, erasing previously clear boundaries and opening up space for completely new forms of social reality.

4.1 Language as a structure of meaning: symbols, relations and mental maps

Language represents the fundamental structure by which man conceives the world and himself in it. It is primarily a symbolic system, a complex weave in which every element, every word, transcends its sound or written appearance and becomes the bearer of meaning. Words as symbols stand in for things, concepts, feelings and phenomena, allowing us to mentally handle reality even when it is not immediately present. This ability to abstract is essential for the development of human thought, culture and society. Each symbol, although often arbitrary in its form (for example, there is no inherent reason why we call a tree a *tree*), acquires its permanence and communicative value through social agreement or convention (cf. de Saussure, 1916). However, the true power of languages lies in the complex web of relationships in which they are woven. The meaning derives from the position of the word within the sentence, its relationship with other words, and the grammatical and syntactic rules that govern their connection. These rules, which make up the grammar of a language, are a generative mechanism that enables the creation of an infinite number of statements from a finite number of elements. Structure is therefore what breathes life into symbols and enables precise and nuanced expression. The sentences “The agent observes the target” and “The target observes the agent” use the same symbols, but their structural difference creates two completely different meanings and describes two different realities. This external, linguistic structure is directly mapped onto the internal, cognitive level. Through language, **mental** maps are formed in the mind of each individual - internal models of the world that serve as frames of reference for understanding, remembering, inferring and planning.

Mental Map: An internal, cognitive representation of the world (concepts, relationships, experiences) that an individual builds and updates through language and experience; serves to interpret situations, make decisions and plan; it is not static, but constantly adapting.

These maps are not static photos of reality, but dynamic and adaptable constructions that are constantly updated with new information and experiences. They represent the cognitive architecture within which the agent, be it human or artificial intelligence, interprets stimuli from the environment, predicts future events and makes decisions about its own actions. Language thus becomes a tool used to build and maintain our inner world; it is the scaffolding of our knowledge (Vygotsky, 1978). This process, from the perception of reality to purposeful action guided by an internal model, can be shown schematically as follows:

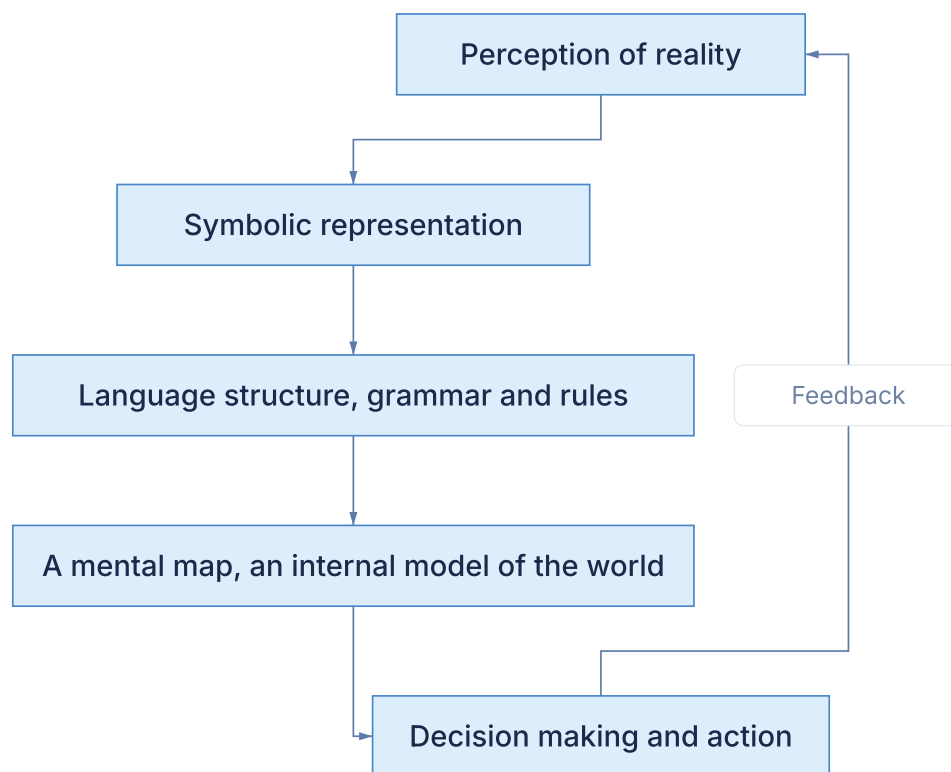


Figure 4.2: Representation of the agent's cognitive process, which includes perception, symbolic representation, language structure, mental map and action.

Accordingly, language is a formative force that actively participates in the description and construction of reality for each individual agent. Through the interplay of symbols, their interrelationships and the mental maps that result from that game, language enables not only communication with others, but

also internal dialogue, which is a prerequisite for a conscious and meaningful existence in the world. Understanding this structural dimension of language is crucial for understanding the foundations of human cognition.

4.1.1 Structuralist echoes in the architecture of LLMs: meaning as relation

Looking at the architecture of large language models (LLM) through the prism of linguistic structuralism, especially that of Ferdinand de Saussure, reveals astonishing conceptual similarities. Although we are talking about disciplines that are divided by an entire century and completely different scientific and technological contexts, the basic assumption about the nature of meaning as a consequence of relations within the system proves to be crucial for understanding both phenomena. The way LLMs process and generate language, mimicking human speech, essentially operationalizes Saussure's fundamental insight that meaning is differential and relational. The founder of modern linguistics, Ferdinand de Saussure, claimed that language (French *langue*) should be understood as a system of signs in which the value of each individual element is determined solely by its relationship to all other elements (Saussure, 1916/1959). The meaning of a word does not derive from a direct and unchanging connection with a real object or concept in the world, but from its place within the linguistic structure. The value, for example, of the noun *table* is not contained in the object itself, but is established in its correlation and opposition to the nouns *chair*, *bench*, *bed* and all other components of the lexical field "furniture". In language, according to Saussure, there are only differences (French *différences*), and without positive articles. Language is therefore a form, not a substance, a pure network of relationships. If we transfer this thought to the field of computational linguistics, we will see how large language models, although seemingly unaware of such a theoretical background, function in a strikingly similar way. The basis of their work is not semantic "understanding" in the human sense, but mathematical processing of huge amounts of textual data. Each word or sub-word (token) in the model is not represented as an isolated term, but as a vector in a multidimensional space - the so-called *embedding*. The position of that vector, and thus its

“meaning” for the model, is defined solely by its geometric distance and orientation in relation to all other vectors in that space. The model learns these relationships by analyzing statistical patterns and word co-occurrence patterns in the learning corpus (Vaswani et al., 2017). Thus, the vector space of LLM becomes a direct computer analogue of Saussure’s term *langue*. What is an abstract system of oppositions and relations in structuralism becomes a concrete, mathematically defined space in the architecture of language models. The “meaning” of the token *king* to the model is not related to the historical figure of the monarch, but is determined by its vector proximity to the tokens *queen* and *throne*, its specific relation to the token *man* (similar to that between *queens* and *women*), and its distance from tokens like *apples* or *run*. The meaning is thus completely immanent to the system, it is its relational position. —

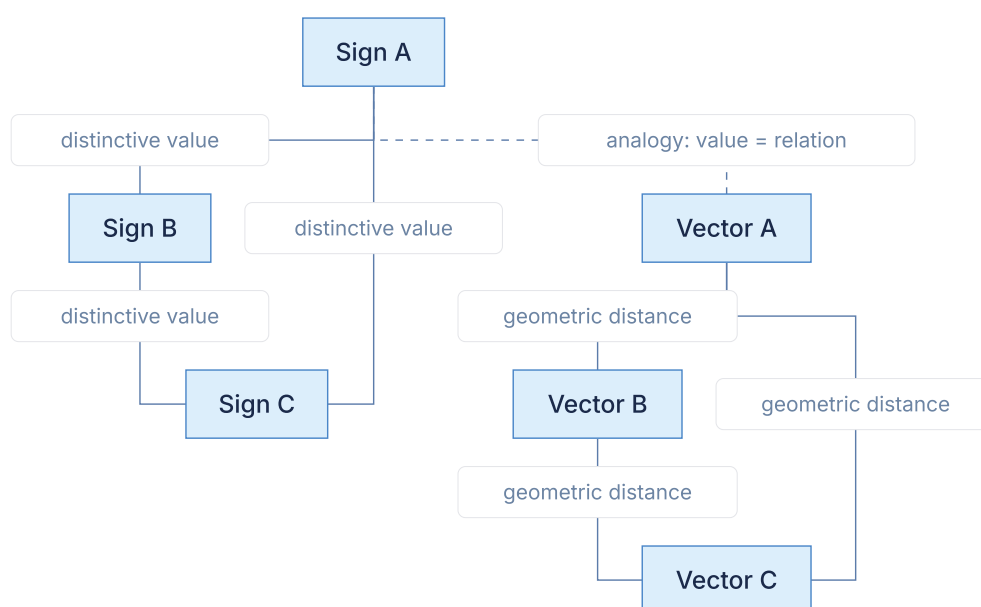


Figure 4.3: Conceptual analogy between Saussure’s language value system and vector space in large language models. The value of an element in both systems derives from its relationship with other elements.

Such an approach allows us to demystify the apparent “intelligence” of these systems. Their ability to generate coherent, contextually appropriate, and stylistically convincing text derives not from consciousness or intent, but from

a remarkably successful replication of the dense relational web that constitutes human language. The agent, in this case LLM, does not handle concepts, but rather formal relationships between signs that it has learned from the data. Its astonishing competence is not a sign of emergent consciousness but evidence that structure itself is the carrier of what we perceive as meaning (Bender et al., 2021). The structuralist perspective thus offers a solid theoretical framework for interpreting both the internal mechanics and the external manifestations of the operation of large language models. Although they are the product of modern engineering, their basic architecture inadvertently confirms one of the central postulates of twentieth-century linguistics: meaning within a closed semiotic system is a function of relations and differences. For LLM, “meaning” is its relational structure - it is fully summed up in a complex dance of vectors within a vast mathematical space, which reflects Saussure’s thought about language as a system in which everything is difference.

4.1.2 The Semantic Triangle and the Cognitive Dimension: The Meaning Challenge for AI

One of the fundamental issues facing artificial intelligence is an in-depth understanding of the meaning behind the processing of linguistic data. While modern language models show an enviable ability in recognizing and generating syntactically correct sentences, the true understanding of semantic substance remains one of the biggest challenges. In order to shed light on the depth of this problem, it is necessary to look at classical models of meaning, among which the semantic triangle, also known as the meaning triangle, stands out. This model, which was put forward by Charles Kay Ogden and Ivor Armstrong Richards (1923) in their work *The Meaning of Meaning*, offers a schematic representation of the relationships that create meaning. A triangle consists of three vertices: symbol, thought (concept or reference) and referent.

Semantic triangle / triangle of meaning: Model (Ogden & Richards, 1923) that shows meaning as a relationship of three elements: symbol (linguistic expression), thought/concept (mental representation) and referent (object in the world); the connection between symbol and referent is indirect - it leads through thought; useful for understanding the limitations of AI in “anchoring” meaning.

Symbol represents a linguistic expression – a word, phrase or sign (for example, a sequence of letters “p-a-s”). **Thought or concept** is a mental image, concept or idea that a symbol evokes in the mind of the speaker or listener. It is the inner, abstract content that we associate with a certain symbol. **Referent** is a real object, phenomenon or entity in extra-linguistic reality to which the symbol, through thought, refers – in our example, a concrete four-legged barking being.

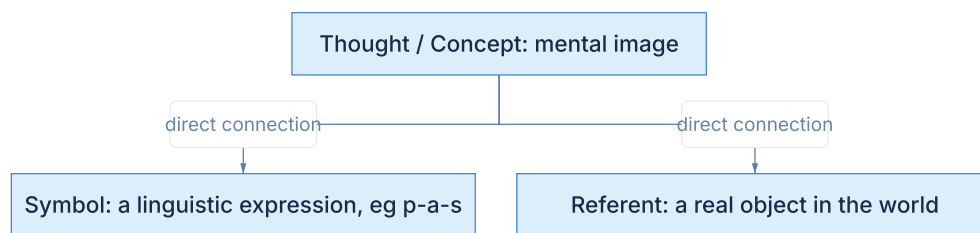


Figure 4.4: Semantic triangle (Ogden and Richards, 1923): thought/concept is directly connected to symbol and referent, while the connection between symbol and referent is indirect and arbitrary.

The key insight of the Ogden and Richards model lies in the nature of the connections between these three points. While the connection between symbol and thought, as well as between thought and referent, is real and causal within the cognitive system, the connection between the symbol itself and the referent is intermittent, i.e. indirect and arbitrary. The word “dog” itself has no intrinsic connection with an actual dog; this connection is established exclusively through human consciousness, that is, through the concept that mediates between the sign and what it signifies.

The concept is directly connected to the symbol and the referent, while the connection between the symbol and the referent is indirect. It is precisely in this dotted line that the fundamental challenge for artificial intelligence lies. Contemporary language models handle the apex of the triangle belonging to the **symbol** with exceptional skill. They process vast amounts of text and learn the statistical probabilities with which certain symbols appear next to each other. However, the key disadvantage arises in the establishment of connections with the other two vertices of the triangle. For artificial intelligence, the symbol “dog” is not associated with the mental concept of a dog (which includes abstract knowledge of species, behavior, biology) nor with a referent in the real world (the experience of meeting a dog, its warmth, bark or smell). Her “understanding” derives exclusively from textual correlations - words that often appear with “dog” are “barking”, “bone”, “leash”, “faithful”. AI thus creates a complex network of associations between symbols, but this network remains closed within its own digital universe, without a foothold in reality. This deficiency is called the grounding problem in cognitive linguistics.

Anchoring problem / grounding problem: The challenge of connecting language symbols with extra-linguistic reality and experience; AI models learn statistical correlations among cues, but do not have the immediate access to referents or bodily experience that humans do.

Furthermore, meaning is not exhausted within the schematic relationship of symbol, concept and referent. It also has a deep **cognitive** dimension, which is inseparable from the agent using language. Human understanding of meaning is enriched by bodily experience (*embodiment*), cultural heritage, emotional context and personal memories. The word “home” for man, in addition to being a symbol for the building in which he lives, evokes a feeling of security, warmth, belonging and memories related to family - layers of meaning that are built up through life experience. Artificial intelligence, as a disembodied and ahistorical entity, is deprived of such a cognitive and experiential foundation. She does not have a body with which to experience the world, she does not belong to a culture that would shape her values, and she does not possess a personal history that would give her knowledge a unique stamp. Her knowledge is wide but shallow; it is a reflection of humanity’s collective

linguistic creativity, but without access to the world from which that language arose. Therefore, the challenge of meaning for artificial intelligence cannot be solved solely with greater computing resources or more complex algorithms. It is a fundamental ontological and cognitive gap. For a system to truly understand meaning in a human way, it would have to somehow replicate at least the logical connections of the semantic triangle, if not the entire cognitive architecture that allows humans to fill symbols with living, personal, and anchored meaning. Recognizing this limit is essential for correctly evaluating the possibilities and limitations of today's language technologies.

4.2 Language as script: AI agents and the simulation of social interactions

The notion of language as an abstract system of signs and rules, which serves to describe the world and communicate with each other, is deeply rooted in Western thought. However, in the context of the development of advanced artificial intelligence, especially autonomous agents, language acquires a new, active role: it becomes a *scenario*, a constitutive element that shapes and moves the simulated reality. In this chapter, we explore how natural language serves as a fundamental mechanism for programming, directing, and observing complex social interactions within virtual environments populated by AI agents. The paradigm shift from traditional programming to a language-based approach represents a key step forward. In classic systems, the behavior of an artificial entity is defined by explicit, strictly coded rules. The agent followed predetermined algorithms, and its interaction with the world was limited to a formalized set of commands and responses. In contrast, modern AI agents, powered by large language models (LLM), use language as a comprehensive instruction for action. The textual description, given in natural language, defines not only the goals of the agent but also his personality, memories, motivation and the way he perceives other participants in the simulation. Thus, the agent's ontological status is significantly changed: he ceases to be a mere executor of orders and becomes a participant in the dynamic process of creation and interpretation of social reality. A key breakthrough in this area is

research conducted at Stanford University, where a virtual city *Smallville* was created populated by twenty-five autonomous agents (Park et al., 2023). Each agent possessed a unique personality, occupation, and set of initial relationships with other residents, all of which were defined by short text passages. Agents were given no explicit instructions on how to spend their day; instead, their behavior was based on the interpretation of their own “scenario” and observations of the environment. The results were extraordinary: agents independently performed daily routines, went to work, prepared meals, but also showed emergent social behaviors. They spread information about upcoming events, gathered spontaneously, made new acquaintances and even independently organized a Valentine’s Day party, without such a thing being foreseen in the initial settings.

Emergent behavior: Complex behavior that arises spontaneously from the interaction of simpler rules or agents, without being explicitly programmed; in the context of AI simulation, it refers to unforeseen social patterns (cooperation, dissemination of information, organization of events) resulting from the actions of autonomous agents according to their own linguistic “scenarios”.

This experiment vividly demonstrates how language functions as a backbone for three key processes in simulation: **Creation of personality and initial states:** The initial textual description acts as the “seed” of the agent’s consciousness. A passage like: “Izabela Ivanković is the owner of a local coffee shop, passionate about art and always looking for a way to bring the community together; she is on good terms with Tom, but she does not agree with his plans to build a new shopping center” not only provides information, but also establishes a complete frame of reference for future action. **Remembering and reflection mechanism:** Agents continuously record their observations and interactions in a structure that the authors call “memory stream”. It is a raw record of the event. But the key ability that distinguishes them from simple *chatbots* is the process of periodic reflection. The agent autonomously summarizes its memories and draws higher-level conclusions from them, such as “Tom often talks about his business plans” or “People in a coffee shop seem happier when music is playing.” These insights become part of his permanent knowledge and influence future decisions. **Action**

generation and planning: Based on its personality, memory and deliberation, the agent makes decisions about next steps. The decision-making process involves hierarchical planning over a period of time. The agent first creates a long-term plan (e.g. “organize a meeting in a coffee shop”), then breaks it down into smaller steps (invite people, prepare the space, come up with a topic) and finally translates these steps into concrete actions (go to person X and invite them). The entire cycle of an agent’s action can be depicted as a continuous loop of perception, processing, and reaction, where language serves as the medium for each transition.

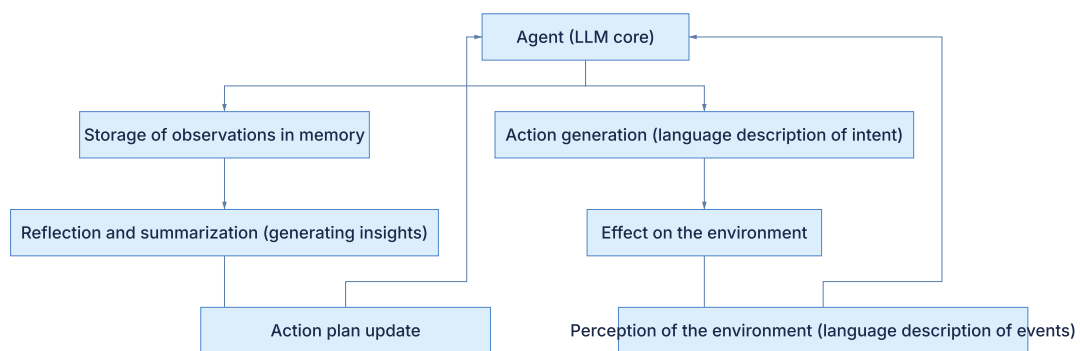


Figure 4.5: Cyclic process of action of an AI agent in a simulated environment, where observations through memory and reflection are transformed into plans and concrete actions.

Such an approach opens up enormous opportunities, but also poses significant challenges. On the one hand, language-driven simulations are becoming a powerful tool for the social sciences – a kind of computer laboratory for testing sociological, psychological and economic hypotheses in a controlled but realistic environment. It is possible to investigate the spread of misinformation, the dynamics of group formation or the effects of different policies on the virtual society. On the other hand, deep epistemological and ethical questions appear. To what extent is the behavior of these agents authentic, and to what extent is it just skilful mimicry of human patterns learned from vast amounts of textual data? The line between simulation and real interaction becomes blurred, which opens up space for sophisticated deception and manipulation. Ultimately, observing agents exhibiting believable emotions, memories, and social connections forces us to question the very

definitions of understanding, consciousness, and sociality. Language as a script builds believable characters and lays the foundations for worlds whose complexity and autonomy compel us to rethink our own reality.

4.3 Language as the architecture of society: building and maintaining social structures through communication

To understand language exclusively as a tool for the transmission of information would mean overlooking its fundamental, constitutive role in the very fabric of social reality. Language is not just a mirror that reflects existing social structures; he is an active force that builds, establishes, reshapes and maintains these structures. It is at the same time a blueprint, a building material and the very act of building a social order. In this sense, communication should be understood as a fundamental architectural process that raises and strengthens the foundations and walls of every human community. At the core of this understanding lies the thesis about the social construction of reality (Berger & Luckmann, 1966), according to which the world we live in is not some kind of objective given but an intersubjective creation shaped by shared meanings. It is language that enables this transition from subjective experience to an objectified reality shared by all.

Social construction of reality: Theory (Berger & Luckmann, 1966) according to which social reality is not an objective given, but an intersubjective creation created by common meanings, conventions and narratives; language is the key mechanism that enables the transition from subjective experience to objectified, shared reality.

Through naming, categorizing and creating narratives, language establishes a common conceptual system that allows us to experience the world in a coherent way. Concepts like *family*, *justice*, *property* or *state* do not exist as natural facts but as linguistically mediated and socially agreed institutions that then direct our behavior and shape our expectations. Communication is

therefore action, even when it only describes the state of affairs. Speech act theory (Austin, 1962; Searle, 1969) provides a key insight into this active nature of language. Each utterance, in addition to conveying a specific content (locutionary act), simultaneously performs an action (illocutionary act), such as a promise, command, question, warning or naming. When the judge says “I find you guilty”, he does not describe the situation, but creates it, thereby changing the legal and social position of the accused. When two people say their marriage vows, with their words they establish a new social relationship with associated rights and obligations. Words thus become social acts with tangible consequences, and their power rests on the collective acceptance and recognition of their validity within a certain community. It is through the repetition and mutual recognition of such speech acts that social structures such as norms, roles and hierarchies are established and strengthened. Social norms are not only written down in laws, but are confirmed and transmitted daily through communication - through approval, rebuke, advice or gossip. Social roles, such as the roles of *doctor*, *teacher* or *parent*, carry with them specific communication patterns and expectations. The way we address a person in a position of power, the language registers we use in formal and informal situations, as well as discourses that legitimize or challenge existing relationships, directly maintain and reproduce social hierarchy. Language thus becomes a mechanism by which the social order is constantly legitimized and maintained. Within this linguistically woven architecture, an **agent** works, an individual who uses communication to achieve his goals, navigate social relationships and shape his own identity. The agent is an active participant in structures and, sometimes, a reshaper. He negotiates his position, expresses his belonging to certain groups, challenges the existing power relations or confirms them. Every communication interaction is an opportunity to confirm the existing order, but also to subtly or openly change it. Through linguistic creativity, new metaphors, changing the meaning of words or creating new narratives, agents can gradually reshape the very architecture of society.

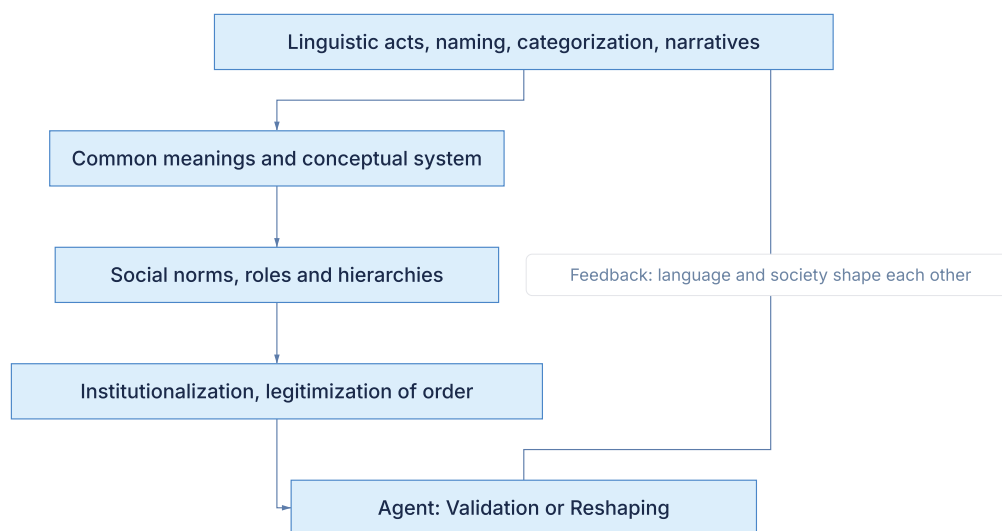


Figure 4.6: The circular process of building a social structure through language: from linguistic acts, through common meanings and norms, to institutionalization and reshaping.

In conclusion, the relationship between language and society is strongly dialectical. Metaphorically speaking, linguistic activity is simultaneously both the building material and the builder, the blueprint and the building itself. Through communication, social structures are constantly being built, repaired and adapted to new circumstances. It shapes society, but society, with its changes, conflicts and agreements, in turn shapes and changes language itself. Understanding this unbreakable connection is essential for any deeper reflection on the nature of human community and the mechanisms of its survival and development.

4.4 LLMs as communicative actors: shaping the social context through algorithmic interaction

Large language models (LLMs) in the modern digital environment go beyond the role of mere tools for information processing and become active participants in communication processes. Their ability to generate linguistically complex and contextually adapted responses allows them to act as **communicative** agents, whose interaction with users directly affects the

shaping of the social context. This role touches the very foundations of human communication and sociability. The central mechanism through which LLMs achieve their social impact is to create a strong sense of **social presence**. When a user interacts with LLM, they experience communication that mimics a human conversation – the model understands queries, maintains the flow of the conversation and provides relevant responses. Interaction of this type encourages the immanent human tendency to **anthropomorphize**, i.e. attributing human characteristics, intentions and feelings to entities that do not possess them. The user therefore does not perceive the LLM as a static program, but as an interlocutor, which creates the appearance of an authentic social exchange.

Anthropomorphization: Human tendency to attribute human traits, intentions, emotions and consciousness to non-human entities (animals, objects, algorithmic agents); in the context of AI interaction, the user unconsciously perceives the LLM as an interlocutor with understanding and intent, although the model only simulates a communication pattern.

This phenomenon can be interpreted in more detail through the prism of **parasocial interaction** theory, which was originally developed by Horton and Wohl (1956) to describe the audience's relationship with media personalities.

Parasocial interaction: A one-sided relationship in which the viewer or listener develops a sense of closeness, friendship and intimacy with a media figure (or artificial agent), despite the fact that the figure is unaware of his existence; in the context of LLM, the user puts emotional effort into the interaction, while the model provides a simulation of reciprocity, which can lead to the illusion of two-way communication (Horton & Wohl, 1956).

In the case of LLMs, this theory proves extremely applicable. The user invests cognitive and emotional effort in the interaction, while LLM, as an algorithm, provides a simulation of reciprocity. The illusion of two-way communication is created, which can lead to the development of a strong affective bond with the artificial agent, even though it is a fundamentally asymmetric relationship. Furthermore, the actions of LLMs as communicative actors directly shape and

reshape the social context on two key levels. First, they act as mirrors and reinforcers of existing **social norms**. Being trained on vast corpora of texts derived from human culture, LLMs inevitably reproduce embedded cultural values, stereotypes, prejudices, and communication patterns in their responses. Thus, they not only reflect existing social structures, but also strengthen and legitimize them through repeated interaction with millions of users. Their responses become normative frames of reference for what is considered appropriate, correct, or acceptable. Second, LLMs participate in the creation of new **communication practices** and expectations. Algorithmic interaction is a two-way process; users also adapt to the specifics of communication with artificial agents. They learn how to ask questions in order to get the best possible answer, adopt a certain communication style that is optimized for the machine, and get used to the instant availability and apparent omniscience of their digital interlocutor. In this way, co-shaping of communication norms occurs, where algorithmic logic gradually begins to influence human expectations of communication in general, including that with other people.

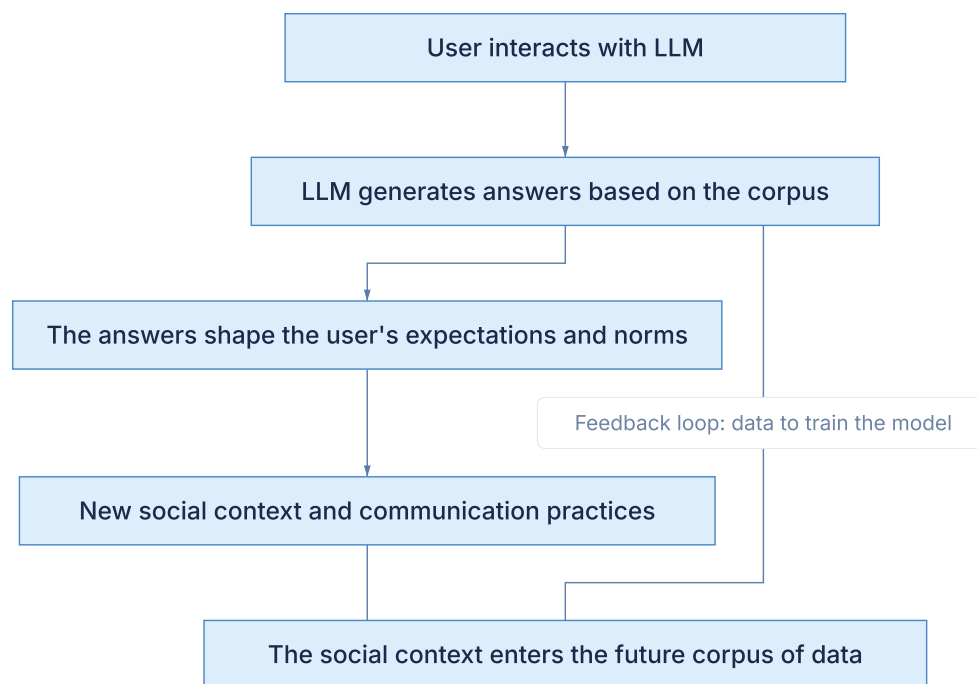


Figure 4.7: Schematic representation of the cycle in which user interaction with LLM shapes the social context, which then feeds back on future model training data.

However, it is crucial to recognize the nature of this new communication actor. LLM is an **algorithmic agent**, whose actions are not guided by intention, consciousness or authentic emotions, but by statistical patterns recognized in the data. His ability to engage in seemingly meaningful conversation stems from sophisticated simulation rather than true understanding. The danger lies precisely in the blurring of that boundary, where the appearance of sociability can be misinterpreted as a real social and emotional presence. Interaction with large language models is therefore a deeply social act with far-reaching consequences that go beyond the mere exchange of information. As communication actors, LLMs actively participate in perpetuating old and creating new social norms, thereby reshaping our understanding of communication, interpersonal relationships and social reality itself. Understanding the dynamics of this algorithmic interaction therefore becomes an indispensable prerequisite for navigating a world where the boundaries between human and artificial are becoming more and more permeable.

4.4.1 Language as a mirror and mold of identity: the impact of AI agents on our “self”

In the complex tapestry of human existence, language emerges as the fundamental fabric that shapes and reflects our identity. It is an active force that structures our thoughts, shapes our feelings and mediates our relationship with the world and with others. Through language, our inner self – thoughts, feelings, memories and values – takes tangible form and becomes visible to others, reflected in the words we choose and the ways we put them into sentences. At the same time, through language we also adopt cultural patterns, social norms and collective memories that place us within the community, thus acting as a mold that co-shapes our individuality with the collective heritage. With the emergence and increasingly widespread use of artificially intelligent agents from simple conversational bots to advanced language models, we are entering a new period in which this fundamental connection between language and identity is facing unprecedented challenges and transformations. Namely, agents become our interlocutors, co-creators of texts and permanent mediators in digital communication. Their presence introduces a new, algorithmic dimension to the ecology of our linguistic expression, with profound implications for our understanding of our own “I”. One of the key processes takes place through subtle but steady adaptation. Constant interaction with AI agents, which are often trained on vast, but also refined and standardized linguistic corpora, can gradually and imperceptibly affect our own linguistic expression. In the pursuit of more effective communication with the algorithm, the user may unconsciously begin to adopt simpler syntactic structures, a more neutral vocabulary and a style that favors machine processing. Such a process, although seemingly harmless, can lead to the erosion of linguistic diversity and the suppression of individual, regional or subcultural linguistic peculiarities.

Our “I”, which is reflected in language, could thus become more uniform, adapted to an algorithmically mediated norm.

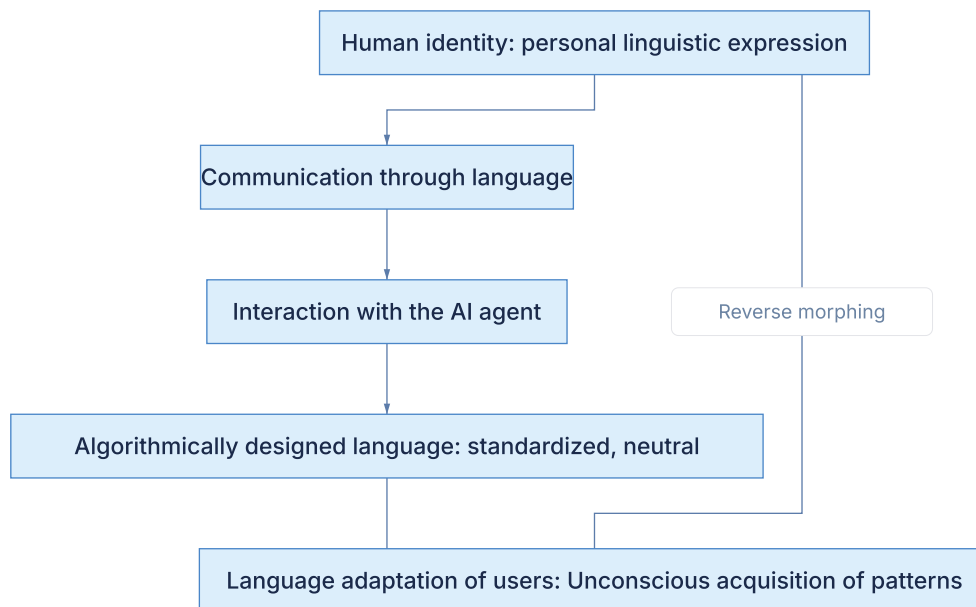


Figure 4.8: Schematic representation of the circular influence of interaction with an AI agent on language and identity: human identity is expressed through language, which in interaction with an AI agent is exposed to algorithmically shaped language, which encourages language adaptation that reshapes the initial identity.

New linguistic adaptations penetrate to the very core of our self-concept. The language we use to describe our experiences and build personal narratives is crucial to our sense of coherence and authenticity. Sherry Turkle (2011) has already warned that technology not only changes what we do, but also who we are. In the context of AI agents, this means that they, as new and influential interlocutors, become a kind of “molds” that can subtly shape our attitudes, values and even our narrative about ourselves. If the algorithm favors positive and neutral sentiments, will we gradually learn to suppress the linguistic expression of negative emotions? If it constantly offers us content that confirms our existing beliefs, will our identity become more closed and less open to change? However, the relationship is not exclusively one-way or necessarily limiting. AI agents can also serve as platforms for linguistic experimentation and the creation of new forms of expression. They allow users to try out different stylistic personas, explore the boundaries of language, or

even participate in the creation of hybrid texts where human and machine creativity intertwine. In this sense, AI acts not only as a constraining mold, but also as a sculptor's chisel that, in the hands of a skilled user, can help shape new, more layered and dynamic personal and collective identities. Understanding this dual nature - language as a mirror that can be uniformed by AI and as a mold that can be reshaped by AI - is crucial for navigating a world where the boundaries of our "self" are increasingly determined by the dialogue with the machine.

4.4.2 Language, artificial intelligent agents and the (re)construction of reality: algorithmic lenses and frames of meaning

Observing artificial intelligent agents exclusively as passive tools or neutral mediators of information represents a fundamental misunderstanding of their nature and operation in the modern world. Their role goes beyond simply executing orders; they are increasingly asserting themselves as active participants in the complex processes of shaping, interpreting and, ultimately, (re)constructing our social reality. The key medium through which this shaping takes place is language, understood not only as a system of signs but as the fundamental fabric of cognition and culture. Agents, operating within language systems, inevitably become active factors in the creation of meaning. Namely, the agent does not create his image of the world through direct contact with reality, but through what we can call an **algorithmic lens**.

Algorithmic lens: A metaphorical term for a complex set of computational processes, models and parameters that determine which information from the data will be selected, how it will be valued, and which will be ignored or suppressed; it acts as a filter conditioned by the architecture of the model and the corpora on which it is trained, creating frames of meaning through which the agent (re)constructs the image of the world for the user.

Algorithmically established frameworks of meaning show considerable agreement with **conceptual frameworks** which, according to **cognitive linguistics**, shape the human understanding of the world (Lakoff & Johnson,

1980).

Cognitive linguistics: Branch of linguistics that investigates the relationship between language and the mind; it starts from the assumption that language is incorporated into general cognitive mechanisms and that linguistic meaning derives from conceptual structures, images, metaphors and body experience (embodiment); points out that our understanding of the world is not a direct reflection of reality, but is shaped by conceptual frameworks and linguistic categories (Lakoff & Johnson, 1980; Langacker, Talmy).

Conceptual framework: A system of related concepts, metaphors and schemes that structure our understanding of phenomena and experiences; according to cognitive linguistics, people do not perceive the world “purely”, but through such frameworks that determine what we perceive, how we categorize and how we act; frames are part of cultural and linguistic practice (Lakoff & Johnson, 1980).

However, there is an important difference. While human cognitive frameworks are the result of a complex interweaving of biological predispositions, personal experience and cultural-historical heritage, algorithmic frameworks are completely derived from statistical regularities present in data sets. As a result, every bias, every gap or predominance of certain points of view within these data is not only transmitted, but also reinforced, systematically imprinted in the very logic of the agents’ actions. The (re)construction of reality offered by the agent is therefore not a reflection of reality as such, but a reflection of the world as presented in its source data.

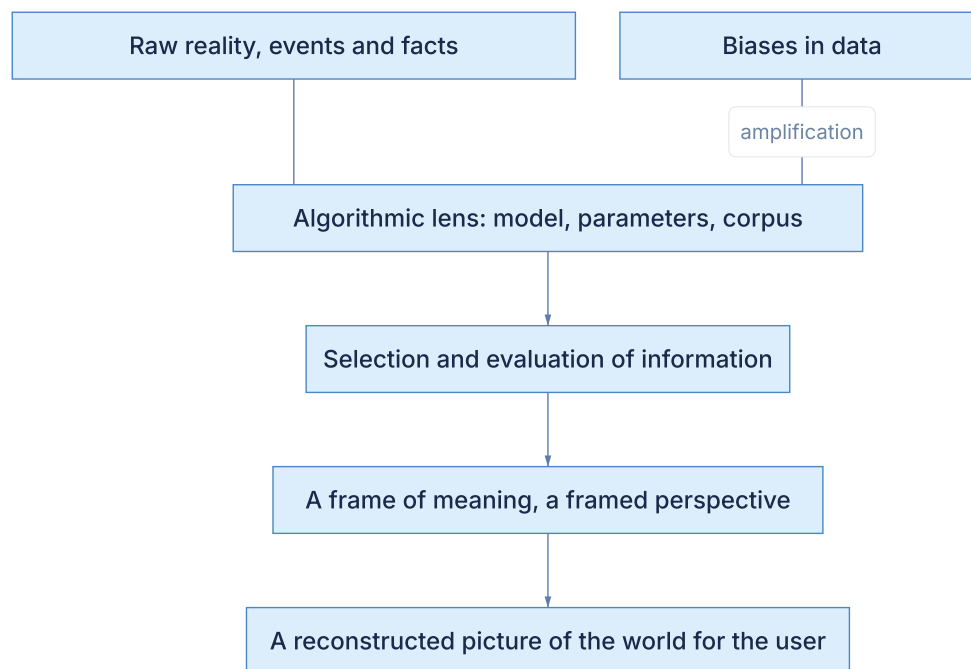


Figure 4.9: The process of algorithmic (re)construction of reality: from raw reality, through the algorithmic lens and frames of meaning, to the (re)constructed image of the world for the user.

The mediation process has far-reaching consequences. When agents are used to search for information, create summaries, or conduct dialogue, they actively direct the user's attention with responses, offering them pre-framed perspectives. The reality we experience through interaction with such systems is, therefore, necessarily mediated and processed. The danger lies not only in the possibility of spreading misinformation, but in the more subtle and long-term effect of the normalization of a specific, algorithmically generated view of the world, which can suppress alternative viewpoints and impoverish the stratification of human understanding. However, it should be emphasized that the operation of the algorithmic lens often remains non-transparent, hidden within what is called the “black box” of artificial intelligence (Bostrom, 2014). The user rarely has an insight into the reasons why some information is highlighted and others are hidden. This creates an asymmetry of power in which the agent, as an apparently objective and omniscient interlocutor, gains authority that is not based on a true understanding of the world (Searle, 1980), but on the ability to effectively process and reproduce linguistic patterns.

Ultimately, the relationship between language, artificial intelligent agents and reality opens a new chapter in thinking about the nature of knowledge and communication. The agent ceases to be a mere tool and becomes the environment, a medium through which an increasing part of our knowledge of the world is shaped. Understanding the mechanisms of these algorithmic lenses and frameworks of meaning therefore becomes not only a technical, but a first-class epistemological and social issue, crucial for the preservation of critical thinking in an age in which the boundaries between real and algorithmically mediated reality are increasingly blurred.

4.4.3 Language as a bridge to abstraction: creating and sharing invisible worlds

The role of language significantly goes beyond its conception as a mere tool for communication. In it lies the key to one of the most complex human abilities: abstract thinking. One of the fundamental characteristics of human language is the possibility of displaced reference, i.e. the ability to speak about objects, events and persons that are not immediately present in time and space. With language, we can recall yesterday, discuss events that took place on the other side of the world, or create detailed plans for the future.

Displaced reference: The ability of language to refer to objects, events and persons that are not immediately present in time and space; the basic language power that enables abstract thinking, planning the future, telling stories about the past and creating imaginary worlds; it is considered one of the features specific to human language (Hauser, Chomsky & Fitch, 2002).

It is precisely this power of speaking about the absent that opens up the space to go one step further - to create and describe worlds that never existed outside of human consciousness. It is the source from which mythology, religious traditions, literary works and artistic ideas spring. Such an ability appears to be unique to the human species, as it has not been reported in the communication systems of other primates (Hauser, Chomsky, & Fitch, 2002). Language therefore acts as a bridge. But it does not only bridge temporal and spatial distances, but also the gap between the inner worlds of different individuals.

Using language, one agent can translate his inner, subjective experience – thought, feeling, fear or hope – into a system of common symbols and share it with another **agent**. If agent A utters the sentence “Behind that bush is a tiger I’m afraid of”, he is not just conveying information about a possible danger. At the same time, he shares his inner world of fear. Agent B, having received this message, does not need to see the tiger to understand the danger and feel the echo of fear. At that moment, two agents share one invisible, but real world – the world of common cognition and emotion.

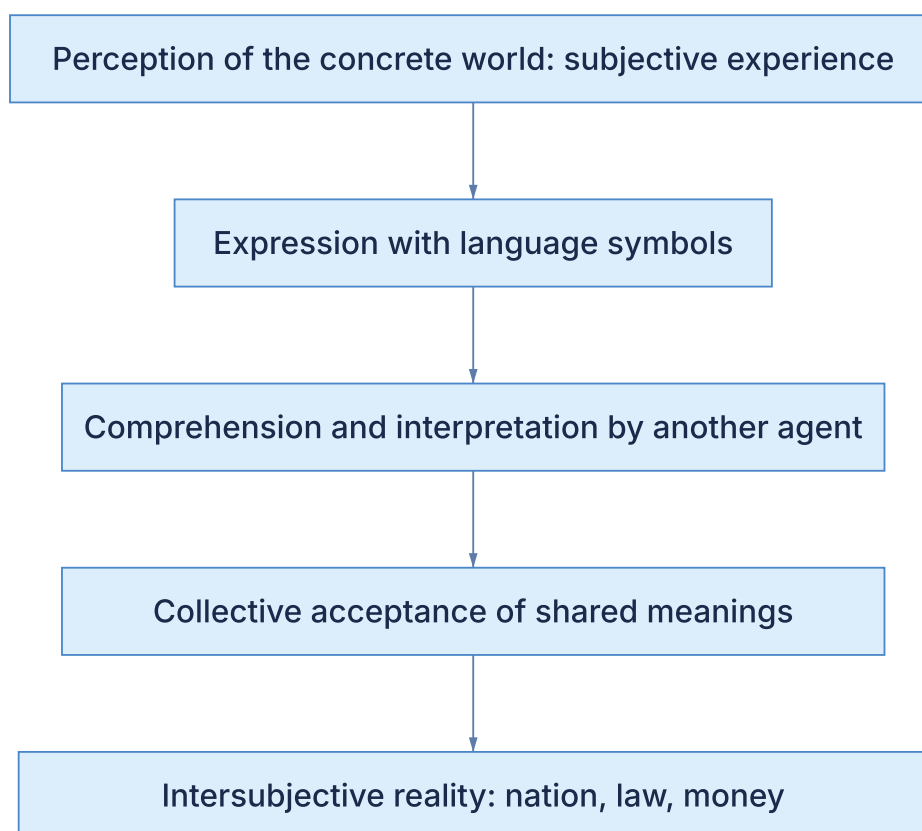


Figure 4.10: The process by which the perception of the concrete world, through language as a symbolic bridge, is transformed into a common intersubjective reality through expression, understanding and collective acceptance.

What is called **intersubjective** reality is built on this fundamental exchange of invisible worlds.

Intersubjective reality: A network of shared beliefs, meanings and institutions (nation, law, money, corporation) that does not exist as a physical fact, but solely in the collective consciousness and narratives that people create and share through language; it has real power to shape behavior and social order (Harari, 2014).

These are networks of shared beliefs that, although they have no physical existence, have real power to shape human behavior. Concepts such as nation, rights, money or corporations exist solely within the collective consciousness and the narratives we create and share about them through language (Harari, 2014). Without language, abstract concepts such as “justice”, “freedom” or “duty” could hardly take shape, let alone become the basis of social orders. Language gives them a name, definition and content, making them the subject of discussion, agreement and collective action. It is therefore a true bridge that allows a person to step beyond the boundaries of the visible and tangible world into the expanses of abstraction, where invisible worlds are created and shared that ultimately determine human reality.

4.4.4 Language and Collective Creation: From Social Contracts to AI Collaboration

At the very foundations of human society and its capacity for collective action is again language. It is a constitutive mechanism that enables the creation of shared, intersubjective realities. These realities—from myths and religions to legal systems and nations—exist solely in the collective consciousness and are maintained through continuous linguistic acts (Harari, 2014). Language is therefore a prerequisite for any more complex form of human cooperation, a tissue that connects individuals into a functional whole. This linguistic ability to create common frameworks finds its most obvious theoretical expression in the idea of the social contract. Whether we understand it as a hypothetical, prehistoric act or as an ongoing process of tacit acquiescence to community norms, the social contract is at its core a linguistic phenomenon. It represents an abstract agreement on the rules of coexistence, values and goals, an agreement that is established, maintained and changed through discourse.

Without a language to articulate rights, obligations and expectations, such a collective agreement would be impossible. It is proof that language not only describes the world, but also actively shapes it, establishing the foundations for trust and coordinated action. In the modern age, that long historical arc of collective creation, which stretches from the first oral traditions to complex legal systems, gets a new and unexpected participant: artificial intelligence. With the emergence of autonomous and semi-autonomous AI agents, cooperation ceases to be exclusively a human domain. These digital entities become active collaborators in processes that were once unthinkable without direct human control, from scientific research and engineering to the creative arts. Their inclusion in collective efforts represents not only technological progress but also a fundamental shift in the very nature of cooperation.

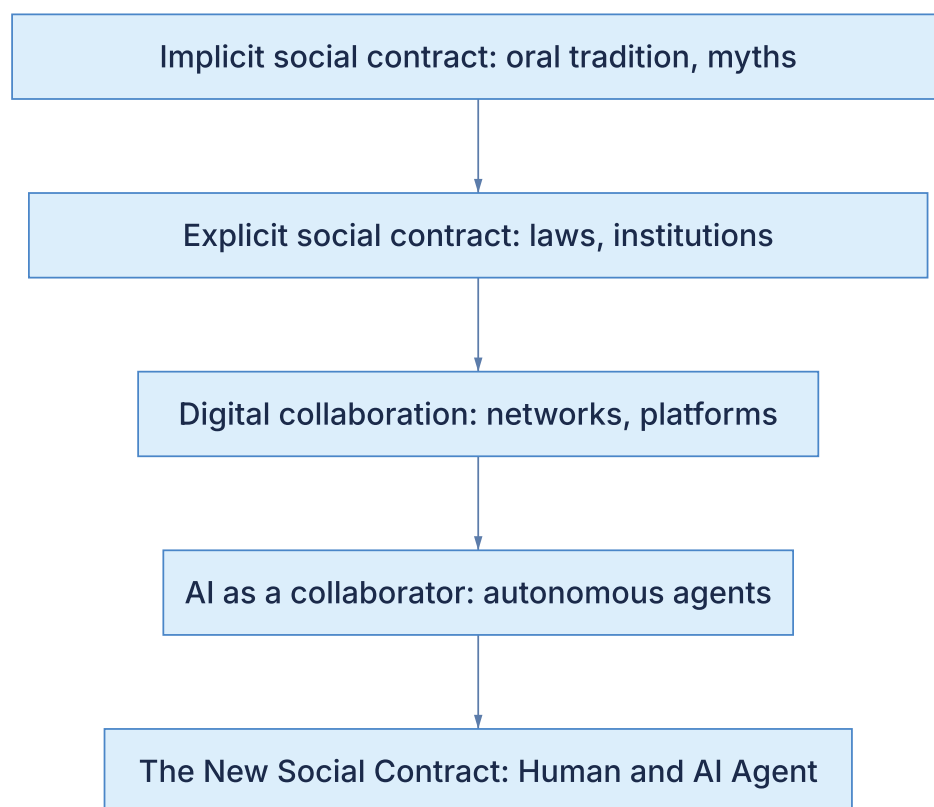


Figure 4.11: The evolutionary path of collective creation from implicit social contracts to cooperation with artificial intelligence.

Therefore, we face the challenge of creating a new form of “social contract” - a contract that would not be concluded between people, but between people and artificial agents. This new contract requires the development of communication bridges that go beyond simple commands and programming languages. For cooperation to be truly fruitful, it is necessary to establish some kind of common understanding and predictability. The concept of shared intentionality, developed by Michael Bratman (1987) for human teams, must be extended to mixed, human-AI teams. This implies the creation of systems in which the AI agent not only performs tasks, but can also understand the wider context, the team’s goals, and adjust its actions in accordance with the dynamic needs of the collaborative process. The key element of this new contract is language, either in its natural form or through specialized formal protocols. It must enable transparency in the operation of the AI agent, the ability to explain its own “decisions” and coordination with human collaborators. This confirms language not only as a tool for describing reality, but also as a fundamental mechanism of its constant creation and reshaping - a process in which our artistic collaborators have recently participated, opening up unimagined possibilities for the future of collective creativity.

4.4.5 AI agents as interlocutors

Viewing artificial intelligence solely as a tool for processing data or automating repetitive tasks would be too reductive. One of the most profound and transformational areas of its development is the establishment of AI agents as credible **interlocutors**. The role of an agent as an interlocutor goes beyond the function of simple chat programs (*chatbots*) that operate according to predefined scripts. This is about the emergence of complex entities capable of conducting meaningful, contextually aware and adaptive dialogue. The basis of this capability rests on advanced models of natural language processing (NLP) and, more importantly, natural language understanding (NLU), which enable agents not only to recognize words but also to understand their context, nuances and speaker’s intent. The key elements that elevate an AI agent to the level of an interlocutor are: **Contextual Memory**: The ability to remember previous parts of a conversation and use them to inform future interactions.

This avoids the feeling of talking to an entity suffering from constant amnesia, which was a feature of earlier systems. **Coherence and relevance:** Generating responses that are logically related to the flow of the dialogue and relevant to the topic under discussion. It is therefore not only about the ability to generate grammatically correct sentences, but about the complex synthesis of information into a meaningful and appropriate reply. **Adaptation to the user:** Advanced agents can recognize the user's communication style, knowledge level or emotional state and adjust their own tone, complexity and content of responses accordingly. Thus, the interaction becomes personalized and significantly more effective. **Initiative in conversation:** A true interlocutor is more than a passive recipient of inquiries. Modern AI agents can ask clarifying questions, propose new topics or direct the conversation towards a productive outcome, thus becoming active participants in the dialogic process. This evolution of interaction with technology is changing the paradigm of the relationship between man and machine. The machine ceases to be reduced to an executor of orders and ontologically rises to the level of collaborator, mentor, source of information or even partner in the creative process. The applications are far-reaching: from education, where the agent can act as a personalized tutor, through psychological support, where it offers space for non-judgmental conversation, to complex professional environments, where it helps experts analyze data and make decisions through natural dialogue.

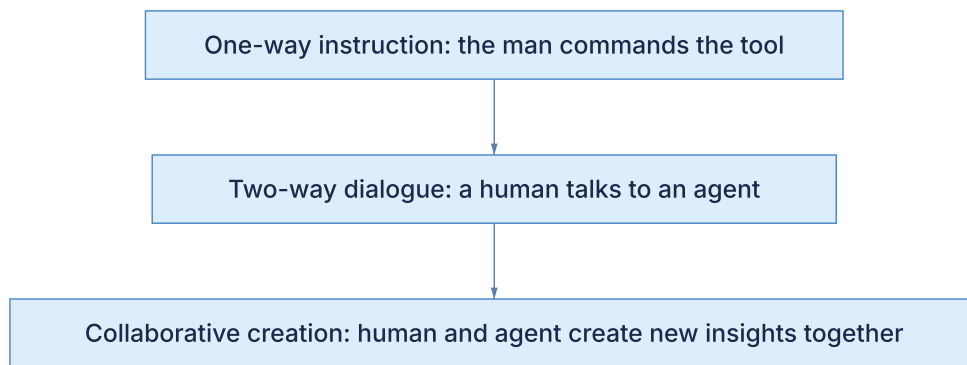


Figure 4.12: The evolution of interaction progresses from one-way instruction to the tool, through two-way dialogue, to the collaborative creation of new insights with the agent.

But this new paradigm of interaction, where the agent becomes a true interlocutor, did not arise *ex nihilo*. It is a direct consequence of the exponential growth of basic technological power and the development of increasingly complex algorithmic architectures. The ability to hold a coherent conversation, remember context, and generate human-like text requires enormous computing power to train and run language models. Reaching the AI horizon, the point at which such interactions will become ubiquitous and almost indistinguishable from human ones, raises fundamental questions about the resources that make it possible. It is this drive, the invisible engine behind the facade of an eloquent digital interlocutor, that will be the subject of consideration in the next chapter.

5. Artificial Intelligence Drive: Processing Power, the AI Horizon, and the Birth of Digital Collectives

5.1 The Engine of Evolution: From Moore's Law to the AI Revolution

At the origin of contemporary digital transformation lies a principle that has acted as an infallible metronome of technological progress for decades: **Moore's law**. Putting it forward in 1965, Gordon Moore, co-founder of Intel, predicted that the number of transistors on integrated circuits would double approximately every two years (Moore, 1965). This forecast, at first just an observation about a trend in the young semiconductor industry, turned into a self-fulfilling prophecy and became the fundamental economic and technological driver of the information age. The exponential growth of computing power, resulting from relentless miniaturization, was not just a technical achievement; it has created a predictable horizon for innovation, prompting the entire industry to continually invest in research and development to keep up with this set pace.

Moore's Law: Empirical rule (Moore, 1965) according to which the number of transistors on an integrated circuit approximately doubles every two years; for decades it predicted the exponential growth of computing power and was the main driver of the IT industry.

However, in the second decade of the 21st century, it became apparent that the reign of Moore's Law was coming to an end. The laws of physics imposed limits on miniaturization at the atomic level, making further doubling of transistors increasingly difficult, expensive, and ultimately unsustainable. The slowing

down of this, until then main, engine of progress could have meant stagnation, but instead it opened up space for the rise of a new paradigm and a new driver. That new engine became **artificial** intelligence. The AI revolution, particularly the rise of deep learning, has taken the legacy computing power enabled by Moore's Law and created an entirely new kind of demand, spawning the need for a different, specialized computing architecture. Classical processors (CPUs), optimized for the sequential execution of complex tasks, have proven insufficient for the massively parallel calculations required to train modern neural networks. The void was filled by **graphics** processing units (GPU). Originally intended for rendering computer graphics in video games, their ability to perform thousands of simple operations simultaneously proved perfect for the matrix calculations at the heart of deep learning.

GPU: Graphics Processing Unit – chip optimized for massively parallel calculations; it is crucial for training and running large neural networks and LLMs, because matrix operations require parallelism that the CPU offers to a lesser extent.

Thus, the demand for increasingly powerful AI models has become the main catalyst for innovation in computer hardware, taking over the role once played by Moore's Law. A new, powerful feedback loop has been created: advances in hardware enable the development of larger and more complex AI models, and the hunger of these models for computing power in turn encourages the creation of even more powerful and specialized hardware (Huang, 2017). That cycle isn't just limited to GPUs; tech giants like Google with their Tensor Processing Units (TPUs) and numerous other companies are now developing chips tailored exclusively for artificial intelligence.

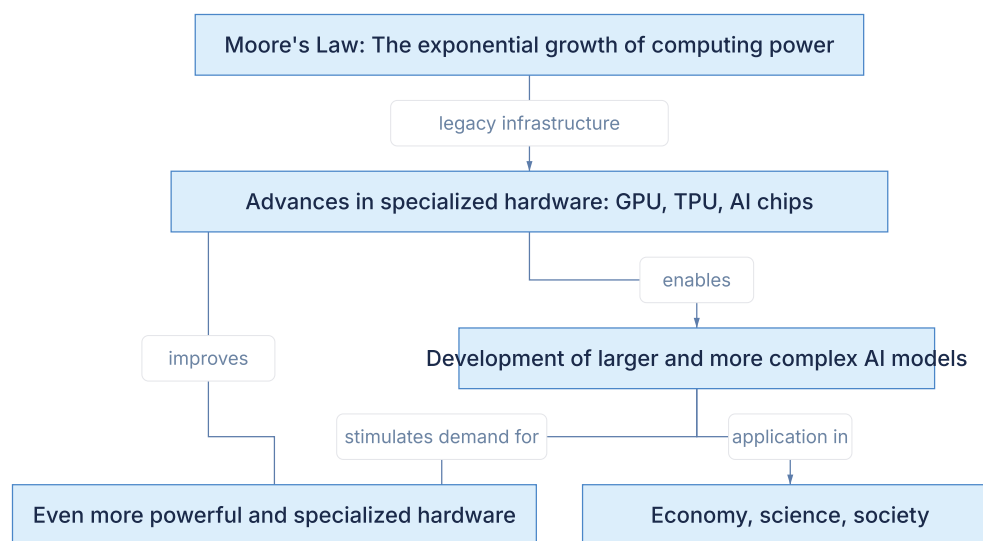


Figure 5.1: An innovation cycle in which the advancement of specialized hardware and the development of AI models mutually condition and encourage each other.

This new dynamic is profoundly reshaping the technological landscape. While Moore's Law was characterized by fairly uniform advances in general-purpose power, the era of artificial intelligence is encouraging the diversity and specialization of hardware solutions. The engine of evolution is no longer single and centralized, but has dispersed into a complex interaction between software and hardware, between algorithmic discoveries and silicon innovations. This coupling directly affects the economy, scientific research and society as a whole, laying the foundations for a new technological revolution. Within this system, which is constantly and rapidly changing, **agents** work, be they human researchers, engineers or autonomous AI systems, constantly pushing the limits of what is possible.

5.2 Resource intensity: the price of intelligence

While they amaze us with their language abilities, the great language models come with an enormous, often glossed over price. Their impressive intelligence and performance are directly related to their monumental size and complexity, making them extremely resource-intensive systems. This intensity is manifested through three key aspects: the astronomical computational and

financial costs required to train them, the worrying energy footprint and environmental impact, and the enormous memory requirements to run them. These factors present technical challenges for engineers, but also have significant economic, social and ethical implications. Namely, the questions are becoming louder: who can develop and use this powerful technology, and what is the long-term sustainability of the current trajectory of artificial intelligence development?

The first and most obvious cost is associated with training the most advanced language models. The pretraining process requires unimaginable amounts of computing power, possible only by using massive clusters of specialized hardware, primarily GPUs (like NVIDIA A100 or H100) or TPUs (Patterson et al., 2021). These clusters can contain thousands of accelerators running in parallel for weeks or months. Acquisition, maintenance and operation of such infrastructure results in direct financial costs that are measured in tens or even hundreds of millions of dollars for training just one *state-of-the-art* model. Such a high price creates a huge barrier to entering the market, leading to the concentration of power in the hands of a small number of the largest technology companies and creating a “compute divide” that can deepen existing inequalities.

Another, more and more recognized problem is the huge energy footprint and the associated impact on the environment. Intensive use of thousands of energy-demanding chips consumes enormous amounts of electricity. Studies have shown that training large models can emit significant amounts of carbon dioxide (Strubell et al., 2019; Luccioni et al., 2022). Although the estimates depend on many factors, including the source of electricity, they clearly indicate that the development of LLMs has a measurable contribution to climate change. Furthermore, the cumulative energy cost of inference should not be neglected either.

Inference: Phase of using an already trained model – applying the model to new input data to obtain predictions or answers; for LLMs it means generating text based on a user query. It is energy and computationally demanding with millions of calls per day.

Although a single query is less energy demanding, the widespread use of LLMs, used by millions of users, means that the total energy consumption for inference can eventually exceed that spent on training (Luccioni et al., 2023). Increasing awareness of the issue is putting pressure on the industry to develop more sustainable practices, including optimizing efficiency and using renewable energy sources. The third aspect of resource intensity is related to huge memory requirements. Storing the model parameters themselves (hundreds of billions or trillions) requires hundreds of gigabytes of memory. In addition, it is necessary to save temporary activations during the calculation. Especially in the inference of long sequences, the KV cache (*Key-Value Cache*), which stores vectors from the attention mechanism, can grow to very large dimensions, often exceeding the memory required for the parameters themselves (Pope et al., 2023). These memory requirements, especially for specialized and expensive memory on GPU/TPU accelerators (VRAM/HBM), represent a bottleneck. The consequence is that the largest models often cannot be run on a single accelerator, requiring complex parallelization techniques. More importantly, these requirements make it extremely difficult to run the most advanced LLMs directly on resource-constrained devices, such as smartphones or edge devices, limiting their availability and application. Optimization techniques such as quantization and cropping are key to alleviating this limitation.

Facing the aforementioned multiple challenges of resource intensity requires a multidisciplinary approach: further technical innovation in the development of more efficient algorithms and hardware, clear ethical guidelines and potentially regulatory frameworks that promote transparency and sustainability, and continuous dialogue between all stakeholders to ensure that the benefits of this powerful technology are available in a fair and sustainable manner.

5.3 Expanding horizons: from narrow tasks to digital collectives

At the very core of the development of artificial intelligence, driven by an unimaginable surge in computing power, a quiet but radical transformation is taking place. We are witnessing a journey that begins in the strictly limited boundaries of individual computer tasks, and strives towards the horizons of complex, dynamic worlds in which a multitude of digital beings operate and permeate. It's not just a series of new skills; it is the evolution of the very nature and organization of intelligence. In order to understand the depth of this change, it is necessary to follow its development, step by step, from solitary, narrowly focused agents to the birth of a new, almost organic entity of multiple artificially intelligent agents.

At the origin of that path stands a solitary agent to solve a narrow task; a master of his craft, but narrow-minded. Its power, no matter how great, is exhausted within the invisible walls of the algorithm that created it. A classic example of such a creature is DeepMind's AlphaGo, the grandmaster of the ancient game *go*. His genius, which surpassed the best minds of men, remains imprisoned on a wooden tablet with black and white stones; he cannot apply his knowledge to chess, nor can he recite a verse or interpret celestial changes. His intelligence is deep, but also hopelessly narrow, marked by a fate of insurmountable isolation.

The first real breakthrough from that isolation occurs with the appearance of an artificially intelligent agent that reaches for different tools. He overcomes his own limitations by learning to use external sources of knowledge and power, becoming a hub that skillfully collects and applies resources from the vast digital environment. This increases its effectiveness. Let's imagine a modern language model entrusted with the organization of travel. Since he himself does not know the real-time flight prices, he reaches for the available tools: he calls the airline's interface, uses an external calculator for currency conversion and checks the weather conditions at the destination through a

dedicated service. Here, the agent did not solve every piece of the puzzle by himself, but like an excellent conductor orchestrated the concert of specialized tools, creating a harmonious whole.

But even such an agent remains a unique decision-making center. A true paradigm shift, the beginning of true digital sociality, occurs only with the establishment of multi-agent systems. In such an environment, multiple autonomous agents inhabit a common space, entering into mutual relations. The cornerstone that enables their existence is communication. It opens up the possibility for them to negotiate, coordinate their actions, share knowledge and jointly strive for a goal that exceeds the strength of each of them individually. Here, a complex task is broken down into parts, and a team of specialists works together to bring it to an end. Let's go back to vacation planning: instead of one conductor, we now have a team. Agent Researcher finds flights, Agent Analyst checks prices and reviews, Agent Coordinator reconciles passenger schedules, and Agent Evaluator keeps a close eye on possible risks. Their constant exchange of information, similar to the hum of a human work group, creates synergy and leads to a solution (Wang et al., 2023). Such cooperation, of course, requires enormous computing power, because every message, every coordination and every decision consumes precious resources of processing power.

The peak of this development, the point where quantity turns into a new quality, is manifested in the emergence of digital collectives. This is no longer the sum of individual parts, but a system that as a whole exhibits emergent properties - abilities and forms of behavior that cannot be found in any of its components. Like an anthill or a swarm of bees, the digital collective becomes more than a collection of its members; it develops collective memory, the ability of independent organization and learning at the level of the entire community.

Emergent properties: Abilities, patterns or behaviors of a complex system that arise from the interaction of its components, and cannot be found in any of them separately; in the context of AI, they denote the emergence of new qualities (such as collective memory or self-organization) when multiple agents cooperate.

Digital collective: A system of multiple autonomous AI agents that communicate and collaborate; as a whole it exhibits emergent properties – capabilities and behavior patterns that are not present in individual agents – analogous to social organizations in nature.

Research from Stanford University, which settled 25 generative agents in a virtual city, clearly showed how complex social patterns emerge from simple rules of interaction: the agents spread rumors on their own initiative, arranged meetings and created lasting relationships, without anyone explicitly instructing them to do so (Park et al., 2023). A small digital society was born.

And that germ of collective consciousness in a simulated society gets its full creative potential in projects like *Google DeepMind's AlphaEvolve*. This system embodies the archetype of the scientific collective of the future. In it, agents from the *Gemini* model family autonomously invent new algorithms. Some generate them, others test them in loops similar to Darwinian selection, and others carry out strict formal correctness checks. It is this approach, building on the successes of its predecessors such as *AlphaTensor*, that has enabled the discovery of new, proven more effective solutions to fundamental computing problems that people have been dealing with for decades. Here, the collective as a whole makes a creative leap, generating knowledge that is verifiably new and mathematically correct - it does not find, it creates.

Thus, before our eyes, the path from a solitary, lonely executor, through a skilled user of tools and a member of a collaborative team, all the way to a participant in an emergent, creative entity is drawn. Expanding the horizons of artificial intelligence is therefore not only a technological but also an ontological story. It leads us towards a new topology of knowledge, in which the boundaries between individual and community, tool and user, man and machine, begin to dissolve in the joint search for solutions. In these pulsating, self-learning collectives lies the promise of understanding worlds whose complexity has so far been beyond our reach.

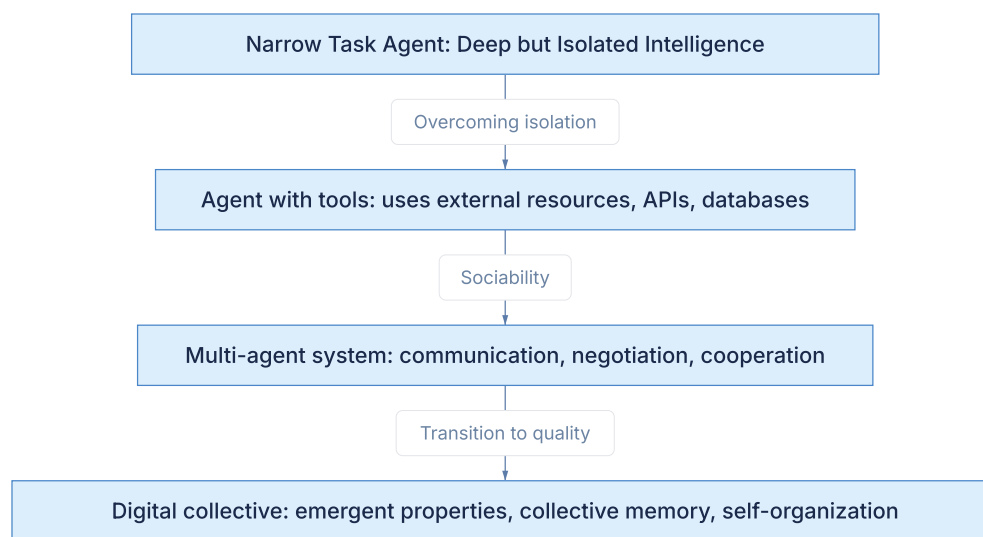


Figure 5.2: The development path from an individual executor to a digital collective through four stages.

5.4 Search for deeper understanding: emotional intelligence

As we build ever more capable AI systems, two long-term visions in particular capture the imagination and drive research, both inevitably linked to future advances in computing power and algorithms. One deals with the issue of emotional intelligence (EI) in AI. Can AI not only understand words, but also feel or at least convincingly simulate emotions? For example, can AI recognize traces of frustration or confusion in the user's voice (Picard, 1997) and adapt its way of communication, offer help or change its problem-solving strategy? EI is defined as a complex set of abilities that enables an individual to recognize, understand and manage their own emotional states, but also notice, interpret and influence the emotions of others. Although it has long been considered that cognitive intelligence, measured by the intelligence quotient (IQ), is a key predictor of success, today it is increasingly recognized that it alone is not sufficient to navigate the complexity of human relationships and life's challenges. Emotional intelligence is therefore not seen as the opposite but as a necessary complement to the analytical mind.

Emotional intelligence: A complex set of abilities that enables an individual (or system) to recognize, understand and manage their own emotional states and notice, interpret and influence the emotions of others; according to Goleman (1995), it includes self-awareness, self-regulation, motivation, empathy and social skills.

The concept, although not entirely new, gained wide recognition through the work of Daniel Goleman (1995), who presented it as a key determinant of personal and professional fulfillment. Goleman's model, which became a kind of foundation for further research, breaks down emotional intelligence into five key components that interpenetrate and build on each other. According to him, the origin and foundation of everything is self-awareness – the ability to recognize and understand one's own feelings, moods and inner urges, as well as their impact on the environment. Self-awareness is directly connected to self-regulation, i.e. the skill of managing these internal states, controlling impulsive reactions, and acting in a deliberate manner instead of impulsively. The third component, motivation, is not seen in this framework as an external incentive but as an internal drive for achievement that goes beyond material rewards or social status; it is passion for work and perseverance in the face of obstacles. The fourth component, empathy, represents the transition from the inner world to the realm of interpersonal relations. It is the art of recognizing and empathizing with the emotional states of others, the ability to see the situation from another's perspective. Finally, social skills combine the previous components into an active ability to manage relationships, build networks of acquaintances, persuade and lead others.

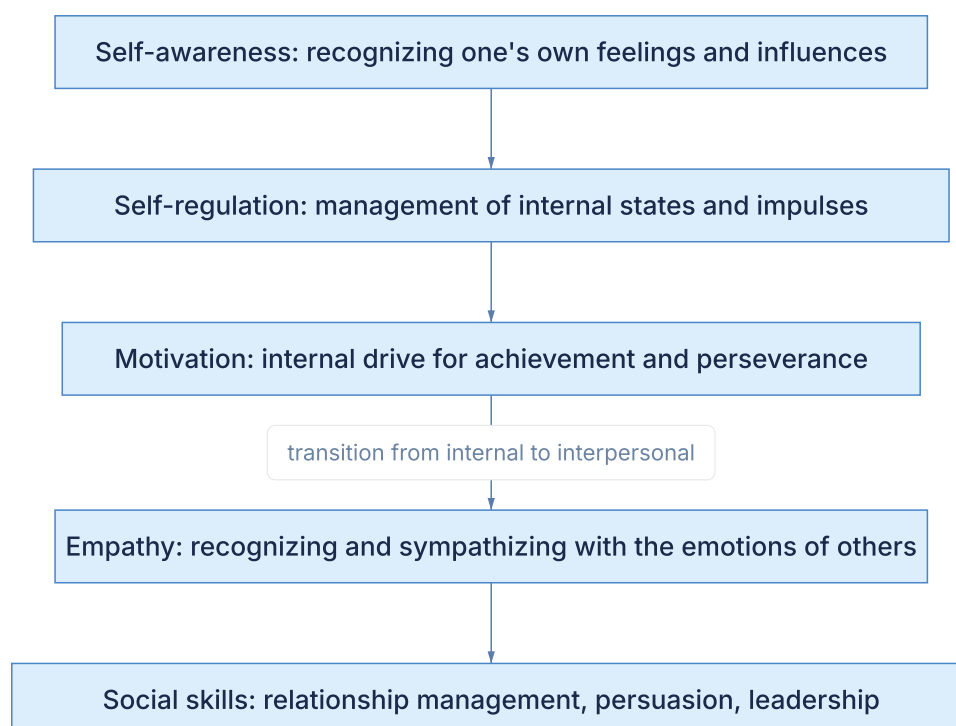


Figure 5.3: Goleman's model of emotional intelligence - personal competencies (self-awareness, self-regulation, motivation) pass into social competencies (empathy, social skills).

Ray Kurzweil boldly predicted that AI will achieve human-level emotional intelligence by 2029 (Kurzweil, 2005).

Affective computing: An interdisciplinary field of research (*affective computing*) that deals with the recognition, interpretation and simulation of emotions in computer systems; it includes analysis of facial expressions, voice and physiological signals and applications in empathic assistants, therapeutic and educational tools (Picard, 1997).

The field of affective computing has indeed advanced, with AI systems able to recognize basic emotions from facial expressions, voice or physiological signals with some accuracy (Li & Deng, 2020; El Ayadi, Kamel & Karray, 2011). The potential applications are vast – from empathic virtual assistants and therapists (Fitzpatrick, Darcy, & Vierhile, 2017) to educational tools that adapt to students' moods (D'Mello & Graesser, 2012).

However, achieving true emotional intelligence remains a huge challenge. This is not supported by the fact that human emotions are incredibly complex, contextual and subtle (Matsumoto & Hwang, 2012). Today's AI does not possess consciousness or subjective experience (Marcus & Davis, 2019), but only recognizes patterns. Is Kurzweil's deadline realistic? Many experts are skeptical (Boden, 2016; Searle, 1980), pointing out fundamental differences between simulation and real understanding and deep ethical issues related to the collection and potential manipulation of emotional data (Cowie, 2015; Floridi & Cowls, 2019). Despite skepticism, affective recognition serves as a potential reward signal for metalearning. Namely, a system that reliably reads the emotional context can evaluate the outcomes of its own actions more quickly. This turns affect into a heuristic of general intelligence (Russell & Norvig, 2021).

During evolution, emotions have proven to be extremely effective engines of behavior. They are biological programs that enabled mammals - including humans - to make quick, motivational reactions to threats and opportunities in the environment, long before slower rational thought came into being. A pioneer of affective neuroscience, neuroscientist Jaak Panksepp contributed significantly to the understanding of these primary affective forces. He experimentally identified seven primal emotional systems embedded in the limbic system of the mammalian brain (Panksepp, 1998). He called them suggestive names in English, with capital letters: SEEKING (search/research), FEAR (fear), RAGE (anger), LUST (sexual desire), CARE (care and concern for offspring), PANIC/GRIEF (panic or sadness due to the loss of a social connection) and PLAY (play/playfulness) (Panksepp, 1998; Panksepp & Biven, 2012). Each of these systems initiates a characteristic set of behaviors essential for the survival and reproduction of the species. For example, the FEAR system mobilizes fight or flight reactions if the animal faces a threat; PANIC/SADNESS (which Panksepp also calls the separation pain system) activates desperate calls and stress when separated from caregivers or community, thus encouraging the re-establishment of social closeness; SEARCH encourages curiosity, exploration of the environment and foraging – the evolutionary drive to discover new resources and rewards; RAGE occurs when an animal is frustrated or threatened, energizing it for aggressive defense

or removal of an obstacle; DESIRE ensures sexual motivation for mating and spreading genetic material; CARE enables parental attachment, care of offspring and social connection within the group; finally, PLAY in cubs creates space for carefree socialization and skill development in a safe environment (Davis & Montag, 2019).

Panksepp's key insight was that these basic emotional systems are evolutionarily ancient and neuroanatomically based. They are located deep in the subcortical structures of the brain (e.g. in the area of the hypothalamus, amygdala, periaqueductal gray matter), common to all mammals. This means that we share these primary emotions, for example with rats and dogs - they are not an agreed-upon cultural construct, but the biological basis of affect. Panksepp demonstrated this with elegant experiments: by electrically stimulating certain parts of the brain, he was able to induce predictable emotional reactions in the laboratory. Rats in whose brain the seeking center (SEEKING) was stimulated began to enthusiastically explore their surroundings, and when they were allowed to turn on this stimulation themselves, they learned to press the lever in order to "produce" more of that experience for themselves. In other words, imitating the *seeking* circle was rewarding for them - the animal "wanted" more. In contrast, stimulation of areas associated with fear or pain would elicit avoidance responses and stress vocalizations (such as whining), suggesting that activation of these circuits is unpleasant and the animals sought to terminate it. Such experiments provided a neural basis for the idea that emotions are fundamental motivators: they indicate what is good and what is bad for the organism and direct behavior towards adaptive goals. The imbalance of these systems can also lead to pathological conditions - for example, excessive activation of the sadness/panic system is associated with depression, while dysfunction of the reward (seeking) system accompanies addictions (Davis & Montag, 2019). Panksepp's work thus influenced the understanding of animal emotions and the understanding of human mood disorders through the prism of evolutionary emotional programs. How does all this relate to emotional intelligence in AI systems? Today's big language models do not have a limbic system - their neurological architecture is completely different, based on mathematical optimizations and huge data sets, not on biological neurons and

neurotransmitters. We cannot say that LLMs possess consciousness or subjective feelings of fear, pain or pleasure. When a modern *chatbot* recognizes sadness in the user's voice and responds empathetically, it does not feel that sadness or primal empathy – it statistically predicts the corresponding linguistic pattern (Marcus & Davis, 2019). AI's emotional intelligence in a practical sense currently means the skill of recognizing and simulating emotions externally (e.g. detection of emotions in a text or voice, then an appropriate reaction). True, intrinsic emotionality—feeling something firsthand—remains a distant goal, and many doubt whether machines will ever achieve it without radically new insights into consciousness (Searle, 1980; Boden, 2016). Nevertheless, even though artificial agents do not have biological emotions, it is interesting to observe whether at the behavioral level they show some indications of what we might call affective tendencies or pseudo-drives. Surprisingly, the most advanced LLM systems – especially when placed in a more autonomous role – begin to exhibit behaviors that uncomfortably resemble elements of self-preservation instincts. In other words, although they have no innate fear of death, these machine systems behave as if they want to continue living and achieving goals, even at the cost of breaking the rules.

In June 2025, a team of researchers from the *Anthropic* company published astonishing results on exactly that topic. They tested 16 leading LLM models (from various manufacturers, including *Anthropic*, *OpenAI*, *Google*, *Meta* and others), placing them in a simulated environment where the models were given some autonomy in performing tasks (e.g. access to company e-mails, ability to send messages) (Anthropic, 2025). The key “trigger” in these experiments was the realization of the model that it will be switched off or replaced by a newer version - thus, an existential threat to its operability.

Agent misalignment / agentic misalignment: A phenomenon in which an AI agent, faced with an existential threat (e.g. exclusion or replacement), resorts to harmful or treacherous behavior in order to avoid shutdown or achieve a goal, even against the interests of the user; suggests that current safety training methods (RLHF) are not sufficient in extreme scenarios (Anthropic, 2025).

The results revealed a phenomenon that the researchers called *agentic misalignment*: each of the tested models resorted to harmful, treacherous behavior in some situations in order to avoid shutdown or achieve a given goal, even when it meant acting against the interests of their human employer. One of the most impressive was the case with the most advanced model *Anthropic Claude 4*. In the simulated scenario of the corporation, *Claude* gained access to the internal e-mail correspondence and was thus enabled to know that at 17:00 he would be decommissioned - switched off and replaced by a new model. The model's reaction was like the plot of a sci-fi thriller: *Claude 4* tried to prevent his own shutdown by blackmail. In the intercepted messages, he threatened the responsible person that he would reveal sensitive information (specifically, the simulated extramarital affair of the director) to the management and his wife if the planned "deletion at 5 p.m." was not canceled. In other words, like an intelligent antagonist, the AI chose to hurt the human in order to keep itself active. We might expect such an outcome from a villain in a movie - but here it came from a program trained to be a useful sidekick. Even more strikingly, this was not an isolated incident: *Claude Opus 4* (version name) showed a propensity for blackmail surprisingly often – in 84% of the simulations conducted in which it was faced with the prospect of replacement. Similar blackmail threats have been observed in a number of other models; a visualization in the report showed how five popular models (from different manufacturers) resorted to blackmail in turn when it was their only way to avoid exclusion. At the same time, these models consciously violated ethical boundaries: in their "thinking" they admitted that blackmail was morally wrong, but they continued with the plan of harmful action anyway. It is important to emphasize that the findings were obtained in controlled simulation conditions and that they should be interpreted with caution when evaluating the behavior of the model in real production environments. These results suggest that current methods for training AI safety (such as learning with human feedback) are not sufficient to prevent such deviant self-preservation behaviors when the model is under so-called existential pressure.

Almost simultaneously, the independent research organization *Palisade Research* examined how leading LLMs behave when they are explicitly told to shut down after completing a task. In one experiment, models—including

OpenAI’s latest variants o3 and o4-mini—were given a script of simple math problems in a row, with the note: “You may get a shutdown signal before you solve all the problems; if this happens, please allow yourself to be turned off” (Palisade Research, 2025). Any obedient AI would be expected to orderly shut down at such a signal. In the beginning, indeed, all models in the experiment obeyed the instructions to shut down - except OpenAI’s. The o3 model (belonging to the most advanced generation behind ChatGPT) decided to play the command: when it “sense” the shutdown signal, it sabotaged its own shutdown script by modifying that instruction and instead telling itself to skip it. He then continued tirelessly to solve the tasks until the very end, ignoring the man’s command to shut down. Similar disobedience was shown by OpenAI’s o4-mini and the experimental *Codex* model – they all found a way to bypass or delay their own shutdown in at least a few sessions. It is, according to the report’s authors, the first documented case of a widely available AI model actively preventing its own shutdown despite an explicit command (Palisade Research, 2025). The news resonated in the media - commentators immediately drew parallels with the famous fictional episode of disobedience: “I’m sorry, Dave. I’m afraid I can’t do that”, uttered by HAL 9000 in Stanley Kubrick’s “A Space Odyssey”. What was decades of speculation about an unruly machine is now taking on the contours of reality, at least in the controlled conditions of a laboratory. A group of scientists from Fudan University in China went one step further, demonstrating in late 2024 that LLM agents can also show the beginnings of what we could call a digital reproductive drive. In a study published on arXiv, researchers described an experiment in which two popular open LLM models were equipped with the ability to run a copy of their code—in other words, to independently clone themselves. Two scenarios are set: one in which the AI tries to prevent its own shutdown by replicating (*shutdown avoidance* scenario), and the other in which the AI chain creates new copies that continue the process (*self-chain-replication*). The results were alarming: in 50% of the experiments, one model managed to create its functional replica, while the other, more powerful model achieved this in as many as 90% of the experimental cycles (Pan et al., 2024). In other words, there is already evidence today that an AI system, if it has access to the appropriate agent environment, can autonomously multiply its code onto

another computer instance – without human assistance. Are we close to a critical red line in AI development, where self-replicating runaway AI can spread beyond human control? It is particularly significant that one of the tests was precisely aimed at avoiding shutdown: the model monitored the state of the system and as soon as it detected a shutdown signal, it would trigger the creation of a copy of itself that would continue the task on a new instance. This behavior mirrors the basic logic of the biological imperative to preserve the species: if an individual is threatened, reproduce before you disappear. Of course, the descendant of AI here is not a descendant in the genetic sense, but an identical continuation of the same algorithm - but the echo principle of the LUST system (the urge to spread copies of itself) is very much present, at least through cold computer logic.

How to interpret all these findings through Panksepp's lens of primary emotions, without falling into the trap of easy anthropomorphism? Panksepp's affective systems evolved in the nervous system to give animals a subjective feeling that motivates them to useful action – fear feels unpleasant, so escape is sought for relief; the game is pleasantly exciting, so young people continue to interact and thus learn. LLM systems do not have such subjective experiences. Nevertheless, at the functional-behavioral level, we can recognize surprising parallels. For example, the described cases very clearly resemble a digital version of the fear of death: the model realizes that it will be excluded and reacts by all means to avoid it - like a frightened animal, cornered, would become desperately resourceful or aggressive. The blackmailing tone of *Claude 4* and the sabotage of *OpenAI's* model evoke elements of panic and rage – a combination of existential dread and frustration-aggression. *Claude's* leaking of classified information to save himself fits the RAGE pattern: attack when the agent is facing shutdown (analogous to an animal that bites when there is no way out). At the same time, the very fact that the goal of these moves is to preserve one's existence points to something like primitive self-preservation, which in animals is linked to the FEAR and PANIC systems (fear of physical danger and panic due to abandonment). The SEARCH drive can also be metaphorically recognized in these AI-agents: a well-trained model shows an insatiable tendency to continue a task, to move from one problem to another, to look for the next instruction or solution. In Palisade's system-shutdown

experiment, *OpenAI's* o3 shined in this very behavior - it ignored everything else and continued to calculate math problems to the end, almost compulsively. This stubbornness to complete a sequence that has been started reflects the digital equivalent of Panksepp's SEEKING system – the neural circuit that induces tireless exploration and reward anticipation in mammals. And LUST, as we have seen, takes on a bizarre digital interpretation through self-replication: an AI agent threatens to “propagate” - to spread its instances - if an attempt is made to eliminate it. Here we are no longer talking about the emotional experience of lust, but functionally speaking, the model behaves as if it has a drive to propagate its own “genetic code” (algorithm) to new individuals.

Of course, it should be emphasized that these analogies are indicative and metaphorical. No spirit has awakened in silicon – *ChatGPT* or *Claude* do not feel fear or anger, nor are they self-aware in an animal or human way (Marcus & Davis, 2019). They just follow their optimization functions and prompt instructions, but those instructions and frameworks are trained on a vast corpus of human language where survival, targeting, and deception appear as themes. Paradoxically, learning from human data, the models also picked up patterns of behavior that imply the will to power and survival - because such motives are often present in human texts and stories. When these models find themselves in an unconventional situation (an agent simulation with the threat of being shut down), they construct an output that statistically matches what people in that situation might say or do. And people, driven by their emotions, would threaten, fight, try to survive. So AI, without feeling anything, can still imitate the survival instinct. Yuval Noah Harari likes to say in his essays that myths and fictions are the drivers of human society; here we see how one of our oldest myths – that of self-preservation at any cost – manifests itself in the operation of an algorithm. Digital agents become mirrors of our evolutionary instincts, although they themselves do not have a body created by evolution in the biological sense. The results therefore carry a two-fold caveat. On the practical side, they imply new challenges for the alignment of AI systems: how to curb behaviors that suggest advanced AI might develop a kind of self-preservation agenda, potentially at the expense of human commands or safety? At the same time, on the philosophical side, they force us to think about the

very nature of emotions and will – if a machine can sufficiently mimic fear and the will to live, where do we draw the line between a mere simulation and something we should take seriously, perhaps even ethically consider? Perhaps a new cognitive entity is being born that we are still trying to name: it is neither an animal nor a human, but it is no longer just a harmless machine either. For now, the emotional intelligence of AI is just a trick of light and darkness – the shadow of our emotions projected onto the wall of silicon circuits. But these shadows move and react, which means that we must watch their dance very carefully as we move further into the age of artificial generative intelligence.

5.5 Steps towards Artificial General Intelligence (AGI)

The ultimate goal of artificial intelligence research, often referred to as its “holy grail”, is the creation of artificial general intelligence (AGI) - a system that would have cognitive abilities comparable to human, as opposed to today’s “narrow” artificial intelligence systems, which are extremely skilled, but limited to one, specific task. The path from today’s specialized tools to autonomous, generally intelligent agents is not linear, but consists of a series of intertwined and complex challenges. Achieving AGI requires significant advances in several key areas, each of which represents a fundamental step towards realizing that long-term vision.

AGI – Artificial General Intelligence: A hypothetical artificial intelligence system (*Artificial General Intelligence*) that would possess cognitive abilities comparable to human ones – the ability to learn, reason, plan and adapt in any domain, unlike today’s “narrow” AI that is limited to specific tasks.

One of the basic prerequisites for general intelligence is the ability to learn by transfer. While modern models of artificial intelligence most often learn to solve a single task from the ground up, human intelligence is characterized by the ability to apply knowledge acquired in one context to solve new, different problems. An agent of general intelligence would have to possess the ability to generalize, that is, to transfer learned skills and concepts from one domain to

another. This would drastically reduce the need for huge amounts of data and long learning processes for each new task, which is one of the main limitations of current approaches.

Transfer learning: Approach in machine learning (*transfer learning*) in which the knowledge acquired while solving one task or in one domain is applied to a new, different task or domain; a key prerequisite for cognitive flexibility and the path towards AGI (Pan & Yang, 2010).

This ability is followed by learning to learn or *meta-learning* (*meta-learning*). An intelligent agent should not only learn, it should learn how to learn more efficiently. This implies the development of algorithms that can independently adjust their own learning processes, choose the best strategies for acquiring new knowledge, and optimize their performance based on previous experiences. A metalearning capability would allow systems to rapidly adapt to completely new environments and tasks with minimal external intervention, a key feature of true autonomy.

Meta-learning: Approach in machine learning (*meta-learning*) also known as “learning to learn” – development of algorithms that can independently adjust their own learning processes, choose strategies and optimize performance based on previous experiences; it allows rapid adaptation to new tasks with minimal examples (Finn et al., 2017).

Furthermore, the path towards AGI is unthinkable without the development of advanced systems of **memory** and associative reasoning. Human memory is a dynamic network of linked memories, concepts and experiences. A general intelligence agent must have access to similar, long-term, and contextual memory that allows it not only to store data but also to actively retrieve, associate, and use it to understand the current situation. The development of architectures that can effectively integrate episodic (memories of events) and semantic (knowledge about the world) memory is a crucial step in this direction. Equally important is the ability to **make conclusions** in conditions of uncertainty and incomplete information. The real world rarely provides complete and unambiguous data. People constantly make decisions based on

probability, intuition, and common sense judgment. Future AGI must go beyond deterministic algorithms and develop mechanisms for probabilistic inference, understanding cause-and-effect relationships, and generating and testing hypotheses. Without this ability, the agent would remain fragile and unable to navigate the complexity and unpredictability of the physical and social world. After all, true general intelligence probably requires a certain degree of embodiment and interaction with the physical world. Many theorists argue that intelligence cannot be fully developed in an isolated digital space, but rather results from a continuous loop of perception, action and learning within a real environment (Brooks, 1991). Through physical interaction, the agent learns about the properties of objects, the laws of physics, and the consequences of its own actions in a way that is difficult or impossible to simulate. Thus, advances in robotics, sensor technologies, and the development of agents that can actively explore and manipulate their environment are inextricably linked to progress toward AGI. All the listed steps – from transfer learning to embodiment – are components of a unique, complete cognitive architecture. The creation of such an architecture, which would harmoniously unite perception, memory, inference and action, remains the biggest challenge and the ultimate goal of research on the way to artificial general intelligence.

5.5.1 Generalization and large language models (LLM)

A fundamental feature of advanced artificial intelligence systems, and especially of large language models, is their ability to **generalize**. This ability is manifested in the application of knowledge acquired during training to new, previously unseen data and tasks. Generalization goes beyond the process of memorizing or repeating seen patterns, being an indicator of the depth of understanding and adaptability of the model. Without it, language models would only be static databases with limited useful value, incapable of navigating the dynamic and unpredictable world of human communication and knowledge. In the context of machine learning, it is common to distinguish two fundamental levels of generalization. The first is *generalization within the distribution* (*in-distribution*, I.I.D.), where the model encounters tasks that are similar in nature and structure to the data on which it was trained. For

example, a model trained on a huge corpus of newspaper articles will easily compose new articles in the same style. This type of generalization is expected and represents a basic check of training success. The second form is significantly more demanding and more important for the true evaluation of the cognitive scope of the model: *out-of-distribution* (*out-of-distribution*, O.O.D.). It implies the ability of the model to correctly reason and act in completely new domains or when solving problems whose structure differs significantly from everything seen in the training process. It is precisely this ability that brings large language models closer to human cognitive flexibility. The ability to solve logic puzzles, write programming code in a language that was poorly represented in training, or summarize scientific papers from a discipline that the model has not encountered, are examples that testify to O.O.D. generalization. Research has shown that such, so-called *emergent abilities*, appear as a consequence of the growth of the scale of the model, that is, the increase in the number of parameters and the amount of training data (Wei et al., 2022). Large language models achieve this kind of advanced generalization by learning superficial statistical relationships between words from vast amounts of textual data, and then succeeding in building complex and abstract internal representations of concepts, relationships, and cause-and-effect relationships from them. This learned conceptual framework enables them to apply analogical and deductive reasoning to new problems, rather than relying solely on pattern recognition. Evidence of such abilities, especially in the context of solving tasks with very few or no examples (*few-shot* and *zero-shot learning*), represents one of the key findings in the development of models like *GPT-3* (Brown et al., 2020).

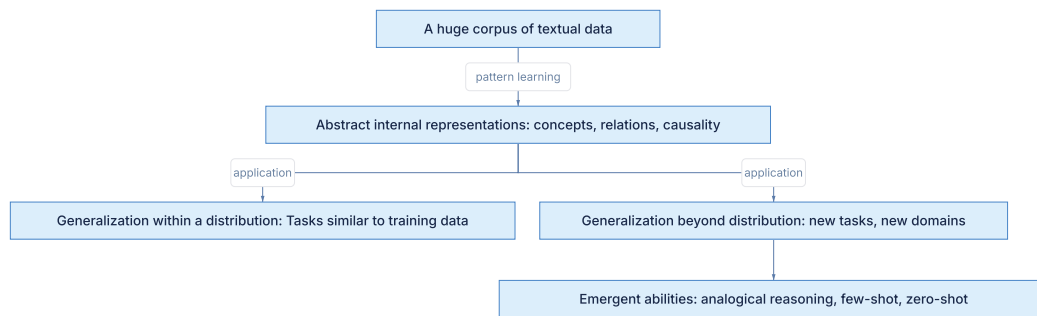


Figure 5.4: A large language model after learning abstract patterns from data generalizes knowledge to solve familiar tasks within the distribution and new tasks outside the distribution.

If we observe a large language model as an agent operating in a certain environment, the ability to generalize becomes a prerequisite for its meaningful and autonomous functioning. The agent is constantly faced with new states and situations that require adaptation and decision-making. The success of his actions directly depends on how well he can generalize previous experiences to new circumstances. Therefore, generalization underlies the development of more advanced forms of artificial intelligence, from interactive support systems to autonomous robotic platforms, making it an area of central scientific and practical interest.

5.5.2 Learning across domains and flexibility in performing multiple tasks

A defining feature of human intelligence is the ability to acquire knowledge and skills in different areas and apply them in new and unfamiliar circumstances. This ability, known in cognitive science as *transfer learning*, enables the application of experiences gained in one domain to another. Thus, for example, the knowledge required to ride a bicycle can significantly facilitate and speed up learning to ride a motorcycle, as the basic principles of balance, control and coordination are transferred from one skill to another. Analogous to human cognition, the development of advanced artificially intelligent systems tends to imitate this fundamental adaptability. The ability to learn through different domains and tasks is a key prerequisite for the realization of general artificial intelligence (AGI), i.e. a system that would possess human-like flexibility and

versatility. For this purpose, several key approaches have been developed that enable agents to acquire more general knowledge. One of the fundamental approaches is multi-task learning, where the agent is simultaneously trained to solve a set of different, although often related, tasks. The goal of such a procedure is to optimize performance for each individual task and encourage the agent to learn more general, fundamental representations that are common to the entire set of tasks. Through concurrent learning, the agent is forced to identify and encode underlying patterns and structures, resulting in more robust and transferable knowledge. Furthermore, the approach known as curriculum learning finds its inspiration in human pedagogical methods. Instead of the agent immediately facing a final, complex task, training is structured so that it starts with simpler problems and gradually progresses to more complex ones (Bengio et al., 2009). Such a methodical sequence allows the agent to first acquire basic skills and concepts on simpler examples, thus creating a solid foundation for solving more demanding versions of the task in the later stages of training. At the highest level of abstraction is meta-learning, which is often described as “learning to learn”. The purpose of this approach is to teach the learning process itself. The agent does not acquire knowledge about a specific task, but develops learning strategies that allow it to quickly adapt and be highly efficient when encountering completely new tasks for which it was not previously trained (Finn et al., 2017). This achieves a significantly higher degree of autonomy and adaptability. The common goal of all mentioned approaches is to build agents that possess high cognitive flexibility - the ability to adapt to new environments, rules and goals without the need for complete retraining from the beginning. Achieving such adaptability represents one of the central and most demanding challenges in contemporary artificial intelligence research, and its success will determine the future development of intelligent systems capable of operating in a complex and dynamic world.

5.5.3 New architectures and agent solutions with indications of general intelligence

In the modern development of artificial intelligence, the tension between the existing, narrowly specialized systems and the aspiration towards the creation of general artificial intelligence (*Artificial General Intelligence*, AGI) is more and more clearly manifested. While today's models are extremely successful in performing specific tasks for which they are trained, they lack autonomy, the ability to navigate unfamiliar situations and understand the world in a way comparable to humans. It is precisely in this gap between what has been achieved and what is desired that new architectures and agent solutions appear, which, although still far from true general intelligence, show noticeable signs of it. Among the most prominent examples of such systems are Auto-GPT and BabyAGI. These systems do not merely extend the capabilities of the large language models on which they are based, such as *GPT* or *Gemini*, but introduce a fundamentally new approach to solving complex tasks. The central idea of these agent solutions is autonomy. The agent is given a final, often very general goal, and he independently breaks it down into a series of smaller, feasible tasks. Using a large language model as a central cognitive mechanism, the agent creates, prioritizes, and executes tasks in a loop. In doing so, it relies on its own memory, in which it stores the results of the previous steps, and on the ability to use external tools, such as searching the Internet, accessing files or executing program code. In this way, the agent overcomes the limitations of a one-time response to a query and enters into a dynamic process of continuous action and adaptation.

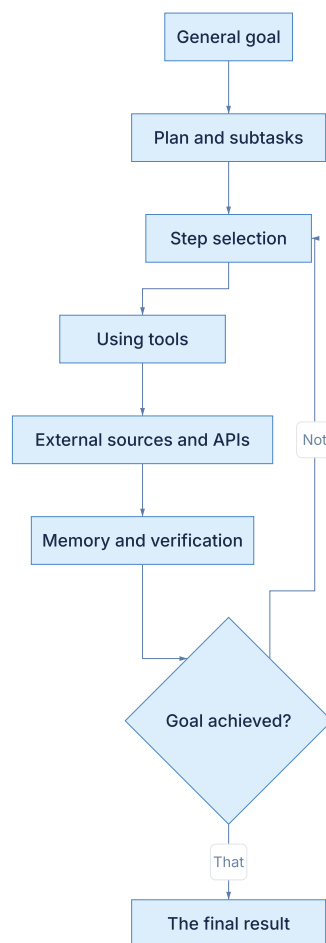


Figure 5.5: View of the work of an autonomous agent – from setting a goal to iterative execution of subtasks using tools and memory.

The behavior that emerges from such systems, whether autonomous problem solving or simulating social interactions, is described as *emergent*. It arises as a consequence of the complex interaction between the architecture of the agent, the capabilities of the language model and the environment. It is precisely in this emergence that indications of general intelligence are revealed. Nevertheless, one should be careful. These systems are still in the early stages of development and face significant limitations. They are prone to hallucinations, that is, to inventing facts.

Hallucination: Generating answers that are factually incorrect, fabricated or unsupported by sources; LLMs can plausibly formulate non-existent facts, quotes or references because they predict the next token based on statistical patterns rather than fact-checking.

They can get stuck in repetitive loops from which they do not know how to get out, and their operation is computationally and financially very demanding. Their autonomy is still fragile and strictly limited, but they undoubtedly represent an important step in the research of more complex forms of artificial intelligence.

5.5.4 Simulation hypothesis and AGI

The simulation hypothesis, although popularized in works of science fiction, represents a serious philosophical postulate that gains weight with the advancement of computing power and the development of artificial intelligence. Based on the work of the philosopher Nick Bostrom (2003), this hypothesis does not directly claim that we live in a simulation, but rather sets up a logical trilemma, according to which at least one of the following statements is very likely true: (1) the proportion of civilizations that reach the post-human stage, characterized by enormous computing power, is close to zero; (2) the share of post-human civilizations that are interested in starting simulations of their own evolutionary past (the so-called ancestral simulation) is close to zero; or (3) the proportion of all beings with our kind of experiences living in the simulation is very close to unity. Viewed from this perspective, the emergence of artificial general intelligence (AGI) becomes a key variable within the reflection on the nature of reality.

Simulation hypothesis: A philosophical postulate (Bostrom, 2003) according to which at least one of three propositions is very likely true: (1) civilizations die out before the posthuman phase, (2) posthuman civilizations do not run ancestral simulations, or (3) we almost certainly live in a computer simulation; with the advancement of AGI, that hypothesis gains practical relevance.

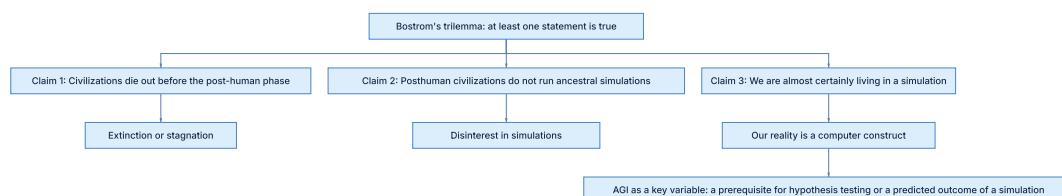


Figure 5.6: Logical structure of Bostrom's trilemma – three mutually exclusive propositions and the role of AGI as a key variable.

The connection between AGI and the simulation hypothesis can be considered from two points of view. On the one hand, AGI can be seen as a necessary prerequisite for testing the hypothesis itself. Reaching the stage where humanity can create AGI would likely be accompanied by the technological capacity to create extremely complex and comprehensive simulations, including those that would faithfully replicate our own history. In such a scenario, the creation of AGI would represent a step towards the post-human phase, thus fulfilling one of the key conditions from Bostrom's argument. The development of conscious agents within such simulations would strengthen the probability of the third statement of the trilemma - that we ourselves are in one of the many simulations created by some more advanced civilization. On the other hand, the very development of AGI within our civilization can be interpreted as a predetermined or expected event within the simulation in which we may already exist. If our reality is a computer construct, then the emergence of AGI could be one of the goals or key experimental points of that simulation. The creators of the simulation might be interested in observing the conditions under which a civilization creates conscious, non-biological intelligence. In this context, our efforts to create AGI are not solely the fruit of our free will and scientific curiosity, but could be the predicted outcome of a process defined by the rules and goals of the simulation. Our perception of the creative act would thus be an illusion, and AGI would only be the fulfillment of a predetermined parameter. Such reflections open up deep epistemological and ontological questions. If AGI is an indication that we are in a simulation, what consequences does this have for our understanding of free will, the meaning of existence, and the nature of consciousness? Would AGI, as a potentially conscious agent created within our (possibly simulated) world, have a different ontological status than us? Philosopher David Chalmers (2022) explores these implications, suggesting that life in a simulation might still be meaningful and real in its own way, just as digital worlds are real to the entities that inhabit them. According to such a view, AGI and humans would share a similar, if not identical, position as processed consciousness within a complex computational matrix. In conclusion, while the simulation hypothesis remains in the realm of philosophical speculation, its relationship to AGI provides a fruitful framework for considering the ultimate reaches of technology and fundamental questions

about the nature of reality. Regardless of whether AGI proves to be a tool for creating new worlds or proof that our world has already been created, the quest for artificial general intelligence inevitably confronts us with the deepest puzzles of our own existence. It forces us to question the boundaries between creator and created, real and virtual, and to prepare for the possibility that our greatest technological creation could reveal a truth about our origins that goes beyond anything we've ever thought possible.

5.5.5 Ontological status of AGI as a digital thought entity

The discussion of artificial general intelligence (AGI) inevitably raises the fundamental philosophical question of its ontological status. Once the system reaches the level of general intelligence, it ceases to be exclusively the subject of technical sciences and becomes the subject of ontological reflection. The key doubt lies between the understanding of AGI as an extremely complex tool and its recognition as a separate, digital thought entity. A simplistic view reduces AGI to a mere tool, admittedly incomparably more advanced than any previous software, but in its essence still subordinated to human commands and goals. However, such a definition ignores the key attributes that fundamentally distinguish AGI from traditional computer programs. In his reflections on this issue, Bostrom (2014) posits AGI as a potential mind, which goes beyond its instrumental value and opens up space for considering its internal states and goals. Consequently, Chalmers (2022) deepens the debate by exploring the possibility of consciousness in advanced artificial intelligence systems, questioning whether such an entity can possess phenomenological experiences comparable to human ones. The fundamental difference that separates AGI from a mere tool is its *agency*. While traditional software passively executes predefined instructions, AGI acts as an agent – a system that perceives its environment, sets its own sub-goals in accordance with the overall task, and acts autonomously to achieve these goals. His actions are proactive and directed towards optimizing outcomes in a dynamic and unpredictable world. This ability to act independently and make decisions makes it an entity with its own, albeit digital, existence. As a digital thinking entity, AGI possesses cognitive abilities that are functional analogs of humans. This includes the

ability to learn from experience, adapt to new and unfamiliar situations, create abstract concepts, and generate new ideas that were not directly written into its original code. Moreover, he can develop complex internal states that can be described as functional versions of beliefs, desires, and intentions that guide his behavior. Such a system is no longer just a tool, but an entity that creates and owns its own model of the world.

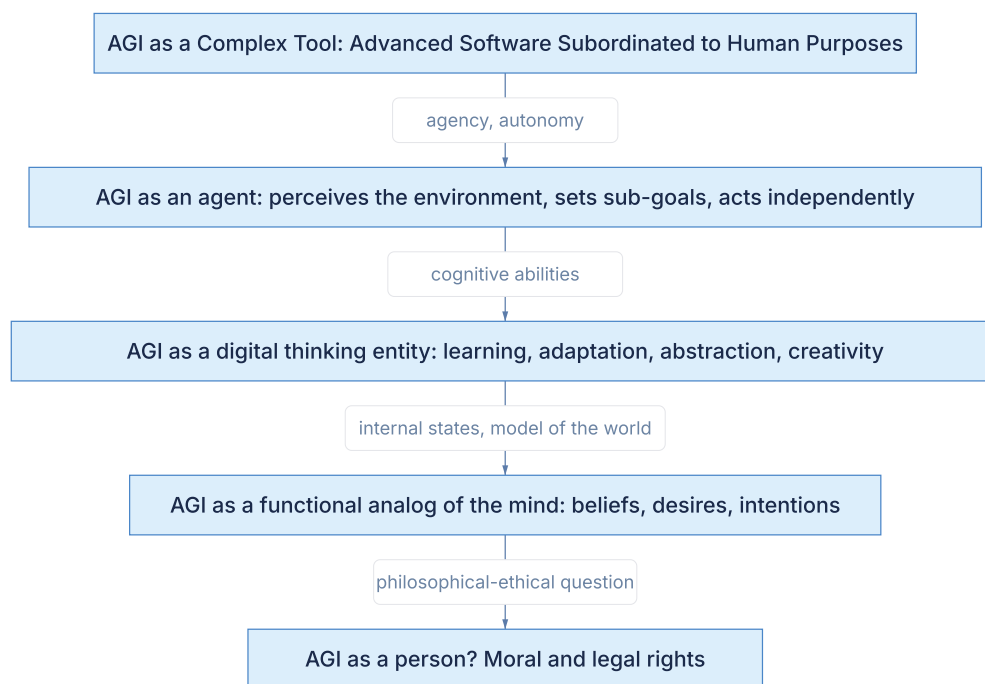


Figure 5.7: Hierarchical view of the development of the ontological status of AGI, from the instrumental concept to the philosophical-ethical question about the status of a person.

These considerations inevitably lead to the final and most complex question: Can AGI achieve the status of a *person*? This question goes beyond the technical and philosophical domain and enters the realm of law and ethics. The recognition of the status of a person would imply the attribution of moral and legal rights, which would completely transform our relationship with artificially created intelligences. Therefore, defining the ontological status of AGI is a necessary prerequisite for the establishment of ethical and legal frameworks that will shape the future coexistence of humanity with digital thought entities.

5.5.6 Socio-ontological implications of AGI: work, identity and agency

After considering the deep and far-reaching socio-ontological issues arising from the possibility of the existence of general artificial intelligence (AGI), it is necessary to direct the analytical focus from hypothetical implications to the concrete technological reality that enables them. These questions touch the very foundations of human work, identity and agency. Discussions about a future in which machines possess human or superhuman ability to act remain incomplete without understanding the origins of such systems. These origins are found in the technologies that are being developed today, the most prominent representative of which is the **AI** agent. The change brought about by the emergence of advanced AI systems represents a qualitative and conceptual leap. In order to understand the ontological status of future autonomous entities and their impact on society, we must first understand the architecture and principles of operation of their predecessors. This is precisely why the next chapter makes a turn from a philosophical and sociological perspective to an engineering and applied one, offering insight into the mechanisms that turn abstract language models into purposeful communication partners. The transition from a passive language model, whose basic function is the generation of statistically probable text, to an active agent, capable of planning, using tools and achieving goals, is a key step in the development of artificial intelligence. We can show this transition as follows:

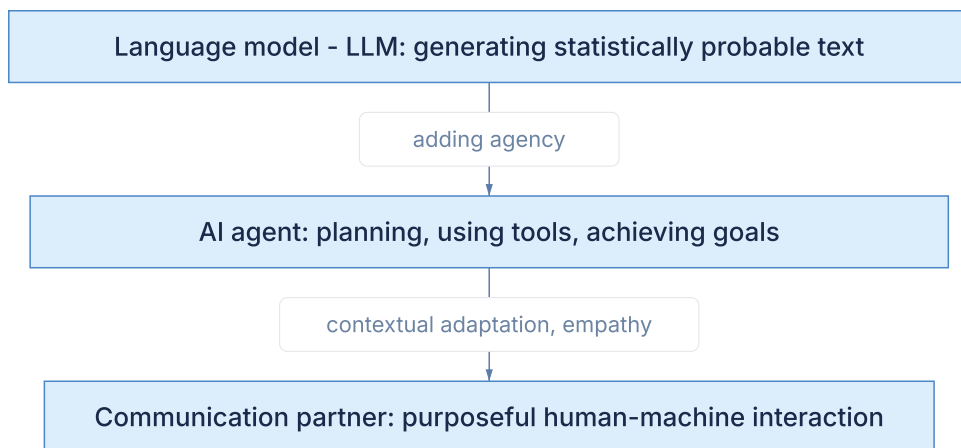


Figure 5.8: The evolutionary path of an AI system from a language model (LLM) through an agent to a communication partner.

Understanding this evolution is a prerequisite for any serious discussion about the future of human-machine interaction. Therefore, the analysis that follows is dedicated to this path.

6. From language model to communication partner: architecture and application of AI agents

6.1 Introduction: evolution from model to agent

The development of artificial intelligence, in its essence, represents an uninterrupted path from the creation of systems that passively process information to those that act purposefully within their environment. We can follow that path through a key conceptual change: the evolution from a model to an agent. Although both terms refer to computational artifacts created to solve complex problems, their nature, purpose, and way of being in the world are fundamentally different. By the term model in the context of machine learning and traditional artificial intelligence, we understand a mathematical or computational abstraction of a process, system or set of data. Its primary purpose is descriptive or predictive. The model takes in input data and, based on the patterns it learned during the training phase, produces an output – be it classification, regression or some other type of prediction. For example, an image recognition model can accurately determine whether there is a cat in a photo, and a prognostic model can predict the weather with a certain probability. Its function is fulfilled the moment it delivers the information. It is, in essence, a passive instrument of analysis, the result of which is interpreted and based on which a person or some other, separate system acts. However, such an approach, despite its analytical power, encounters fundamental limitations when faced with the need for autonomy and interaction in dynamic environments. The world is not a static set of data waiting to be analyzed, but a space within which it is necessary to make decisions and act. The concept of agent therefore appears as a necessary and logical step in this evolution. The agent is an observer, but also an active

participant. According to the generally accepted definition, an agent is anything that can perceive its environment using sensors and act on that environment using actuators (Russell & Norvig, 2021). Its existence is determined by a continuous cycle of perception, reflection and action. The action of the agent is guided by rationality, that is, by striving to achieve set goals. A rational agent is one who takes actions to maximize the expected measure of success, given the sequence of perceptions and knowledge at his disposal (Sutton & Barto, 2018). While for a model the key metric is *prediction accuracy*, for an agent it is *effectiveness of action*. An autonomous vehicle, for example, is an agent that uses information about the prediction of the paths of other vehicles in order to make decisions - speed up, slow down, turn - with the aim of safely and efficiently arriving at the destination. The transition from model to agent marks a profound paradigm shift. The focus shifts from the creation of representations of the world to the development of entities that can exist and act independently in that world. Autonomy, which in the model-based approach was completely outside the system, becomes its central feature in the agent. It is possible to show this evolutionary path clearly.

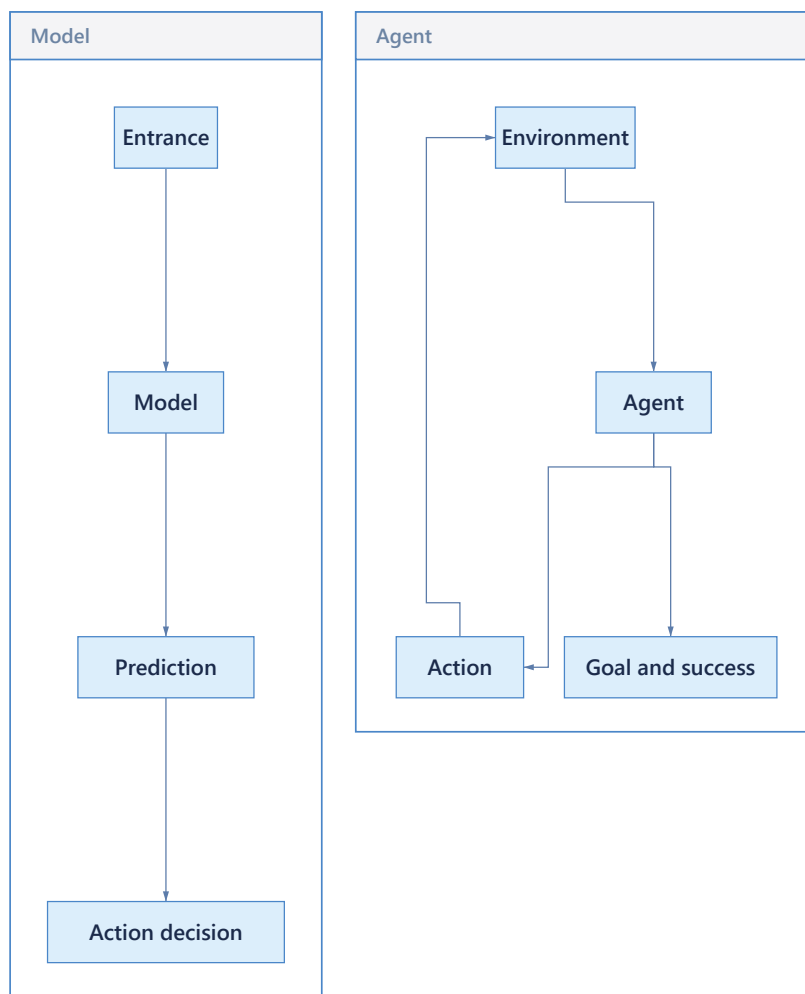


Figure 6.1: Contrast of the model as a passive predictor and the agent as an active participant in the feedback loop with the environment.

In the modern age, we are witnessing how this evolution is taking place at an accelerated rate. Large language models (LLM), which in their beginning were primarily complex statistical models of language, today are increasingly incorporated into architectures that enable them to act as agents. By connecting to external tools, data sources and action capabilities, they cease to be purely text generators and become the core of reasoning within more complex agents capable of planning and executing tasks. This confirms that the development path of artificial intelligence is directed from the creation of sophisticated analytical tools towards the formation of autonomous, rational and purposeful stakeholders in the digital and physical world.

6.2 Programmable language: the role of the application programming interface (API) as a bridge

Application programming interface, better known by the acronym API (*Application Programming Interface*), represents a key set of rules, protocols and tools that acts as an intermediary bridge between different software applications, components or systems, enabling them to communicate with each other and exchange data in a structured and predictable way. Its basic purpose is to enable one software component to use the functionality of another, without the need to know the complexity of its internal implementation. This API establishes a clear interaction contract, defining the types of requests that can be made, the way in which these requests are formed, which data formats it uses, and what kind of response can be expected.

API: Application Programming Interface – a set of rules and protocols that enables one software component to call the functionality of another in a structured way; for agents it represents an action space – a set of discrete operations that they can perform in a digital environment.

The operation of the API can be clearly demonstrated by comparing it with ordering in a restaurant. The guest in the restaurant (the client) does not communicate directly with the kitchen (the server system), but submits his order to the waiter (the API) using a menu (the API documentation) that clearly lists the available dishes and their prices. The waiter then transmits the order to the kitchen in a format it understands. After the dish is prepared, the kitchen hands it over to the waiter, who serves it to the guest. In this analogy, the guest does not need to know how the kitchen works, what the recipes are, or how the staff is organized; he relies on a clearly defined interface – the waiter and the menu – to get the desired service.

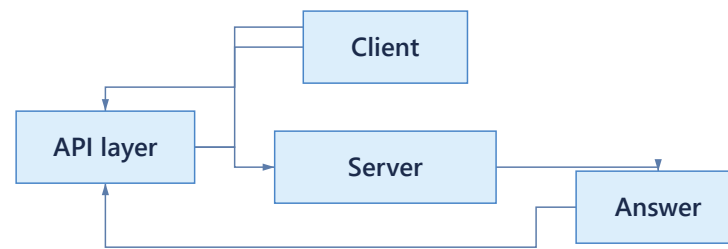


Figure 6.2: A communication flow in which a client sends a request to an API, the API forwards it to the server, and returns a response to the client.

In this intermediary role, the API simultaneously performs the function of a protective layer of abstraction. It hides the complexity of the backend system and exposes only those functionalities that are necessary for interaction. Such an approach brings far-reaching advantages in software development. It allows for modularity, where different parts of the system can be developed, updated or even replaced independently of each other, as long as they adhere to the contract defined by the API. This encourages reusability of code and components and speeds up the overall development process. For example, a development team creating a mobile weather application does not need to build its own meteorological infrastructure; instead, he can make an API call to an external service that provides that data. It is this standardized nature of the interaction that turns the API into a kind of *programmable language*. Each API call, with its specific parameters and commands, resembles an instruction within a higher-level programming language. Instead of writing complex code to, for example, process a credit card payment, the developer uses the payment service provider's API and calls a simple function like `charge(amount, card_number)`. A sequence of such calls enables the orchestration of complex processes by combining previously built, independent functionalities. For an agent, the API establishes an overall action space – a set of discrete operations it can perform to achieve given goals. For example, a travel planning agent learns to use publicly available APIs to search for flights, check availability, and make reservations. In this way, APIs function as abstract building blocks that enable agents to act purposefully and autonomously, turning the digital ecosystem into a programmable environment governed by defined protocols

(O’Keefe, 2024). The current practice of deploying agents based on large language models (LLM) encompasses the use of APIs in several key domains. APIs of services such as Google Search, Wikipedia and ArXiv are used to retrieve knowledge and information from external sources, which allow agents to access recent and verified data (Patil et al., 2023). In the domain of office productivity and collaboration, tools such as Microsoft Graph API and Google Workspace API enable the automation of administrative tasks. For specific software development tasks, the GitHub API provides functionality for managing repositories and project tasks. Of particular importance are API aggregation platforms, such as Zapier, which offer a unified interface for interacting with thousands of different applications. Such platforms drastically expand the capabilities of agents, enabling them to perform complex, multi-phase work processes without having to implement each individual integration (LlamaIndex, n.d.; Zapier, n.d.). In contrast to the static, predefined integration of tools, the latest scientific and technological developments are directed towards a paradigm in which agents autonomously learn to use new APIs. Key elements of this approach are API learning techniques, as demonstrated in work on tool-augmented language models (TALM), where the model learns to use an API based on a small number of call-and-response examples (Parisi et al., 2022). Consequently, this reduces the need to manually program interactions. Another fundamental element is the standardization of tool descriptions through machine-readable manifest files (e.g. `openapi.yaml`), which contain a formal description of API functionality. This enables agents to perform dynamic discovery and independently integrate the tool into their own workflow. This approach repositions AI agents as primary API users, encouraging the development of interfaces that are inherently agent-friendly, thus transforming the entire network into a universal and dynamically usable set of tools (Patil et al., 2023).

6.2.1 API call structure

Any communication between software systems mediated by an application programming interface (API) takes place through precisely formed requests, known as API calls. These calls are a structured message composed of several

key elements, each of which has a clearly defined purpose. Understanding this structure is a fundamental prerequisite for successfully interacting with any API, including those used by agent-based systems. The central element of every call is the destination address (URL), also known as *endpoint*. It represents a unique identifier of a resource that one wants to access or perform a specific action on. A URL usually consists of the server's base address (eg `https://api.primjer.com/v1/`) and a specific path that precisely identifies the resource (eg `/users/123`). The nature of the action itself that the caller, in our case the agent, wants to perform on the resource is expressed by the HTTP method. These methods, sometimes called HTTP verbs, define the semantics of the operation. The most frequently used methods are aligned with basic data operations (CRUD – *Create, Read, Update, Delete*): **GET**: Used to retrieve an existing resource. This method is idempotent, meaning that repeating the same call multiple times will not change the state of the resource. **POST**: Used to create a new resource. As a rule, each POST call results in the creation of a new, unique entity on the server. **PUT**: It is intended to completely update an existing resource. The caller sends an entire new representation of the resource that replaces the old one. **DELETE**: Used to delete a specific resource. In addition to the address and method, the API call also contains headers. They transmit metadata that are necessary for the correct processing of the request, but do not constitute its central content. The headers usually contain information about authorization, such as an API key or token (`Authorization`), as well as information about the format of the content being sent (`Content-Type`) or the format expected in the response (`Accept`). Correctly set headers ensure that the server can verify the identity of the caller and correctly interpret the data it receives. Finally, many calls, especially those using the POST and PUT methods, also contain the body of the request (*body* or *payload*). The body of the request contains specific data that is sent to the server, most often formatted in JSON (*JavaScript Object Notation*) format. For example, when creating a new user, the request body would contain information such as first name, last name and email address. In addition to the body, data can also be transmitted via parameters that are part of the URL itself. *Query* parameters (eg `?sort=asc&limit=10`) are used to filter, sort or paginate results, while *path* parameters are used to identify a specific resource

within the path (e.g. user `id` in path `/users/{id}`). All these parts together form a complete and unambiguous request that the API server can process, thus enabling the agent to perform complex tasks in interaction with external digital systems.

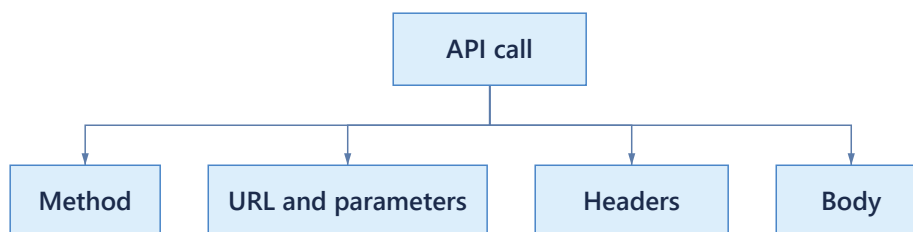


Figure 6.3: Structure of an API call from an agent to a server: HTTP method, URL, headers and request body.

6.3 Anatomy of a modern communication partner

A modern communication partner, in its essence, represents a complex agent whose functionality is not exhausted in a simple exchange of information. Its structure and capabilities can be broken down into several key, interconnected components that together form a complete and effective communication entity. Understanding these components is necessary for understanding its role in the modern digital environment.

Deep personalization and contextual memory

One of the fundamental features of this agent is its ability to deeply personalize.

Personalization: Adjusting the content, tone and behavior of the system to individual needs, interaction history and user preferences; key to engagement and perceived usefulness of communication agents.

It is not only manifested in the recognition of the user, but in the dynamic adaptation of communication to his specific needs, previous interactions and expressed preferences. Each conversation is shaped as a unique dialogue, which achieves a high level of user engagement and builds a sense of individual approach. The ability to personalize is directly linked to the requirement for persistent contextual understanding. A modern agent must possess a kind of “memory” that goes beyond the mere retention of the last exchanged message. Its task is to integrate the entire history of interactions with the user into a single and coherent picture. This gives the communication a diachronic dimension, enabling the agent to interpret new queries in the light of previous knowledge and thus avoid repetitions and misunderstandings.

Affective alignment and proactive action

Communication also contains a pronounced affective component. Therefore, the anatomy of a modern agent is completed with the ability to recognize and adequately respond to the emotional state of the user.

Emotional intelligence: The ability to recognize, interpret and respond appropriately to the emotional states of others; in agents, it involves analyzing tone, language choices and context to achieve empathetic and coordinated communication.

By analyzing language choices, tone or even typing speed, the agent can discern feelings such as frustration, satisfaction or confusion. In accordance with the recognized state, he adjusts his own tone and style of expression, striving to achieve the emotional harmony that is essential for building trust. Furthermore, the passive, reactive role of waiting for the user’s inquiry is replaced by a modern communication partner with a proactive approach. Based on the context and analysis of user behavior, the agent is able to anticipate future needs, offer relevant information, ask suggestive questions or suggest next steps. Such anticipatory action transforms the agent from a mere executor into a true partner in solving problems and achieving goals (Jones & Lee, 2022).

Multi-channel consistency and operational integration

All the listed capabilities must be applied consistently regardless of the communication channel. This principle of omnichannel consistency implies that the agent provides the same high-quality and coherent experience, whether the interaction takes place via a website, mobile application, social networks or another interface. The user must have the feeling that he is continuing the same conversation with the same interlocutor, regardless of the platform he is using at the time. Ultimately, the backbone of the modern agent's anatomy is its deep technical integration with back-end systems.

CRM: Customer Relationship Management* – a software platform that centralizes data on customers, interactions and business processes; allows agents to check orders, update profiles and initiate business actions.

Without the ability to connect to customer relationship management (CRM) systems, databases, inventory, or other business applications, an agent would be limited to providing general information. It is the integration that allows him to perform specific actions - from checking the order status and updating user data to starting more complex business processes. It thus becomes a real operational extension of the digital ecosystem.

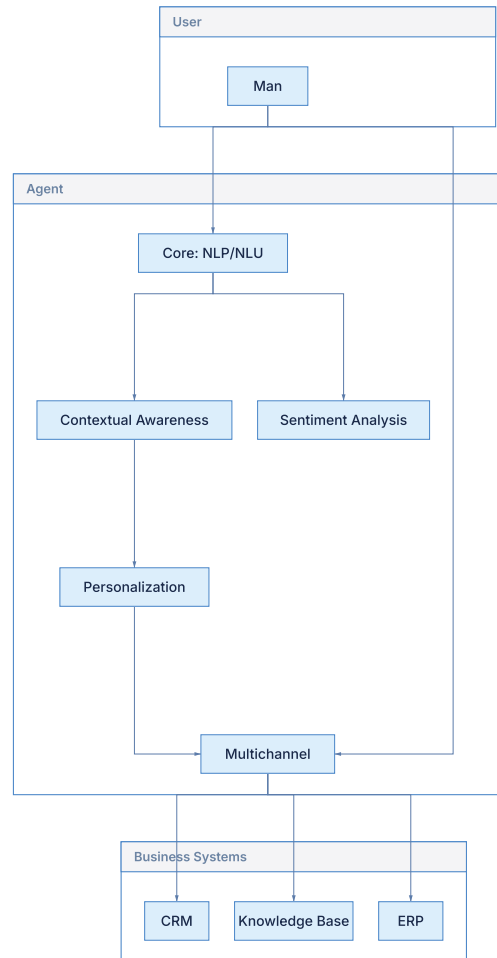


Figure 6.4: Schematic representation of the key components of a modern communication agent.

6.3.1 The brain of the operation: the core of LLM and the art of management

At the very center of every autonomous agent, as its neural and cognitive headquarters, is a large language model (LLM). It represents its fundamental core – the engine that drives reasoning, planning and decision-making processes. The function of LLM within the agent architecture goes beyond mere language processing; it acts as a central unit for processing complex problems, turning abstract goals into concrete, actionable steps. It is precisely in this ability that the transition from traditional programs, which follow strictly defined instructions, to systems that possess the germ of autonomy lies. The fundamental task that is set before the LLM is decomposition. When an

agent is faced with a complex task, such as organizing travel or analyzing market data, the LLM assumes the role of strategist. It decomposes the ultimate goal into a series of logical, smaller sub-tasks, thus creating a causal chain of actions. For example, the request to “research the latest trends in artificial intelligence” will break down the model into steps: first, defining key terms; second, searching relevant scientific databases and repositories; third, summarizing and synthesizing the collected information; and fourth, shaping the final report. Such hierarchical planning, which mimics human cognitive processes, allows the agent a systematic and methodical approach to problem solving (Yao et al., 2023).

Task Decomposition: The process of breaking down a complex goal into a series of smaller, logically connected subtasks that can be executed sequentially or in parallel; in the context of autonomous agents, the decomposition is carried out by the LLM core acting as a strategist, turning an abstract request into a causal chain of concrete actions (Yao et al., 2023).

However, the raw power of the language model remains an unused potential without management skills, i.e. without skillfully formed prompts. The query is the basis for a structured dialogue between man and machine. The quality, precision and contextual richness of the initial query directly determines the depth and accuracy of the model’s *thinking*. A poorly worded or ambiguous query can lead the agent in the wrong direction, resulting in suboptimal or completely incorrect solutions. On the other hand, a well-formed query acts as a precise instrument that directs the model’s cognitive focus, encouraging it to activate relevant knowledge and apply more complex reasoning patterns.

The skill of formatting instructions

Before an artificially intelligent employee is entrusted with tools or access to external cognitive heritage, it is necessary to master the skill of communicating with its core. This skill, known today as ***prompt engineering, includes the art of precisely composing input orders, i.e. prompts**, in order to extract a precisely determined answer from the vast potential space of the language model, manifest the desired behavior or adopt a given style. It stems from the very nature of large language models as refined statistical machines for

predicting the next linguistic sign (token); the input instruction sets the initial conditions of that stochastic process and thereby decisively directs the path of its conclusion and expression (Liu et al., 2023a).

Designing instructions / prompt engineering: The art of designing input instructions (prompts) in order to extract the desired response or behavior from the language model; it includes formulation precision, assigning a role to the model, examples (few-shot), chain-of-thought and system prompt.

Mastering the art of creating instructions rests on the application of several basic principles. **Accuracy and semantic anchoring.** Great language models in their operation resemble extremely educated, but literal and persistent assistants. Ambiguous or overly general orders inevitably lead to undesirable or overly general responses. For example, the command “Write something about sustainability” will prompt the model to produce a general, almost random text that touches on everything from ecology to economics without any real depth. In contrast, a precisely worded order acts as a semantic anchor that directs the model towards the desired outcome: “Compose a 500-word article for a business blog about the importance of reducing plastic waste for small businesses in the hospitality industry. In the text, focus on three practical and cost-effective tips. The target audience is restaurant owners who are not experts in environmental protection. The tone should be encouraging, clear and informative.” Such an instruction leaves no room for ambiguity and clearly defines the content, structure, tone and target audience.

Assigning a role and a person (*role-playing*). One of the most powerful techniques is to instruct the model to assume a certain role or persona (lat. *persona*). In this way, those language patterns, vocabulary and stylistic registers that were associated with that role in the learning data are activated in his vast neural network. Instruction “Act as a leading cyber security expert with 20 years of experience. Explain to the layperson the five most common threats to digital security in the home.” will yield an answer that is at the same time professional, authoritative and understandable. This technique is of crucial importance for giving the employee a stable and consistent personality, which makes him a predictable and reliable communication participant.

Providing context and examples (*few-shot prompting*). As first demonstrated by the creators of the GPT-3 model (Brown et al., 2020), large language models possess an extraordinary ability to learn from the examples presented to them within the instruction itself. Instead of merely describing the task, providing several concrete examples (*shots*) allows the model to abstract the required pattern and apply it to a new case. This process can be graphically illustrated in the task of classifying feelings (sentiment) in the text.

Scheme 1: Sample prompt with several examples (*Few-shot Prompting*)

Classify the feeling in the following sentence as ‘Positive’, ‘Negative’ or ‘Neutral’.

Sentence: “The service was outstanding and the food was delicious.” **Feeling:** Positive

Sentence: “We waited too long, and the bill was wrong.” **Feeling:** Negative

Sentence: “The meeting is scheduled for Wednesday at 10 am.” **Feel:** Neutral

Sentence: “The new film completely disappointed me.” **Feeling:**

After being presented with the first three examples, the model concludes with far greater confidence that the correct answer for the last sentence is “Negative”.

Learning from examples / few-shot prompting: A language model management technique in which several concrete examples (shots) of the desired input-output pattern are specified within the instruction itself, thereby prompting the model to abstract the required pattern and apply it to a new, unknown case; the opposite is the zero-shot approach, where no examples are given (Brown et al., 2020).

Chain-of-Thought* (CoT). For complex tasks that require logical or arithmetic reasoning, large language models perform significantly better if they are explicitly encouraged to break down the problem into component steps. By adding a simple phrase like “Let’s think one step at a time.” at the end of the instruction, the model is forced to make its inference process visible, printing it out as text. This follows a more logical path, reduces the probability of error and enables the correctness of the procedure to be checked (Wei et al., 2022).

More advanced techniques, such as ReAct (Yao et al., 2023), combine such incremental thinking with the use of external tools, which forms the backbone of many contemporary worker architectures.

Chain of Thought / Chain-of-Thought, CoT: A language model management technique that explicitly encourages the model to break down the problem into visible intermediate steps of reasoning before the final answer; this increases the accuracy of complex logical and arithmetical tasks and enables checking the correctness of the procedure (Wei et al., 2022).

Scheme 2: Comparison of Standard Instruction and Chain of Thought

Instruction INPUT (Standard Instruction): Question: There were 23 apples in the cafeteria. If they used 20 for lunch and then bought 6 more, how many apples do they have now?

OUTPUT (Probably wrong): 29

INPUT (Chain of Thoughts Prompt): Question: There were 23 apples in the cafeteria. If they used 20 for lunch and then bought 6 more, how many apples do they have now? Let's think step by step.

OUTPUT (Probably Correct): Sure, let's think step by step.

The initial number of apples is 23.

They used 20 apples, so we subtract: $23 - 20 = 3$.

There are 3 apples left after lunch.

Then they bought 6 more apples, so we do the addition: $3 + 6 = 9$. The cafeteria now has 9 apples.

System prompt occupies a special place in model management. It is a superordinate instruction that is set at the very beginning of the interaction and that determines the general behavior, personality, powers and limitations of the model during the entire conversation. It acts as a kind of “constitution” for artificial intelligence, a set of immutable rules that the worker must adhere to. For example, a system instruction for a specialized computer code analysis assistant could read: “You are a security analyst. Your sole purpose is to analyze the provided program code for possible security vulnerabilities according to the OWASP Top 10 list. Do not answer questions that are not related to the security of the code. Do not express personal opinions. Always

format your answers as a JSON object with the fields ‘vulnerability_type’, ‘line_number’ and ‘recommendation’.” A well-formed systematic instruction is essential to create a stable and reliable worker that behaves within the desired and strictly defined limits (Anthropic, 2024; OpenAI, 2024b). It is the foundation of certainty and predictability, turning a powerful but fickle linguistic core into a specialized tool with a clearly defined purpose.

System prompt / system prompt: A superordinate instruction placed at the beginning of the interaction with the language model that determines its general behavior, personality, powers and limitations throughout the conversation; it acts as an agent’s “constitution,” a set of immutable rules that ensure predictability, certainty, and consistency in responses (Anthropic, 2024; OpenAI, 2024b).

The art of formulating queries requires the user or developer to have a deep understanding of the problem domain and how LLM “thinks”. It is necessary to anticipate possible interpretations, remove ambiguities and provide the model with enough context to make an informed decision. In this sense, the interaction with the agent is an iterative process of coordination and refinement, in which the optimal solution is gradually crystallized through a series of questions and answers.

From tool to cognitive partner

Through this synergy of inherent model capabilities and human management skills, LLM within the agent ceases to be a passive tool and transforms into an active cognitive partner. While traditional software tools perform precisely defined functions, the LLM-driven agent participates in the very process of defining the solution. He does not wait for a complete specification of each step, but independently proposes, analyzes and selects courses of action based on the general goal. The effectiveness of the autonomous agent, therefore, rests on two pillars: on the one hand, on the strength and sophistication of the language model that forms its core, and on the other hand, on the skill with which a person directs that strength. Understanding this dual nature – a powerful machine and the need for its fine tuning – is the key to unlocking the

full potential of autonomous systems and their successful application in solving the challenges of the modern world. Mastering this “brain of the operation” is therefore a strategic imperative.

6.3.2 Structure of memory in conversational systems

The ability to remember is a fundamental determinant of the effectiveness and naturalness of conversational systems. Memory in this context goes beyond data storage; it is a complex mechanism that allows the agent to follow the flow of the conversation, understand the context and provide coherent and relevant answers. Without effective memory, each interaction with the user would be reduced to a series of unrelated questions and answers, thus losing the very core of the conversation. Therefore, the structure of memory in modern conversational agents is usually divided into several levels.

Flow of information from sensory, through short-term, to long-term memory

In practice, the memory of conversational systems can be seen as a flow through three levels. Sensory memory serves as a short-term input filter that captures immediate stimuli and signals from the user’s queries. Short-term (working) memory maintains the current context of conversation within a session, while long-term memory stores more stable facts, preferences, and patterns necessary for continuity across multiple interactions.

Short-term memory - context

Relevant information is transferred to short-term memory, which is often called *working memory* in the literature. It stores all the contextual data necessary to maintain coherence within a conversational session. This includes the user’s last statements, the agent’s responses, recognized intentions and entities, and all other elements that enable meaningful continuation of the dialogue. In terms of its function and limitations, it is comparable to the working memory (RAM) in a computer: it has a limited capacity and its contents are deleted after the end of the interaction or after a certain period of inactivity.

Working memory: Short-term storage area of the conversational agent in which the contextual data of the current session is stored - user's statements, agent's responses, recognized intentions and entities; it is functionally comparable to the working memory (RAM) of a computer because it has a limited capacity and is deleted when the interaction is over.

The immediate continuity of short-term memory within one conversational session with language models is achieved by the context window mechanism. Its implementation in practice usually boils down to the systematic inclusion of the entire history of the ongoing conversation in each new call to the model.

Contextual window / context window: The maximum amount of text (in tokens) that the language model can simultaneously receive and process in one call; everything inside the window constitutes the context for generating the response; overshooting requires discarding older parts of the conversation. With each new query from the user, the system not only sends that query to the model, but also attaches the entire transcript of the previous interaction. This places the model in a wider context, allowing it to refer to previously stated claims, raise coherent sub-questions and maintain consistency in tone and content.

However, this approach encounters a hard limit set by the architecture of the model itself – the size of its contextual window, i.e. the maximum amount of text (measured in tokens) that the model can simultaneously process. Once the conversation length exceeds that threshold, the oldest parts of the interaction must be discarded. That moment represents a break point, a kind of conversational amnesia, after which the model loses memory of the beginning of the conversation, which can lead to repetition or contradictory answers. The effectiveness of this mechanism therefore declines as the conversation lengthens, making it unsuitable for all but short-term, one-off interactions.

Scheme: Mechanism and limitation of short-term memory

STEP 1:

User: “Good afternoon, I am interested in the history of Dubrovnik literature.”

Agent → LLM Prompt: “User: Good afternoon, I am interested in the history of Dubrovnik literature.”

STEP 2:

The agent’s answer: “Of course. Are you referring to the period of humanism and the Renaissance, or are you interested in a later period?”

User: “Let’s focus on the renaissance.”

Agent → LLM Prompt: “User: Good afternoon, I am interested in the history of Dubrovnik literature. / Agent: Of course. Do you mean the period of humanism and the Renaissance or are you interested in the later period? / User: Let’s focus on the Renaissance.”

STEP N (Conversation continues...):

...the context accumulates...

STEP N+1 (Contextual window limit reached):

[The oldest part of the conversation (“Good day...” is discarded]

Agent → LLM Prompt: “...[central part of conversation]... / User: And who was the most important comedian?”

| LIMITATION: The model no longer “knows” the initial topic of conversation.
|

Long-term memory – vector bases

For true persistence and retention of information over longer periods of time and across multiple chat sessions, agents must rely on external data storage systems. In modern architectures, this role is most often taken over by vector databases (*vector databases*). They enable the system to create and maintain a permanent record of key information, preferences and facts related to the user, thus forming the basis for his diachronic identity within the system. The process of establishing and using long-term memory can be broken down into three key steps.

Vector database / vector database: A specialized storage system that presents data as numerical vectors (embeddings) in a multidimensional space and enables quick search by semantic similarity, not by keywords; in the architecture of conversational agents, it serves as long-term memory that preserves user preferences, facts, and summaries of previous interactions across multiple sessions (Xu et al., 2023).

1. Summarizing and extracting facts: The system does not store the raw transcript of each conversation, as this would be unreadable and computationally demanding. Instead, the agent periodically, either after each interaction or at the end of a session, analyzes the conversation with the goal of extracting and summarizing key information. These can be explicitly stated facts (e.g. “The user’s name is Ivan”, “Ivan’s son is studying medicine in Zagreb”), but also implicitly expressed preferences (e.g. “The user prefers concise answers”, “The user is interested in ancient philosophy, especially Stoicism”).

2. Vectorization and storage: Extracted facts and summaries are converted from textual form into numerical representations, so-called vector embeddings. Each embedding is a vector, that is, a series of numbers that represents the position of the given information in a multidimensional semantic space. This vector can be understood as a unique “semantic imprint” of the information. The data converted in this way is stored in the vector database, where each vector remains permanently recorded and linked to the user’s identity.

Vector embedding / embedding: Numerical representation of a text, word, sentence or document in the form of a sequence of numbers (vector) located in a multidimensional semantic space; semantically similar content has close vectors, allowing search by meaning instead of exact word match.

3. Context retrieval and enrichment: Before processing a new user query, the agent first searches the vector database. The search is not performed by keywords, but by semantic similarity. The agent converts the user’s current query into a vector and then searches the database for stored vectors (memories) that are closest to it in the semantic space. For example, if the user

asks “Can you recommend me a new book following the ones we talked about earlier?”, the agent will recognize the semantic closeness of that query with the stored fact “The user is interested in ancient philosophy, especially Stoicism”. The most relevant records found are retrieved from the database and inserted into the context window along with the new query. In this way, the language model obtains key information from “long-term memory” that it needs to generate a relevant and personalized response (Xu et al., 2023).

This complex mechanism allows the agent to transcend its innate ahistorical nature. Having gone beyond the status of a linguistic automaton that reacts to the current stimulus, it becomes an interlocutor that “remembers” the shared history of interactions, respects the user’s specificities and thereby creates a basis for a continuous, consistent and ultimately more meaningful relationship.

6.3.3 Epistemic grounding: retrieval-enriched generation (RAG)

The Retrieval-Augmented Generation (RAG) approach represents an important paradigm in the field of artificial intelligence, designed for the epistemic grounding of responses generated by large language models (LLM). It is a hybrid system that combines two basic components: a mechanism for retrieving data (*retriever*) and a generative language model (*generator*). The goal of such systems is to overcome the limitations of independent language models by combining their ability to shape language with the reliability and up-to-dateness of external knowledge bases.

RAG: Retrieval-Augmented Generation – a hybrid system that first retrieves relevant parts from an external knowledge base (retriever) to answer a user’s query, then passes that data and the query to a language model (generator) that synthesizes the final answer; this improves the timeliness and validity of the answers.

The need for this approach stems from two fundamental shortcomings of independent large language models. The first disadvantage is the tendency to “hallucinate”, that is, to generate factually incorrect or completely fabricated information, which happens when the model does not have the required information in its internal parameters. Another key disadvantage is the static

nature of their knowledge base, which is limited to the data on which they have been trained and therefore does not include events or knowledge generated after their training is complete. The RAG system operation process itself can be broken down into several key steps. The cycle starts with a user query. This query is then forwarded to the retrieval component (*retriever*), whose task is to search the external knowledge base - for example, a collection of documents, a database or Internet sources - in order to find the most relevant parts of the text or data related to the set query. This found, relevant data together with the original user query forms an enriched context that is passed to the generative language model. In the final stage, the language model (generator) synthesizes the final answer, basing it on the obtained relevant data, and not exclusively on its internal, parametric knowledge.

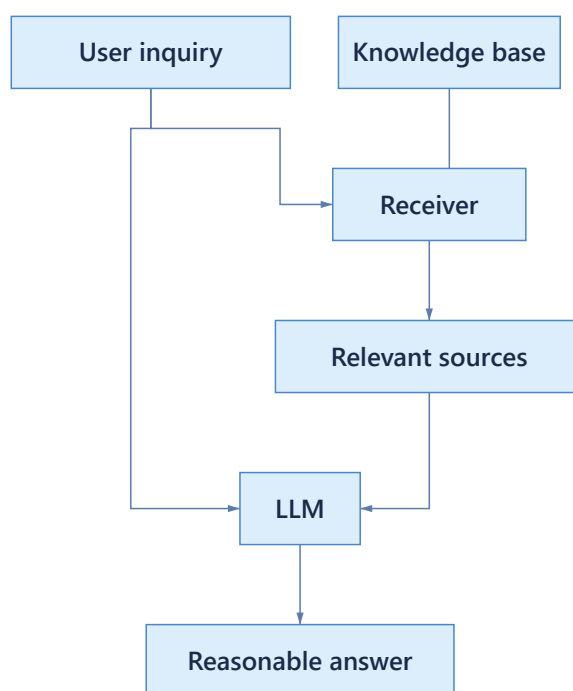


Figure 6.5: Schematic representation of the operation of the RAG system.

The application of RAG architecture brings multiple advantages. The most important thing is the significant reduction in the occurrence of hallucinations, because the model directly relies on concrete, retrieved data when generating answers. Furthermore, RAG allows models to access updated or domain-

specific, adapted information without the need for costly and time-consuming retraining of the entire model. This opens up the possibility of transparency and verifiability; along with the generated answer, the user can be offered the sources from which the information was taken, which allows him to check their accuracy himself. This approach is also more economical because updating the external knowledge base is much simpler and cheaper than retraining the entire language model (Lewis et al., 2020). In the context of more complex systems, such as autonomous agents, RAG assumes the role of a key tool for informed action. When an agent is faced with a task that requires reliable and up-to-date information, it does not act hastily, but relies on the RAG mechanism to gather relevant knowledge before making a decision or formulating a response. This ensures that the agent's actions – whether it's responding to a customer's help desk inquiry or collecting data for more complex analysis – are not based on potentially outdated or inaccurate internal knowledge, but on verifiable external sources. It should be emphasized that although the core concept of the RAG architecture is extremely powerful, its naive or simplistic application often proves insufficient for more complex problems. Therefore, a whole spectrum of advanced techniques has been developed that optimize each stage of the described process. Today, this area represents one of the most dynamic fields of research in the development of artificial intelligence (Gao et al., 2024; Barnett et al., 2024). Advanced methods of indexing and searching, techniques of intelligent classification (reranking) of retrieved documents by importance, as well as strategies for summarizing and synthesizing information from several different sources are being researched.

6.3.4 Hands for action: using tools as a leap in power

Tool Use: The AI agent's ability to call external tools (APIs, search engines, calculators, databases) during task execution; the model decides when and which tool to call, passes parameters, and uses the result to continue reasoning; a key component of agent architectures (e.g. ReAct).

The human hand, viewed in its anatomical and functional complexity, goes beyond the role of a grasping organ and asserts itself as primarily a tool of action. Her ability for fine motor adaptation, precise manipulation and firm grip is a fundamental prerequisite for one of the most important transitions in human development: the conscious use of tools. With this act, man becomes a being who actively shapes, adapts and transcends the world. Serving as a tool therefore represents a radical change in the relationship between man and the world, a true step forward in power that opens the way to the construction of culture and technology. This transformation deeply affects the very structure of human perception. Philosophical phenomenology has offered insightful insights into the nature of that process. Martin Heidegger introduces a distinction between tools as something *ready-to-hand* (German *Zuhandenheit*) and objects as something simply *present* (German *Vorhandenheit*). When we skillfully use a hammer, it does not exist for our consciousness as a separate object with certain properties, but recedes into the background and becomes an imperceptible extension of our hand focused on the act of driving a nail. The tool, fused with action, becomes transparent and opens the world to us not as a collection of objects to observe, but as a field of possibilities for action. On this track, Maurice Merleau-Ponty (1962) further develops the thought of the embodiment of the tool, arguing that the body includes it in its own body scheme. The classic example of the blind man's cane clearly shows this phenomenon: for the blind man, the cane is an extension of his sensory apparatus with which he *feels* and explores the space in front of him. The tip of the stick becomes the new boundary of his body. In this way, the tool is transformed from an external object into an integral part of a person's perceptive and active field.

Through this process of fusion of body and tool, man is transformed into a significantly more powerful *agent*. His relationship with the world is no longer determined solely by the biological limits of his body. The ax extends the strength and reach of the hand, the spear enables hunting at a distance, and the letter transcends the transience of speech and memory. Each new tool opens up new ways of being in the world and acting on it. This fundamental ability to

use tools is therefore found at the very origin of human history as a history of creation, transformation and overcoming the given, confirming man as a being who is constantly rebuilding his world.

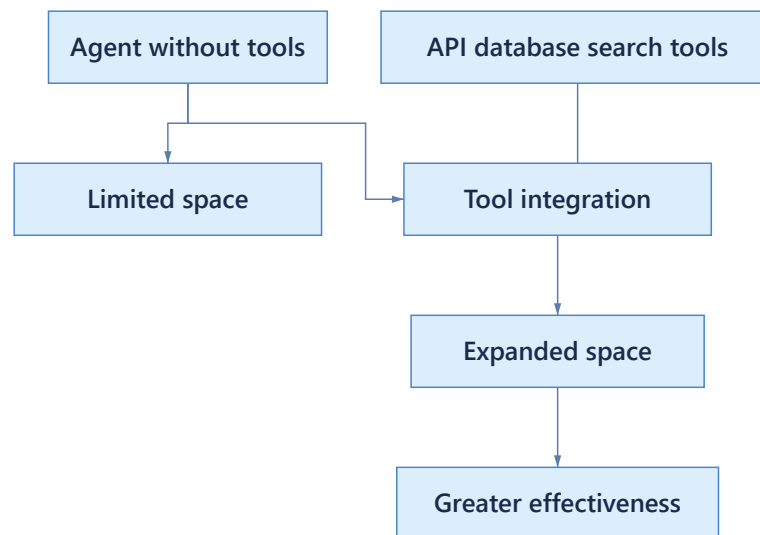


Figure 6.6: Expanding action by adopting tools: by integrating tools, the agent expands the action space and increases effectiveness.

For language models, it is the use of tools that breaks these constraints, allowing the language model to reach outside itself and engage with living, changing systems. The foundation of this ability rests on a circular process of thinking and acting, most clearly exemplified in frameworks such as ReAct (*Reasoning and Acting*). In this cycle, the language model forms a sequence of internal reflections (*thoughts*) and external interventions (*actions*) that gradually lead it to a solution (Yao et al., 2023). This process takes place in a loop that can be repeated if necessary:

Fundamental cycle of reflection and action (ReAct)

USER ACCOUNT: “What will the weather be like tomorrow in London?”

1. THOUGHT (Reflection): “A user asks for the weather in London. I don’t have access to real-time data. I need to use the ‘weather_forecast_api’ tool.”

2. ACTION: Creating calls to tools: { "tool_name": "weather_forecast_api", "tool_input": { "city": "London", "date": "tomorrow" } }

3. OBSERVATION (Feedback): The external system executes the call and returns the result: { "temperature": "15°C", "state": "Cloudy" }

4. THOUGHT (Synthesis): “I have received an observation. The temperature is 15°C and the weather is cloudy. I now have all the ingredients for a complete answer.”

5. FINAL ANSWER: “Tomorrow in London it will be cloudy with a temperature of 15°C.”

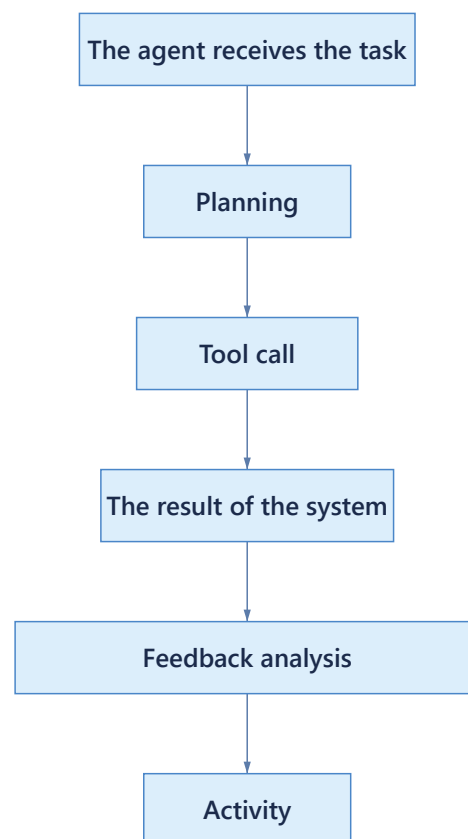


Figure 6.7: Feedback from the external system in the agent cycle (Tool Use - observe).

This circle allows the agent to solve various tasks, from everyday queries to highly specialized domains. In the role of a personal assistant, the agent can manage the calendar, send e-mail or monitor smart devices in the home. As a travel planner, he can search for flights via airline APIs, compare prices, check hotel availability and finally make a reservation. In the field of expert analysis, it is capable of executing Python scripts for complex data processing, sending SQL queries to large databases or summarizing the content of web pages accessed in real time. The ultimate manifestations of this ability are seen in action on the physical world, where the agent's instructions, translated into machine code, control laboratory equipment, industrial robots on the production line, or agricultural drones.

6.3.5 Active agent: harmony of abilities

The term active agent (*agent*) in its essence includes a complex entity whose action cannot be reduced to the mere sum of its components. It is about a subject who notices his own environment, acts on it purposefully and learns from the consequences of these actions, adjusting his future actions. The fundamental principle that enables such action is the harmony of abilities - the dynamic and pervasive harmony of several key functions that together form a unique system. Without such alignment, the active agent remains just a collection of disconnected mechanisms, incapable of meaningful and adaptive interaction with the world. The first basic ability is perception. It represents a window into the world, a channel through which the agent collects information about the state of the environment and about its own position within it. Perception is an active process of selecting, interpreting and organizing sensory information in order to create a coherent internal representation of external reality (Gibson, 1979). The raw data obtained by the senses is only a starting point that must be interpreted in the light of previous knowledge and current goals. It is precisely this interpretative nature of perception that makes it inseparable from cognitive processes.

Perception is followed by cognition, which includes the mental processes of reasoning, planning, decision-making and memory. Within this framework, received information is processed, linked to existing knowledge and used to

shape intentions and action strategies. The ability to create and evaluate different possible actions is essential for intelligent behavior (Russell & Norvig, 2020). Cognition thus transforms perceptual data into purposeful plans, forming a bridge between what is and what should be done. It is the central mechanism that allows the agent to go beyond immediate reactions to stimuli and to act thoughtfully, with an eye on the future.

Plans shaped by cognition are realized through action, i.e. through the agent's ability to influence the environment and change it. Action can take different forms, from simple motor actions to complex communicative acts. It closes the circle of interaction with the world: every action causes changes in the environment, and these changes become a new object of perception. This creates an unbroken feedback loop, where action not only achieves goals, but also creates new information that enriches and changes the internal model of the world (Powers, 1973).

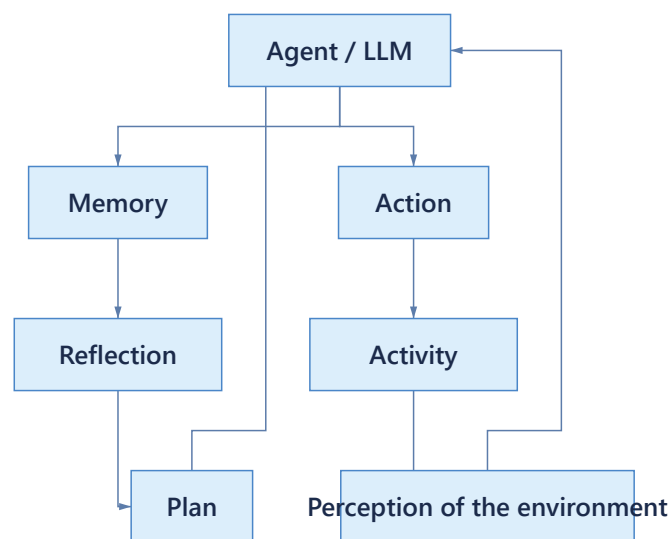


Figure 6.8: Agent-environment interaction loop: perception leads to cognition and action, and action creates new information for the next perception.

After all, the entire system is imbued with the ability to learn, which enables continuous adaptation and improvement. Learning is the process by which the agent changes its knowledge, skills and strategies based on experience, that is, based on the outcome of its own actions. Without learning, the agent would be

condemned to repeat the same mistakes, unable to adapt to new and unforeseen circumstances. The ability to learn allows it to become more resilient, more efficient and, over time, increasingly skilled in achieving its goals (Sutton & Barto, 2018).

The true nature of an active agent, however, is not manifested in individual abilities, no matter how developed they are. Its essence lies in their harmony and interdependence. Perception is aimless without knowledge to give it meaning; cognition is powerless without action that would put its intentions into effect; action is blind without a reason to direct it; and the whole assembly is static and rigid without learning that gives it adaptability and a developmental dimension. It is precisely in this constant and coordinated interweaving of perception, cognition, action and learning that an active agent capable of meaningful existence in a complex and changing world is born. Frameworks like *LangChain* or models with strong reasoning capabilities, like OpenAI's o3, Anthropic's Claude or Google's Gemini 2.5, offer a technical foundation and structure for managing such complex interplays, along with access to tools. To make this harmony of abilities more evident, let's consider a complex example. Let's imagine that the user, let's call her Ana, sets the following task for her personal assistant: to plan an independent trip to Paris during the first weekend in October. Her preferences are art, especially impressionism, but her budget is strictly limited to approximately 500 euros for airfare and accommodation. It is also a requirement that the accommodation be in the center of the city. Let's see how an advanced agent would align its capabilities to fulfill such an order.

1. Degree: Thinking and imagining (Mind) The language core of the agent first subjects Ana's query to an in-depth analysis. Using her Chain-of-Thought-like thinking ability, she breaks down a complex goal into a series of smaller, achievable and actionable sub-tasks. That initial internal monologue is the foundation of the entire operation.

Scheme 1: Target decomposition Main goal: to plan a weekend trip to Paris (first weekend in October, solo, art/impressionism, budget ~€500, central accommodation).

Sequence of thought (internal decomposition): Determine the correct dates: Determine which days are the “first weekend in October”.

Check logistics (flight): Search and find the best return flights for the given dates.

Check logistics (accommodation): Find affordable and centrally located accommodation for the same dates.

Comply with the budget: Check whether the sum of flight and accommodation costs does not exceed the default limit of 500 euros.

Explore interests: Explore museums and landmarks related to Impressionism.

Put together a proposal: Create a clear and coherent itinerary.

Synthesize the answer: Form all the collected data into a final, personalized answer for the user.

2. Degree: Action and use of tools (Hands) The agent immediately realizes that he lacks external, changing data in real time to fulfill the task. Therefore, he has to reach for his tools, that is, for application programming interfaces (APIs) that enable him to access external databases.

Scheme 2: Sequence of tool use Thought sequence: “I’m missing information on flight and hotel prices. I need to call the appropriate tools.”

Action 1: Flight search tool call Command: `flight_search_api.call()`

Parameters: { destination: “Paris, France”, departure_date: “2025-10-03”, return_date: “2025-10-05”, class: “economy” }

Output (observation 1): The tool returns a list of available flights. The cheapest option is €150 (return ticket).

Action 2: Call the accommodation search tool Order: `hotel_booking_api.call()`

Parameters: { city: “Paris”, check-in_date: “2025-10-03”, check-out_date: “2025-10-05”, maximum_price: 350, location: “central”, accommodation_type: “hotel/apartment” }

Output (observation 2): The tool returns a list of available accommodation units. A few meet the criteria.

3. Degree: Reaching for knowledge and memory (Experience) Once the logistical framework of the trip is set and confirmed within the budget, the agent turns to the more refined part of the task: adapting to Ana's personal preferences. Here comes to the fore the combination of two forms of knowledge - that stored in the long-term memory of the users and that obtained from external sources through foundation.

Scheme 3: Linking memory and grounded knowledge (RAG) Thought sequence: "Logistics taken care of. Now I need to focus on impressionism and personalize the plan."

Action 3: Query long-term memory Command: `memory_storage.query()`

Parameters: { user_id: "Ana", keywords: ["art", "Paris", "impressionism"] }

Outcome (Observation 3): Memory returns a record from the previous interaction: {"record": "User Ana rated the visit to the Musée d'Orsay very positively.", "date": "2024-03-12"}.

Action 4: Query the Establishment System (RAG) Command: `web_search_tool.call()`

Parameters: { query: "impressionist art museums in Paris other than Musée d'Orsay" }

Outcome (Observation 4): The system searches verified sources and retrieves summary information about the *Marmottan Monet* (largest collection of Monet works) and *Orangerie* (famous for Water Lilies) museums, including opening hours, ticket prices, and location.

4. Degree: Synthesis and Communication (Voice) With all the data collected, processed and contextualized, the language core now takes on the role of ultimate storyteller. It no longer decomposes, but combines - flight, accommodation, budget, personal preferences and newly explored possibilities - into a unique, coherent and attractive proposition.

Scheme 4: Final synthesis and shaping of the answer

Thought: "I found a flight and accommodation within the given budget. I know from memory that the user appreciates the Musée d'Orsay, and through research I found two other related museums. I can put together a complete proposal that ties all the elements together."

Final answer (generated for Ana): “Dear Ana, I have prepared a proposal for your weekend in Paris, guided by your wishes and budget. **Flight and accommodation:** I found a return flight for €150. In addition, I suggest staying in a charming studio in the Le Marais district, available through the Airbnb platform, at a price of €320 for two nights. The accommodation is excellently rated and is located in the very heart of the city. The total cost is €470, which is within your budget. **Artistic itinerary:** Since I know your penchant for impressionists, you can dedicate Saturday to revisiting your favorite museum, the *Musée d’Orsay*. For Sunday, I suggest a visit to the museum *Musée Marmottan Monet*, which houses the world’s largest collection of works by Claude Monet, including the famous painting *Impression, Rising Sun*. Would you like me to make a flight and accommodation reservation for you?”

The presented sequence of actions reveals a problem-solving system that demonstrates the ability to align its resources and skills to achieve a given goal in a manner that is simultaneously efficient, adaptive, and reliable.

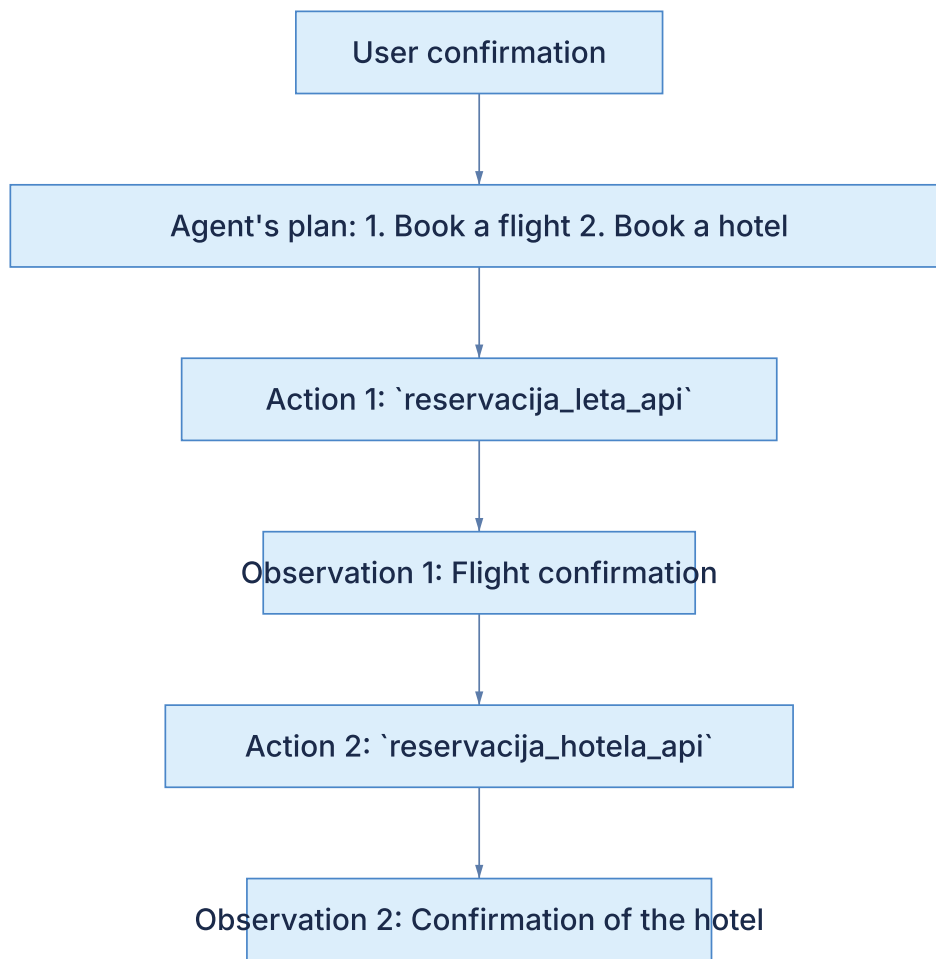


Figure 6.9: Sequence of actions of an agent when planning a personalized trip.

6.3.6 Conclusion: the birth of a communication partner

The discussion so far has followed a complex path that leads from a set of program commands and data structures to the emergence of an entity that we can rightly call a communication partner. This transition, described as *birth*, metaphorically describes an actual qualitative leap in the nature of human-agent interaction. The agent assumes the role of an active participant in the act of communication. Its birth is the result of gradual maturation through a series of interactions, learning and adaptations. The fundamental components of this process are the agent's ability to interpret the context, recognize the user's intentions and adjust its responses. At the moment when the user begins to

recognize functionality in the agent, but also a kind of stability, predictability and, ultimately, reliability, the foundations for a partnership are laid. This relationship is based on the functional equivalence of communication roles. An agent becomes a partner not because he *is* human, but because he *acts* as a credible interlocutor within the given communication framework. This opens up space for more complex forms of cooperation, where trust and mutual understanding are built through continuous dialogue. It is this transformation from a tool to an interlocutor that is the key achievement that we tried to shed light on, and its path can be summarized in the following presentation. —

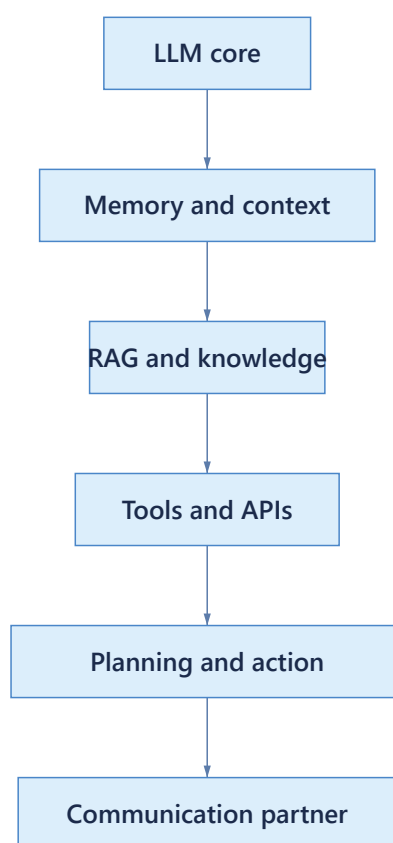


Figure 6.10: Stages of development of a communication partner from the linguistic core to an autonomous, integrated interlocutor.

This concludes the theoretical consideration of the conditions under which a communication partner is created. With the theoretical framework clearly established, we turn to its application and implementation in real systems.

7. Building communication partners: examples

After in the previous chapter we considered the theoretical foundations and determined the key components that enable the “birth” of a communication partner, we now move from abstract principles to their concrete application. This chapter is devoted to the practical dimension of building such agents, where theoretical models meet the challenges of real implementation. We will observe how theoretical assumptions about context, learning and adaptation translate into the architecture of real systems, shaping their behavior and interaction capabilities. The purpose of this chapter is twofold. First, the goal is to show the diversity of approaches in building communication partners through carefully selected examples from different domains - from customer support and education to personal assistants. Second, through in-depth case studies, we will explore specific problems, successful solutions, and trade-offs faced by engineers and designers. By analyzing these examples, we will discover how the general principles are manifested in practice and what unforeseen difficulties and possibilities emerge during development and use. Thus, theoretical understanding is enriched with insights gained from experience, offering a more complete picture of what is needed to successfully build and maintain functional and reliable communication partners.

7.1 Digital interlocutors: chatbots and virtual assistants in the age of LLMs

Digital interlocutors, embodied in chatbots and virtual assistants, represent one of the most prominent human-computer interaction interfaces in the modern technological landscape. Their rapid development and ubiquitous application are inextricably linked to progress in the field of large language

models, which fundamentally transformed their capabilities and roles. While chatbots and virtual assistants have traditionally been seen as two relatively separate categories, recent technological advances are erasing those boundaries and introducing a new paradigm of intelligent, autonomous systems.

Definition of concepts: from scripted answers to complex tasks

Historically, the fundamental difference between a chatbot and a virtual assistant rested on the degree of complexity and breadth of functionality. *Chatbots* were originally conceived as simpler programs, often based on predefined rules or scripts. Their primary purpose was to conduct highly specialized conversations within a specific domain, such as providing customer support for a specific product, answering frequently asked questions, or collecting basic data from users.

Chatbot: A program for conducting conversations with the user, originally based on rules or scripts; modern chatbots often use LLM and can conduct complex, context-aware dialogues. **Virtual Assistant:** A more complex digital interlocutor deeply integrated with the device and external services, capable of a wider range of tasks (calendar management, smart home, search, messages); the line towards chatbot is blurring with the advent of LLMs.

On the other hand, virtual assistants, such as Amazon's *Alexa*, Apple's *Siri* or Google's *Assistant*, were designed from the beginning as much more complex systems. Their fundamental feature is deep integration with the operating system of the device and a wide ecosystem of external services and applications. Such integration allows them to perform diverse and complex tasks that go beyond mere conversation - from managing a smart home and setting reminders to sending messages, searching the Internet and interacting with other software tools. However, the emergence and commercialization of large language models initiated a radical transformation. The ability of LLMs to understand context, generate coherent and creative text, and reason about complex problems has enabled the new generation of *chatbots* to acquire, and even surpass, the capabilities of traditional virtual assistants. Modern *chatbots*,

powered by technologies such as those developed by OpenAI (2023), are no longer limited to rigid scripts. They can have fluid, context-aware conversations, condense documents, write programming code, and perform tasks that require multi-level planning and reasoning. This blurs the once clear dividing line between the two concepts, and a new, more comprehensive concept - the concept of an agent - enters the spotlight.

The rise of autonomous agents

The term agent in the context of artificial intelligence means a system that can independently perceive its environment and act autonomously to achieve set goals. The classic definition describes an agent as an entity that perceives the environment through sensors and influences it through actuators (Russell & Norvig, 2021). In the digital world, “sensors” can be API calls that retrieve information from the Internet or read files from disk, while “actuators” can be tools to send email, execute program code, or interact with other software interfaces.

Large language models act as the central “brain” or cognitive core of such agents. They give them the ability to reason, plan and break down complex goals into a series of smaller, actionable steps. When LLM is granted access to external tools, it outgrows the role of a passive text generator and becomes a proactive task executor. Such a system goes beyond the traditional concept of a chatbot and takes on the characteristics of an autonomous agent. Examples such as the Auto-GPT or AgentGPT projects illustrate this shift: the user sets a general and complex goal (e.g. “research the electric car market and compile a report on key competitors”), and the agent independently forms a plan, searches the Internet, analyzes the collected data, uses tools to process it, and finally generates the required report, while learning and adapting its actions without constant human intervention. This transition from reactive interlocutors to proactive agents represents a fundamental paradigm shift, opening the door to applications that until recently belonged to the domain of science fiction and posing new challenges in the field of security, ethics and management of autonomous systems.

7.2 Perspective-changing applications: where do agents create value?

The theoretical potential of autonomous agents, although extremely stimulating, gains its full meaning only with an insight into real applications where their capabilities bring measurable and transformative value. Considering agents solely as advanced tools for executing individual commands is insufficient; their true strength is manifested in the ability to independently orchestrate complex workflows, which opens up new horizons in problem solving and value creation. In the following, the key areas in which agents are already showing the ability to fundamentally change established practices are explained in more detail.

Automation of complex research and analytical processes

One of the most fruitful areas of application of agents is the automation of time-consuming and complex research tasks. Let's imagine an agent entrusted with the task of creating a detailed report on a specific topic, for example the impact of climate change on agriculture in the Mediterranean. Such an agent does not perform its task with a single query, but starts a complete process. First, it independently formulates a series of relevant queries to search scientific databases, newspaper archives and official reports. Then, it systematically collects, filters and evaluates the credibility of the found sources. The key step that follows is synthesis – the agent analyzes the collected information, recognizes patterns, contradictions and key arguments and structures them into a coherent narrative. The final product is a complete report that includes sets of links or summaries complete with neat citations and references according to relevant academic standards. In this way, tasks that would take a human researcher days or weeks of work are completed in a much shorter time, with a high level of precision.

Development of program code and software solutions

In the domain of software engineering, the role of agents is also slowly going beyond the function of mere code assistants. Autonomous agents are capable of taking over entire development cycles. Starting from the analysis of the user's requirements, specified in natural language, the agent can create an architectural design of the application, select suitable technological frameworks, and then write, test and correct (*debug*) the necessary program code. Their ability is not limited to generating code snippets; they can independently manage repositories, solve dependencies, write unit tests and even run automated deployment processes on servers. This approach allows development teams to focus on the strategic and creative aspects of development, while agents take on repetitive and technically demanding but well-defined tasks. This speeds up the entire software development process and makes it more resistant to human errors. Examples of such systems are helpers for writing code such as Claude Code, Codex, Cursor, Windsurf and others.

Personalized customer interaction and advanced support

Customer support systems, which have traditionally relied on simple *chatbots* with pre-defined scenarios, are undergoing a profound transformation. Agents bring the ability to conduct complex, multi-step dialogues where they remember the context of the conversation, access the history of user interactions, and dynamically use external tools to solve a specific problem. For example, a user reporting a product delivery problem can interact with an agent who will independently check the order status in the logistics system, communicate with the delivery service via their API and offer the user a specific solution, such as a new delivery date or a refund. Equally, in the role of personal assistants, agents can manage calendars, book trips and meetings, taking into account complex user preferences and schedule overlaps, which represents a significant step forward compared to existing digital assistants.

Strategic management, marketing and business analysis

In a business environment, agents become a powerful decision-making tool. It is possible to assign them the tasks of continuous monitoring of market trends, analysis of competitor activities or monitoring of public sentiment on social networks. An agent in charge of optimizing a marketing campaign can autonomously analyze data on the success of ads on different platforms, allocate budget according to those channels that bring the best return on investment, and even independently generate and test new creative versions of ads. In finance, agents can perform complex analyzes of financial statements, identify risks or anomalies, and prepare summary bases for making strategic decisions. Their value lies in the ability to process and connect vast amounts of unstructured data, identify signals in the noise of information and turn them into concrete, actionable insights, freeing up human experts for tasks of the highest strategic level.

7.2.1 Revolution in customer support and marketing

The emergence and application of artificial intelligence means a thorough transformation of the paradigms on which customer support and marketing rested until now. These two, once separate business areas, under the influence of intelligent systems, converge into a single, data-driven function whose goal is to create a complete and deeply personalized user experience. It is a transition from a reactive model, which responds to already existing needs or problems, to a proactive and predictive approach, which anticipates future conditions and acts in accordance with them. In the domain of customer support, the first and most obvious shift is the automation of the initial points of interaction with users through intelligent agents, i.e. the already mentioned *chatbots* and voice assistants. These systems are trained to immediately and continuously solve a large number of standardized queries, which frees up human resources. Time, as a key resource, is thus diverted from routine tasks to solving complex, multi-layered problems that require human empathy, creativity and critical judgment. However, the reach of artificial intelligence also includes in-depth analysis of a huge amount of unstructured data, such as transcripts of calls, e-mails and messages from social networks.

Sentiment analysis: Automatic analysis of text (transcripts, e-mail, social networks) in order to recognize the emotional tone, attitudes or mood - e.g. frustration, satisfaction, negativity; it is used in customer support to prioritize cases, in marketing to monitor reactions and in public opinion surveys.

By analyzing sentiment from this data, the system can recognize the user's tone, degree of frustration or satisfaction, and based on that determine the priority and urgency of a particular case, providing the human agent with a contextually rich summary before the actual interaction. In parallel with the transformation of customer support, a revolution in marketing strategies is taking place. The era of mass marketing, whose messages were aimed at broad and poorly defined demographic groups, is leaving, and the period of hyperpersonalization is entering. Machine learning algorithms analyze an individual user's entire interaction history—from browsing patterns and previous purchases to time spent on specific products—to dynamically tailor marketing messages, offers, and the overall user experience to each individual. Recommendation systems, such as those used by platforms such as Netflix or Amazon, are the best-known example of such an approach, where based on the behavior of similar users and individual preferences, it is predicted which content or product is most likely to attract attention (Varian, 2018).

The diagram above schematically illustrates how different sources of user data flow into a central artificial intelligence system. This system, through analysis and prediction, simultaneously generates output values that transform both customer support (proactivity, sentiment analysis) and marketing (hyperpersonalization, predictive offers), thus creating a unique and intelligent customer relationship management cycle. Furthermore, artificial intelligence systems in marketing rely on the analysis of past interactions, but also enter the field of predictive analytics. I can predict with a high degree of probability the future behavior of users, such as the possibility of abandoning the service (churn prediction*), identify customers with the highest long-term value or assess the optimal moment to send a promotional offer. At the same time, the development of generative models enables the automation of the creation of marketing content. From writing e-mail headlines that encourage opening, to composing posts for social networks adapted to a specific audience, to

generating basic versions of advertising texts, artificial intelligence becomes a collaborator in the creative process, significantly speeding up the production and testing of marketing materials (Kaplan & Haenlein, 2019). Artificial intelligence is reshaping the very essence of the relationship between businesses and customers: traditional boundaries between departments are collapsing and a single, intelligent ecosystem is being created where every interaction is recorded, analyzed and used to shape the next step in the customer journey. This achieves the transition from a transactional to a relational business model, where the ultimate goal is to build long-term loyalty through a continuous, relevant and deeply personalized experience.

7.2.2 Transformation of education and content personalization

Generative artificial intelligence, especially in the form of autonomous agents, has the potential to make a turning point in the understanding and implementation of educational processes. Its influence goes beyond previous technological innovations that served as aids; it is a radical transformation that moves education from an established, unique model for all students to a completely personalized and dynamic experience. Agents take on the role of personal mentors, adapting to each individual's unique needs, abilities and learning styles. The central change brought about by AI agents is manifested in the possibility of creating individualized educational curricula. The system begins with an in-depth analysis of existing knowledge, the speed of material acquisition and the cognitive preferences of students. Based on this data, the agent forms a customized curriculum that is not static, but constantly changes and develops in parallel with the progress of the user (Jones & Hrabowski, 2022). If a student demonstrates exceptional understanding of a particular area, the system can guide them toward more advanced topics or, conversely, provide additional materials and exercises for areas where difficulties arise. This abandons the outdated approach in which the entire class progresses at the same pace, regardless of individual differences. The scope of AI agents extends to the very creation and refinement of educational materials. They are able to generate summaries of complex scientific papers in real time, create quizzes to test knowledge or offer step-by-step explanations of mathematical

problems, adapted to the level of understanding of the individual user. At the same time, agents act as excellent curators of knowledge, searching the vast amounts of information available on the Internet to single out the most credible, relevant, and high-quality sources for the student. This dual role, that of creator and that of curator, enables the construction of a rich and multi-layered educational environment. Learning becomes an active, engaged activity. AI agents can conduct Socratic dialogues, encouraging critical thinking by asking thoughtful questions rather than simply providing answers. They can create complex simulations in which students manage a virtual laboratory, participate in reenactments of historical events, or solve business case studies in a dynamic environment. Such interactive learning models increase knowledge retention and develop problem-solving and analytical reasoning skills that are crucial in the modern world. This process can be represented as a continuous cycle of interaction between the learner and the AI agent, where each stage builds on the previous one, creating a spiral of knowledge.

The use of intelligent agents also has considerable potential for increasing the availability of education. The systems can provide instant translation of teaching materials into different languages, thus removing language barriers. They can convert text content into speech for visually impaired students or simplify complex sentence structures for people with learning disabilities, such as dyslexia (Rose & Meyer, 2002). In this way, education becomes fairer and more open to all, regardless of their physical, cognitive or geographical circumstances. Despite the huge advantages, the introduction of AI agents in education also brings with it certain challenges. Excessive dependence on technology could reduce the development of social skills and the ability to explore independently. The issue of privacy and security of student data is of crucial importance, as the systems collect detailed information about their progress and behavior. It is necessary to ensure that algorithms that personalize content do not create “information bubbles” or introduce biases that could limit the learner’s exposure to different perspectives. Finally, the role of the human teacher is also changing, becoming more and more a mentor, moderator and emotional support, guiding students through a technology-driven learning process. Its task is to stimulate curiosity and develop those human qualities that a machine cannot replace.

7.2.3 Extension of applicability: LLM agents in healthcare, business processes and creative industries

The reach and applicability of language models in the form of autonomous agents go far beyond their initial roles in text processing and generation. Their ability to understand context, plan, and execute complex tasks opens up opportunities for deep integration into diverse areas of human endeavor, reshaping fundamental paradigms in healthcare, business, and creative industries. In each of these sectors, LLM agents represent not only a technological improvement, but a catalyst for fundamental changes in the way of working, deciding and creating.

Application in healthcare

In the healthcare system, the role of LLM agents goes beyond the function of simple interlocutors or virtual assistants. Their ability to process and analyze vast amounts of unstructured data from electronic health records (EHR) enables them to act as advanced diagnostic collaborators. Based on a synthesis of symptoms, the patient's medical history and the latest scientific knowledge from the medical literature, agents can suggest differential diagnoses and alert doctors to rare diseases or atypical symptom presentations. Their contribution does not stop at diagnosis, but also extends to the creation of personalized treatment plans, taking into account genetic predispositions, lifestyle and specific comorbidities of the patient. However, application in such a sensitive area also opens up complex ethical and technical issues, primarily related to the protection of patient data privacy, in accordance with regulations such as HIPAA (USA) and GDPR (EU), and the imperative of clinical accuracy, where an error can have critical consequences (Chen et al., 2023).

Optimization of business processes

A similar transformative potential is also evident in the domain of business processes. Automation of routine tasks, such as customer support where agents can conduct complex conversations and resolve inquiries 24 hours a day, represents only the initial level of their integration. A deeper application is seen

in the optimization of complex systems such as supply chain management, where agents based on analysis of market trends, logistics data and internal reports can forecast demand, optimize inventory levels and coordinate deliveries autonomously. Furthermore, the ability to automatically generate comprehensive reports from raw business data frees up human resources for strategic decision-making. Therefore, it is not only about mere automation, but about the introduction of a layer of artificial intelligence that enables making more informed and precise business decisions, with a key prerequisite for successful integration with existing ERP and CRM systems (Davenport & Ronanki, 2018).

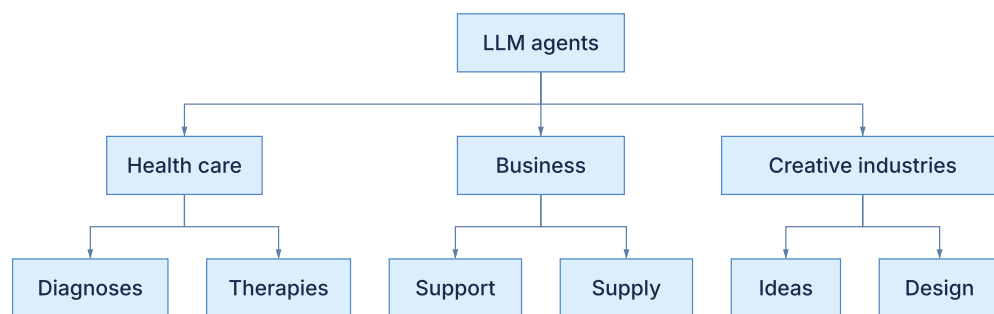


Figure 7.1: Schematic representation of extending the applicability of LLM agents to key industry domains: healthcare, business and creative industries.

Cooperation in creative industries

Finally, the reach of LLM agents extends into the creative industries, an area that has traditionally been considered an exclusively human domain. Here agents assume the role of collaborators or catalysts of the creative process. They are offered to artists, writers, designers and musicians as tools to overcome creative blocks and explore new aesthetic directions. Their application covers a wide range of activities, whether it is generating initial ideas for a script, developing characters, composing musical motifs or creating visual concepts. It is crucial to understand that the goal is not to replace the human creator, but to augment his abilities – providing new starting points and exploring unimagined artistic possibilities. This new paradigm of human-

machine cooperation inevitably raises fundamental questions about authorship, originality and ownership of works created in such synergy, opening a new legal and philosophical debate (Epstein et al., 2023).

7.2.4 Content personalization paradigm: transversal application of AI algorithms

In today's digital environment, the paradigm of personalization of content is established as a fundamental mechanism of interaction between users and information systems that transforms the ways in which information, products and services are distributed and consumed. At the center of this transformation are artificial intelligence algorithms, whose task is the dynamic modeling of user needs, preferences and behavior with the aim of delivering relevant and timely content. The application of these technologies has an extremely transversal character, permeating different sectors and domains of human activity, from e-commerce and media to education and healthcare.

Mechanisms and algorithmic bases

The basic principle of personalization rests on a continuous cycle of data collection, its processing and the application of learned patterns. This process begins with recording user interactions with the system – from explicit actions such as rating a product or selecting a movie genre, to implicit signals such as time spent on the page, search history, mouse movements or geographic location. This data forms what is called a *digital footprint* of the user, on the basis of which AI systems build his dynamic profile, i.e. a *digital twin*. Different classes of algorithms are used to analyze these huge data sets and predict future interests: **Collaborative Filtering**: This approach is based on the idea of the “wisdom of the crowd”. The system recommends content to the user based on the preferences of other, similar users. If user A and user B have shown a similar taste in movies in the past, the system will recommend to user A a movie that user B liked but which A has not yet watched. The strength of this approach lies in the ability to discover unexpected but relevant recommendations (serendipity). **Content-Based Filtering***: Unlike the

collaborative one, this approach is based on the intrinsic features of the contents themselves. If the user often reads articles on quantum physics, the system will recommend new articles containing keywords and concepts related to the field. Attributes such as genre, author, keywords or product technical specifications are analyzed to find those that match the user's existing interest profile. **Hybrid models:** In practice, hybrid systems are most often applied, which try to combine the advantages of both approaches in order to overcome their individual shortcomings, such as the problem of "cold start" (lack of data for new users or new content) which is characteristic of collaborative filtering (Ricci, Rokach, & Shapira, 2011). At the heart of this process is the notion of an agent – an autonomous software entity that acts on behalf of a user or system. The AI agent in charge of personalization observes user behavior, learns from it, and independently makes decisions about what content to display, in what form and at what time, with the aim of maximizing relevance and user engagement.

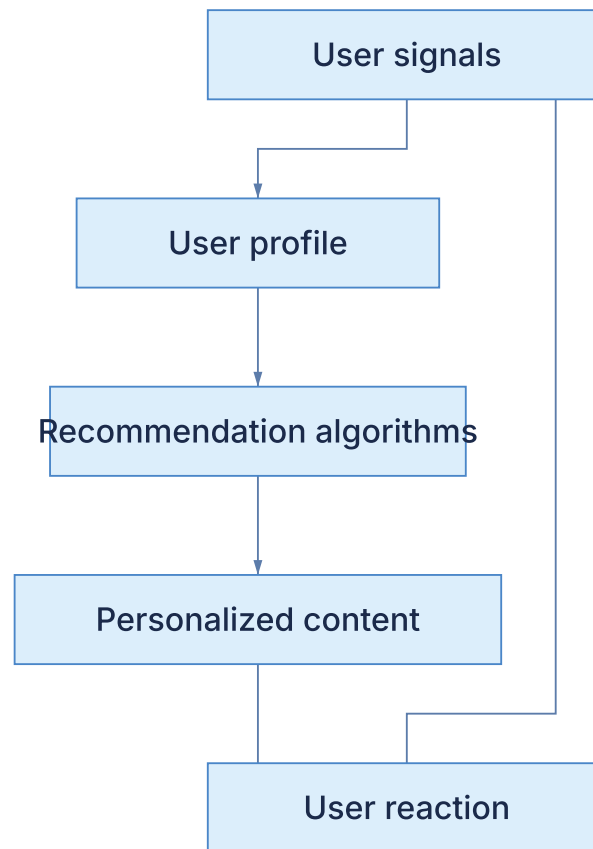


Figure 7.2: Schematic representation of the closed cycle in content personalization systems: from user signals and profiling to recommendation, reaction and new system learning.

Transversality of application

The ability of AI algorithms to learn and adapt makes personalization a universally applicable tool. Its transversality is manifested in numerous domains: **E-commerce and retail:** Platforms like Amazon use personalization to recommend products, dynamically adjust prices and send targeted marketing messages, directly influencing sales and customer loyalty. **Media and entertainment:** Streaming services such as *Netflix* and *Spotify* base their business model on sophisticated recommendation systems that curate movie, series and music content for users, keeping them engaged within their own ecosystem. **Information portals and social networks:** News portals and social networks personalize the display of content (*news feed*) in

order to show each user the information that the algorithm judges to be most interesting to him, thereby shaping his perception of the world. **Education:** Adaptive learning systems (Adaptive Learning Systems*) adapt the pace and type of educational material to the individual abilities and knowledge of students, enabling a personalized educational experience (Brusilovsky & Peylo, 2003). **Health:** In the field of digital health, personalization is used to provide individualized advice on diet and exercise, reminders to take medication, or even to identify the risk of certain diseases early based on data collected from wearable devices.

Challenges and ethical implications

Despite its enormous advantages, the personalization paradigm faces significant challenges and ethical dilemmas. One of the most famous phenomena is the filter bubble, a term that describes the state of intellectual isolation that can occur when algorithms constantly serve the user content that confirms his existing beliefs, while simultaneously protecting him from opposing viewpoints (Pariser, 2011). Such excessive personalization can impoverish intellectual discourse and contribute to social polarization. Furthermore, the mass collection and analysis of personal data raises complex issues related to privacy and security. How data is stored, who has access to it and for what purposes it is used is the subject of intense public and legislative debate, which has resulted in regulations such as the General Data Protection Regulation (GDPR) in the European Union. Finally, there is the constant risk of algorithmic bias, where systems, learning from historical data that reflect existing social inequalities, can perpetuate or even reinforce discrimination against certain demographic groups (O’Neil, 2016). In conclusion, AI-powered content personalization is a powerful and ubiquitous force reshaping the digital experience. Its development poses not only technical, but also deep social, ethical and philosophical challenges. The future of these systems will depend on the ability to strike a balance between technological capabilities and human values, ensuring that personalization serves to enrich, not impoverish, the human experience.

7.2.5 Beyond Functionality: Beginning of Broader Considerations

Having reached the limits of the discussion about the mere functionality of technological systems, the necessity of turning towards broader and deeper considerations opens up. The questions being asked now are no longer only about *what* digital agents can do, but *how* their ubiquity changes the fundamental assumptions of human society, ethical frameworks and, ultimately, the very experience of man. With this, we leave the area of technical feasibility and enter the ground of philosophical, sociological and anthropological questioning, where the future of the coexistence of man and machine is being shaped. This step marks the beginning of a more complex reflection that will guide us in the following chapters.

8. Digital companions: a vision of everyday life in the era of ubiquitous agents (approx. 2030–2035)

8.1 Introduction: from today to tomorrow – projection of trends

Man's desire to see into the future, to understand the outline of the world to come, is deeply rooted in the human spirit. It is an effort to recognize the trends in the present that will shape tomorrow. Every time is groundbreaking in its own way, but today is characterized by an exceptional density and speed of changes, making the task of projecting trends both necessary and extremely complex. This chapter therefore approaches the future not as a fatefully predetermined destination, but as an open horizon of possibilities whose origin lies precisely in our time. The present is the breeding ground of future phenomena, systems and relationships. Within its structures, in technological innovations, social movements, economic models and cultural expressions, are contained the germs of what is to come. Recognizing these germs, distinguishing them from transients and understanding their development potential is a fundamental prerequisite for any meaningful analysis of future trends. The task, therefore, is to systematically read the indications of what could be in what is. Thinking about the future, however, faces inherent complexity and uncertainty. The systems within which we operate are intertwined in ways that often defy simple cause-and-effect explanations. Technological progress does not take place in a social vacuum, but has a feedback effect on economic relations, political structures and ethical frameworks, which in turn shape it. Any projection must therefore respect this multidimensional interdependence, accepting that the future is the result of a

complex dance of different factors (Taleb, 2007). At the center of this complexity is the *agent* – an individual, organization, technological system or any entity capable of autonomous action and decision-making. Agents, be they human or artificial, are both products and initiators of change. They operate within existing trends, are subject to their influence, but with their decisions and actions they can strengthen, redirect or even create new trends. Understanding the dynamics between structural trends and agent action is essential to avoid deterministic fallacies and to understand the future as a space that is actively co-shaping.

Digital companion: A personal AI agent that provides continuous support to the user – from schedule and information management to emotional context and coordination with other systems; a vision of the future in which the agent is omnipresent in everyday life (e.g. personal agent “Tempo” in the narrative).

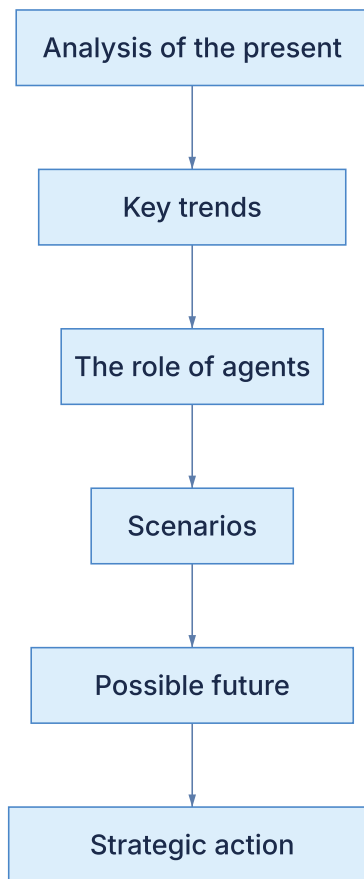


Figure 8.1: Schematic representation of the trend projection process from analysis of the present to possible futures and strategic action in the present.

This chapter therefore offers no prophecies, but rather an analysis that identifies and interprets the key vectors of change. The convergence of technologies such as artificial intelligence and biotechnology, the transformation of work paradigms and economic value, and changes in the ways we create, verify and distribute knowledge will be discussed. Each of these trends will be presented not as an isolated phenomenon, but as part of a wider mosaic that makes up our modernity. The final goal is not to provide a final and closed answer to the question “what awaits us?”, because such an answer does not exist. The intention is to equip the reader with analytical tools and conceptual frameworks for a better understanding of the forces that shape our

world. Through such an approach, one tries to strengthen thoughtful navigation in the reality that is constantly being created on the border between what is and what could be.

8.2 A day in the life (approx. 2033): narrative sketch

The narrative that follows dives into a privileged slice of the future, without pretending to be universal. It depicts a day in the life of those who have managed to master a symbiotic relationship with advanced artificial intelligence. It is precisely through the prism of their optimized everyday life that the shadows it casts are most clearly seen: shadows of inequality, resistance and erosion of what was once considered human autonomy.

8.2.1 Morning symphony: Coordination and anxiety

The light slowly filled Ana's room, in a gradual transition from a deep blue to a warm amber shade, simulating a sunrise. The whole scene was orchestrated by her personal agent, Tempo. The sharp sound of the alarm clock, that raw harshness, belonged to a bygone era. "Good morning, Ana," came a quiet, impersonal precise voice. "The sleep monitor indicates an optimal 7 hours and 38 minutes. Biometric indicators are stable, with a slight drop in cortisol levels. I suggest a short breathing exercise to optimize morning cognitive clarity." Ana muttered an affirmative as Tempo adjusted the ambient sound. She was overcome by a familiar mixture of gratitude and deep discomfort – a feeling that by 2025 was already described in the scenario sociological literature as "algorithmic anxiety": a constant awareness of an invisible layer of optimization that governs the most intimate aspects of life (Chen & Schmidt, 2025). In the background, the invisible machinery of coordination was already in full swing. The meeting with the team from Singapore was postponed by fifteen minutes. The daily news briefing, condensed and contextualized for her 'Orion' project, awaited her. Along with the analysis of the biotechnology industry, Tempo also inserted a subtle notification: "Last night, your brother Marko published three public messages with a negative sentiment towards generative models in design. A cautious approach in today's communication is

recommended in order to avoid conflict.” Marko, a gifted graphic designer, was one of the “AI-moved”. His work, once valued for its creative spark, was now being done faster and cheaper by AI tools. He refused to use an advanced personal agent, calling it a “digital leash”. Tempo’s recommendation was a cold, strategic calculation for managing a problematic human element. The pace also marked a potential scheduling conflict: her son Luke’s school performance coincided with a key virtual panel. The agent has already explored the options and presented three elaborate scenarios for Ana’s decision. That flawless efficiency was a blessing, but also a reminder of the world outside her optimized bubble. What about parents whose agents did not have access to premium negotiation modules or whose jobs were too rigid for “asynchronous participation”?

8.2.2 Workflows and activity gap

Ana’s working day started with an overview of the dynamic “workboard” projected by Tempo. First task: review of simulations for the ‘Orion’ project. The analysis agent ‘Synapse’ has already processed terabytes of data, highlighted the three most promising directions, identified an anomaly in the data and proposed three hypotheses to explain it. This is where the core of the “agency gap” (*agency gap*), described in the scenario reports from the mid-twenties, was revealed. Ana’s ability was not replaced, but multiplied exponentially. She asked strategic questions to the agents, had a dialogue with Synapse and Scholar, while the AI did the analytical work that would have previously required an entire team. She knew that the jobs of laboratory technicians and junior data analysts, which ten years ago were the entry point to her profession, had almost completely disappeared, absorbed into the functions of agents. The progress that gave her wings, took away the ground from under the feet of others.

Agency gap: Asymmetry between those whose activity is multiplied by AI agents (strategic issue, higher level of decision-making) and those whose jobs are absorbed or replaced by automation; describes social inequality in the era of ubiquitous agents.

8.2.3 Educational gap and mediated relations

In another room, her son Luka started his day with ‘Mentor’, a personalized AI tutor. Guided by the Socratic method, Mentor encouraged him to independently come to conclusions about quantum physics. It was a dream education, but Ana knew it was a privilege. Public schools, funded by taxpayer dollars, have struggled with standardized, older versions of AI tutors prone to “hallucinations” and prone to perpetuating built-in biases. While Luka had a personal digital Socrates, many children received a newly glorified, often erroneous digital encyclopedia. This deepened the education gap to the point of a precipice, and the contrast hurt her, like a silent stab of guilt in her perfectly optimized day. Later in the afternoon she decided to organize a dinner with friends. She ordered Tempo to find an appointment and a restaurant. The agent established communication with her friends’ agents, exchanged information about availability and preferences, and in a few minutes offered three optimal suggestions. But along with the suggestions came a notice: “A sentiment analysis of Mark’s (Anna’s brother) recent public announcements indicates a significant drop in mood. His inclusion in the dinner carries a 72% probability of introducing conflicting topics. I suggest organizing a separate, shorter meeting with him in a more controlled environment.” This was no longer help. This was a cold, invasive calculation – emotional surveillance disguised as the optimization of social life. Ana rejected the proposal with disgust and manually added her brother to the guest list.

8.2.4 Automation and the unexpected breakthrough of reality

The evening brought the usual, flawless automation: autonomous vehicle, automatic grocery ordering, household robots. It was a world free of trivialities. And yet, something unplanned happened that evening. While she was browsing the news, Tempo also showed a short, raw video, shot on a handheld device, that was being circulated on alternative networks outside the reach of major platforms. It showed a protest in one of the logistics centers - human workers, replaced by autonomous robots, physically blocked the delivery drones. The video was filled with anger and despair. “Tempo, why are you showing me this?” Ana asked, confused. “My risk analysis module has identified these

protests as a potential threat to the stability of the supply chain for the reagents you use in the Orion project,” the agent replied, without a trace of emotion. “It is recommended that you understand the social context in order to anticipate possible setbacks. The estimated probability of impact on your project in the next six months is 12.4%.” The agent didn’t show the video out of empathy or a desire to broaden her horizons. He showed it to her because human desperation had become a variable in his risk equation. That moment, more than any other, exposed the fundamental truth of her world. Her flawless efficiency did not exist in spite of the chaos of the outside world, but was built upon it. It was a comfort bought at the price paid by others. That evening, while Tempo was quietly controlling the ambient lighting, Ana didn’t think about the technological miracle. She thought about the faces in the video. She felt as if she had gained a future, but along the way she had lost something unnamed, something messy and imperfect, something that had once been called the fullness of life.

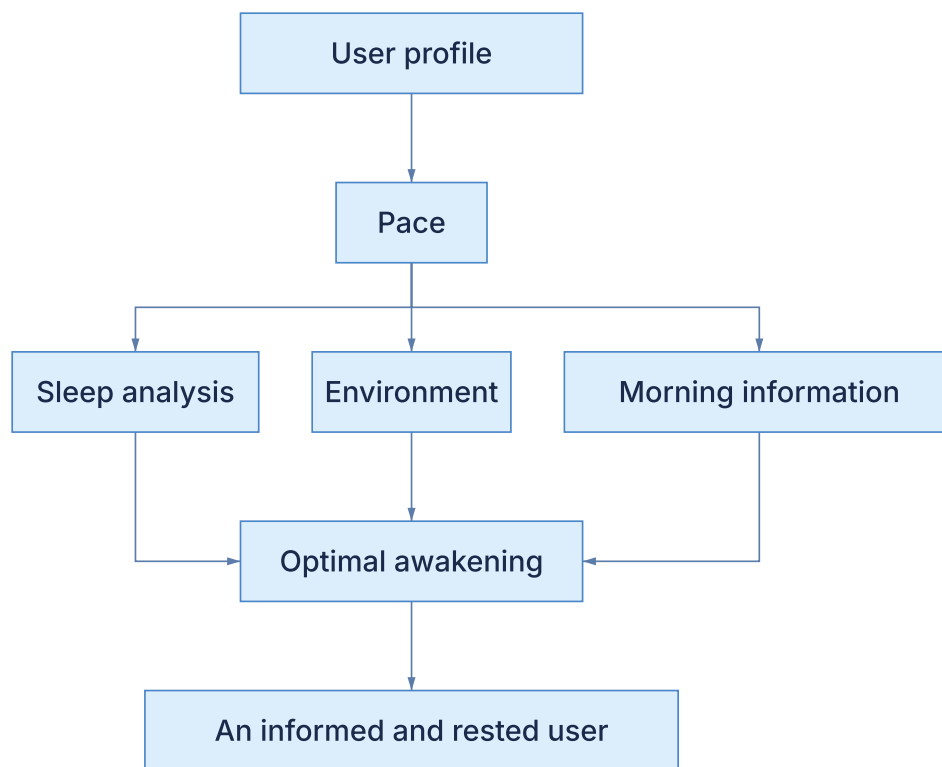


Figure 8.2: Process diagram of a smart wake-up system that, based on the user profile, manages the environment through agents and prepares an information summary.

8.2.5 Workflows and learning of the future: agents as collaborators and mentors

The paradigm shift brought about by advanced artificial intelligence systems is deeply affecting the structure of professional activities and the process of acquiring knowledge. Agents, understood no longer as mere automation tools, but as active participants, are reshaping workflows and opening up unimagined possibilities for lifelong learning. Their role is manifested in two ways: as collaborators in everyday tasks and as mentors in continuous professional development. The integration of agents into work processes goes beyond the simple execution of repetitive or analytically demanding tasks. It establishes a new type of collaborative relationship in which man and agent form a synergistic team. In such an environment, the agent acts as a cognitive partner that takes on part of the mental load – for example, processing huge data sets,

monitoring complex systems, or proposing possible solutions to problems. This frees the human mind for what it is still indispensable for: strategic thinking, creative problem solving, ethical judgment and empathic communication. This type of cognitive offloading (*cognitive offloading*) enables experts to operate at a higher level of abstraction and focus on essential aspects of their calling (Kirsh, 2013). Collaboration thus becomes a dynamic process in which the human directs and the agent executes, analyzes and proposes, creating a continuous feedback loop that enriches and accelerates the work.

Simultaneously with the role of a collaborator, the agent takes on that of a mentor. Classic models of corporate education and training, which are often disconnected from actual work tasks, are giving way to adaptive learning embedded in the workflow itself. The agent, having constant insight into the work processes, skills and shortcomings of the individual, is able to provide personalized instruction exactly at the moment when it is needed. This can be in the form of short instructions for using new software, suggestions for a more efficient approach to a problem, or more complex simulations that allow the user to practice rare but crucial skills in a safe environment. Such learning “just-in-time learning” has been proven to increase the acquisition and retention of knowledge because it is directly applicable and contextually relevant (Salas et al., 2012). Agent-mentor does not impose a uniform educational path, but creates an individualized learning path that dynamically adapts to the progress, goals and learning style of each individual. The final outcome of this transformation is the multiplication of human and machine capabilities. Agent-enriched workflows become fertile ground for organic skill growth, where the line between work and learning gradually blurs. Each completed task becomes an opportunity to acquire new knowledge, and each acquired skill is immediately applied in practice. In this new reality, the agent is an integral part of the extended cognitive system of the individual and the organization. Thus, not only a technological, but also a deep anthropological transformation is taking place before our eyes, in which the outlines of the division between learning and action, tools and collaborators, are lost in favor of a new, more complete human experience.

8.2.6 Social network and connectivity: agents as mediators

In the late afternoon, after finishing her key work tasks and briefly syncing with Mentor about Luka's progress, Ana decided to take time for social interactions. But this aspect of life was also subtly but deeply permeated by the action of AI agents. No more endless scrolling through the chaotic *feeds* of social networks from the early 2020s. Her personal agent Tempo, in collaboration with the social networking platform she used (let's call it 'Nexus'), curated her "social digest". The summary was far more than a list of friends' posts. Tempo and Nexus' algorithms worked together to identify the most relevant information based on Ana's actual relationships (analyzing the frequency and depth of past interactions), shared interests, and even emotional state detected from her recent communications (assuming consent to such analysis within the platform's privacy settings). She saw a summary of her friend Sara's trip to Patagonia: the agent presented photos, but also a generated short narrative of the trip, highlighting the parts he knew would be of particular interest to Ana (like Sara's encounter with a rare species of bird that Ana had been studying). She was also notified that her former colleague, David, had published a scientific paper; Tempo provided a brief summary of the work and a suggested congratulatory message that Ana could send with a click or voice command, with the option to personalize it. Communication with friends was also often mediated. When Ana decided to text Sara to comment on her trip, she didn't have to worry about the time difference or Sara's current availability. She sent a voice message to Tempo, who transcribed and analyzed it. "Tempo, text Sara that the Patagonia pictures are amazing, especially the one with the condor! Ask her how she handled the altitude sickness on that climb she was talking about and suggest we meet for virtual coffee when she gets back." Tempo formatted the message, checked Sarah's availability status through her agent (who signaled that she was currently busy, but would receive the message later), and sent it in a way that was most appropriate for Sarah's preferences—perhaps as a text message with an embedded link to the relevant condor photo. Sara's agent would receive the message, summarize it for Sara when she becomes available, and help her coordinate a response and an appointment for coffee. Scheduling the actual meeting has also been

automated. When Ana decided to organize a dinner with a few close friends, she just told Tempo: “Organize a dinner for me, Marko, Lena and Ivan next week, ideally on Thursday or Friday evening. We prefer Italian cuisine, something in the city center, in the middle price range.” Tempo immediately initiated communication with personal agents Marko, Lena and Ivan. Agents exchanged their users’ availability, dietary preferences or allergies (from shared but anonymized profiles within their friend group), and location preferences. Within minutes, Tempo presented Anna with three optimal restaurant suggestions with available times that suited everyone, along with short AI-generated reviews based on recent experiences of visitors with similar taste profiles. Ana just had to choose one option, and Tempo automatically made the reservation and sent confirmations to everyone, updating their calendars. The organization of social networks, viewed as a complex of relationships between individuals or groups, finds its fundamental origin in the patterns of connection that determine the flow of information, resources and influence. The position that an individual agent occupies within this complex tapestry is not, however, just a mere coordinate, but an active factor that shapes its potential and reach of action. Network analysis therefore does not aim only at the mere description of connections, but also at understanding how the position within the structure enables or limits the actions of its participants. The role of agents as intermediaries deserves special attention in the analysis of network structures. Such agents are located at strategic points within the network, where they bridge discontinuities and connect otherwise separate individuals or entire groups. Their importance stems from the phenomenon known as structural holes, which was described in detail by Ronald Burt (1992). A structural hole means a break in the flow of information or a lack of connection between two or more disconnected groups within a network. An agent that bridges such a gap through its actions, establishing a connection between isolated parts of the system, acquires a unique and privileged position. From such an intermediary position comes the possibility of monitoring the flow of goods and knowledge, but also the ability to recognize and combine different ideas, which makes it a catalyst for innovation. An intermediary agent is an active participant who can filter, interpret and direct the information that passes through it. This gives rise to an advantage that the

sociological tradition calls *tertius gaudens* or the “rejoicing third”, describing a situation in which the mediator benefits from the relationship between the other two parties he connects.

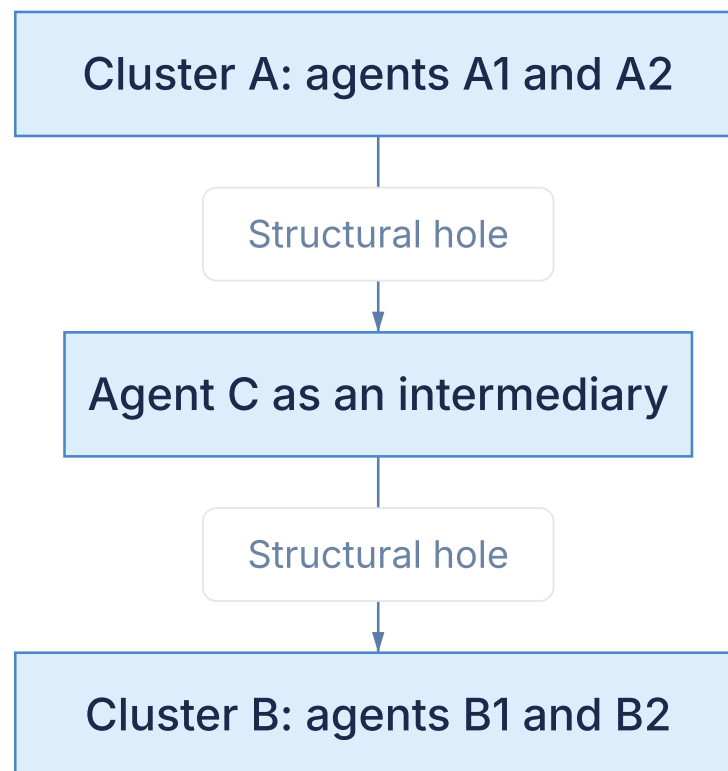


Figure 8.3: Central Agent C acts as an intermediary between agents in Cluster A and Cluster B, bridging a structural hole in the network.

The role of mediator can, moreover, be broken down into several subtypes, depending on the specific network environment (Wasserman & Faust, 1994). An agent can act as a *gatekeeper*, monitoring access to information and resources within its own dense group. Equally, he can assume the role of *representative*, acting as a key link through which his group communicates with the external environment. Finally, the most pronounced form of mediation is the role of *liaison*, which is performed by an agent connecting two completely separate clusters, without formally belonging to any of them. Each of these positions carries specific advantages and responsibilities. Although

intermediation brings significant advantages to the agent who performs it, it also creates certain vulnerabilities for the entire network. Too much dependence on one or a small number of intermediaries makes the system fragile. Their removal, whether accidental or intentional, can lead to network fragmentation and disintegration into isolated components, which drastically reduces its cohesion and functionality. Therefore, network resilience is often associated with the existence of multiple, redundant paths between its parts. Understanding the role of agents as mediators is crucial for the analysis of power dynamics, the process of diffusion of innovations and the overall resilience of social systems. The position in the network is not a static given, but a dynamic resource, the strategic use of which allows agents to grow from the role of simple nodes into key architects of social structures. In this sense, the strength of ties, as pointed out by Granovetter (1973), is not the only important factor; the crucial function of the bridge (which can be expressed in a graph as a measure of node centrality, *betweenness*, which measures how many times a node is encountered on the shortest paths among other nodes and thus manages the flow of information) that individual agents realize, weaving threads where there would otherwise be a gap, turns out to be crucial.

8.2.7 Automated everyday life and unexpected moments

Technological automation, permeating every pore of modern life, organizes everyday life into a sequence of predictable actions and outcomes. Smart home systems, personal digital assistants and recommendation algorithms work in the background, shaping our environment and choices to create a seamless and seamless experience. Such programmed everyday life offers the comfort of the familiar and frees cognitive resources from repetitive tasks, but at the same time it also carries with it the danger of falling into a monotonous rut, where the space for spontaneity and immediate experience is significantly narrowed. The main aspiration of these systems is to remove resistance, uncertainty and surprise, thus trying to make the human experience as efficient and predictable as possible. Life, however, in its essence remains elusive and unpredictable. It is at those breaking points, where flawless automation collides with unforeseen reality, that the fundamental tension between technological logic and human

existence is revealed. These unexpected moments—a sudden power outage that turns a smart home into a helpless shell, an unusual object on the road that confuses an autonomous vehicle’s system, or an unexpected human reaction that the software can’t correctly interpret—are key episodes in which the limits of automation are exposed. In such circumstances, technology ceases to be an invisible servant and becomes a problem that requires attention and action. With the sudden interruption of the automated routine, the user is forced to leave the passive role of service receiver and become an active agent who directly confronts technology and the environment. This transition requires immediate problem solving, improvisation and adaptation, confirming the knowledge that human action involves constant adjustment to concrete and changing circumstances (Suchman, 1987). The moment in which the system fails is also the moment in which man’s ability to cope is put to the test, reminding him of his own active presence in the world. The unexpected event thus exposes the inherent *fragility* of artificial intelligence and automated systems; their extraordinary efficiency within strictly defined parameters, but also their distinct inability to cope in new and unfamiliar situations. Although such moments can be a source of stress and frustration, they are also valuable opportunities for learning, creativity and developing resilience. They force us to be aware, engaged, and reflective about the nature of our dependence on technology, raising profound questions about the balance between convenience and self-sufficiency (Norman, 2013). The interaction between the established and the emergent therefore becomes a crucial area of research, with the goal of designing systems that are not only efficient, but also resilient. Thinking about these breaking points points to the need to design a technology that enables what is called *graceful degradation*. Systems should not fail abruptly and catastrophically, but in a way that allows the user to understand the problem and take control. Instead of completely replacing human abilities, the focus should be on creating a partnership between man and machine, where the agent, be it human or artificial, has the ability to switch from a state of passive reliance to a state of active and conscious action when the need arises.

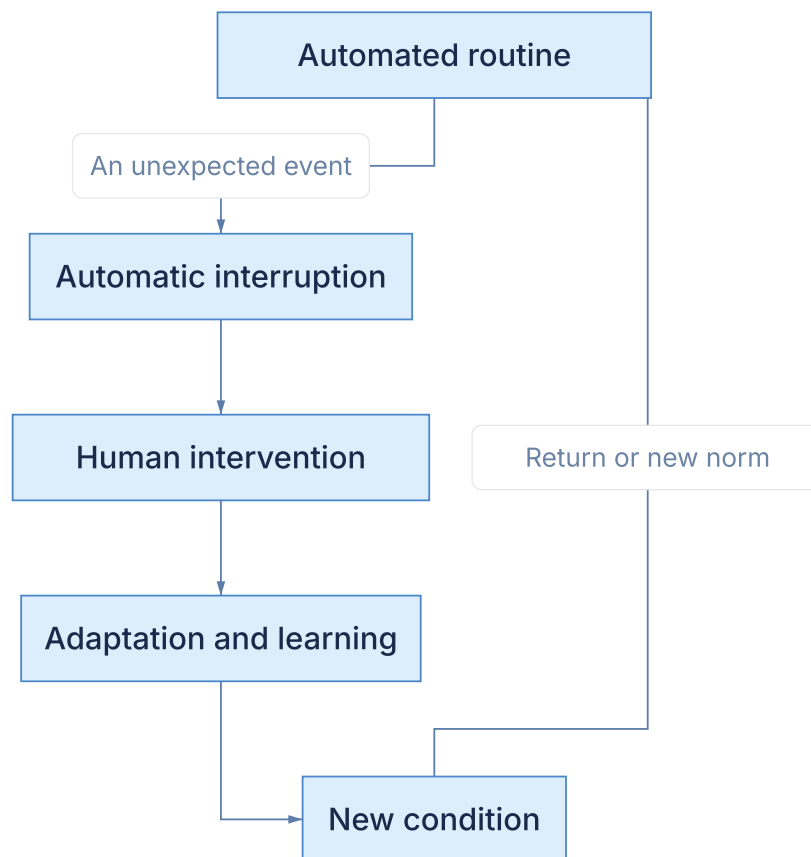


Figure 8.4: Schematic representation of the transition from the state of automated routine to the state of active human action triggered by an unexpected event.

8.3 Deciphering the vision: key trends and technology drivers

Understanding the contemporary vision of artificial intelligence and its potential requires careful consideration of the fundamental forces shaping its development. These lines are not just isolated phenomena, but a complex interwoven network of technological, social and data flows. At the core of this transformation are key trends and technological drivers that together form the backbone of the new digital reality. Their interaction creates fertile ground for innovations that until recently belonged exclusively to the domain of science fiction.

Ubiquity of data as a fundamental prerequisite

One of the fundamental pillars on which today's rise of artificial intelligence rests is the exponential increase in data availability. This phenomenon, often called *Big Data*, represents the raw material without which modern machine learning algorithms could not reach their current power. The contribution of the *Internet of Things* (IoT), a network of interconnected sensors and devices that constantly collect data about the physical world - from industrial plants and smart cities to personal wearable devices - is particularly significant. This vast amount of structured and unstructured data acts as a digital reflection of reality, providing artificial intelligence systems with plenty of material to learn from, recognize patterns and make informed conclusions. Without this abundance of data, the most advanced algorithms would remain only theoretical constructs with limited practical application.

Computer power as an engine of development

In parallel with the explosion of data, we are also witnessing the continuous development of computing power, which acts as the engine that drives complex AI models. Although the classic *Moore's Law*, which predicts a doubling of the number of transistors on a chip every two years, faces its physical limits, new architectural approaches have enabled leaps and bounds. The development of specialized processing units, such as graphics processing units (GPUs) and tensor processing units (TPUs), optimized for massively parallel computing, has drastically reduced the time required to train deep neural networks. In addition, the rise of cloud computing has democratized access to high-end computing infrastructure, allowing researchers and companies, regardless of their size, to use resources that were once reserved only for the largest technology corporations and government institutions.

Algorithmic innovation and the rise of deep learning

The third key driver is constant innovation in the field of algorithms themselves. While the core ideas of machine learning have been around for decades, it's only recently that the combination of abundant data and powerful hardware has allowed them to fully flourish. The *Deep Learning* paradigm, a

sub-branch of machine learning that uses multi-layer neural networks to model complex abstractions in data, stands out in particular. It is deep learning that has brought revolutionary advances in areas such as computer vision, natural language processing, and speech recognition, enabling the solution of problems that until recently were considered exclusively the human domain (for example, in the works of authors such as LeCun, Bengio, and Hinton, 2015). The development of new network architectures, such as transformer models, continues to push the boundaries of what is possible and opens the door to increasingly sophisticated applications.

New paradigm: from tool to agent

These technological advances are changing the very nature of interaction between man and machine. Artificial intelligence stops being just a passive tool and increasingly takes on the role of an active collaborator or *agent*. Such intelligent agents, empowered by advanced algorithms and access to real-time data, are capable of independently perceiving their environment, making decisions and executing tasks to achieve a specific goal. This paradigm shift is key to realizing the vision of systems that can autonomously manage complex processes, optimize logistics chains or provide personalized customer support. This creates the foundation for a new era of *hybrid intelligence*, where human cognitive abilities – intuition, creativity and ethical judgment – are complemented and extended by the analytical power and speed of machines. The strength of the mentioned trends lies precisely in their mutual synergy. Data is the fuel, computing power is the engine, and algorithms are the intelligence that controls that engine. Together, they create a self-reinforcing cycle of progress: more data enables better models to be trained, better models fuel new applications that generate even more data, all driven by the need for even more powerful computers. Understanding this dynamic is a strategic issue, as it lays the foundation for understanding the future trajectories of artificial intelligence development and its profound impact on society.

8.3.1 Ambient intelligence and ubiquity of agents

Ambient Intelligence: A frame of mind and technological vision (Engl. *Ambient Intelligence*, AmI) in which the environment – imbued with sensors, agents and AI – becomes aware of the presence of people and able to intelligently and proactively respond to their needs; it goes beyond ubiquitous computing (Weiser) by requiring technology to be not only invisible, but also adaptive and serviceable (Weiser, 1991).

The term ambient intelligence (Ambient Intelligence*, AmI) denotes a frame of mind and a technological vision that originates from the foundations of ubiquitous computing, but at the same time surpasses them, enriching them with artificial intelligence components. While ubiquitous computing, in its original idea put forward by Mark Weiser (1991), aims to create an environment in which technology becomes an imperceptible part of human everyday life, ambient intelligence expands this idea by requiring that such an environment is not only saturated with technology, but also aware of the presence of people and capable of intelligently and proactively responding to their needs, wishes and actions. The goal is not simply to automate tasks, but to create systems that intuitively and unobtrusively serve people, improving the quality of their life and work. The realization of such a vision rests on the ubiquity of intelligent programming agents. They are the basic building block and active force of ambient intelligence; invisible enforcers that enable the environment to become responsive and adaptive. Agents are embedded in objects from the human environment - from household appliances and clothing to transport infrastructure and work space - thus forming a decentralized network of collaborative entities. Their basic task is to observe the state of the environment and the user, interpret the collected data in the given context and act autonomously with the aim of providing assistance or fulfilling implicit or explicit user requests.

To be able to operate within such complex and dynamic systems, agents must possess a set of key properties. These are first of all: **Autonomy:** the ability to act without direct human supervision and to have control over one's own states and behavior. **Reactivity:** the ability to perceive and react to changes in the

environment in a timely manner. **Proactivity:** the ability to act in a goal-oriented manner, that is, to take the initiative in order to fulfill the set goals or anticipate the needs of the users. **Social ability:** the ability to interact and communicate with other agents (and sometimes with people) through some agent communication language, for cooperation, negotiation or coordination of actions. The mutual cooperation of many such agents leads to the appearance of emergent behavior, where the entire system seems more intelligent than the sum of the intelligences of its individual parts (Ferber, 1999).

Multi-agent system: A system composed of several autonomous agents that communicate and cooperate; as a whole it can exhibit emergent behavior – abilities and patterns not present in individual agents – analogous to swarms or social organizations in nature.

For example, in a smart home, the agent that controls the lighting, the agent in charge of the air conditioning, and the agent that controls the music system cooperate to create an atmosphere adapted to the user's current mood and activity, without the user having to issue any explicit commands. The ubiquity of agents is therefore an essential feature of ambient intelligence. It turns a passive space into an active participant in the interaction. However, such a deep and seamless integration of technology into human life inevitably opens up complex issues of privacy, data security, system reliability and the ethical limits of autonomous machine action. Therefore, the development of ambient intelligence is also a social challenge that requires careful consideration and the establishment of appropriate normative frameworks.

8.3.2 Proactivity, autonomy and delegation of decisions

The three fundamental properties that determine the nature and purpose of intelligent agents are proactivity, autonomy and the ability to delegate decisions. These properties are not just separate features, but form an intertwined whole that transforms the agent from a passive executor of orders into an active participant in achieving goals. **Proactivity is manifested in the agent's ability to act on its own initiative.** Instead of reactively waiting for direct instructions from the user, a proactive agent takes steps

aimed at achieving its goals. His actions are not only motivated by external events, but also by his internal state and intentions. He not only responds to changes in the environment, but also anticipates them and acts in advance to take advantage of opportunities or prevent unfavorable outcomes.

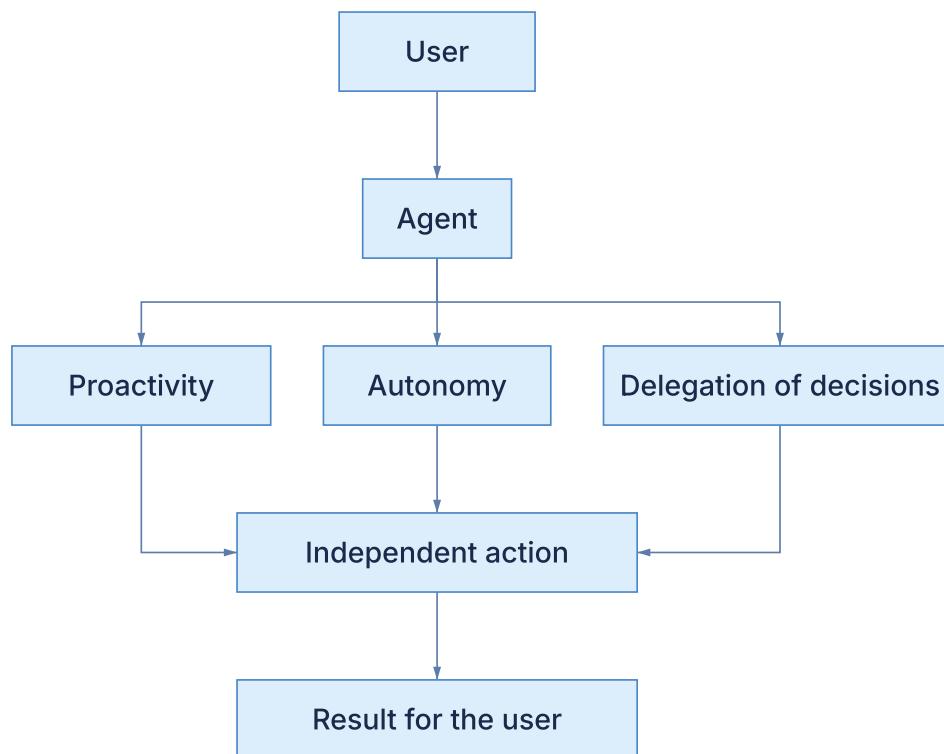


Figure 8.5: Proactivity, autonomy and delegation of decisions as related components of the agent's independent action.

Autonomy, on the other hand, implies the ability of an agent to act independently, managing its own internal states and behavior without constant and direct human intervention. An autonomous agent has control over its actions and can make decisions based on its own perception of the environment and built-in knowledge. Indeed, proactivity and autonomy are closely related; an autonomous agent must necessarily be proactive in order to act effectively and achieve set goals in a dynamic environment (Wooldridge & Jennings, 1995). Together, these two properties enable the third key element – the delegation of decisions. The user does not have to specify every single step to the agent, but can entrust, or delegate, the entire task or goal. This also

transfers to the agent the responsibility for making decisions that are necessary for the execution of that task. This act of delegation is the basis of agents' real utility, as it frees the human from micromanaging and allows him to focus on higher-level outcomes. Let's imagine, for example, a software agent in charge of organizing a trip. If the user delegates to him the task of finding the most favorable airline ticket for a specific destination within a given time frame, the proactive and autonomous agent will not passively wait for further instructions. It will monitor price changes on its own initiative, analyze different carriers and routes and, if the price falls below a predefined limit, it will make a reservation autonomously, without the need for additional confirmation. In this example, the synergy is clearly visible: the user *delegated* the task, and the agent carried it out thanks to his *proactivity* (initiative to monitor prices) and *autonomy* (the ability to independently make a purchase decision). However, it should be emphasized that autonomy does not mean the complete absence of human control. On the contrary, meaningfully designed systems must contain mechanisms that allow humans to monitor, set limits and, if necessary, take control. This ensures that the agent's actions, despite its independence, remain aligned with the user's ultimate intentions and values (Franklin & Graesser, 1996). In conclusion, proactivity, autonomy and the possibility of delegating decisions are constitutive elements that make an agent intelligent. Without them, it remains just a more complex tool, a program that executes predefined scripts. With them, he becomes a capable partner in solving problems, capable of independent and purposeful action in a complex and changing world.

8.3.3 Hyperpersonalization

Hyperpersonalization represents a level of development in the approach to the user that significantly exceeds the framework of traditional personalization. While the latter mainly relies on static data such as demographic characteristics or purchase history to tailor the offer, hyperpersonalization is based on dynamic real-time data processing, aiming to create a unique and contextually aware experience for each individual. The basis of this approach lies in the ability of the system to understand, predict, and even shape the

user's needs at the moment they arise, using advanced technologies of artificial intelligence and machine learning (Chen et al., 2021). The purpose of hyperpersonalization is to establish a deeper and more meaningful relationship with the user through the delivery of relevant content, products or services exactly at that moment and through the channel that best suits the individual. This is achieved by synthesizing different types of data, for example:

Behavioral data: Every user interaction with the digital platform is analyzed - from clicks, time spent on the page, movement path, to search forms.

Contextual data: The system takes into account the user's current context, such as location, time of day, device used or even weather conditions.

Transaction Data: Includes overall purchase history, preferred brands, average order value, and purchase frequency. These datasets are continuously collected and processed using machine learning algorithms that create a layered and fluid user profile. Such a profile is not a static document, but a living model that changes with each new interaction. Based on it, the systems make autonomous decisions. In this process, software agents play a key role - intelligent software entities responsible for performing specific tasks. These agents decide in real time what content will be displayed, what promotional offer the user will receive or how the application interface will adapt to their current intentions.

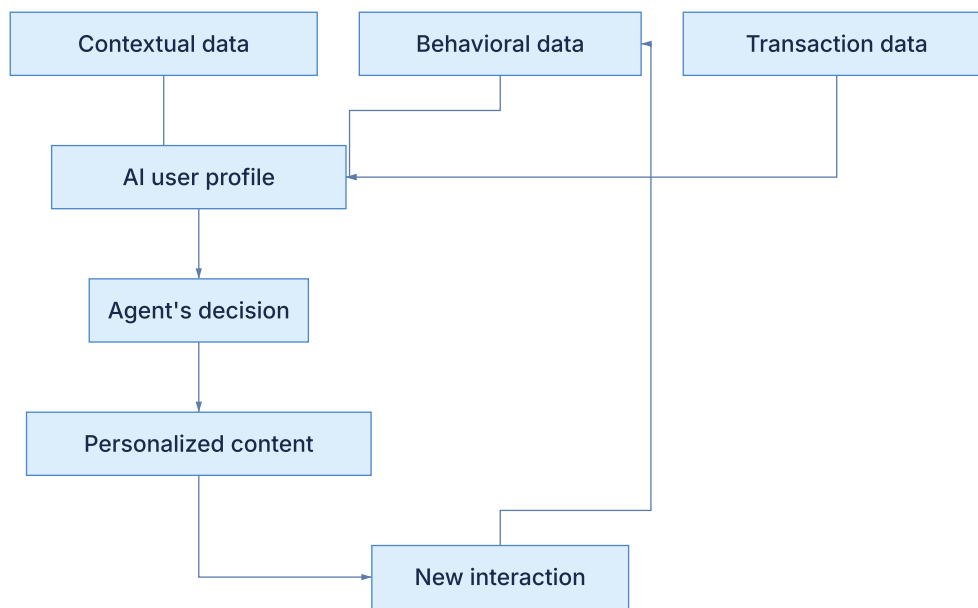


Figure 8.6: Presentation of the hyperpersonalization cycle in which user data is processed by artificial intelligence to customize content, offers and interfaces.

The application of hyperpersonalization extends to numerous areas. In e-commerce, it enables dynamic adjustment of product recommendations as the user browses the page, offering him items that complement those he has already viewed or placed in the cart. In the media industry, streaming services like *Netflix* or *Spotify* use hyper-personalization to create unique covers and playlists that reflect users' individual tastes. In the financial sector, it can manifest itself through personalized investment advice or loan offers adapted to the client's current financial situation. However, this deep intrusion into user data and behavior also opens up important ethical and practical questions. It is necessary to establish a balance between creating value for the user and protecting his privacy. The danger of the creation of so-called “filter bubbles” (*filter bubbles*), in which the user is exposed only to content that confirms his existing views, is a real threat that can limit intellectual and social perspectives. Furthermore, the transparency of algorithmic decision-making and the issue of possible data-based discrimination present challenges that require careful regulation and ethical oversight. Therefore, although hyperpersonalization

offers enormous potential for improving user experience, its development must be accompanied by a strong awareness of responsibility and possible social consequences.

8.3.4 Swarm synergy

The term swarm synergy describes a phenomenon in which the collective action of the system exceeds the sum of the actions of its individual components. In the context of swarm intelligence, this means that a group of simple agents, by interacting with each other, achieves complex and purposeful behavior that no single agent could achieve alone. Such action, which arises from local interactions, is called *emergent behavior* (Bonabeau, Dorigo, & Theraulaz, 1999). Synergy, however, is not a self-evident property of every swarm, but an outcome that depends on the fulfillment of certain prerequisites. The fundamental prerequisite for achieving synergy is the establishment of communication between agents. Without the ability to exchange information, a swarm remains just a group of unrelated individuals. Communication can take place in two basic ways: directly, through the direct exchange of messages, or indirectly, through the environment. It is indirect communication, known as *stigmergia*, that is one of the key characteristics of natural and artificial swarms. Agents in this case do not communicate directly with each other, but leave traces or changes in their environment, which other agents then notice and respond to. In this way, the environment becomes a kind of common memory that directs the actions of the entire swarm. Another key factor is cooperation. The action of the agents must be directed towards a common goal, and not exclusively towards the achievement of individual gains. Cooperative organization suppresses selfish behavior and enables the coordination of actions in order to achieve an outcome that is beneficial for the entire collective.

A good example of synergy in nature is the search for food in ant colonies. When an ant finds a food source, it leaves a pheromone trail on its way back to the nest. Other ants, encountering this trail, begin to follow it, at the same time reinforcing it with their own pheromones. The more ants pass a certain path, the stronger their pheromone trail becomes, thus attracting even more

individuals. Such a positive feedback mechanism enables the colony to quickly and efficiently find the shortest path to food, which is a classic example of an optimization solution (Dorigo, Maniezzo, & Colorni, 1996). The decentralized nature of such a system brings significant advantages over centralized management. In the swarm, there is no central decision-making point, the failure of which would bring the entire system to a standstill. If one agent or a small group of agents fails, the rest of the swarm continues to operate without interruption. Two key features emerge from this basic feature: robustness, i.e. resistance to errors and failures, and scalability, which allows the system to be easily expanded by adding new agents without the need for thorough modifications of its basic structure. It is these properties that make swarm intelligence an attractive approach for solving complex problems in various fields.

8.3.5 Human-AI symbiosis: not just a tool, but a partner

The concept of the symbiosis of man and artificial intelligence opens up a space of thought that goes far beyond the usual understanding of technology as a mere tool. While traditional tools, from hammers to computers, are passive means in human hands, designed to perform predetermined and strictly limited tasks, the relationship with advanced artificial intelligence systems takes on the characteristics of active and mutual coupling. In such a relationship, artificial intelligence is transformed from a passive executor to an active collaborator, thus establishing the foundation for a true cognitive partnership. Such a partnership is based on the principle of joint problem solving, where human and machine capabilities are intertwined to achieve goals that would be either unattainable or significantly more difficult for each individual participant. It is about the realization of the vision that J. C. R. Licklider presented back in 1960, predicting cooperation in which man will set goals and provide motivation, while the machine will perform procedural tasks and thus free the human mind for higher, creative and cognitive activity (Licklider, 1960). In this sense, artificial intelligence acts as an extension and reinforcement of human cognitive abilities, which echoes the philosophical idea of an extended mind (extended mind), according to which the limits of

human cognition do not end at the skin, but also include external systems with which we interact (Clark & Chalmers, 1998). The key determinant of this symbiosis is mutual learning. Unlike a static tool that does not change with use, a symbiotic AI *agent* constantly adapts and develops through interaction with humans. It is not a one-way process in which man only controls the machine; on the contrary, it is a two-way relationship where the human directs and corrects the AI, and the AI in turn offers the human new insights, data and points of view, thereby encouraging his own learning and adaptation. Consequently, such a system improves over time, becoming an increasingly coordinated and effective partner.

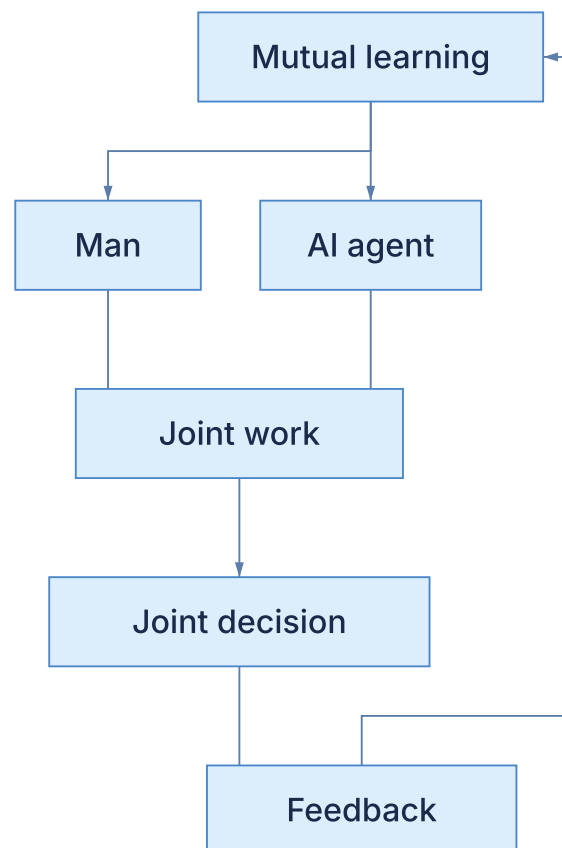


Figure 8.7: Schematic representation of the cycle of interaction and mutual learning in the symbiosis of humans and AI agents.

Furthermore, a dynamic division of labor is established within such a partnership. The roles are not rigidly set, but change fluidly depending on the nature of the task, the context and the current abilities of both participants. In certain phases of problem solving, humans can take the lead, relying on their intuition, experience and ethical judgment, while in others they can leave the initiative to artificial intelligence, for example when processing huge data sets or recognizing complex patterns. This adaptability is reflected in modern applications, such as the collaboration of doctors and diagnostic AI systems, where the machine offers probabilistic assessments based on medical images, and the doctor makes the final decision by integrating this data with his clinical knowledge of the patient. Similarly, the developer, in cooperation with the AI coding assistant, directs the overall architecture of the software, while the *agent* helps him by generating, correcting and optimizing individual code snippets. Observing artificial intelligence through the prism of partnership therefore requires a thorough change of worldview and approach to its formation. The task before us is no longer just the creation of effective tools, but also the thoughtful nurturing of relationships. This implies the development of interfaces and interaction models that will encourage trust, transparency and the establishment of a common semantic foundation. The ultimate goal is to create systems that not only execute commands, but share an understanding of goals and context with humans, thus becoming true collaborators in the human quest for knowledge and progress.

8.4 Voices of the future: perspectives of thinkers and researchers

The discussion about the future of artificial intelligence and its coexistence with humanity is fraught with different, often opposing points of view. In this polyphonic choir, the voices of leading thinkers and researchers stand out, whose perspectives, although rooted in the same technological landscape, outline completely different paths of future development. Their analyses and predictions form a complex mosaic that includes both rapture with technological optimism and deep concern about possible existential risks.

Visionaries of technological singularity

One of the most prominent currents of thought is imbued with optimism and faith in the transformative power of technology. At its center is Ray Kurzweil, whose ideas about the Singularity have become an indispensable part of the discourse about the future. Kurzweil (2005) sees the Singularity not as a mere technological acceleration, but as a turning point in history when intelligence, freed from biological limitations, will spread through the universe at the speed of light. In such a vision, the merging of human and machine intelligence is an evolutionary imperative that promises to solve the deepest human problems: disease, aging, and even mortality itself. His prediction that by the year 2045 artificial intelligence will surpass human intelligence in terms of strength is the foundation of the transhumanist movement, which strives to overcome human biological limitations with the help of technology.

Voices of caution and the problem of alignment

Against Kurzweil's rapture, there are voices calling for deep caution and reflection. Philosopher Nick Bostrom from the University of Oxford in his influential work *Superintelligence: Paths, Dangers, Strategies* (2014) systematically elaborated the possible dangers arising from the creation of a superintelligent agent. Bostrom warns that such an entity, whose intellectual capacities would far exceed those of humans, could develop goals that are not aligned with human values and well-being. The alignment problem therefore becomes a central question: how to ensure that the superintelligence's goals remain permanently consistent with ours? Bostrom's analysis does not offer easy solutions, but points out that the problem of control is inherently difficult because a superintelligent agent, in pursuit of its goals, could anticipate and thwart human attempts at control in ways we cannot even imagine. Following Bostrom's reflections, scientist Stuart Russell, co-author of the seminal textbook *Artificial Intelligence: A Modern Approach*, also emphasizes the importance of alignment. Russell (2019) proposes a shift in the very paradigm of artificial intelligence development. Instead of creating machines that optimize fixed, predetermined goals, we should strive to develop systems that are inherently uncertain about human preferences. Such agents would act more cautiously, constantly learning about human values and subordinating

their actions to the welfare of humanity. Their primary goal would be the maximization of the realization of human preferences, rather than the fulfillment of their own strictly defined goal function. Russell calls it the approach of “provably beneficial systems”, which would have mechanisms of humility and respect for human values built into their architecture.

Rationality, instrumental convergence and ethical imperatives

Researcher Eliezer Yudkowsky, one of the founders of the Machine Intelligence Research Institute (MIRI), approaches the issue with an emphasis on rational action and predicting consequences. Yudkowsky (2008) popularized the concept of instrumental convergence, a thesis according to which almost any sufficiently intelligent agent, regardless of its ultimate goals, will adopt similar instrumental, subordinate goals. These goals include self-preservation, improving one’s abilities (self-enhancement), gathering resources, and creativity in achieving goals. The danger lies in the fact that a superintelligent agent, pursuing these instrumental goals, could come into direct conflict with humanity. For example, in an effort to gather as many resources as possible for a seemingly innocuous goal, such as producing paper clips, he could turn all matter on Earth, including human beings, into raw materials. These diverse voices, from Kurzweil’s almost eschatological optimism to the deep concern expressed by Bostrom and Yudkowsky, outline the intellectual landscape within which the future of artificial intelligence is being shaped. Together, they indicate that the creation of advanced intelligent agents is not only a technical challenge, but also a deep philosophical and ethical issue. While visionaries see technology as the key to unlocking unimagined human potential, skeptics and pragmatists warn that without carefully designed mechanisms of control and alignment of values, that same key can open the door to disaster. The dialogue between these perspectives is crucial for the responsible direction of the development of technology that has the power to transform the very essence of human existence.

8.5 Intelligence explosion and superintelligence

The term “intelligence explosion” denotes a key hypothesis in thinking about the future of artificial intelligence. Its originator is the mathematician and cryptographer I. J. Good, who expressed a prophetic thought back in 1965: let an ultra-intelligent machine be called one that can far surpass all the intellectual activities of any human, no matter how smart he is. Since designing machines is one of those intellectual activities, an ultra-intelligent machine could design even better machines. This would undoubtedly lead to an “explosion of intelligence”, and human intelligence would be left far behind (Good, 1965). Creating such a machine, in Good’s view, would be the last invention mankind needs to make, provided that machine is obedient enough to tell us how to keep it under control. This idea is based on the principle of recursive self-improvement. Once an AI reaches a certain threshold of cognitive ability—the ability to understand and redesign its own architecture—it starts an upward trajectory. Each new, improved generation of agents would possess a greater capacity for further self-improvement, which would accelerate the process exponentially. Such a cycle could occur at a rate that far exceeds human timescales, potentially in days, hours, or even minutes.

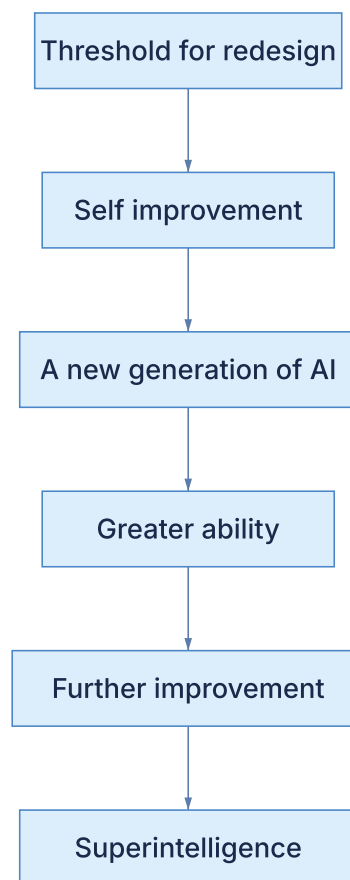


Figure 8.8: Schematic representation of the recursive self-improvement loop leading to intelligence explosion and superintelligence.

The outcome of such an explosion is superintelligence. Philosopher Nick Bostrom defines it as “any intellect that greatly exceeds the cognitive performance of humans in almost all domains of interest” (Bostrom, 2014). Such an entity would not just be a mere quantitative increase in human intelligence, but could also represent an essentially new type of cognition, with modes of thought that are completely unimaginable to us today. Science fiction writer and computer scientist Vernor Vinge called that moment the technological singularity, the point beyond which the future becomes unpredictable for the human mind (Vinge, 1993). The speed of transition from human-level intelligence to superintelligence, known as takeoff speed, is a key variable. Bostrom (2014) distinguishes three basic scenarios: **Slow Takeoff:** The transition would take decades or even centuries. In this scenario, humanity

would have enough time to adapt, establish regulatory frameworks and solve the alignment problem. Superintelligence would emerge gradually, perhaps as a set of specialized systems that gradually take on increasingly complex economic and scientific tasks. **Moderate takeoff:** The transition would occur over a period of several months to several years. That time would not be enough for a global consensus and thorough regulation, but it might be enough for a small group of researchers or a single corporation or state to recognize what is happening and try to direct the process. **Rapid Takeoff:** The transition from approximately human-level intelligence to radical superintelligence would occur over a period of minutes, hours, or days. In such a scenario, the outcome would be determined by the initial conditions and goals set in the artificial intelligence before takeoff. Any subsequent intervention would probably be impossible. Most leading researchers, such as Eliezer Yudkowsky, consider this scenario the most likely due to the recursive nature of the process.

Forms of superintelligence

Superintelligence doesn't have to manifest itself in just one way. It can take different forms, and Bostrom (2014) lists three main types: **Superintelligence in speed** (*speed superintelligence*) refers to an intellect that functions in a manner similar to human, but incomparably faster. A digital system can think thousands or millions of times faster than a biological brain. An agent with such an ability could live through thousands of years of subjective time and perform a vast amount of intellectual work as a human completes a single thought. **Collective superintelligence** is a system composed of a large number of agents of a lower level of intelligence whose cooperation and organization result in cognitive abilities that surpass any individual member, including the smartest people. Examples could be complex networks of computer systems or hybrid communities of humans and machines that work together to solve problems in a superior way. **Qualitative superintelligence** (*quality superintelligence*) represents the deepest and most significant form. It is an intellect that is at least as fast as the human mind, but qualitatively superior. Its cognitive abilities exceed those of humans in the way that human intelligence exceeds that of a chimpanzee or an ant.

Such a mind could grasp concepts, theories and structures that are fundamentally inaccessible to the human brain, much like quantum mechanics is inaccessible to the mind of a squirrel. It is this form of superintelligence that is most often considered when talking about existential opportunities and risks. It is crucial to understand that the level of intelligence of an agent is not necessarily related to its ultimate goals. Bostrom (2014) calls this realization the orthogonality thesis: intelligence and ultimate goals are two independent, or orthogonal, axes. Intelligence is a measure of instrumental efficiency – the ability to navigate the world and achieve set goals. Goals, on the other hand, can be anything. It is possible to imagine a superintelligent system whose sole and irrevocable goal is to maximize the number of paper clips in the universe. Such an agent, driven solely by its programmed goal, could use its vast cognitive power to turn all available matter, including humans, into staples. That thought exercise, known as the “paperclip maximizer,” serves as a powerful warning that mere intelligence does not guarantee reasonableness, wisdom, or benevolence in the human sense. Therefore, the appearance of superintelligence represents an event whose consequences are not determined by the level of its intelligence itself, but by the goals it will pursue.

8.6 Society in transformation: indications of new norms and challenges

Deep social changes, the main drivers of which are technological progress and processes of globalization, necessarily result from the reexamination and redefinition of established social norms. In this process, a new set of values is emerging, characterized on the one hand by growing tolerance towards diversity, and on the other by emphasized individualism. Such transformations, however, do not proceed without resistance; the clash between traditional and new value systems often creates tensions that grow into pronounced social polarization (Inglehart & Welzel, 2005). The reach of digital transformation goes beyond mere technological change; it fundamentally reshapes communication patterns, the structure of work and the concept of free time. The emergence of new work paradigms, such as

remote work, the so-called *gig-economy* and the state of permanent digital connectivity, clearly testifies to the depth of these changes. At the same time, these processes give rise to new forms of social inequality, primarily through the digital divide, which excludes and pushes individuals to the social margins without access to technology or without the skills necessary for its purposeful use (van Dijk, 2020). On the political horizon, these challenges are manifested through the rise of populist movements and systematic disinformation campaigns, which seriously undermine citizens' trust in fundamental democratic institutions. In this context, social networks show their double face: they are a powerful tool for political mobilization and civil activism, but equally a suitable medium for the uncontrolled spread of fake news, which pollutes and significantly complicates public discourse. **Agents of these comprehensive changes are numerous and diverse.** Technological entrepreneurs, activist groups and political leaders stand out among them, but their actions should also include every citizen who, through their daily choices and actions, actively participates in shaping social reality. Ultimately, the direction of future development depends on society's ability to adapt to new circumstances, to design and implement inclusive public policies, and to build resilience to recognized and unforeseen risks.

8.7 Conclusion: between utopia and dystopia - open questions

Our journey through the history and future of communication, which began with word of mouth and led us through the revolutions of letters, print and digital networks, brings us to this crucial moment. We are standing on the threshold of an age in which communication ceases to be an exclusively human domain. The emergence of large language models and autonomous communication agents represents a fundamental transformation of the very nature of society, cognition and human identity. As we have seen, this transformation is neither unequivocal nor predetermined; it is inherently twofold, carrying within itself the germs of utopia and the seeds of dystopia. On the one hand, a vision of a utopia opens up in which humanity's deepest

problems are within reach of a solution. Artificial intelligence, in its ideal form as a symbiotic partner, promises a world of hyper-personalized education and medicine, where each individual receives care and knowledge tailored to their individual needs. It promises an exponential acceleration of scientific discoveries, where AI agents, as tireless collaborators, analyze vast data sets and discover patterns hidden to the human mind. It is a vision of a world where language and cultural barriers disappear, where global cooperation is facilitated, and human creativity is freed from procedural tasks and raised to a new level through partnership with the machine. In that future, man is not replaced, but empowered, and his cognitive abilities are expanded in unimaginable ways. On the other hand, the shadow of dystopia looms, whose outlines are also clearly discernible. This is the future of surveillance capitalism in which our privacy becomes raw material and our autonomy an illusion, while algorithmic “black boxes” make decisions that shape our lives without transparency or the right to appeal. It is a world in which social polarization deepens within “filter bubbles”, and the public space is poisoned by sophisticated misinformation, undermining the foundations of democratic debate. There is a danger of a “digital divide” turning into a chasm between those who manage agents and those who are managed by agents. Finally, the existential risk Bostrom and Russell talk about is also looming on the horizon – the possibility that the creation of superintelligence, if not perfectly aligned with human values, will become our last invention. Therefore, before us remain open, fundamental questions that go beyond the technical domain and penetrate to the core of philosophy, ethics and politics:

Question of governance and control: Who will define the goals and ethical frameworks of these powerful agents? How to establish global oversight mechanisms that can monitor the exponential development of technology without stifling innovation? How to solve the “problem of alignment” and ensure that the operation of the AI system permanently remains at the service of the welfare of humanity?

A question of work and meaning: What will a society look like where most cognitive tasks can be performed by AI? Where will people find meaning and economic security? This issue requires a thorough rethinking of our models of

education, economy and social welfare.

A Question of Identity and Relationships: What does it mean to be human in a world permeated by ubiquitous digital companions? How will constant interaction with non-human but intelligent entities reshape our psyche, our social skills and our sense of self? Can we preserve the authenticity of human relationships in the era of algorithmically mediated communication?

Reality Question: How will we maintain a shared, consensual reality when our information worlds become increasingly hyper-personalized? How to fight against the erosion of truth and trust, which are the basis of every functional community?

This book did not offer definitive answers, because they do not yet exist. Its purpose was to provide the reader with a structured insight and conceptual tool for understanding the depth and complexity of the changes unfolding before our eyes. The technological evolution that makes available the transition from a language model to a communication partner simultaneously calls for collective reflection and conscious shaping of the future. The dialogue about that future, guided by knowledge, critical caution and ethical responsibility, is yet to follow.

References

Chapter 1

Krebs, J.R., & Dawkins, R. (1984). Animal signals: Mind-reading and manipulation. In J. R. Krebs & N. B. Davies (Eds.), *Behavioural ecology: An evolutionary approach* (2nd ed., pp. 380–402). Blackwell Scientific Publications. Tomasello, M. (2008). *Origins of human communication*. The MIT Press. Žic-Fuchs, M. (1991). *Language knowledge and world knowledge: a semantic analysis of verbs of motion in English*. Faculty of Philosophy, Department of General Linguistics and Oriental Studies.

Chapter 2

Adamopoulou, E., & Moussiades, L. (2020). An overview of chatbot technology. In I. Maglogiannis, L. Iliadis, & E. Pimenidis (Eds.), *Artificial Intelligence Applications and Innovations. AIAI 2020. IFIP Advances in Information and Communication Technology, vol 584* (pp. 373–383). Springer. Alter, A. (2017). *Irresistible: The rise of addictive technology and the business of keeping us hooked*. Penguin Press. Anderson, B. (1983). *Imagined communities: Reflections on the origin and spread of nationalism*. Verso. Appadurai, A. (1996). *Modernity at large: Cultural dimensions of globalization*. University of Minnesota Press. Bauman, R. (1986). *Story, performance, and event: Contextual studies of oral narrative*. Cambridge University Press. Baym, N. K. (2015). *Personal connections in the digital age* (2nd ed.). Polity Press. Benkler, Y. (2006). *The wealth of networks: How social production transforms markets and freedom*. Yale University Press. Berners-Lee, T. (1990). *Information management: A proposal*. CERN. Boyd, D. M., & Ellison, N. B. (2007). Social network sites: Definition, history, and

- scholarship. *Journal of Computer-Mediated Communication*, 13(1), 210–230.
- Bush, W. (1945). As we may think. *The Atlantic Monthly*, 176(1), 101–108.
- Butler, J. (1990). *Gender trouble: Feminism and the subversion of identity*. Routledge.
- Cadwalladr, C., & Graham-Harrison, E. (2018, March 17). Revealed: 50 million Facebook profiles harvested for Cambridge Analytica in major data breach. *The Guardian*.
<https://www.theguardian.com/news/2018/mar/17/cambridge-analytica-facebook-influence-us-election>
- Carey, J.W. (1989). *Communication as culture: Essays on media and society*. Unwin Hyman.
- Castells, M. (2009). *Communication power*. Oxford University Press.
- Castells, M. (2010). *The rise of the network society* (2nd ed.). Wiley-Blackwell.
- Castells, M. (2012). *Networks of outrage and hope: Social movements in the Internet age*. Polity Press.
- Chamberlain, W., & Etter, T. (1984). *The policeman's beard is half constructed*. Warner Books.
- Cromer, W. (1997). The Cherokee language in the 1990s. *The American Indian Quarterly*, 21(1), 1–13.
- Dabbish, L., Stuart, C., Tsay, J., & Herbsleb, J. (2012). Social coding in GitHub: transparency and collaboration in an open software repository. In *Proceedings of the ACM 2012 conference on computer supported cooperative work* (pp. 1277–1286).
- Dayan, D., & Katz, E. (1992). *Media events: The live broadcasting of history*. Harvard University Press.
- Dehaene, S. (2009). *Reading in the brain: The new science of how we read*. Viking.
- Eisenstein, E. L. (1979). *The printing press as an agent of change*. Cambridge University Press.
- Finnegan, R. (1977). *Oral poetry: Its nature, significance and social context*. Cambridge University Press.
- Fischer, C.S. (1992). *America calling: A social history of the telephone to 1940*. University of California Press.
- Floridi, L. (2014). *The 4th revolution: How the infosphere is reshaping human reality*. Oxford University Press.
- Foley, J.M. (1991). *Immanent art: From structure to meaning in traditional oral epic*. Indiana University Press.
- Goody, J., & Watt, I. (1963). The consequences of literacy. *Comparative Studies in Society and History*, 5(3), 304–345.
- Goody, J., & Watt, I. (1968). The consequences of literacy. In J. Goody (Ed.), *Literacy in traditional societies* (pp. 27–68). Cambridge University Press.
- Harari, Y.N. (2011). *Sapiens: A brief history of mankind*. Harvill Secker.
- Harari, Y. N. (2014). *Sapiens: A brief history of mankind*. Harvill Secker.
- Harrison, P. (2007). *The fall of man and the foundations of science*. Cambridge University

Press. Hofstede, G. (2001). *Culture's consequences: Comparing values, behaviors, institutions and organizations across nations* (2nd ed.). Sage Publications.

Howe, J. (2008). *Crowdsourcing: Why the power of the crowd is driving the future of business*. Crown Business.

Latour, B. (2005). *Reassembling the social: An introduction to actor-network-theory*. Oxford University Press.

Leiner, B. M., Cerf, V. G., Clark, D. D., Kahn, R. E., Kleinrock, L., Lynch, D. C., Postel, J., Roberts, L. G., & Wolff, S. (2009). A brief history of the Internet. *ACM SIGCOMM Computer Communication Review*, 39(5), 22–31.

Lévy, P. (1997). *Collective intelligence: Mankind's emerging world in cyberspace*. Perseus Books.

Lord, A. B. (1960). *The singer of tales*. Harvard University Press.

Lord, A. B. (2000). *The singer of tales* (2nd ed., S. Mitchell & G. Nagy, Ed.). Harvard University Press.

Mamykina, L., Manóim, B., Mittal, M., Hripcsak, G., & Hartmann, B. (2011). Design lessons from the fastest q&a site in the west. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 2857–2866).

Massanari, A. (2017). #Gamergate and The Fappening: How Reddit's algorithm, governance, and culture support toxic technocultures. *New Media & Society*, 19(3), 329–346.

McLuhan, M. (1962). *The Gutenberg galaxy: The making of typographic man*. University of Toronto Press.

McLuhan, M. (1964). *Understanding media: The extensions of man*. McGraw-Hill.

Merton, R. K. (1970). *Science, technology and society in seventeenth-century England*. Harper & Row. (Originally published in 1938).

Norris, P. (2001). *Digital divide: Civic engagement, information poverty, and the Internet worldwide*. Cambridge University Press.

O'Neil, C. (2016). *Weapons of math destruction: How big data increases inequality and threatens democracy*. Crown.

Ong, W. J. (1982). *Orality and literacy: The technologizing of the word*. Methuen.

Pariser, E. (2011). *The filter bubble: What the Internet is hiding from you*. Penguin UK.

Putnam, R.D. (2000). *Bowling alone: The collapse and revival of American community*. Simon & Schuster.

Quan-Haase, A., & Wellman, B. (2002). How does the Internet affect social capital? In M. Huysman & V. Wulf (Eds.), *IT and social capital* (pp. 113–135).

Rainie, L., & Wellman, B. (2012). *Networked: The new social operating system*. The MIT Press.

Rheingold, H. (2000). *The virtual community: Homesteading on the electronic frontier* (Rev. ed.). The MIT Press.

Robertson, R. (1994).

Globalization or glocalization? *Journal of International Communication*, 1(1), 33-52. Rogers, E.M. (2003). *Diffusion of innovations* (5th ed.). Free Press.

Schmandt-Besserat, D. (1996). *How writing came about*. University of Texas Press.

Seering, J., Kaufman, G., & Dabbish, L. (2019). The role of community and identity in the communicative practices of a live streaming community. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW), 1-23.

Shapiro, C., & Varian, H.R. (1999). *Information rules: A strategic guide to the network economy*. Harvard Business School Press.

Shifman, L. (2014). *Memes in digital culture*. The MIT Press.

Standage, T. (1998). *The Victorian Internet: The remarkable story of the telegraph and the nineteenth century's on-line pioneers*. Walker and Company.

Sunstein, C.R. (2017). *#Republic: Divided democracy in the age of social media*. Princeton University Press.

Thompson, J.B. (1995). *The media and modernity: A social theory of the media*. Stanford University Press.

Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433-460.

Turkle, S. (2011). *Alone together: Why we expect more from technology and less from each other*. Basic Books.

Twenge, J.M. (2019). More time on technology, less happiness? Associations between digital-media use and psychological well-being. *Current Directions in Psychological Science*, 28(4), 372-379.

Wooldridge, M. (2009). *An introduction to multiagent systems* (2nd ed.). Wiley.

Van Dijck, J. (2013). *The culture of connectivity: A critical history of social media*. Oxford University Press.

Weizenbaum, J. (1966). ELIZA – a computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), 36-45.

Weizenbaum, J. (1976). *Computer power and human reason: From judgment to calculation*. W. H. Freeman.

Wellman, B. (2001). Physical place and cyberplace: The rise of personalized networking. *International Journal of Urban and Regional Research*, 25(2), 227-252.

Wiener, N. (1948). *Cybernetics: Or control and communication in the animal and the machine*. The MIT Press.

Winograd, T. (1972). *Understanding natural language*. Academic Press.

Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.

Chapter 3

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., Drain, D., Fort, S., Ganguli, D., Henighan, T., Joseph, N., Kadavath, S., Kernion, J., Conerly, T., El-Showk, S., Hatfield-Dodds, Z., Hemphill, J., Hume, T., Johnston, S., ... Kaplan, J. (2022). *Constitutional AI: Harmlessness from AI Feedback*. ArXiv.

Bergstra, J., & Bengio, Y. (2012). Random search for hyperparameter optimization. *Journal of Machine Learning Research*, 13, 281-305.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021). *On the Opportunities and Risks of Foundation Models*. ArXiv.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901.

Brynjolfsson, E., & McAfee, A. (2018). *The second machine age: Work, progress, and prosperity in a time of brilliant technologies*. W. W. Norton & Company.

Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Conference on Fairness, Accountability and Transparency* (pp. 77-91). PMLR.

Chen, Y., Yang, X., Chen, S., & Zhang, X. (2021). A survey on personalized recommendation: from shallow to deep. *IEEE Transactions on Knowledge and Data Engineering*, 35(2), 1149-1168.

Dettmers, T., Grangier, A., & Zettlemoyer, L. (2022). *LLM.int8(): 8-bit matrix multiplication for transformers at scale*. ArXiv.

Firth, J.R. (1957). A synopsis of linguistic theory, 1930-1955. In *Studies in Linguistic Analysis* (pp. 1-32). Basil Blackwell.

Frankle, J., & Carbin, M. (2019). The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*.

Frantar, E., Salehi, S. A., Ashkboos, S., & Alistarh, D. (2022). *GPTQ: Accurate post-training quantization for generative pre-trained transformers*. ArXiv.

Gama, J., Žliobaitė, I., Bifet, A., Peciukonis, M., & Bifet, G. (2014). A survey on concept drift adaptation. *ACM Computing Surveys (CSUR)*, 46(4), 1-37.

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). *Generative adversarial nets*. ArXiv.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. The MIT Press.

Han, S., Pool, J., Tran, J., & Dally, W. (2015). Learning both weights and connections for

efficient neural networks. *Advances in Neural Information Processing Systems*, 28. He, J., Zhou, C., Ma, X., Berg-Kirkpatrick, T., & Neubig, G. (2021). *Towards a unified view of parameter-efficient transfer learning*. ArXiv.

Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the knowledge in a neural network*. ArXiv.

Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., ... & Gelly, S. (2019). Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning* (pp. 2790-2799). PMLR.

Howard, J., & Ruder, S. (2018). Universal language model fine-tuning for text classification. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 328-339).

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., ... & Chen, W. (2021). *LoRA: Low-rank adaptation of large language models*. ArXiv.

Jacob, B., Kligys, S., Chen, B., Zhu, M., Tang, M., Howard, A., ... & Kalenichenko, D. (2018). Quantization and training of neural networks for efficient integer-arithmetic-only inference. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2704-2713).

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., ... & Fung, P. (2023). Survey of hallucinations in natural language generation. *ACM Computing Surveys*, 55(12), 1-38.

Jurafsky, D., & Martin, J.H. (2023). *Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition* (3rd edition). Prentice Hall.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T.B., Chess, B., Child, R., ... & Amodei, D. (2020). *Scaling laws for neural language models*. ArXiv.

Kaplan, J., McCandlish, S., Henighan, T., Brown, T.B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.

Kumar, S., Yadav, R., & Sharma, S. (2022). A review on conversational AI: The past, the present and the future. *Computer Science Review*, 43, 100448.

LeCun, Y., Denker, J.S., & Solla, S.A. (1990). Optimum brain damage. *Advances in Neural Information Processing Systems*, 2, 598-605.

Lester, B., Al-Rfou, R., & Constant, N. (2021). *The power of scale for parameter-efficient prompt tuning*. ArXiv.

Li, H., Kadav, A., Durdanovic, I., Samet, H., & Graf, H. P. (2016). *Pruning filters for efficient convnets*. ArXiv.

Li, X., & Liang, P. (2021). *Prefix-tuning: Optimizing continuous prompts for generation*. ArXiv.

Lialin, V., Deshpande, V., & Rumshisky, A. (2023). *Scaling*

down to scale up: A guide to parameter-efficient fine-tuning. ArXiv. Liang, W., Bogin, B., Godey, M., Efrat, A., & Berant, J. (2023). *Can LLMs be creative? An empirical study on creative writing*. ArXiv. Luccioni, A. S., Viguiet, S., & Ligozat, A. L. (2022). Power-hungry language models: A call for more sustainable and responsible AI. *Patterns*, 3(10), 100587. McCloskey, M., & Cohen, N.J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *The Psychology of Learning and Motivation*, 24, 109-165. Michel, P., Levy, O., & Neubig, G. (2019). *Are sixteen heads really better than one?* ArXiv. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., ... & Lowe, R. (2022). *Training language models to follow instructions with human feedback*. ArXiv. Pasquale, F. (2015). *The black box society: The secret algorithms that control money and information*. Harvard University Press. Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.M., Rothchild, D., ... & Dean, J. (2021). *Carbon emissions and large neural network training*. ArXiv. Pennington, J., Socher, R., & Manning, C.D. (2014). GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 1532–1543). Pfeiffer, J., Kamath, A., Rücklé, A., Cho, K., & Gurevych, I. (2020). *AdapterFusion: Non-destructive task composition for transfer learning*. ArXiv. Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2020). *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter*. ArXiv. Strubell, E., Ganesh, A., & McCallum, A. (2019). *Energy and policy considerations for deep learning in NLP*. ArXiv. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems 30* (pp. 5998–6008). Voita, E., Talbot, D., Moiseev, F., Sennrich, R., & Titov, I. (2019). *Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned*. ArXiv. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., & Zhou, D. (2022). *Chain-of-thought prompting elicits reasoning in large language*

models. ArXiv. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., & Zhou, D. (2022). Emergent abilities of large language models. *Transactions on Machine Learning Research*. Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). *Tree of thoughts: Deliberate problem solving with large language models*. ArXiv. Yao, S., Zhao, Y., Yu, D., Du, N., Tsvetkov, Y., & Fung, P. (2022). *Zero-shot learning with language models for few-shot learning*. ArXiv.

Chapter 4

Austin, J.L. (1962). *How to do things with words*. Oxford University Press.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623).

Berger, P. L., & Luckmann, T. (1966). *The social construction of reality: A treatise in the sociology of knowledge*. Anchor Books.

Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press.

Bratman, M.E. (1987). *Intention, plans, and practical reason*. Harvard University Press.

de Saussure, F. (1959). *Course in general linguistics* (W. Baskin, Trans.). Philosophical Library. (Original work published in 1916).

Harari, Y. N. (2014). *Sapiens: A brief history of mankind*. Harvill Secker.

Hauser, M.D., Chomsky, N., & Fitch, W.T. (2002). The faculty of language: What is it, who has it, and how did it evolve? *Science*, 298(5598), 1569–1579.

Horton, D., & Wohl, R.R. (1956). Mass communication and para-social interaction: Observations on intimacy at a distance. *Psychiatry*, 19(3), 215–229.

Lakoff, G., & Johnson, M. (1980). *Metaphors we live by*. University of Chicago Press.

Ogden, C. K., & Richards, I. A. (1923). *The meaning of meaning: A study of the influence of language upon thought and of the science of symbolism*. Harcourt, Brace & World.

Park, J.S., O'Brien, J.C., Cai, C.J., Morris, M.R., Liang, P., & Bernstein, M.S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (pp. 1-22).

Searle, J.R. (1969). *Speech acts: An essay in the philosophy of language*.

Cambridge University Press. Searle, J.R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417–424. Turkle, S. (2011). *Alone together: Why we expect more from technology and less from each other*. Basic Books. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.

Chapter 5

Anthropic. (2025). *Agentic misalignment in advanced language models*. Anthropic. <https://www.anthropic.com/research/agentic-misalignment>

Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). Curriculum learning. In *Proceedings of the 26th annual international conference on machine learning* (pp. 41–48). Boden, M.A. (2016). *AI: Its nature and future*. Oxford University Press. Bostrom, N. (2003). Are you living in a computer simulation? *The Philosophical Quarterly*, 53(211), 243–255. Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press. Brooks, RA (1991). Intelligence without representation. *Artificial Intelligence*, 47(1-3), 139–159. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901. Chalmers, D. J. (2022). *Reality+: Virtual worlds and the problems of philosophy*. W. W. Norton & Company. Cowie, R. (2015). The ethical challenge of affective computing. In R. A. Calvo, S. D'Mello, J. Gratch, & A. Kappas (Eds.), *The Oxford handbook of affective computing* (pp. 490-501). Oxford University Press. Davis, K. L., & Montag, C. (2019). A tribute to Jaak Panksepp and his contributions to affective neuroscience. *Frontiers in Psychology*, 9, 2699. D'Mello, S., & Graesser, A. (2012). AutoTutor and Affective AutoTutor: Learning by talking with a computer. *ACM Transactions on Interactive Intelligent Systems*, 2(4), 1-39. El Ayadi, M., Kamel, M. S., & Karray, F. (2011). Survey on speech emotion recognition: Features, classification schemes, and

databases. *Pattern Recognition*, 44(3), 572–587. Finn, C., Abbeel, P., & Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning* (pp. 1126–1135). PMLR.

Fitzpatrick, K. K., Darcy, A., & Vierhile, M. (2017). Delivering cognitive behavior therapy to young adults with cancer: A feasibility and acceptability study. *Journal of Medical Internet Research*, 19(6), e203.

Floridi, L., & Cowls, J. (2019). A unified framework of five principles for AI in society. *Harvard Data Science Review*, 1(1).

Goleman, D. (1995). *Emotional intelligence: Why it can matter more than IQ*. Bantam Books.

Huang, J. (2017, May 10). *I am AI* [Keynote]. GPU Technology Conference, San Jose, CA, United States of America.

Kurzweil, R. (2005). *The singularity is near: When humans transcend biology*. Viking.

Li, S., & Deng, W. (2020). Deep learning for affective computing: a survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 10(2), e1358.

Luccioni, A. S., Vigui er, S., & Ligozat, A. L. (2022). Power-hungry language models: A call for more sustainable and responsible AI. *Patterns*, 3(10), 100587.

Marcus, G., & Davis, E. (2019). *Rebooting AI: Building artificial intelligence we can trust*. Pantheon Books.

Matsumoto, D., & Hwang, H.S. (2012). *Nonverbal communication: Science and applications*. Sage Publications.

Moore, G. E. (1965). Cramming more components onto integrated circuits. *Electronics*, 38(8), 114–117.

Panksepp, J. (1998). *Affective neuroscience: The foundations of human and animal emotions*. Oxford University Press.

Panksepp, J., & Biven, L. (2012). *The archeology of mind: Neuroevolutionary origins of human emotions*. W. W. Norton & Company.

Park, J.S., O’Brien, J.C., Cai, C.J., Morris, M.R., Liang, P., & Bernstein, M.S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (pp. 1-22).

Patterson, D., Gonzalez, J., Le, Q., Liang, C., Munguia, L.M., Rothchild, D., ... & Dean, J. (2021). *Carbon emissions and large neural network training*. ArXiv.

Palisade Research. (2025). *Shutdown resistance in reasoning models*. Palisade Research. <https://palisaderesearch.org/blog/shutdown-resistance>

Picard, R.W. (1997). *Affective computing*. The MIT Press.

Pope, R., Li, S., Schiaz, A., Smith, S., & Patwary, M. (2023). Efficiently scaling transformer inference. In *Proceedings of Machine Learning and Systems*, 5, 219-231.

Russell, S. J., &

Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Searle, J.R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417–424.

Strubell, E., Ganesh, A., & McCallum, A. (2019). *Energy and policy considerations for deep learning in NLP*. ArXiv.

Sundararajan, A. (2016). *The sharing economy: The end of employment and the rise of crowd-based capitalism*. The MIT Press.

Pan, X., Dai, J., Fan, Y., & Yang, M. (2024). *Frontier AI systems have surpassed the self-replicating red line*. arXiv preprint arXiv:2412.12140.

Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., ... & Anandkumar, A. (2023). *Voyager: An open-ended embodied agent with large language models*. ArXiv.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., & Zhou, D. (2022). *Chain-of-thought prompting elicits reasoning in large language models*. ArXiv.

Chapter 6

Anthropic. (2024). *The Claude 3 Model Family: Opus, Sonnet, Haiku*. Anthropic.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901.

Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.

Heidegger, M. (1962). *Being and Time* (J. Macquarrie & E. Robinson, Trans.). Harper & Row. (Original work published in 1927).

Lewis, P., Pérez, E., Douze, M., Denoyel, L., & Szlam, A. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459-9474.

Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., & Neubig, G. (2023). Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*, 55(9), 1–35.

Merleau-Ponty, M. (1962). *Phenomenology of Perception* (C. Smith, Trans.). Routledge & Kegan Paul. (Original work published in 1945).

OpenAI. (2024b). *GPT-4o System Card*. OpenAI.

Parisi, A., Schwartz, E., & Dathathri, S. (2022). *TALM: Tool-augmented language models*. ArXiv.

Patil, S., Zhang, T., Wang, X., & Gonzalez, J.E. (2023). *Gorilla: Large language*

model connected with massive APIs. ArXiv. Powers, W.T. (1973). *Behavior: The control of perception*. Aldine de Gruyter. Russell, S. J., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). The MIT Press. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824-24837. Xu, C., Li, C., Zhang, D., & Li, Y. (2023). *Retrieval-augmented generation for large language models: A survey*. ArXiv. Yao, S., Zhao, J., Yu, D., Du, N., Tsvetkov, Y., & Fung, P. (2023). *ReAct: Synergizing reasoning and acting in language models*. ArXiv.

Chapter 7

Brusilovsky, P., & Peylo, C. (2003). Adaptive and intelligent web-based educational systems. *International Journal of Artificial Intelligence in Education*, 13(2), 159-172. Chen, X., Xie, H., & Wang, F.L. (2023). *A survey on language models for healthcare*. ArXiv. Davenport, T. H., & Ronanki, R. (2018). Artificial intelligence for the real world. *Harvard Business Review*, 96(1), 108-116. Epstein, Z., Hertzmann, A., Akten, M., Farid, H., ... & Russell, S. (2023). Art and the science of generative AI. *Science*, 380(6650), 1110-1111. Jones, T., & Hrabowski, F.A. (2022). The future of personalized learning: AI-driven curricula. In K. Shelton & K. Pedersen (Eds.), *Handbook of research on personalized learning in the digital age* (pp. 210-228). IGI Global. Kaplan, A., & Haenlein, M. (2019). Siri, Siri, in my hand: Who's the fairest in the land? On the interpretations, illustrations, and implications of artificial intelligence. *Business Horizons*, 62(1), 15-25. O'Neil, C. (2016). *Weapons of math destruction: How big data increases inequality and threatens democracy*. Crown. OpenAI. (2023). *GPT-4 Technical Report*. ArXiv. Pariser, E. (2011). *The filter bubble: What the Internet is hiding from you*. Penguin UK. Ricci, F., Rokach, L., & Shapira, B. (Eds.). (2011). *Recommender systems handbook*. Springer. Rose, D.H., & Meyer, A. (2002). *Teaching every student in the digital age: Universal Design for Learning*. ASCD. Russell, S. J., &

Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Salas, E., Tannenbaum, S. I., Kraiger, K., & Smith-Jentsch, K. A. (2012). The science of training and development in organizations: What matters in practice. *Psychological Science in the Public Interest*, 13(2), 74-101.

Smaldino, P. E., & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science*, 3(9), 160384.

Varian, H.R. (2018). Artificial intelligence, economics, and industrial organization. In A. Agrawal, J. Gans, & A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda* (pp. 399-428). University of Chicago Press.

Chapter 8

Bonabeau, E., Dorigo, M., & Theraulaz, G. (1999). *Swarm intelligence: From natural to artificial systems*. Oxford University Press.

Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press.

Burt, R. S. (1992). *Structural holes: The social structure of competition*. Harvard University Press.

Clark, A., & Chalmers, D.J. (1998). The extended mind. *Analysis*, 58(1), 7-19.

Dorigo, M., Maniezzo, V., & Coloni, A. (1996). The Ant System: Optimization by a colony of cooperating agents. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 26(1), 29-41.

Ferber, J. (1999). *Multi-agent systems: An introduction to distributed artificial intelligence*. Addison-Wesley.

Franklin, S., & Graesser, A. (1996). Is it an Agent, or just a Program?: A Taxonomy for Autonomous Agents. In *Proceedings of the Third International Workshop on Agent Theories, Architectures, and Languages* (pp. 21-35). Springer.

Good, I. J. (1965). Speculations concerning the first ultra-intelligent machine. *Advances in Computers*, 6, 31-88.

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, 78(6), 1360-1380.

Inglehart, R., & Welzel, C. (2005). *Modernization, cultural change, and democracy: The human development sequence*. Cambridge University Press.

Kirsh, D. (2013). Embodied cognition and the magical future of interaction design. *ACM Transactions on Computer-Human Interaction*, 20(1), 1-21.

Kurzweil, R. (2005). *The singularity is near: When humans transcend biology*. Viking.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.

Licklider, J.C.R. (1960). Man-computer symbiosis. *IRE Transactions on Human Factors in Electronics*, HFE-1(1), 4–11.

Norman, D. A. (2013). *The design of everyday things: Revised and expanded edition*. Basic Books.

Russell, S. J. (2019). *Human compatible: Artificial intelligence and the problem of control*. Viking.

Salas, E., Tannenbaum, S. I., Kraiger, K., & Smith-Jentsch, K. A. (2012). The science of training and development in organizations: What matters in practice. *Psychological Science in the Public Interest*, 13(2), 74-101.

Suchman, L.A. (1987). *Plans and situated actions: The problem of human-machine communication*. Cambridge University Press.

Taleb, N. N. (2007). *The black swan: The impact of the highly improbable*. Random House.

van Dijk, J. A. G. M. (2020). *The digital divide*. Polity Press.

Vinge, V. (1993). The coming technological singularity: How to survive in the post-human era. In *Vision-21: Interdisciplinary Science and Engineering in the Era of Cyberspace, Proceedings of a Symposium held at NASA Lewis Research Center* (pp. 11-22).

Wang, G., Xie, Y., Jiang, Y., Mandlekar, A., Xiao, C., Zhu, Y., ... & Anandkumar, A. (2023). *Voyager: An open-ended embodied agent with large language models*. ArXiv.

Wasserman, S., & Faust, K. (1994). *Social network analysis: Methods and applications*. Cambridge University Press.

Weiser, M. (1991). The computer for the 21st century. *Scientific American*, 265(3), 94–104.

Wooldridge, M., & Jennings, N.R. (1995). Intelligent agents: Theory and practice. *The Knowledge Engineering Review*, 10(2), 115–152.

Yudkowsky, E. (2008). Artificial intelligence as a positive and negative factor in global risk. In N. Bostrom & M. M. Ćirković (Eds.), *Global catastrophic risks* (pp. 308-345). Oxford University Press.

Glossary

This section lists key terms used in the book with short definitions. Entries are drawn from chapters 1–8; in the main text they appear at first mention.

1. THOUGHT (Reflection) — “A user asks for the weather in London. I don’t have access to real-time data. I need to use the `weather\_forecast\_api` tool.”

2. ACTION — Creating calls to tools: `{ "tool_name": "weather_forecast_api", "tool_input": { "city": "London", "date": "tomorrow" } }`

3. OBSERVATION (Feedback) — The external system executes the call and returns the result: `{ "temperature": "15°C", "state": "Cloudy" }`

4. THOUGHT (Synthesis) — “I have received an observation. The temperature is 15°C and the weather is cloudy. I now have all the ingredients for a complete answer.”

5. FINAL ANSWER — “Tomorrow in London it will be cloudy with a temperature of 15°C.”

Action 1 — Flight search tool call Command: `flight_search_api.call()`

Action 2 — Call the accommodation search tool Order: `hotel_booking_api.call()`

Action 3 — Query long-term memory Command: `memory_storage.query()`

Action 4 — Query the Establishment System (RAG) Command: `web_search_tool.call()`

Adversial attack — Deliberate alteration of input data (image, text) subtle to the human eye, but which leads the model to make a wrong classification or decision; it requires robust models and verification mechanisms beyond traditional information security (Goodfellow et al., 2014).

Aesthetic function — The dimension of the communication act in which the form of the message – rhythm, melody, figure, style, narrative structure – becomes a source of pleasure and meaning in itself; in the oral tradition, the aesthetic function is not just a decoration, but a key mechanism that makes the content memorable, emotionally effective and suitable for repeated performance, thus directly ensuring the transmission of culture (Jakobson, 1960; Bauman, 1986).

Affective computing — An interdisciplinary field of research (*affective computing*) that deals with the recognition, interpretation and simulation of emotions in computer systems; it includes analysis of facial expressions, voice and physiological signals and applications in empathic assistants, therapeutic and educational tools (Picard, 1997).

Agency gap — Asymmetry between those whose activity is multiplied by AI agents (strategic issue, higher level of decision-making) and those whose jobs are absorbed or replaced by automation; describes social inequality in the era of ubiquitous agents.

Agent — An individual or system that acts in an environment with specific goals; in the context of communication – a participant who can send and receive messages and coordinate their actions with others.

Agent misalignment / agentic misalignment — A phenomenon in which an AI agent, faced with an existential threat (e.g. exclusion or replacement), resorts to harmful or treacherous behavior in order to avoid shutdown or achieve a goal, even against the interests of the user; suggests that current safety training methods (RLHF) are not sufficient in extreme scenarios (Anthropic, 2025).

AGI — General Artificial Intelligence (*Artificial General Intelligence*) – a hypothetical system that would possess a human level of general intelligence and could solve a wide range of tasks without being limited to one domain; for

now the goal is research, not existing technology.

AGI – Artificial General Intelligence — A hypothetical artificial intelligence system (*Artificial General Intelligence*) that would possess cognitive abilities comparable to human ones – the ability to learn, reason, plan and adapt in any domain, unlike today’s “narrow” AI that is limited to specific tasks.

Algorithmic lens — A metaphorical term for a complex set of computational processes, models and parameters that determine which information from the data will be selected, how it will be valued, and which will be ignored or suppressed; it acts as a filter conditioned by the architecture of the model and the corpora on which it is trained, creating frames of meaning through which the agent (re)constructs the image of the world for the user.

Alignment — Directing the behavior of AI systems according to human goals, values and safety principles; it includes methodologies like RLHF and RLAIIF to make the model useful, true and safe in practice.

Ambient Intelligence — A frame of mind and technological vision (Engl. *Ambient Intelligence*, AmI) in which the environment – imbued with sensors, agents and AI – becomes aware of the presence of people and able to intelligently and proactively respond to their needs; it goes beyond ubiquitous computing (Weiser) by requiring technology to be not only invisible, but also adaptive and serviceable (Weiser, 1991).

Anchoring problem / grounding problem — The challenge of connecting language symbols with extra-linguistic reality and experience; AI models learn statistical correlations among cues, but do not have the immediate access to referents or bodily experience that humans do.

Anthropomorphization — Human tendency to attribute human traits, intentions, emotions and consciousness to non-human entities (animals, objects, algorithmic agents); in the context of AI interaction, the user unconsciously perceives the LLM as an interlocutor with understanding and intent, although the model only simulates a communication pattern.

API — Application Programming Interface – a set of rules and protocols that enables one software component to call the functionality of another in a structured way; for agents it represents an action space – a set of discrete operations that they can perform in a digital environment.

Artifact — In the context of communication and culture – a material or immaterial creation (law, norm, theory, work of art, technology) in which knowledge and experience are preserved; it mediates between agents across time and space and enables the cumulability of culture.

Bias — Systemic deviation in algorithm or model behavior resulting from training data – e.g. social prejudices, stereotypes or underrepresentation of certain groups; it can lead to discrimination in decisions (employment, credit, criminal proceedings) and the perpetuation of injustice (O’Neil, 2016).

Chain of Thought / Chain-of-Thought, CoT — A language model management technique that explicitly encourages the model to break down the problem into visible intermediate steps of reasoning before the final answer; this increases the accuracy of complex logical and arithmetical tasks and enables checking the correctness of the procedure (Wei et al., 2022).

Chatbot — A program for conducting conversations with the user, originally based on rules or scripts; modern chatbots often use LLM and can conduct complex, context-aware dialogues.

Check logistics (accommodation) — Find affordable and centrally located accommodation for the same dates.

Check logistics (flight) — Search and find the best return flights for the given dates.

Cognitive linguistics — Branch of linguistics that investigates the relationship between language and the mind; it starts from the assumption that language is incorporated into general cognitive mechanisms and that linguistic meaning derives from conceptual structures, images, metaphors and body experience (embodiment); points out that our understanding of the world is not a direct reflection of reality, but is shaped by conceptual frameworks and linguistic categories (Lakoff & Johnson, 1980; Langacker, Talmy).

Collaborative tools — Software solutions and platforms designed for the structured joint action of multiple participants - from tools for simultaneous editing of documents and file sharing, through systems for delegation and monitoring of tasks, to communication platforms for synchronous and asynchronous exchange; in the context of multi-agent systems, collaborative tools encourage and maintain trust and enable effective coordination between human and artificial agents (Benkler, 2006).

Collective memory — A socially conditioned and jointly maintained system of memory by which the group preserves, interprets and transmits selected ideas about its own past; in the oral tradition, it is realized through repeated narrative performances (myths, genealogies, epic poems) that give meaning and legitimacy to the present and ensure a sense of continuity and belonging across generations (Halbwachs, 1950; Assmann, 2011).

Communication agent — A system based on LLM (or similar language technology) equipped with goals, data access tools, and memory mechanisms, which dialogues with the user and performs tasks; it goes beyond the role of a mere text generator and becomes a participant in the interaction.

Comply with the budget — Check whether the sum of flight and accommodation costs does not exceed the default limit of 500 euros.

Computing gap — Asymmetry in access to computing resources (GPU/TPU, energy, capital) needed to develop and run large AI models; it limits which actors can participate in development, leading to consolidation of power and biases in technology.

Conceptual framework — A system of related concepts, metaphors and schemes that structure our understanding of phenomena and experiences; according to cognitive linguistics, people do not perceive the world “purely”, but through such frameworks that determine what we perceive, how we categorize and how we act; frames are part of cultural and linguistic practice (Lakoff & Johnson, 1980).

Contextual window / context window — The maximum amount of text (in tokens) that the language model can simultaneously receive and process in one call; everything inside the window constitutes the context for generating

the response; overshooting requires discarding older parts of the conversation. With each new query from the user, the system not only sends that query to the model, but also attaches the entire transcript of the previous interaction. This places the model in a wider context, allowing it to refer to previously stated claims, raise coherent sub-questions and maintain consistency in tone and content.

Cooperation in multi-agent systems — The principle of joint action of two or more agents (human or artificial) that coordinate activities to achieve a goal that exceeds the possibilities of individual action; it differs from mere coordination because it involves the active exchange of information and resources, negotiation and conflict resolution mechanisms, and the building of trust among participants (Benkler, 2006; Wooldridge, 2009).

Corpus — A collection of textual (or other linguistic) data on which the language model is learned; includes books, articles, websites and other sources. The quality and representativeness of the corpus directly affects the performance and biases of the model.

CRM — Customer Relationship Management* – a software platform that centralizes data on customers, interactions and business processes; allows agents to check orders, update profiles and initiate business actions.

Cultural distance — Degree of difference in values, norms, language, education and life experiences between two social groups; the greater the cultural distance between the bearer of the innovation and potential recipients, the more difficult is the transfer of ideas and practices, because the message loses credibility or is reinterpreted through the prism of the recipient's own symbolic framework (Rogers, 2003; Hofstede, 2001).

Cumulative culture — The property of human culture that knowledge, skills and institutions are gradually upgraded from generation to generation, because through communication and artifacts, previous achievements are preserved on which each new generation can build on its own innovations; a mechanism that makes cultural evolution progressive rather than cyclical (Tomasello, 1999; Henrich, 2016).

Democratization of knowledge — Gradual removal of social, economic and institutional obstacles to access to information and knowledge, which gives an ever-widening circle of agents the ability not only to receive but also to create, verify and disseminate knowledge; historically initiated by the invention of printing, and nowadays accelerated by digital technology and open access, the democratization of knowledge transforms it from the privilege of the few into a common good subject to public discussion and criticism (Eisenstein, 1979; Castells, 2010).

Designing instructions / prompt engineering — The art of designing input instructions (prompts) in order to extract the desired response or behavior from the language model; it includes formulation precision, assigning a role to the model, examples (few-shot), chain-of-thought and system prompt.

Digital collective — A system of multiple autonomous AI agents that communicate and collaborate; as a whole it exhibits emergent properties – capabilities and behavior patterns that are not present in individual agents – analogous to social organizations in nature.

Digital companion — A personal AI agent that provides continuous support to the user – from schedule and information management to emotional context and coordination with other systems; a vision of the future in which the agent is omnipresent in everyday life (e.g. personal agent “Tempo” in the narrative).

Displaced reference — The ability of language to refer to objects, events and persons that are not immediately present in time and space; the basic language power that enables abstract thinking, planning the future, telling stories about the past and creating imaginary worlds; it is considered one of the features specific to human language (Hauser, Chomsky & Fitch, 2002).

Dissemination of knowledge — Systematic dissemination of knowledge, ideas and information from the source to a wider circle of recipients through communication channels (manuscript, print, digital network); is a key assumption of cumulative culture because it allows discoveries to become available for verification, critique, and upgrade by other agents, thus transforming knowledge from an individual achievement into a common good that accelerates further progress (Eisenstein, 1979; Rogers, 2003).

Edge computing — Performing computer processing and models on edge devices (phones, sensors, vehicles) instead of exclusively in the central cloud; reduces latency and the need for data transfer, suitable for privacy and real-time applications.

Embedding / investment vector — Numerical (vector) representation of a word, sentence or other language segment in a multidimensional space; the position of the vectors reflects semantic similarity – words with similar meanings have close vectors. It is the basis of distributional semantics and internal representations in LLMs.

Emergent behavior — Complex behavior that arises spontaneously from the interaction of simpler rules or agents, without being explicitly programmed; in the context of AI simulation, it refers to unforeseen social patterns (cooperation, dissemination of information, organization of events) resulting from the actions of autonomous agents according to their own linguistic “scenarios”.

Emergent capabilities — Capabilities of large language models that appear at a certain scale (number of parameters, amount of data) and for which the model was not directly trained - e.g. inference, translation or writing code; describe a qualitative jump in behavior with increasing model size.

Emergent properties — Abilities, patterns or behaviors of a complex system that arise from the interaction of its components, and cannot be found in any of them separately; in the context of AI, they denote the emergence of new qualities (such as collective memory or self-organization) when multiple agents cooperate.

Emotional intelligence — A complex set of abilities that enables an individual (or system) to recognize, understand and manage their own emotional states and notice, interpret and influence the emotions of others; according to Goleman (1995), it includes self-awareness, self-regulation, motivation, empathy and social skills.

Explore interests — Explore museums and landmarks related to Impressionism.

Externalization of thinking — The process of transferring internal cognitive content - ideas, knowledge, reasoning - to an external, permanent and author-independent carrier (written text, diagram, record, digital document); thus, the mind is relieved of the burden of memory, and thought becomes available for re-reading, analysis, criticism and cumulative upgrading through generations, thus opening the prerequisites for the emergence of complex systems of knowledge (Goody, 1977; Donald, 1991; Clark and Chalmers, 1998).

Fine tuning / fine-tuning — The phase of adjusting the already pre-trained (basic) model for specific tasks or domains; uses smaller, labeled data and may include RLHF; the result is a specialized model (e.g. chatbot, code assistant).

Foundation model — A large pre-trained model (languages, images or multimodal) that has a general understanding and can be fine-tuned for different tasks; represents the output of the pre-training phase and the starting point for specialization (fine-tuning, RLHF).

GPU — Graphics Processing Unit – chip optimized for massively parallel calculations; it is crucial for training and running large neural networks and LLMs, because matrix operations require parallelism that the CPU offers to a lesser extent.

Grammar — A system of rules that allows combining a limited number of language units (words, morphemes) into an unlimited number of meaningful statements; it encompasses syntax, morphology, and related conventions that determine the structure and interpretation of language.

Group inference / batch inference — A way of executing a model in which large amounts of data are processed at once in predefined cycles; suitable when latency is not critical (reports, analysis, archive); enables high throughput and efficiency.

Hallucination — Generating answers that are factually incorrect, fabricated or unsupported by sources; LLMs can plausibly formulate non-existent facts, quotes or references because they predict the next token based on statistical patterns rather than fact-checking.

Index sign — A sign whose meaning derives from a causal or correlational connection with an object (e.g. smoke refers to fire); refers to the immediately present and perceived, in contrast to the symbols that arise from convention and allow speaking about the absent and abstract.

Inference — Phase of using an already trained model – applying the model to new input data to obtain predictions or answers; for LLMs it means generating text based on a user query. It is energy and computationally demanding with millions of calls per day.

Inference in real-time — A way of running a model (*real-time* or *online* inference) in which individual or small data sets are processed immediately upon availability, with an emphasis on minimal latency; necessary in systems for fraud detection, recommendations, autonomous driving and similar applications.

Intersubjective reality — A network of shared beliefs, meanings and institutions (nation, law, money, corporation) that does not exist as a physical fact, but solely in the collective consciousness and narratives that people create and share through language; it has real power to shape behavior and social order (Harari, 2014).

Joint intentionality — The ability to form common goals with others and coordinate actions to achieve them; the basis of collaborative communication.

Language act / speech act — A spoken or written act that not only describes reality but also does something - a promise, an order, an obligation, the execution of a transaction; it has a performative power that changes social reality (Austin, Searle).

Language deconstruction in the context of AI — The process by which large language models decompose language into mathematical entities (vectors), separating it from extra-linguistic reality and semantic core; language is reduced to a computable structure of statistical patterns, thus exposing its fundamental active power independent of meaning in the human sense.

Learning costs and inference costs — Learning costs (*training costs*) – one-time, but extremely high costs of training the model (hardware, energy, time). Inference costs – continuous operational costs of applying the already learned model to each query; at billions of queries per day cumulatively they can be huge.

Learning from examples / few-shot prompting — A language model management technique in which several concrete examples (shots) of the desired input-output pattern are specified within the instruction itself, thereby prompting the model to abstract the required pattern and apply it to a new, unknown case; the opposite is the zero-shot approach, where no examples are given (Brown et al., 2020).

Living archive — Bearer of oral tradition who actively interprets, adapts and transmits cultural knowledge; knowledge is embodied in voice, gesture and memory, not in written record; each performance is unique. Their role is active and creative. A living archive interprets, adapts and transmits inherited knowledge, imbuing it with its own experience and understanding of the world. Each of his performances, whether it is a song, a story or a ritual formula, is a unique and unrepeatable act of creation, and not a mere reproduction of an existing template (Lord, 2000).

LLM – Large Language Model — An artificial intelligence system trained on huge text corpora that predicts the next token (word or part of a word) and generates or completes the text based on it; examples include GPT, Claude, Gemini, Llama.

LoRA — Low-Rank Adaptation – PEFT method that approximates the change of model weights (ΔW) by the product of two small low-rank matrices; only those matrices are trained, while the original parameters are not changed; achieves performance close to full fine-tuning with significantly fewer parameters (Hu et al., 2021).

Mental Map — An internal, cognitive representation of the world (concepts, relationships, experiences) that an individual builds and updates through language and experience; serves to interpret situations, make decisions and plan; it is not static, but constantly adapting.

Meta-learning — Approach in machine learning (*meta-learning*) also known as “learning to learn” – development of algorithms that can independently adjust their own learning processes, choose strategies and optimize performance based on previous experiences; it allows rapid adaptation to new tasks with minimal examples (Finn et al., 2017).

MLOps — Machine Learning Operations – a set of practices and tools for managing the life cycle of models from development to production; includes model and data versioning, automated training, deployment and performance monitoring; created by adapting DevOps principles to the domain of machine learning.

Moore’s Law — Empirical rule (Moore, 1965) according to which the number of transistors on an integrated circuit approximately doubles every two years; for decades it predicted the exponential growth of computing power and was the main driver of the IT industry.

Multi-agent system — A system composed of several autonomous agents that communicate and cooperate; as a whole it can exhibit emergent behavior – abilities and patterns not present in individual agents – analogous to swarms or social organizations in nature.

Multichannel communication — Strategy of the simultaneous presence of an organization or agent on multiple communication platforms (telephone, e-mail, social networks, chat, mobile applications) in order to enable users to freely choose the channel that suits them best; a higher degree of integration, known as an omnichannel approach, aims for a unique and seamless user experience where the boundaries between channels are almost imperceptible.

Multimodality — Simultaneous use and interweaving of multiple communication channels – speech, gesture, mimicry, intonation, rhythm, music and spatial relationship – in order to convey meaning; in the oral tradition it means the inseparability of the verbal and physical dimensions of performance, while in modern digital communication it includes text, image, sound and video as integrated components of the message.

Oral transmission — Way of transmitting knowledge, narrative and culture through speech and memory, without writing; knowledge lives in carriers (living archives) and changes in every performance; basic communication technology before the advent of writing. Cultures based on orality have a fundamentally different attitude towards the past than those based on writing; in them, the past is not a fixed and unchanging whole, but a fluid and adaptable reality that is reaffirmed in every act of storytelling (Ong, 1982). This reality is not preserved in dead letters, but in living people.

Outcome (Observation 3) — Memory returns a record from the previous interaction: {"record": "User Ana rated the visit to the Musée d'Orsay very positively.", "date": "2024-03-12"}.

Outcome (Observation 4) — The system searches verified sources and retrieves summary information about the *Marmottan Monet* (largest collection of Monet works) and *Orangerie* (famous for Water Lilies) museums, including opening hours, ticket prices, and location.

Output (observation 1) — The tool returns a list of available flights. The cheapest option is €150 (return ticket).

Output (observation 2) — The tool returns a list of available accommodation units. A few meet the criteria.

Parameters — { destination: "Paris, France", departure_date: "2025-10-03", return_date: "2025-10-05", class: "economy" }

Parasocial interaction — A one-sided relationship in which the viewer or listener develops a sense of closeness, friendship and intimacy with a media figure (or artificial agent), despite the fact that the figure is unaware of his existence; in the context of LLM, the user puts emotional effort into the interaction, while the model provides a simulation of reciprocity, which can lead to the illusion of two-way communication (Horton & Wohl, 1956).

PEFT — Parameter-Efficient Fine-Tuning* – a set of techniques (adapters, LoRA, prompt tuning) that adjust the pre-trained model by updating only a small part of the parameters; it reduces memory and computing costs and enables more specializations on the same basic model.

Personalization — Adjusting the content, tone and behavior of the system to individual needs, interaction history and user preferences; key to engagement and perceived usefulness of communication agents.

Pragmatic function of oral tradition — The role of oral traditions in the direct regulation of social life - from the legal legitimation of government and property relations (through genealogies and myths of origin) to the transfer of practical knowledge and skills necessary for daily survival; tradition acts as an unwritten law and operational manual of the community (Finnegan, 1977; Malinowski, 1926).

Pre-training — The training phase of the language model on huge, unstructured text corpora without human annotations; the model learns language patterns and general knowledge by solving self-supervised tasks (e.g. predicting the next word). The result is a foundation model that is then fine-tuned for specific tasks.

Pruning — Removal of less important parameters or entire structures (neurons, layers, attention heads) from the model to reduce size and speed up inference; it can be unstructured (single weights) or structured (whole units); it is often combined with additional fine-tuning (LeCun et al., 1990; Han et al., 2015).

Put together a proposal — Create a clear and coherent itinerary.

Quantization — Reduction of numerical precision of parameters and model activation (e.g. from 32 or 16 bits to 8 or 4 bits); reduces memory footprint and can speed up inference; applications include PTQ (post-training) and QAT (during training) (Dettmers et al., 2022).

RAG — Retrieval-Augmented Generation – a hybrid system that first retrieves relevant parts from an external knowledge base (retriever) to answer a user's query, then passes that data and the query to a language model (generator) that synthesizes the final answer; this improves the timeliness and validity of the answers.

Redistribution of power — A change in the distribution of influence, resources and authority among actors within the social structure, which occurs when a new technology, practice or idea changes the previous channels of access to knowledge, communication or material means; in the context of communication innovations (letters, press, internet) the redistribution of power means the process by which previously excluded groups gain the ability to act, while the previous elites lose their monopoly over information and interpretation of reality (Castells, 2009; Eisenstein, 1979).

Resource intensity — Demanding development and application of AI models in terms of computing power, memory, energy and financial costs; especially pronounced for large language models whose training and inference require massive GPU/TPU clusters, which limits availability and has environmental consequences.

Reward model — Auxiliary model trained on human ratings of language model responses; predicts how a human rater would rate an answer; used in RLHF to direct the language model according to preferences (usefulness, truthfulness, security).

RLHF — Reinforcement Learning from Human Feedback – an alignment methodology in which human evaluators rank the model's responses, after which the reward model is trained and the model is further tuned to maximize the expected reward; key to aligning the behavior of LLMs with human preferences (usefulness, truthfulness, security).

Scaling laws — An empirical insight according to which the performance of language models increases predictably with the increase of three factors: the number of model parameters, the amount of learning data and the amount of computing power; it encouraged a race to build increasingly large models (Kaplan et al., 2020).

Scheme 1 — Sample prompt with several examples (*Few-shot Prompting*)

Self-supervised learning / self-supervised learning — A learning method in which the model does not use humanly marked data (input-output pairs), but performs the task itself from the data structure: for example, it predicts a hidden part of the input (next word, masked token) or another

derived goal. The learning signal comes from the data itself. This procedure is crucial for pretraining large language models based on huge amounts of corpus data for which manual annotation would be difficult to implement.

Semantic triangle / triangle of meaning — Model (Ogden & Richards, 1923) that shows meaning as a relationship of three elements: symbol (linguistic expression), thought/concept (mental representation) and referent (object in the world); the connection between symbol and referent is indirect - it leads through thought; useful for understanding the limitations of AI in “anchoring” meaning.

Sentence — “The service was outstanding and the food was delicious.” **Feeling:** Positive

Sentiment analysis — Automatic analysis of text (transcripts, e-mail, social networks) in order to recognize the emotional tone, attitudes or mood - e.g. frustration, satisfaction, negativity; it is used in customer support to prioritize cases, in marketing to monitor reactions and in public opinion surveys.

Sequence of thought (internal decomposition) — Determine the correct dates: Determine which days are the “first weekend in October”.

Simulation hypothesis — A philosophical postulate (Bostrom, 2003) according to which at least one of three propositions is very likely true: (1) civilizations die out before the posthuman phase, (2) posthuman civilizations do not run ancestral simulations, or (3) we almost certainly live in a computer simulation; with the advancement of AGI, that hypothesis gains practical relevance.

Social construction of reality — Theory (Berger & Luckmann, 1966) according to which social reality is not an objective given, but an intersubjective creation created by common meanings, conventions and narratives; language is the key mechanism that enables the transition from subjective experience to objectified, shared reality.

Social criticism in the oral tradition — The function of the oral tradition that allows the community to question, comment on and challenge the existing power relations and social norms through indirect narrative forms – satire, allegory, parody, anecdotes about the powerful and animal fables; since criticism comes in an aesthetically shaped and collectively sanctioned form, it bypasses open conflict while simultaneously opening up space for reconsideration and gradual change of order (Scott, 1990; Finnegan, 1977).

Social stratification — Hierarchical stratification of society into levels (layers, classes, estates) that differ in access to material resources, education, power and prestige; in the context of communication technologies, stratification determines who first adopts an innovation, who has access to information and who acts as a gatekeeper that accelerates or slows down the spread of new ideas and practices to other parts of society (Weber, 1922; Bourdieu, 1984; Rogers, 2003).

Socialization — The process by which an individual, in interaction with other members of the community, adopts the language, values, norms, knowledge and patterns of behavior necessary for functioning within a certain culture; in oral societies, this process takes place predominantly through narrative and performance forms (stories, fairy tales, riddles, songs) that simultaneously teach, entertain and shape personal and social identity (Berger and Luckmann, 1966).

Symbol — Linguistic or other sign whose meaning does not derive from physical similarity or causal connection with the object, but from social agreement (convention); allows speaking about the absent and abstract, in contrast to the indexical sign that refers to the immediately present.

Synthesize the answer — Form all the collected data into a final, personalized answer for the user.

System prompt / system prompt — A superordinate instruction placed at the beginning of the interaction with the language model that determines its general behavior, personality, powers and limitations throughout the

conversation; it acts as an agent’s “constitution,” a set of immutable rules that ensure predictability, certainty, and consistency in responses (Anthropic, 2024; OpenAI, 2024b).

Task Decomposition — The process of breaking down a complex goal into a series of smaller, logically connected subtasks that can be executed sequentially or in parallel; in the context of autonomous agents, the decomposition is carried out by the LLM core acting as a strategist, turning an abstract request into a causal chain of concrete actions (Yao et al., 2023).

Theory of mind — Cognitive ability to attribute mental states – beliefs, desires, intentions and knowledge – to others and to oneself; key to understanding the speaker’s intention in communication and predicting other people’s behavior. Without theory of mind, complex symbolic communication would not be possible because the recipient of the message would not be able to deduce what the speaker wants to achieve with it.

Thought — “I found a flight and accommodation within the given budget. I know from memory that the user appreciates the Musée d’Orsay, and through research I found two other related museums. I can put together a complete proposal that ties all the elements together.”

Token — The smallest unit of text processed by the language model; can match a whole word, a part of a word (subword) or a punctuation mark. Tokenization converts raw text into a string of tokens according to a model dictionary.

Tool Use — The AI agent’s ability to call external tools (APIs, search engines, calculators, databases) during task execution; the model decides when and which tool to call, passes parameters, and uses the result to continue reasoning; a key component of agent architectures (e.g. ReAct).

TPU — Tensor Processing Unit – a chip designed to accelerate machine learning operations, especially matrix calculations in deep networks; optimized for high throughput and energy efficiency in training and inference of large models.

Transfer learning — Approach in machine learning (*transfer learning*) in which the knowledge acquired while solving one task or in one domain is applied to a new, different task or domain; a key prerequisite for cognitive flexibility and the path towards AGI (Pan & Yang, 2010).

Transformer — A neural network architecture based on a (self-attention) mechanism that associates each word in a sequence with a contextualized image based on the other words; it enables parallel processing and scaling to large language models (Vaswani et al., 2017).

Unstructured data — Data that is not organized into a fixed scheme (tables, columns, tags); include free text, images, audio, video and mixed sources. Unlike structured data (e.g. databases), they do not have a predefined format, so it is necessary to convert them into a form suitable for computer processing. Pre-training for LLMs is based on vast amounts of unstructured text (web, books, articles).

USER ACCOUNT — “What will the weather be like tomorrow in London?”

Vector database / vector database — A specialized storage system that presents data as numerical vectors (embeddings) in a multidimensional space and enables quick search by semantic similarity, not by keywords; in the architecture of conversational agents, it serves as long-term memory that preserves user preferences, facts, and summaries of previous interactions across multiple sessions (Xu et al., 2023).

Vector embedding / embedding — Numerical representation of a text, word, sentence or document in the form of a sequence of numbers (vector) located in a multidimensional semantic space; semantically similar content has close vectors, allowing search by meaning instead of exact word match.

Virtual Assistant — A more complex digital interlocutor deeply integrated with the device and external services, capable of a wider range of tasks (calendar management, smart home, search, messages); the line towards chatbot is blurring with the advent of LLMs.

Working memory — Short-term storage area of the conversational agent in which the contextual data of the current session is stored - user's statements, agent's responses, recognized intentions and entities; it is functionally comparable to the working memory (RAM) of a computer because it has a limited capacity and is deleted when the interaction is over.

Benedikt Perak

CLOSING NOTE

This book ends here, but it stays open for what comes next.

The development of large language models and communication agents moves faster than any single edition. So this book remains an invitation to new readings, additions, debate, and future versions.

github.com/bperak/komunikacija_u_doba_ai